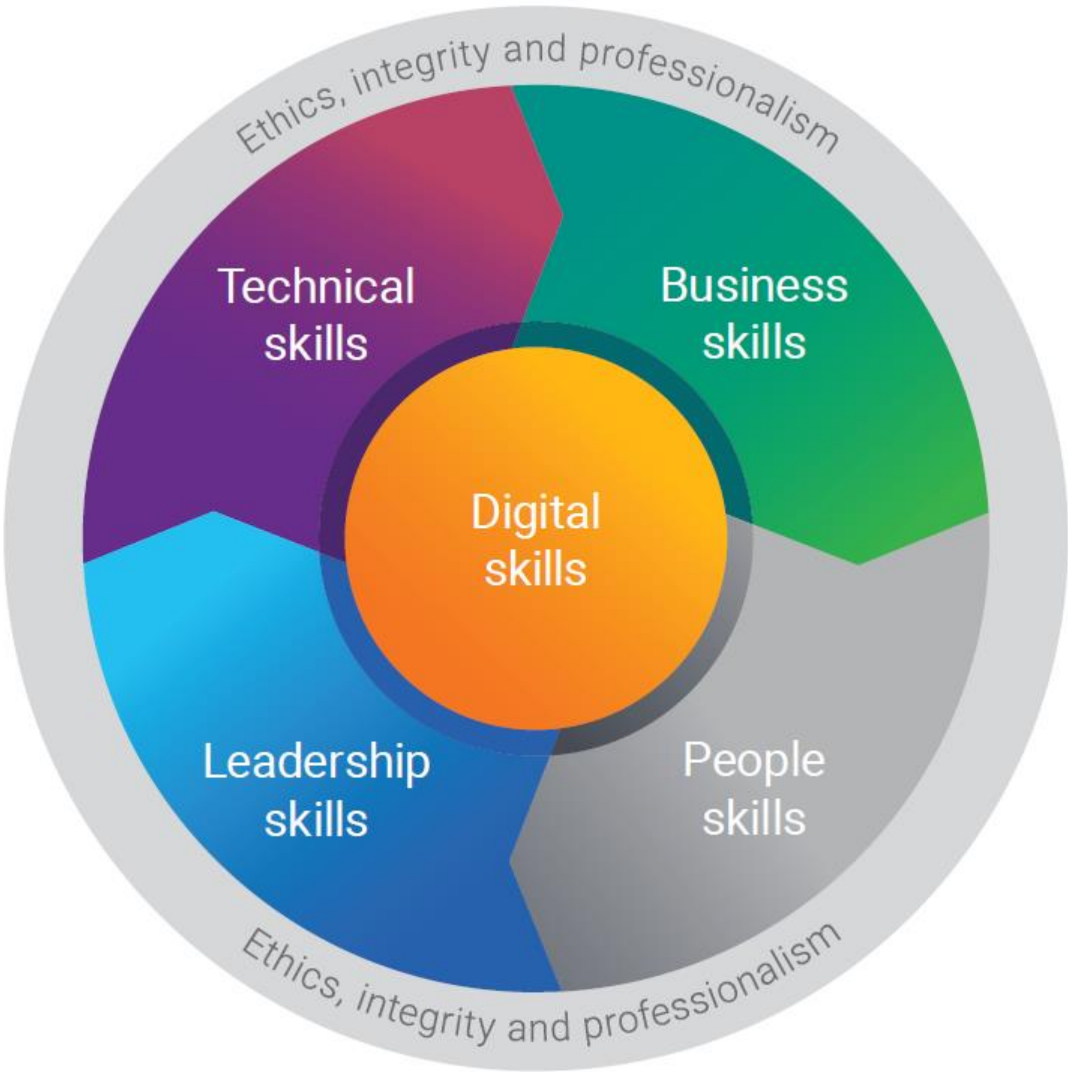
Strategic Case Study Examination

Skills Workbook

2022 Version

Towards the professional designation

CGMA™



About the Association of International Certified Professional Accountants

The Association of International Certified Professional Accountants (the Association) is the most influential body of professional accountants, combining the strengths of the American Institute of CPAs (**AICPA**) and the Chartered Institute of Management Accountants (**CIMA**) to power opportunity, trust and prosperity for people, businesses and economies worldwide. It represents 650,000 members and students in public and management accounting and advocates for the public interest and business sustainability on current and emerging issues. With broad reach, rigor and resources, the Association advances the reputation, employability and quality of CPAs, CGMAs and accounting and finance professionals globally.

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Chapter 1
The exam and sustainability/ESG accounting

1. The Strategic Case Study Assessment Blueprint/Syllabus

Your assessment will be based on a pre-seen case study of about 30 pages. This pre-seen case study describes and industry and a firm within that industry. The industry is real but the firm is fictitious. The assessment will assume that you work for the firm in the industry and that you have read the pre-seen and understand its business model and its business ecosystem. The SCS assessment is primarily based on the final three examination syllabi of the CGMA/CIMA curriculum. The 'assessment blueprint' for the SCS assessment shows the competences you will be examined on.

Core Activity	Assessment outcome 'I can….	Weighting
Develop business strategy	• Evaluate strategic options (digital and otherwise) • Recommend strategic decisions (digital and otherwise) • Evaluate potential acquisitions and divestment opportunities • Recommend responses to opportunities and threats arising from digital technologies	15-25%
Evaluate business ecosystem and business environment	• Select and apply suitable strategic analytical tools • Conduct an analysis of stakeholder needs and recommend appropriate responses • Recommend appropriate responses to changes in the business ecosystem • Recommend KPIs that encourage sound strategic management • Recommend responses to economic, political and currency risks	15-25%

Core Activity	Assessment outcome 'I can….	Weighting
Recommend financing strategies	• Recommend suitable sources of finance • Recommend dividend policy • Recommend and apply business valuation models	15-25%
Evaluate and mitigate risk	• Evaluate risks and recommend responses and can maintain the corporate risk register • Identify ethical dilemmas and recommend suitable responses • Evaluate and mitigate cyber risks • Recommend internal controls	15-25%
Recommend and maintain a sound control environment	• Apply internal audit resources • Recommend appropriate controls and evaluate the implications of compliance failures • Recommend responses to the threats arising from poor governance	15-25%

However, this blueprint is deceptively brief. The examiner has stated that the assessment will include questions on the underpinning theories and learning behind these Assessment Outcomes.

Using this Skills Workbook

The purpose of this Skills Workbook is to provide you with summaries of that underpinning knowledge and Activities that give examples of the sorts of tasks you may be set for each area of knowledge. All Activities have solutions at the end of the Chapter in which they are set.

Passing this exam requires you to provide sensible analysis and workable suggestions to business problems. It is not a test of how much theory you have memorized. Each Chapter in this Skills Workbook summarises the technical content of the CIMA/CGMA syllabus on each topic. But it is the Activities that you need to focus on and to see the sorts of questions on each topic and how much, or little, of the theoretical content from the Chapter can be used to answer them. **Reviewing the activities in each chapter is essential to understanding how to pass this examination.**

The Skills Workbook is also a reference book that will be used on your program to support the lectures.

This Skills Workbook is written to be worked through from the beginning to the end. This means that Activities in later chapters will include learning from earlier chapters.

2. Business Models and Business Ecosystems

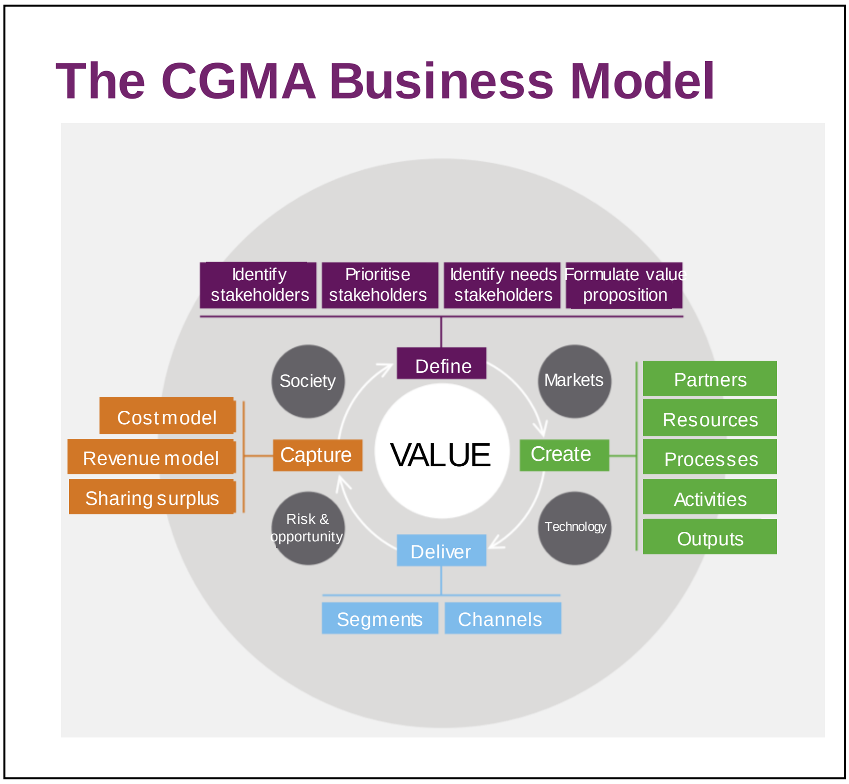
CGMA defines a business model as 'show[ing] how an organization defines, creates, delivers and captures value for, with, and to its key stakeholders in a consistent and coherent manner'

What is meant by value?

Value means 'worth to people'. But CGMA goes beyond the simple notion that value = profit.

- Value = benefit to the customer/user. Without creating this the firm can't expect many sales whilst a not-for-profit organization exists solely to give value to their clients, members, or sponsors
- Value = benefit to society. If an organization gives no benefit to society, or actually causes harm to society, it will eventually be outlawed and forced to cease its business
- Value = benefit to employees. Giving pay, training. Security, personal development and so on are essential to retaining the key resources that manage the business and provide its services and goods

- Value = financial returns to investors. This is nearer the conventional accounting view of value and it remains important in the CGMA model. Without this the business will not gain capital and the management responsible for the business will be replaced.



The four activities of value creation
These are core of the Business Model

1. **Define value**
   This means to establish what value the organization will seek to create. This means to identify the stakeholders whom the organization impacts on, identify which are the more important and less important, and formulate a value proposition for them according to their needs and requirements. As simple examples:
   - investors: decide the growth, future profits and risks of the business and the balance of dividend versus capital growth
   - customers: decide the target customers and the sorts of benefits they will get from the products and services of the firm
   - society: identify near neighbors and interested groups and establish a Corporate Social Responsibility policy to benefit them

2. **Create value**
   This means to use the resources, supply-chain networks, processes and activities of the organization to deliver value. The value created by the organization relies on the *outputs* it generates, but there are also *outcomes* such as wealthier investors, happier customers, or better society. These include the following examples
   - The products and services the firm makes
   - Environmental emissions such as waste, carbon gases, polluted water
   - Jobs
   - Research findings

3. **Deliver value**

This determines how the value is received. The model refers to segments and channels

- Customer segments and routes to market (e-commerce, agents, own stores and so on)
- Beneficiaries of charitable donations and how distributed
- Socialization of research findings

4. Capture value

The management arrangements for ensuring that the business maximizes the financial value it creates. The common management accounting techniques of revenue and cost management are indicated. But also, the notion of how the captured surplus (excess of revenue over cost) will be shares between investors, staff and the rest of society. This could include

- Dividend policy for shareholders
- Capital structure and the amount of surplus being distributed as interest payments
- Payment and bonus systems for management and staff
- External distribution such as charitable giving, research funding, pensions and so on

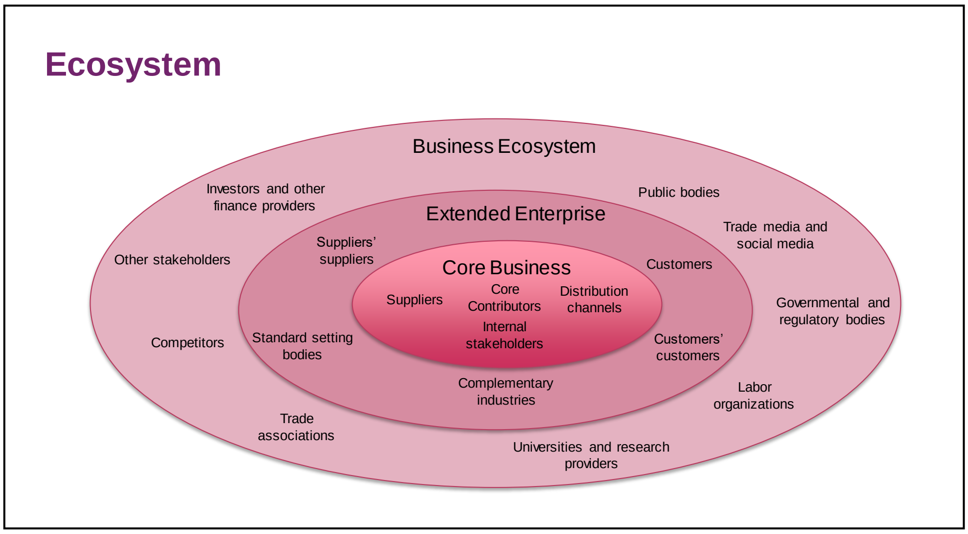
The business ecosystem

A key point about value creation is that it needs to be *sustainable.* That means the organization needs to create value in a way which can continue into the long-term, and also be able to overcome unexpected bad times. Hence the term *sustainable business model*.

The SCS syllabus assessment blueprint refers to the ‘business ecosystem’. This CGMA business model shows this as society, markets, technology, and risk and opportunity. The term is somewhat imprecise and a broader ecosystem can be pictured.

The business must achieve and maintain its ‘fit’ within this organizational ecosystem if it is to sustain.

Simply put, CGMA is employing an *organic metaphor* for business of the sort an environmental scientist might use to understand the survival, or extinction, of a species of plant or animal through time. The organism, such as a species of animal, lives in a jungle with food and water which sustain it. There are other animals of its species, and animals of other species, with which it co-exists. There are also predators. This organism affects its ecosystem by how it behaves and also it may adapt so that it can survive better. CGMA says that it is helpful to think of business in the same way.



Sustainability/ ESG Accounting
The focus on sustainability came from concerns over the effects of business on the natural environment which culminated in the release of Sustainability Accounting Guidelines by the 2002 Earth Summit. There has been a growth businesses reporting aspects of their sustainability in various ways, such as in Corporate Social Responsibility reports.

The scope of social responsibility has widened beyond the green environment to embrace other aspects of the ecosystem such as the pay and treatment of labor, social and health impacts of the product and so on and also the accountability of the business. These are sometimes referred to as ESG methods of accounting (Environmental, Social, and Governance).

Attempts to rationalize and systematize these accounts has led to the development of a number of alternative frameworks for sustainability accounting including:

- International Integrated Reporting Initiative Council (IIRIC)
- Global Reporting Initiative (GRI)
- Sustainability Accounting Standards Board (SSAB)

You are required to have an understating of IIRC and GRI.

3. The Integrated Reporting Framework

This was examined in November 2020. CGMA has been a major exponent of corporations adopting Integrated Reporting (<IR>) and senior figures in CGMA and AICPA hold senior positions in the International Integrated Reporting Council (IIRC) that oversees the development and implementation of the principles and standards of <IR> globally.

More details of IIRC can be found here

Integrated Reporting <IR>

'New approach to corporate reporting founded on integrated thinking, which helps demonstrate interconnectivity of strategy, strategic objectives, performance, risk and incentives and helps to identify sources of value creation'

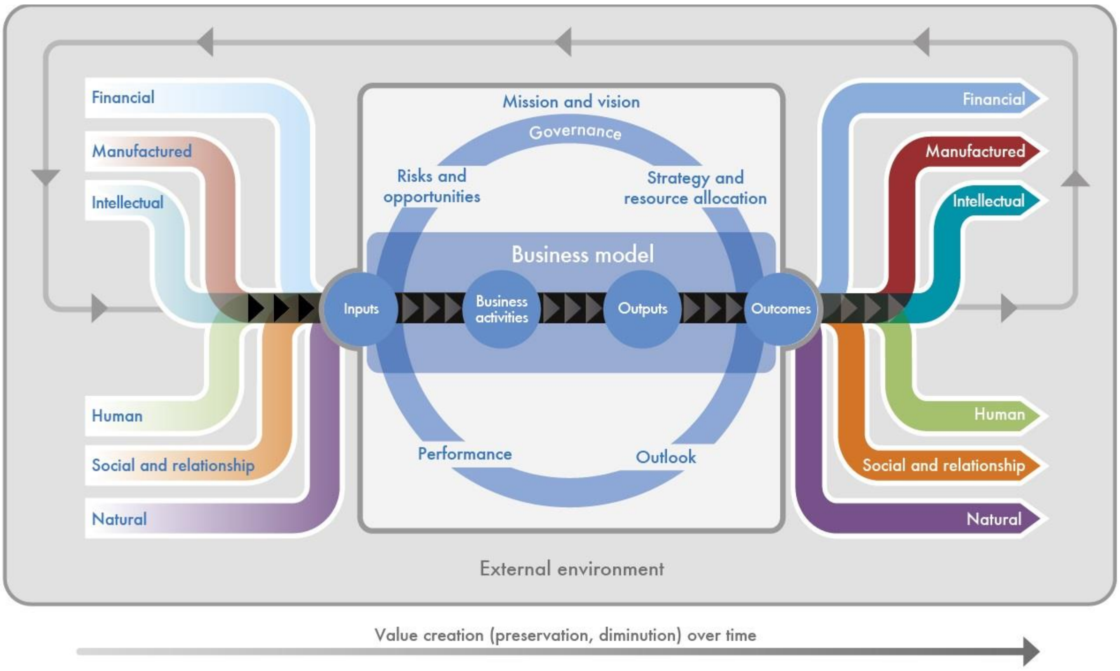
Focused on *sustainability* of the business

Key purposes of <IR>

1. Creation of value
   - Encourage better understanding of business environment (opportunities and risks)
   - Focus on the longer-term
   - Demonstrates management understands interconnections between environmental factors and the business
   - Greater transparency in its corporate governance
2. Telling the story to win trust and secure reputation
   - for investors
   - for employees
   - for other stakeholders

https://integratedreporting.org/
The connection between <IR> and the CGMA business model is clear in the following diagram.

<IR> is a layer of corporate governance that compels businesses to report on their business model, their relations with stakeholders, and the factors impacting on that model and the trust the business enjoys.



The Value Creation Process (IIRC)

The <IR> framework is based on a concept of the business model, and it adopts an *inventory approach* to measuring sustainability. This means that it seeks to base sustainability reporting on the impact of the business's activities on the stock if the *six capitals* in its framework. There have been several assessment tasks on these six capitals.

The Six Capitals of <IR>

Instead of using the term 'value' the <IR> framework put the emphasis on capital creation and destruction.

In conventional accounting, a firm that uses its fixed capital of machines to produce a year's worth of product needs to deduct an amount for the wear and tear on the machine from the profits on the sale of the product. This shows that the value of the business is increased or reduced by the impact on its capital. A firm that makes its profits by exhausting its machinery and not replacing it will not be sustainable.

Integrated Reporting <IR>

- **The six capitals of <IR>**

These are a measure of sustainable value

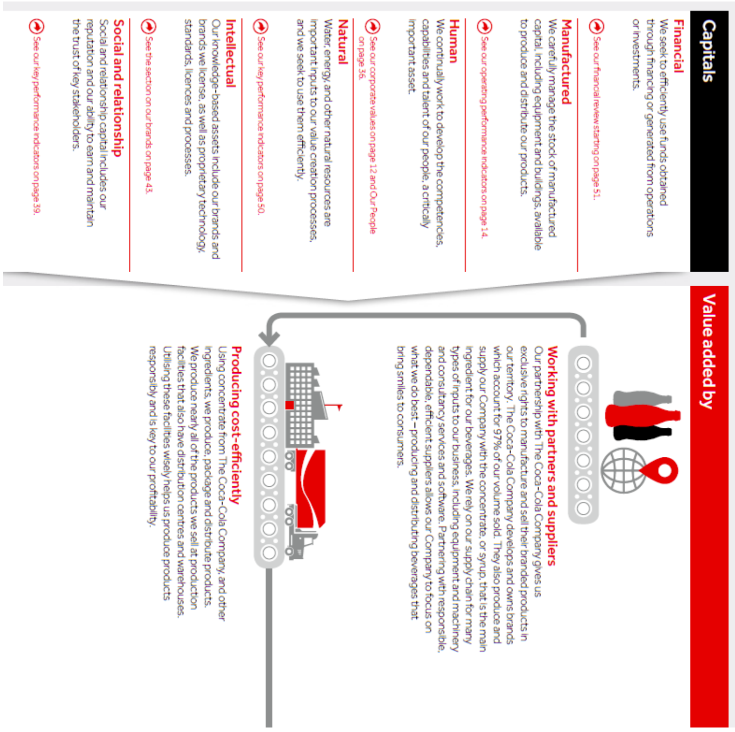
1. **Financial**
   - Funds available to firm from operations and financing
2. **Manufactured**
   - Manufactured physical objects used in value creation (whether owned by the firm or not)
3. **Human**
   - Skills, motivation, alignment with organisational goals
4. **Intellectual**
   - Patents, copyrights, brands, systems, procedures and know-how
5. **Social and relationship**
   - Relations with key institutions, stakeholder groups, shared norms and values, trust and confidence, and its social license to operate
6. **Natural**
   - Renewable and non-renewable natural elements, and the eco-system, used as inputs by the firm now or in the past or future, and impact of firm on them

<IR> generalizes this analysis of capital maintenance to include a broader range of capitals.

Over time if the business improves the values of a capital is can be thought of as increasing value. If it reduces their value, it is destroying value.

Example: Coca Cola Company
The following is an extract from the 2015 Integrated Report of the Coca Cola Hellenic Bottling Company

It depicts the 6 capitals and the value creating and sharing activities of the business



These capitals are not all measured in financial terms. The Integrated Report contains statement of how the firm complies with the principles and of its impact on the capitals during the year. This integrated presentation is regarded as better than stating financial, risk, and Corporate Social Responsibility in separate reports. The actual <IR> reporting principles and content are as follows

Integrated Reporting <IR>

- **Seven principles of <IR>**

1. **Strategic focus and future orientation**
   - How the strategy will deliver value
   - Impact of the strategy on the 6 capitals
2. **Connectivity of information**
   - Show interdependencies and relations between the things that firm uses to create value
3. **Stakeholder relationships**
   - How well it understands the needs of the stakeholders it depends on
   - Nature and quality of its relationship with the stakeholders
4. **Materiality**
   - Report should disclose information about matters that will materially affect its ability to create value in the future
5. **Conciseness**
6. **Reliability and completeness**
   - Contains all material matters in one place
   - An error-free and balanced account
7. **Consistency and comparability**
   - Over time
   - With other firms eg by using the International Integrated Reporting Framework

Integrated Reporting <IR>

- **Eight content elements of <IR>**

1. **Organisational overview and external environment**
   - What it does and how it does it
   - Circumstances under which it operates
2. **Governance**
   - How its governance structure supports its ability to deliver value in short, medium and long-term
3. **Business model**
4. **Risks and opportunities**
   - What can affect its ability to make value
   - How it is dealing with them
5. **Strategy and resource allocation**
   - Where it wants to go
   - How it intends to get there
6. **Performance**
   - How well it has achieved its goals
   - Impact on the 6 capitals
7. **Outlook**
   - Challenges and uncertainties and their implications for the business model and future performance
8. **Basis of preparation and presentation**
   - How it decides what to include
   - How it evaluates them

Past questions on <IR>

- Presentation in the <IR> of a leisure park that releases its performing dolphins after public outcry against the cruelty of keeping them in captivity (narrative explanation of strategic reasons for release and likely impact on future earnings, detailed account of how they were released and the costs if this, statement that it reduces Natural Capital but increases Social and Relationship capital)

- Impact on environmental reporting (natural capital) of decision by leisure park to build hotels in its parks rather than leaving visitors to rely on external hotels (transparency requires inclusion in report, most environmental impact such as water use, emissions and so on are merely transferred from existing hotels, could refer to reduced pollution due to guests no longer commuting to and from the park to their hotel each day, give reassurance/KPIs that hotels are environmentally efficient, report despoiling of green land in park to provide hotel and roads, emissions during construction phase)

- Value of reality television show publishing an environmental report (natural capital) for a new show format on an unspoiled island as a way to reduce short term threats (threats of disruption from environmentalists, advertisers withdrawing support, reputation damage: benefits are it shows firm cares, shows low damage, consistent with transparency. Drawbacks are it may confirm damage, will be disputed, and it's an emotive area where perceptions count more than facts)

- Impact on the reporting of Social and Relationship Capital by a science and technology park from its establishing a competition for new innovation proposals, leading to award of loans and premises for the winners

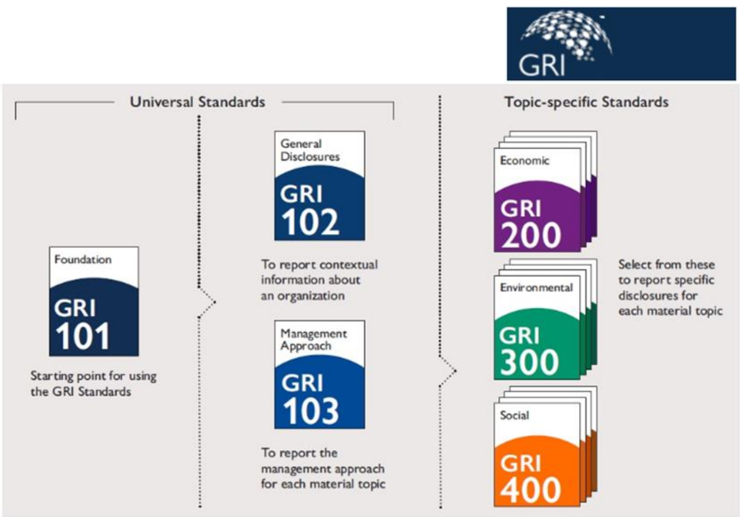
4. Global Reporting Initiative (GRI)

Key Point – examination questions

The GRI has also been the subject of examination questions. These have included requirements to suggest key performance indicators for GRI, and commenting on the principles of GRI

The GRI is another independent standard setting body, set up by the United Nations (UN) that aims to improve the reporting of the *sustainability* of the firm's business model and to help third parties to assess the environmental impact from a business and its supply chain.
The GRI requires firms to explain their performance in terms of its impact on the 'triple bottom line' of the business: Profit, Planet, and People

Global Reporting Initiative (GRI)



The GRI Standards (Version 4.0 released 2016) are arranged as

- Principles for report content: what the report should include
- Principles for report quality: how well things should be presented
- Reporting guidelines: defining general disclosures and specific categories and aspects related to the three Standards (Environmental, Economic and Social impacts)

GRI Principles for Defining Report Content

Stakeholder inclusiveness
- The organization should identify its stakeholders and explain how it has responded to their reasonable expectations and interests

Sustainability context
- The report should present the organizations performance in the wider context of sustainability

Materiality
- The report should cover Aspects that
  - Reflect the organization's significant environmental, economic and social impacts
  - Substantively influence the assessments and decisions of stakeholders

Completeness
- The report should include coverage of material Aspects and their Boundaries sufficient to reflect significant environmental, economic and social impacts, and to enable stakeholders to assess the organization's performance in the reporting period

The GRI has been criticized for its use of the Materiality criteria. Its critics complain that this allows management to ignore some important factors affecting its business model, or some significant impacts of its business model, by giving them latitude to claim that 'nobody really cares about these things and would not change their behavior towards the firm if we did report it (e.g. wouldn't cease to invest, work for, or buy from the firm)'

GRI Principles for Defining Report Quality

Balance
- The report should reflect positive and negative aspects of the organization's performance to enable a reasoned assessment of overall performance

Comparability
- The organisation should select, compile and report information consistently (to allow stakeholders to see changes and trends through time)

Accuracy
- Should be sufficiently accurate and detailed for stakeholders to assess the organization's performance

Timeliness
- Reported on a regular schedule so that information is available in time for stakeholders to make informed decisions

Clarity
- Information should be available in a manner that is understandable and accessible to stakeholders using the report

Reliability
- Organization should gather, report, compile, analyze, and disclose information and processes …in a way that can be subject to examination and that establishes the quality and materiality of the information
- Stakeholders should have confidence that a report can be checked to establish the veracity of its content

Prior to 2016 the GRI Guidelines focused mainly of encouraging firms to report metrics (performance measures) for environmental emissions. But since the release of the GRI 4.0 Standards in 2016 it has encouraged the firm to make a wider range of disclosures

GRI General Standard Disclosures

TABLE 3: REQUIRED GENERAL STANDARD DISCLOSURES		
General Standard Disclosures	'In accordance' – Core (This information should be disclosed in all cases)	'In accordance' – Comprehensive (This information should be disclosed in all cases)
Strategy and Analysis	G4-1	G4-1, G4-2
Organizational Profile	G4-3 to G4-16	G4-3 to G4-16
Identified Material Aspects and Boundaries	G4-17 to G4-23	G4-17 to G4-23
Stakeholder Engagement	G4-24 to G4-27	G4-24 to G4-27
Report Profile	G4-28 to G4-33	G4-28 to G4-33
Governance	G4-34	G4-34 G4-35 to G4-55(*)
Ethics and Integrity	G4-56	G4-56 G4-57 to G4-58(*)
General Standard Disclosures for Sectors	*Required, if available for the organization's sector(*)*	*Required, if available for the organization's sector(*)*

You will recognize the close similarity between the concepts behind the Standard Disclosures in the GRI and the Eight Content Elements of <IR>

But GRI does not refer to the Six Capitals. Instead it requires firms to report their impacts on the Three Categories of Economic, Environmental, and Social. It further subdivides each of these Categories into component Aspects and requires firms to report under each

Here is an example from the Coca Cola Hellenic Bottling Company referred to earlier which reports under both the <IR> and GRI frameworks. Note that this half a page from a 35-page report

This section records how the firm deals with economic threats from its ecosystem

Indirect economic impacts			
103-1 103-2 103-3	Explanation of the material topic and its Boundary Explanation of management approach and its components Evaluation of the management approach	14, 24-54.	Refer to the "Business model", "Our communities", "Our customers" and "Partners in Efficiency" sections of the 2018 IAR.
203-1	Development and impact (current or expected) on local communities and economies of infrastructure investments and services supported	14-15 34-39	Refer to the "Our business model" and "Our communities" sections of the 2018 IAR. In 2018, we invested €7.9 million in our communities, which is an increase of 7% compared to 2017. The biggest proportion of this investment was allocated to our flagship #YouthEmpowered programme, engaging more than 64,000 young participants, joined by more than 750 of our people as mentors, and partnering with more than 380 local non-governmental organisations. In 2018, our employees volunteered more than 34,000 hours in both free time and work time in support of strategic community programmes with 21% taking part in volunteer initiatives.
203-2	Understanding and describing significant indirect economic impacts, including the extent of impacts	34-39, 14, 15, 83	As a business, our primary contribution to the communities we are part of is through our core business activities, which generate income for employees, pay our suppliers and contribute to the public good through e.g. the payment of state taxes. In 2018, a total of €328 million in tax-related liabilities was recorded in our IFRS accounts. During the year, we continued to consolidate our infrastructure to improve efficiency across the organisation. These projects can have an impact on local economies, when they involve the closing or consolidation of facilities. Within the European Union, the Coca-Cola System supports over 500,000 direct and indirect jobs throughout our value chain and contributes approximately €27.4 billion in value added annually (2017, 2018 data).

Past questions on GRI

- Alternative energy consultancy adopting GRI reporting: what to include in report under headings of Stakeholder Inclusiveness and Sustainability Context and how the internal audit department might contribute to assuring Reliability and Balance.
- Extent to which a pharmaceutical firm's CSR reporting satisfies the GRI principles for content on stakeholder inclusiveness, sustainability context, materiality, and completeness

The breadth and complexity of the disclosures means that most firms that adopt GRI (e.g. Coca Cola, Johnson and Johnson) post their reports online with clickable links that allow the reader to drill down to find statements of compliance with each standard

https://www.jnj.com/caring/gri-reporting
http://www.coca-colacompany.com/stories/gri-index

ACTIVITY 1.1

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context
Runn is a major business in the Courier, Express and Parcel industry (CEP). It is very similar to firms like FedEx, DHL and UPS. Runn delivers over 9 million packages a day to over 210 countries using 1,600 depots, 400,000 employees, 104,000 vehicles and 208 cargo aircraft.

Task
Runn's CFO has requested the following:

- What impact does Runn have on the 6 capitals referred to in Integrated Reporting and should Runn adopt Integrated Reporting?

  [sub task (a) = 55%]
- What four KPIs could Runn use to monitor its impacts on 'People' and 'Planet'?

  [sub task (b) = 45%]

ACTIVITY 1.2

Context
Pixlwizz develops and sells video games. It is in the same industry as PlayStation and Xbox. Like real world game developer Activision Blizzard it has suffered social media criticisms and leaks about it harassment of staff. One staff member told journalists 'Pixlwizz has a culture of fear. Managers bully staff and demand extra unpaid work so that games can be developed and launched quickly and cheaply. Our profits are huge and there is no need for this behaviour'

This has led to a series of 2 hour protests at its three international offices.

Task
Pixlwizz's CFO has asked for your opinion on the following:

- How will this staff unhappiness affect the six capitals in our 202x integrated report?

[sub task = 34%]

Solutions to Activities

1.1 Integrated Reporting at Runn Couriers
What impact does Runn have on the 6 capitals referred to in Integrated Reporting and should Runn adopt Integrated Reporting? (55%)
Approach: Note that this sub section has 55% of the total marks, so it requires about 33 minutes to answer it. BUT the task has two elements: impact on the 6 capitals AND whether Runn should adopt <IR> or not. Keep them separate in your answer!

Impact of Runn on the Six Capitals of <IR>
This impact is a measure of the 'value added' by Runn in a year and across a wide range of outcomes. It seeks to supplement the measurement of Runn's value creation solely in terms of financial profit. Value is created when a capital is increased or when, overall, the net value of Runn's capitals increase. Management attention should focus on maintaining and enhancing each of the capitals.

<u>Financial Capital</u>
This is the funds and sources of funds available to Runn. Runn is profitable and this can increase the financial capital by increasing equity and by making Runn an attractive investment proposition if it issues new shares or debt. Runn increased financial capital by G$1,218m of equity in 2019 (*data in pre-seen material G$7,994m – G$6,776m)* and G$958m of debt

<u>Manufactured Capital</u>
These are the produced assets that belong to Runn, such as planes, vans, trucks, depots and IT equipment. These increased G$2,691m in 2019 (*data in pre-seen material).* This shows capital maintenance by Runn which means it is keeping its assets adequate to meet its growing business and that it is replacing them as needed This is a good sign for the future capacity of Runn

<u>Intellectual Capital</u>
This is the knowledge and know-how of Runn. This might include patents over the development of IT systems, such as the rise in intangible assets of G$146m in 2019 (*data in pre-seen material).* Some of this will not be registered in the financial reports due to the conventions of financial reporting, such as improvements in brand equity through promotion. This would also include relations with clients and the knowledge developed from Runn's use of Big Data.

<u>Human Capital</u>
This refers to the staff of Runn, their morale, quality of training and the culture of Runn. Runn has approximately 400,000 employees globally in a variety of roles. It is expected that the quality of human capital will be increased by good recruiting and training and development

<u>Social and Relationship Capital</u>
This is the relationship of Runn with society and, through this, the maintenance of its 'license to operate' Runn provides a valuable service that enhances the convenience and choice of its customers through facilitating commerce and delivering parcels to households and firms at convenient times. It is likely that its services improve the quality of life of time-poor or isolated persons who do not have the time the visit stores and other vendors. However, Runn's social capital could be reduced by the unsightliness of its depots and hubs, and the impacts of its couriers on congestion and road safety.

<u>Natural Capital</u>

This is the impact of Runn on natural environment and natural resources. Runn's impact is strongly negative. The packaging it uses, particularly plastics, create emissions during production and pollution after disposal. The card and paper are produced at the expense of trees, which would otherwise act to capture carbon. The trucks and vans run on fossil fuels and emit carbon gases and other toxins which deplete natural resources and harm the natural environment, such as destroying clean air and promoting climate change. Runn may argue that some of this damage to natural capital is less than the alterative harm caused by firms and households conducting their own deliveries and collections

Adopting Integrated Reporting
Integrated Reporting is an approach to sustainability accounting developed by the International Integrated Reporting Council

Benefits to Runn of adopting <IR> include:
<u>Improved communication with stakeholders</u>
Runn will be able to tell more of its story to stakeholders under <IR> than it can at present using conventional financial reporting. Also, the information will all be in one document rather than spread out across separate documents (CSR, Strategic Insights, House Magazines and so forth) and sections of its web site. This should help build up the trust and commitment of its stakeholders and result in better employment relations, better customer retention and so on.

<u>Improved robustness of Runn's business model</u>
In order to report the elements of the Integrated Report it is necessary for Runn to ensure it conducts the underlying activities. Therefore, management will need to put in processes to assess its external environment, risks and opportunities. It will also need to ensure its corporate governance is robust, that its business model is properly understood, and that it has business strategies to deal with the factors it has identified

Runn does not seem to have identified its reliance on unsustainable elements of its business model, such as its environmental impacts and reliance on carbon energy and plastic packaging. Identifying and fixing these would help it be more sustainable and robust.

<u>Improved share price</u>
The combined effect of better communication, and a more robust business model should be to improve the share price of Runn. This is because the share price is affected by market forecasts of the present value of the free cash flows of Runn (the Efficient Markets Hypothesis).

The drawbacks of adopting <IR> include:
<u>Exposes weaknesses in Runn's business model</u>
If Runn reported honestly it would need to reveal the high carbon impact it has and its lack of arrangements to deal with the risks of this. This could harm its reputation and lead to a loss of business and a fall in incomes. This could harm its share price.

<u>Increases costs of management</u>
Implementing the additional systems described above, such as environmental analysis, stakeholder engagement, and risk management will take up management time and increase operating costs. Given that the trend to urgent deliveries is squeezing Runn's profit margins already these extra costs will be an unwelcome further drain on profits. These extra costs and lower profits would harm the share price of Runn.

<u>Increases the costs of reporting</u>
Integrated Reporting requires the preparation of a broader report, extra KPIs, and the employment of staff, or consultants, able to initiate <IR> and construct these reports. These

extra costs might not be recovered in improved earnings or reduced costs or risks. Therefore, they might reduce the share price.

Conclusion:
Runn should avoid adopting <IR> at the present time. There is work to do before IR could be adopted without exposing Runn to reputation risk. Management should consider implementing

What four KPIs could Runn use to monitor its impacts on 'People' and 'Planet'? (44%)
Approach: Key Performance Indicators (KPIs) will be covered in more detail in a later Chapter. They are introduced here to illustrate the use of the GRI.

KPIs for People and Planet
KPIs enable management to implement strategies and to monitor whether they have been carried out and are effective. Having KPIs for People and Planet will enable Runn to improve public perception of its Corporate Social Responsibility (CSR)

<u>Donations to charitable causes</u>
This could be measured in G$ per year

This is a 'People' KPI. These donations could include money to fund micro-enterprises, education, or community facilities in the countries where Runn operates. These donations will bring good but also may help develop industries in the countries that could become future clients. Runn should aim to increase this metric each year

<u>Hours of volunteering by Runn management and staff</u>
This could be measured by hours per year
This is a 'People' KPI. Volunteering in schemes such as education, improving peoples' environments, helping the needy and so on does direct social good. Also, Runn may find that closer association with the people in the countries it serves, and closer association between Runn staff from different roles and levels in Runn, will help to improve understanding and teamwork. Runn should aim to increase this metric each year.

<u>Carbon emissions</u>
This could be measured as tons per month
This is a 'Planet' KPI. Carbon emissions are a contributor to atmospheric pollution, the creation of micro-climates in cities, and to global warming and climate change. Reducing carbon emissions is consistent with Runn playing its part in helping governments meet their obligations under regulations such as the Paris Accord. It also can represent a reduction in energy costs for Runn because reducing carbon emissions may require shorter journeys for its fleet, or insulation of premises to conserve energy (heating or cooling). Runn should aim to reduce this metric over time.

<u>Reduction in non-recyclable packaging</u>
This could be measured in tons of non-recyclable packaging per month.
This is a 'Planet' KPI. Some packaging materials may be introduced by Runn, such as plastic envelopes, packaging tape, plastic wrap, strapping and so on. Runn could aim to reduce these, or to arrange for their eventual collection and recycling. Customers will also be the source of this non-recyclable packaging, such as the plastic air cushions and protectors inside boxes, plastic cartons and so on. Runn should work with customers to develop ways of reducing these to reduce the amount of environmental waste and contamination. Runn should aim to reduce this metric each month.

1.2 Impact of staff unhappiness at Pixlwizz on its Integrated Report

How will this staff unhappiness affect the six capitals in our 2021 integrated report? (34%)

Impact of staff unhappiness on integrated report

Integrated Reporting, <IR> is an approach to ESG reporting, Environmental, Social and Governance, that reports the value added by the business against 6 capitals.

<u>Reduction in human capital</u>
This reports skills, motivation and alignment with organisational goals.
The discontent suggest a deterioration in motivation and alignment and, potentially, loss of key staff and difficulties in recruitment may reduce the skill base of PW.

<u>Reduction in intellectual capital</u>
This is PW's stock of patents, copyrights, brands, systems, procedures and know-how.
The gaming business is fast-moving and this know-how is generated by its developers. De-motivated staff, and loss of developers, will reduce this capital over time if franchises fail to keep up with rivals, and the programming and gaming quality of PW games falls.

<u>Reduction in social and relationship capital</u>
This reports relations with key institutions, stakeholder groups, shared norms and values, trust and confidence, and its social license to operate

This may not be affected unless Pixlwizz become seen by the public as exploitative of its staff. This could lead to criticism, and for players and other stakeholders to move away from engagement with a toxic firm.

<u>Increase in financial capital</u>
This means funds available to firm from operations and financing.
Low pay and low staffing would reduce costs and so, at least in the short run this increase free cash flows and so improves the amount of retained earnings. One of the benefits of the holistic approach of IR is that its transparency enables stakeholders to see that this improvement may be offset by the reductions above and therefore that it threatens the sustainability of PW.

Chapter 2
Goals of the Organization

1. What are Business Goals?

The pre-seen material for the SCS Assessment will feature the organization that it will be your task to give advice to. It is vital to understand the objectives of the entity in order to give appropriate advice

Levels of business goals

There are several *levels of objectives*

- *Fundamental goals and objectives*: the people or groups for whose benefit the organization exists and the key things the organization will do to deliver value to them. For example, if it exists for shareholders its fundamental goals will be financial
- *Secondary or enabling objectives*: these are the steps on the road to reach the fundamental goals. For example, the sales target of a sales department is an objective for that department. But it is intended to play a part in achieving a financial objective.

Impact on the assessment

Management accounting techniques approach decisions from an economic perspective. This perspective assumes the goal to be maximizing profit or maximizing shareholder value.

But this is only an assumption brought across from the analysis of a business provided by economics. It seems to ignore that selfish economic behavior by management might disrupt the ecosystem on which the business depends by alienating key stakeholders. For example, consider the long-term impacts of a business model that raised profits by using cheap exploited labor in the supply chain, or the sustainability of a pharmaceutical firm that sought to increase profits by massive increase of the drugs essential to patients' life and welfare.

The SCS Assessment frequently features the following:

- Business organizations where management or owners are not solely interested in financial return
- Tasks where you need to balance the impact of the business decisions on the financial and non-financial goals of the business, one task featured a technology that could reduce violent assaults against staff and requested an evaluation of it. Candidates were criticized for undue focus on the financial costs and benefits at the expense of recognizing management's duty of care to its staff.
- Tasks involving business problems where you have to recognize the need for a negotiated decision because the economically optional course of action will not get sufficient support from amongst the decision makers
- Problems where you need to recommend or justify a course of action that gives long term benefits rather than maximizing short term financial benefits

2. Vision, Mission and Values

Vision Mission and Values

Vision
- A statement of what the organization hopes to achieve or become
- A description of a future world

Mission
- A statement of what the organization intends to do now to help realize its vision

Values
- A moral direction for the organization
- Explains its behaviour and how its staff are expected to behave

A survey of academic literature shows that these three terms are not used in a consistent way. In particular Vision and Mission seem to be used interchangeably.
Be prepared to adapt to the way the examiner presents them in the pre-seen material for your assessment.

Exam example
In the February 2020 pre-seen for a manufacturer of coffee pods (similar to Nespresso) the following appeared:

Exam example of Vision, Mission and Values

Vision
A cup of joy in every household

Mission
To succeed by meeting our customers' needs for excellent coffee and our shareholders' needs for a strong return on their investment

Values
1. Deliver the highest possible coffee experience to customers
2. Maintain high standards of ethics and professionalism in all our work
3. Improve the welfare of all our coffee growers
4. Treat our planet with respect

Roles of Vision, Mission, and Values
You need to be able to interpret and apply VMV in several ways:

Roles of Vision, Mission and Values

- Starting point for strategy
- Guide to business behavior
- Benchmark to assess strategic options against
- Device to gain support for the business
- Basis for KPIs

- Starting point for strategy
  Developing a strategy involves the firm progressing its mission in a way that is consistent with its values. Also, the VMV may be an expression of the goals of key stakeholders, in the coffee example the goals of customers and shareholders are specifically mentioned in the mission, and the expectations of staff, customers and society seem to be included in the Values.

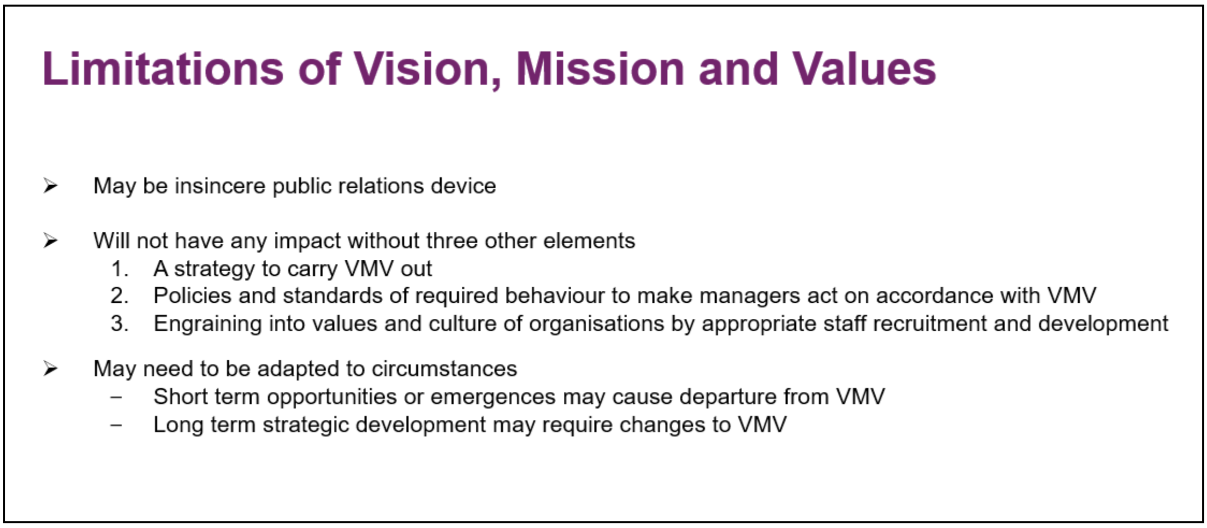
- Guide to business behavior
  The Values are in effect an ethical code for the business. A firm should have codes of conduct to support each and staff should be educated and trained in them. This can have the effect of creating a particular *organizational culture* of values, beliefs and behaviors that characterize the business and which may be hard to change

- Benchmark to assess strategic options against
  Faced with new strategic options the Board should refer to the VMV to answer the question 'is this really the business we are in?' This is termed the criterion of 'suitability' which will be returned to in a later chapter. Accepting lines of business that are not consistent with VMV risks failure if it conflicts with the expectations of key stakeholders, or the values that the firm has operated by in the past.

- Device to gain support for the business
  VMV are usually published by the organization. This means they have a publicity role, and are used to gain support from other members of the ecosystem. They can be attractive, inspirational, and trust-building. For this reason, there is a danger that organizations will write attractive-sounding VMV that are not backed up by the decisions or behavior of the organization. If this is the case the VMV are useless.

- Basis for KPIs
  Each of the commitments in the VMV should be the subject of KPIs otherwise they will not be implemented. You can expect questions that ask you to set KPIs for the business in the pre-seen material related to its VMV. KPIs are discussed more in a later Chapter.

Limitations of Vision, Mission, and Values
Several recent real exams have featured tasks requiring candidates to evaluate whether VFM is relevant consideration in taking a strategic decision or appropriate as a control device to make strategy happen.



ACTIVITY 2.1

Context
Runn is a major business in the Courier, Express and Parcel industry (CEP). It is very similar to firms like FedEx, DHL and UPS. Runn has the following VMV

Vision
'to be the World's favorite delivery company'
Mission
Link businesses and individuals in a responsible and sustainable manner
Values
- Runn cares deeply about its customers
- Runn cares deeply about its stakeholders
- Runn honors its commitments

Task
Runn's has been invited to tender to provide parcel and package delivery and collection services to the armed forces of Kayland, a friendly country. This will involve putting Runn staff in Kayland army camps to provide 24/7 service to the army staff there. The Runn staff will be provided with accommodation and food by the Kayland army.
Runn's CFO has requested your evaluation of the following
- To what extent is this contract consistent with our Vision, Mission, and Values?

[sub task (a) = 25%]

ACTIVITY 2.2

Context
Pixlwizz develops and sells video games. It is in the same industry as PlayStation and Xbox. Pixlwizz has the following VFM

Vision
Always to be the best game developer in the real and the Virtual Worlds

Mission
Attract the talented people who create virtual worlds that enable gamers to immerse themselves in exciting realities. Pixlwizz creates exciting places to work and to play.

Values
Imagination: creating places in which gamers wish to spend their free time

Innovation: set trends across the industry because we strive to surprise and excite gamers with every new release

Commitment: (to be) the place where talented people wish to work…creative staff passionate about developing new ideas and leading the way in creating new forms of gameplay

Teamwork: cooperation between all workers across all levels (through) talented creative people who can count on the support of colleagues who manage the space in which they put their talents to work

Task
Pixlwizz has fallen behind rivals in the development of new technologies, including VR/AR and 4D games.

The Managing Director of the Northland division, where such research takes place, has left Pixlwizz. In her resignation letter she makes some serious allegations against her treatment and Pixlwizz's policies towards the Northland division, including:

'<u>Lack of funding:</u>
The Northland division relies on income from the sale of games to fund our costs. The revenue we receive from sales in W$ has declined. We had to reduce our research work

<u>Lack of support:</u>
The vision, mission and values of Pixlwizz stress the value of innovation through commitment and teamwork. But my division is required to meet strict sales and profit targets and that has forced me to reduce staff numbers and demand a lot from those who remain.

<u>Lack of recognition:</u>
My division develops the core technologies on which Pixlwizz games depend. However, in the recent 202x employee awards the recognition went to sales teams and games artists and designers. I and my staff conclude that we do not have a future in a firm that doesn't recognize our efforts and contribution'.

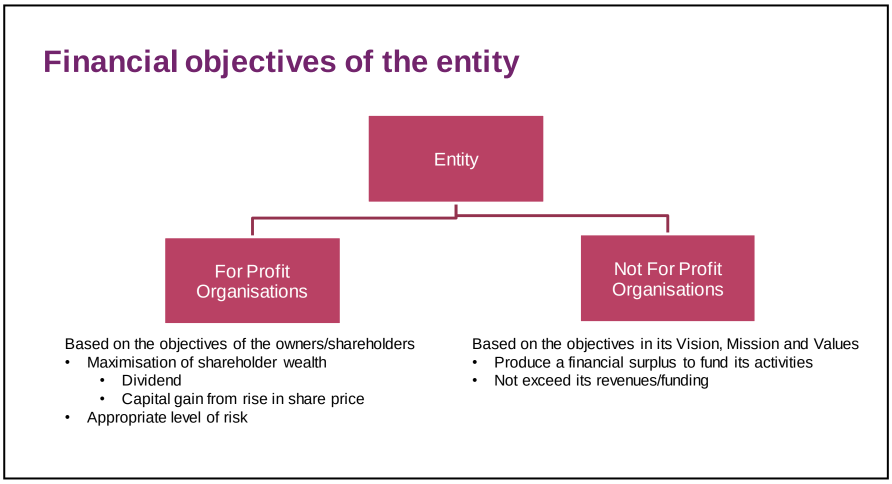
The CFO of Pixlwizz has asked you:

- Should our core values of commitment and teamwork have been sufficient to ensure we had a successful program of research and development?

[sub task = 34%]

3. Financial and Non-Financial Goals

Financial objectives
There are several potential financial objectives:



1. *Maximization of shareholder wealth*
   Shareholder value is assumed to mean the discounted present value of the free cash flows of the entity. Financial economics assumes that the market value of a firm is equal to the present value of its future cash flows. Therefore, maximizing shareholder wealth means maximizing the share price.
   - This takes the view that management do (or should) run the business solely in the interests of the shareholders
   - Encourages management to take a *long-term view* of the development of the firm rather than focusing on annual profit

- It assumes that the cost of equity of the business, which is part of the discount rate that is applied to the future cash flows, represents the best alternative return available to shareholders
- Therefore, a firm is maximizing shareholder returns if its future cash flows have a positive value after discounting at the cost of equity
- Maximizing shareholder value requires that management ensure its earnings stay in line with expectations or, better, exceed market expectations
- It provides the rationale for approaches such as Net Present Value (NPV). Projects with a positive NPV should be adopted because the total market value of the business will increase by the amount of the NPV
- Profit is used instead of cash flow in many corporate objectives and management seek to show rising profits or improvements in other measures (e.g. Earnings Per Share, Return on Capital Employed). Profit-based measures are not strictly appropriate because they are measures of *profit not cash*, and they are assessments of the *past not future*.
- Increases in the share price are called *capital growth*. Having investors such as growth funds and venture capitalists will cause management to focus on this objective
- Some management focus on generating short-term capital growth through cost-cutting or aggressive market strategies, often to boost the value of the business in order to sell it. This is an example of something called ‘ *short-termism*

2. *Dividend*
   - Many investors rely on the cash income from the organization and are less interested in the rise in the share price because they do not intend to sell their shares
   - These investors are often called *institutional investors* by which is meant pension funds and others which have obligations to make cash payments to their investors
   - Other examples can be particular shareholders, such as directors and family investors of smaller firms, who rely on divided for their income
   - Where shares are not traded on a market, such as for a family-owned firm, capital growth is unimportant (unless management is preparing the firm for listing on the market or sale) because investors cannot sell their shares easily.
   - These investors are also likely to be wary of the risks involved in maximizing growth and/or shareholder value. The terms *blue-chip share* or *cash cow investment* is used to denote firms that produce stable profits and which avoid risks
   - Delivering stable/growing dividends requires management to focus on annual earnings and cash flows and on having a suitable *dividend policy*
   - This can be a cause of *short-termism* in decision making in which investments are only undertaken providing they have short *payback periods*
   - Recognition of the need for short term income rather than long term shareholder value compromises the usefulness of techniques such as net present value. Management instead look at profit/cash impact and payback period
     Financial economics utilizes a *dividend irrelevance hypothesis* to argue that maximizing share price is consistent with the cash needs of investors because investors can sell some shares if they need cash. This may overlook real world considerations and behavior

3. *Appropriate level of risk*
   - Investors do not want to lose the wealth they have invested in the business, say from a big fall in its share price, nor to lose their income
   - Risk here refers to volatility in the financial returns from an investment. In effect volatility in earnings and volatility in its share price and/or dividend
   - The level of risk that investors will accept is called their *risk appetite*

- Risk appetite varies between investors. We would expect a venture capitalist or very rich individual to have a higher risk appetite than an institutional investor or a shareholder in the family firm
- Low risk appetites can lead to cautious business strategies
- Financial economics argues that the level of risk appetite is accounted for in the cost of capital used to discount cash flows for investment appraisal and in establishing the market value of the business

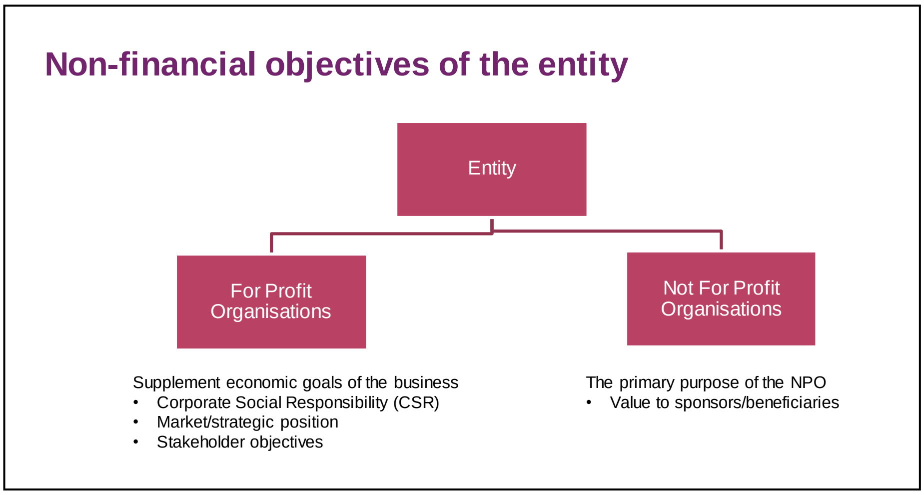
4. *Create a financial surplus*
    - Generally, this objective applies to a Not-For-Profit Organization (NPO) that donates the surplus from trading activities to, say, fund research or help the needy
    - May also apply to NPOs that accumulate surpluses over a few years to fund a socially beneficial capital investment in a later year

5. *Stay within budget/break even*
    - This would also be an NPO that needs to show financial prudence to its stakeholders
    - This organization seeks to maximize the attainment of its non-financial objectives and will see the need to live within its financial means as a constraint (e.g. a state-funded health system)

Non-financial objectives of profit-seeking firms
Profit seeking firms may have non-financial objectives too. These are often expressed in their Vision, Mission, and Values and may refer to fundamental business ethics, or to the firm's Corporate Social Responsibilities.

This inevitably triggers a *means/ends debate*. Does the car manufacturer actually exist to reach the ends of making customers happy, helping assure road safety, reducing pollution, and promoting a fair and diverse working environment? Or is the manufacturer really just interested in the *ends* of making profits and all these other worthy things are merely *means* to the end of making a more sustainable profit?

This cannot easily be answered and for SCS Assessment purposes it probably does not matter. The fact that the firm has non-financial decisions means that in setting its strategy and pursuing its financial goals its management will need to *strike some kind of balance* with the non-financial objectives it espouses.



1. *Corporate Social Responsibility (CSR)*
   Typically involves commitments to natural environment, ethical treatment of labor, responsible products, ethical business practices, and outreach to the community.
   Management pursue these for a variety of reasons:
   - Economic: increases sales, wins contracts, avoids legislation, and attracts ethical investment funds
   - Reduces risk: consistent with good health and safety and avoiding legal action. In the long term in ensures the business model is not outlawed
   - Builds trust and essential stakeholder relations: supports brand proposition, attracts and retain staff, builds relations with regulators and government, maintains license to operate in the community
   - Improves innovation and learning: open dialogue with stakeholders such as suppliers, customers, and interest groups helps suggest better products and processes and identifies market opportunities. Staff involvement in CSR can help make them better managers and more sensitive to diversity

   The need for a *sustainable business model* develops the role of CSR further. Being a CSR firm means caring and nurturing the ecosystem on which the business depends. This means that it is essential to the long-term survival and success of the business. For example, a business that boosts its cash flows by underpaying staff and avoiding taxation will lead to the decline of the ecosystem on which it depends by impoverishing its markets, depleting social infrastructure, and causing the appointment of radical governments that may bring in harsh new tax regimes.

2. *Market/strategic position*
   Firms may wish to pursue goals designed to affect their reputation. These include sector-leading innovation, diverse business portfolio, particular brand values, independence from larger organizations, focus on particular business segments, maintaining traditions and so on.

3. *Stakeholder objectives*
   Profit seeking organizations may not be profit maximizing organizations.
   - Smaller family-owned businesses may balance profit against the work/life balance of its staff and owners
   - Firms may allow managers to indulge their personal objectives
   - Firms may deliberately seek representation from employees, the community and so on

Non-financial objectives of NPOs
There is no easy classification of these. They are all specific to the NPO. AICPA-CIMA is an NPO, but its objectives do not resemble the objectives of international aid agencies like The Red Cross or Red Crescent, and neither are they similar to those of a charity established to preserve old railway engines. Yet all three are NPOs.

The best that can be said is that their objectives will be to *add value to their sponsors and/or beneficiaries*

This means that deciding or identifying the objectives of an NPO will be difficult because:
- Objectives become a matter of frequent debate between different sponsors and managers
- They are qualitative and harder to quantify that the financial targets of firms
- There are traditions of stakeholder engagement which mean the objectives will change as the stakeholders' change
- The objectives often conflict because they are decided on as a compromise between the demands of different stakeholder groups

- They may be constrained in what they can do by the terms under which the NPO was established and financed

4. Personal goals of management

Conventional economics refers to 'the objectives of the organization'. This is a simplification. Organizations do not have objectives. It is the managers and people within them that have the objectives. These may be selfish objectives, designed to benefit themselves, or their personal missions to make the world a better place. There is no reason to believe that these would automatically be the same as the goals of the investors or sponsors.

- In the SCS Assessment you should take care to recognize that different characters in the case will have different goals. These are *stakeholders* and we will cover *stakeholder analysis* later.
- It is also an issue for *corporate governance* because many systems of governance, and the incentive systems for managers, try to strengthen the control over management by investors in order to make managers follow business/economic goals rather than their personal agendas.

Three areas show this problem clearly

1. *Growth of the business*
    - This is usually understood to refer to growth in sales revenue. This has a relationship to another objective, *growth of market share*
    - It can also mean growth in profit or growth in net assets, but these are less common
    - There is concern that growth of the business may be driven more by *managerial ambition* and less by *investor considerations.* This is because a larger firm is not necessarily one that provides better shareholder wealth or dividend
    - Higher sales may not mean higher profits if the sales are gained by cutting prices or by expensive marketing and sales initiatives
    - Growing the business requires more capital investment which will deplete cash flows and could also lead to a lower rate of profitability if the investments grow by more than the profits
    - An argument can be made that growth will lead to higher long-term earnings through increasing market power and removing risk
    - Economic analysis reveals that growth leads to higher managerial pay and power, regardless of its impact on the investors' return. This is an example of the *principal agent problem* in which the needs of the investors (principals) are overridden by the interests of managers (agents) because of weaknesses in investor control over management

2. *Risk appetite*

    Another *principal agent* problem arises in relation to risk appetite. This means the attitude to the balance between upside risk v downside risk. Risk seeking behaviour is a gambler's mentality whereas risk averse behavior is a banker's mentality of seeking low-risk options. There are two interpretations to this:

    Managers may be more risk averse than principals because they are more heavily invested in the business than the investors. A business failure will expose the manager to a stigma that will harm their careers and also, they will lose their jobs.). Investors will be less affected because they have diversified their investments across many firms and they are also anonymous.

    Another view of the principal agent problem might be that managers would be less risk averse than investors because they are able to change jobs before the profits fall, and they lose very little personal investment

3. *Impact of managerial payment systems*
   The type of payment system can encourage managers to make decisions that are not consistent with the interests of the shareholders:
   - *Short-termism* is encouraged where pay or bonuses are based on annual or quarterly measures of profit or sales or other performance measure. The manager will sacrifice the long-term earnings of the business for their short-term bonuses. For example, by boosting profits by cancelling promotional or business development expenditure *risk-seeking behavior* may be encouraged by share options or bonuses. The manager with share options can only benefit if the earnings of the business rise unexpectedly. But they paid nothing for their options and so lose nothing if the share price falls. This will tempt them to take risks to increase earnings. The shareholder has a different attitude to risk because they would lose wealth if the share price fell.

ACTIVITY 2.3

This activity is indicative of the sorts of assessment tasks in the SCS assessment. It would constitute a task component worth 18 marks.

Context
Novak is a multinational business that develops, manufactures, and sells prescription pharmaceuticals. Novak has a statement of mission and corporate values that emphasizes its commitment to 'extend and enhance human lives' as well as maintain the highest standards of ethics and customer safety. It also promises to provide employees with a 'safe and secure working environment in which all feel valued and respected'. It undertakes to 'be commercially responsible, recognizing that sustainable profits fund cutting-edge research activities in addition to rewarding shareholders for their investment'. Elsewhere it refers to giving investors 'a realistic return on investment'.

Task
You have been called into the office of Randal Edwards: Director of Finance for Novak Pharmaceuticals
"I must say that the family is pretty much divided on whether to do this or not. We don't seem to be getting to a decision one-way or the other.

"The advisory and regulatory bodies' changes to prescription policies are forcing us to reconsider whether to continue with a lot of our present drug development. We are also considering the discontinuation of several of the drugs we sell because the financial returns are very poor and many are loss-making, especially the orphan drugs [drugs that treat very rare patient conditions and for which there are unlikely to be effective alternative treatments available].

Before we do this there is something bothering me:
- What will be the effect of such rationalization on our financial and non-financial objectives?

[sub task (a) = 55%]

Solutions to Activities

2.1 Courier firm tendering for contract with army

To what extent is this contract consistent with our Vision, Mission, and Values?

Runn's Vision, Mission, and Values are an expression of how Runn will behave and how it sees its place in the business world. These statements may reflect the culture of Runn, and therefore any disruption to this culture will lead to confusion, staff losses and so on. They may also have been considered by investors, clients and staff when they chose to engage with Runn and these stakeholders may be disappointed or perturbed by a departure from them

The fit of the Kayland Army contract with the Vision, Mission, and Values of Runn can be assessed as follows:

<u>Consistent with vision of delivery</u>

Runn wishes to be 'the World's favorite delivery company'. The Army contract is still delivery (and collection) and therefore is consistent with this

There may be some conflict with 'favorite' from this contract because presumably enemies of Kayland, and people with reservations about armies and war, might dislike Runn for its undertaking of this contract.

<u>Potentially responsible and sustainable</u>

It would be possible for Runn to undertake this contract in a responsible and sustainable manner

This could include actions to reduce carbon and noise emissions, the payment of good wages, and the maintenance of safe operating procedures.

<u>Allows Runn to show it honors its commitments</u>

Providing reliable support to the Army is a high-stakes responsibility, particularly in times of emergency or conflict. Engaging with this contract will help test and demonstrate the ability of Runn to take on and to honor such serious commitments.

However, there are several ways in which the Army contract may be unsuitable for Runn:

<u>Not linking businesses and individuals</u>

The Mission of Runn does not refer to governmental entities. It could be argued that an Army is not a 'business' because it is not involved in selling its services. This would mean the contract is outside the Mission of Runn.

<u>Stakeholder objections</u>

There is a danger that some ethical investors or employees will object to involvement in military projects, in the way that Google staff objected to Google using its mapping software to assist the US Military and led to Google dropping the contract

<u>Potential risk for employees</u>

Runn states a value of 'caring deeply about its stakeholders'. This would imply caring deeply about staff. Posting staff in Kayland Army bases exposes them to risk. Also, the 24/7 availability and army accommodation and food may be unpleasant for Runn's embedded staff. This may not be consistent with caring for stakeholders.

Conclusion

There are several aspects of this tender that are not consistent with Runn's VMV. Management should either decline the invitation, or take care to consult with key stakeholders, especially investors and staff, before undertaking it.

2.2 Adequacy of VFM to ensure R&D in a games developer

Should our core values of commitment and teamwork have been sufficient to ensure we had a successful program of research and development?

Should core values have ensured successful R&D?

The vision, mission, and values of Pixlwizz (PW) could be part of the control systems of the business.

Should reflect and maintain the culture

The core values are published in the annual statement and probably elsewhere such as on the website and in messaging to staff. This means some of the culture of PW, meaning its norms of behaviour and expectations, are contained in these values and are expected of staff. Managers and staff who are not willing to show commitment and teamwork should avoid working for PW and, if they do not exhibit these values, should be encouraged to improve or leave. This could have helped ensure adequate R&D improvements.

Policies should be consistent with values

Usually, policies will be used to implement the values. For example they would be a criteria that would be assessed during appointment interviews and in staff reviews. They may be the basis of rewards such as bonuses for exceptional effort. They may also be the basis of some KPIs. These policies should reinforce the role of values in control.

BUT

Negligent to rely on statement of values alone

Values may motivate the right behaviour but they are not strong enough to assure it. For example a motivated team may still not be effective if it lacks leadership or resources. The management of PW should not assume that things were going will merely because the right values were in place. There should have been checks on progress and on outputs too.

May lack implementation without policies and performance standards

Managers and staff need guidance in how to implement values, and there should be measures to ensure and check they do so. The values are very general with words like ‘passionate’, ‘support’ and so on. There is a lot of interpretation left to the individual and it. These terms need to be reinforced through policies, such as appointment selection criteria, included in training and induction programs, and monitored in reviews, including the use of peer group assessment of whether the individual exhibits these values. For example the comments made in the resignation letter about the COO could suggest that he was able to ignore these values without penalty and, perhaps, was not aware that they were important.

In conflict with financial goals

The VMV of PW contains no reference to making a financial return for investors, which should be a principal goal of any listed company. There will be financial goals and this raises the problems than some values will conflict with financial goals. The resignation letter seems to point to budget constraints and the need to meet targets as a reason for making decisions that undermined some of the teamwork and commitment values of PW. The R&D, and the values that support it, were damaged by this trade-off.

2.3 Novak rationalizing its portfolio of drugs
What will be the effect of such rationalization on our financial and non-financial objectives?
Approach: the 'and' in this task shows the examiner requires a separate treatment of the two classes of objectives.

Discussing the impact on financial objectives requires an evaluation of the rationalization program on shareholder value or 'rewarding shareholders for their investment' and 'offering investors a realistic return'. The non-financial objectives are the remaining elements in Novak's mission and corporate values

In the examination candidates have access to the pre-seen material and would be expected to know of the existence of Novak's mission and values, and where to find them for reference, whilst answering this question

Effect on financial objectives of Novak
Novak is considering curtailing much of its drug development activities, and to cease selling several drugs, in particular certain orphan drugs. This will have impacts on short- and long-term financial results and therefore on the share price

<u>Potential impacts on cash flows</u>
This rationalization will have beneficial impacts on short-term financial results if it involves ceasing expenditure on research and stopping making loss-making drugs. But short-term results could be reduced if cost must be paid to cancel contracts with testing laboratories, to compensate redundant staff, or to refund sponsoring organizations that have helped to finance drug research.

Long-term cash flows may be reduced if the rationalization of the development portfolio leads to a lack of new drugs to replace the present cash cows as their patents, or effectiveness, reduce. There may also be more complex effects such as the loss of complementary sales of other drugs if Novak drops a loss-making drug which, if taken, requires the patient to take other Novak drugs to counter the side-effects of the original drug.

<u>Impact on return to shareholders</u>
Novak refers to financial objectives such as creating wealth to give a realistic return to shareholders, and rewarding shareholders for their investment. These statements are vague.

The assumption made of investor motives is that they seek to maximize their wealth which means that they seek to get the maximum present value of future cash flows from their investment in Novak. The notion of a 'realistic return on investment' may be taken to refer to the cost of capital of Novak, that means the minimum rate of return acceptable to investors for the risks of investing in Novak.

It is generally assumed that the present value of future cash flows will be reflected in the market value of the shares of Novak. Therefore, the effects of rationalization on shareholders may be adverse, despite the rise in short-term profits if it leads to a fall in the share price. The impact of reduced future cash flows due to reductions in the numbers of new drugs coming to market may be one cause of this.

<u>Loss of R&D</u>
The share price might also fall because investors will conclude that the management of Novak has exhausted profitable research opportunities, or perhaps anticipates other problems such as with cash flow, and therefore it will anticipate a decline in the business. This is speculative behavior

Impact on other financial objectives
Novak also has an aim to ‘retain funds that can be used to finance the development of improved medicines. The initial effect of cancelling research will be to release funds that would otherwise have been spent on research.

Novak needs to decide what to do with these extra funds. If they are invested in alternative research projects, such as nanotechnologies, wearables, apps and so on, then the funds will have been retained (*student note: technically they have been spent and not retained because Novak treats its research spending as a cost each year).* But if the funds saved are distributed as dividends this would conflict with the attainment of its financial objective to retain funds although it would be consistent with improving shareholder return, at least in the short run.

Effect of rationalization on non-financial objectives of Novak
The non-financial objectives of Novak include the following:
- Extend and enhance human lives
- Achieve high standards in ethics
- Providing a good working environment for staff

Rationalization may create conflicts with these

Potential conflicts with objective to enhance and extend lives
Cancelling research and development may lead to the loss of medicines that could otherwise have enhanced lives. Ceasing the supply of existing medicines, in particular orphan drugs for which there may be no substitutes, will also conflict with this non-financial objective.

However, if the purpose of the rationalization is to re-direct the funds into researching better ways to extend lives, such as DNA editing or stem cell research, this may be compatible and a better way to achieve the non-financial objective

Conflict with high standards of ethics
The decision to cease supplying drugs on which sick people depend will raise ethical concerns and criticism. To avoid these Novak would need to conduct an impact assessment for each drug, assessing the availability of substitutes and the impact on human lives of withdrawing it.

Conflict with commitments to workforce
Cancellation of research will cause loss of jobs at Novak. This seems to conflict with the welfare of staff. It is essential that any job losses are handled fairly by selecting candidates without prejudice to maintain diversity.

Staff should also be offered generous compensation to help them to maintain their living standards while looking for new jobs

Adverse effect on Novak’s reputation
It could make the company appear to be greedy and not care for the patients who are dependent on Novak’s drugs

Chapter 3
Stakeholder Analysis and Strategy

1. Stakeholder Analysis

The SCS assessment requires you to be able to ‘conduct an analysis of stakeholder needs and recommend appropriate responses’

Questions on this topic appear very frequently in the real exam.

Definition
Stakeholders are:
‘those persons and organizations that have an interest in the strategy of the organization’
The term ‘stakeholder interest’ is used loosely to mean several things:

1. Their interests are affected by what the organization does. For example, they are employees who need the jobs, or investors who need the profits

2. They have legitimate claims that require management to take them into consideration when decisions are made. For example, management may want to consider the impact of the decision on future generations.

Classifications of stakeholders

Internal stakeholders
- Management
- Employees

External stakeholders
- Governmental bodies
- Civil society

Connected stakeholders
- Shareholders
- Banks and other financiers
- Customers
- Suppliers

Assessing stakeholder power and interest
For tasks involving stakeholders it is required that you do three things:

1. Identify a relevant stakeholder related to the situation in the task
2. Identify how the proposed strategy will affect the *interest* of the stakeholder
3. Identify the power of the stakeholder to influence the adoption or success of the strategy

Interest and Power

Stakeholder	Interest	Power
Employees	• Job security • Pay and conditions • Involvement	• Support management • Strike/disrupt • Leave
Managers	• Autonomy/challenge • Pay and profit share • Career progression	• Make decisions • Lead their teams • Take responsibility
Investors	• Financial return • Security of investment	• Appoint board • Buy/sell shares
Suppliers	• Revenues (price and quantity) • Consultation	• R&D support/investment • Decline to supply

Interest and Power

Stakeholder	Interest	Power
• Customers & clients	• Quality • Fair price • Service	• Purchase • Not purchase • Publicity and comment
• Government bodies	• Tax revenues • Employment • Contribution to the economy and life of the country	• Legislation • Licenses • Taxation • Buying the product
• Local community	• Quiet life and clean air • Employment and jobs	• Influence politicians • Direct action
• Interest groups	• Promoting and protecting the things they believe are important	• Influencing public debate • Direct action

Importance of Stakeholders
Stakeholders are an important part of the ecosystem of the business.

Stakeholder engagement is a key principle in sustainable business. For example, if the business doesn't ask its customers what they want how can it expect to sell very much? If it doesn't engage with government how can it expect to avoid unfavorable legislation?

Importance of stakeholders

1. Able to exert power that will affect the ability of the organisation to meet its strategic goals
2. Organisation has a duty or obligation to consult with them and to meet their needs
3. Stakeholder engagement may make the organisation more successful in meeting its strategic goals

2. Stakeholder Strategy

Purpose of stakeholder strategy
The literature on stakeholders seems to two different interpretations of a ‘stakeholder strategy’

1. *Engagement strategy*: this states that the purpose of business strategy is to meet the requirements of stakeholders by delivering value to them. This fits with the ecological metaphor for business which states that a sustainable business is one that evolves to cope with its ecosystem and which sustains that ecosystem. This seems to be the perspective of the CGMA Business Model.

2. *Manipulation strategy*: this seems to see stakeholders as either supporters to be enlisted, or resisters to be overcome (‘backers’ or ‘blockers’) and the purpose of stakeholder strategy being to manipulate them to help make the strategy successful.

Stakeholder mapping
CGMA candidates learn about one particular approach to stakeholder mapping: Mendelow’s

Stakeholder mapping (Mendelow)

Mendelow’s power/interest matrix

	Low Interest	*High Interest*
Low Power	Minimal Effort	Keep Informed
High Power	Keep Satisfied	Key Players

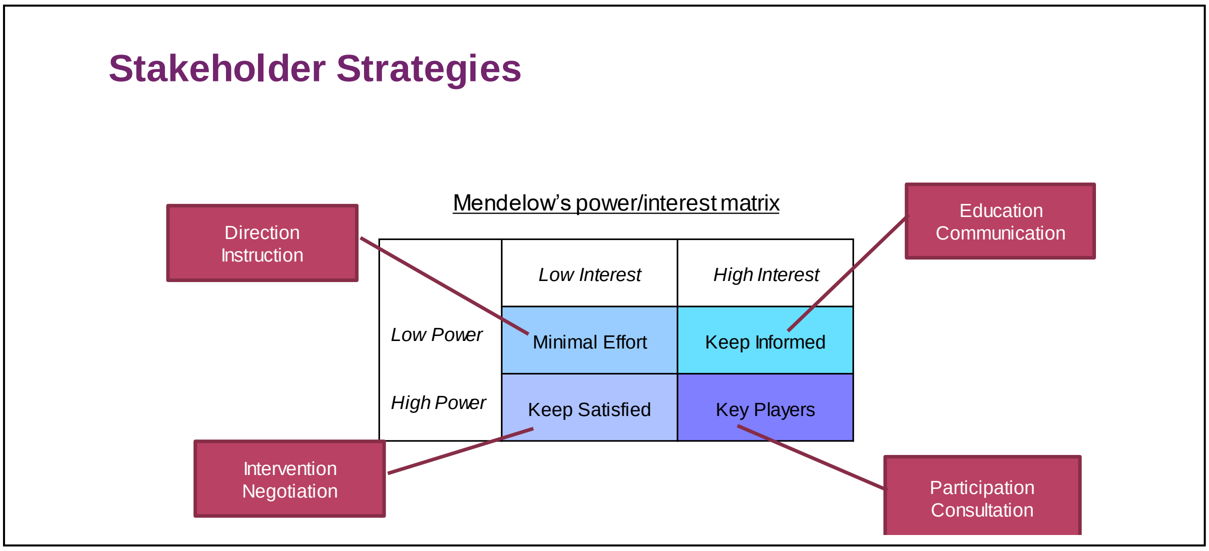
Matrix. However, do not overestimate the importance of this framework. You are not required to use it to answer every question on stakeholders in the assessment.

Stakeholder mapping has several stages:
1. *Assess/analyze the stakeholders*
   This means to scale their power and their interest and assign them to one of the quadrants

2. *Develop appropriate strategies (or value propositions) for the most important ones*
   The Mendelow Matrix advises greater attention should be paid to the Key Players because they have the greater power, but also because the strategy of the business has the greatest impact on their interests

In the SCS Assessment this requires you to make some assumptions about the power and the interests of individuals or groups in the pre-seen material and/or the task material

Strategies for stakeholders
Developments on the Mendelow Matrix has suggested types of strategy to deal with stakeholders in each quadrant.



- *Minimal effort*: this is a strategy of minimal engagement due to the limited power of these stakeholders and the fact that they are probably indifferent to the strategy. For example, day laborers or a supplier of consumables to your office. Management simply directs and instructs them.

- *Keep informed*: a strategy of limited engagement and restricted to 'telling' the stakeholders rather than 'asking and responding' to their needs. For example, a restructuring of the business that requires administrative staff to take on new roles and for some to lose their jobs. If the staff are not represented by trade unions or protected by legislation, they have very little power and therefore communication of the change, and education in the reasons for it and in the new tasks will be sufficient engagement.

- *Keep satisfied*: this represents a strategy of increased engagement in which management shows a willingness to modify or change its strategies to suit these stakeholders better. This is due to their high power. The purpose is to keep them in the 'low interest' zone by removing or modifying the strategy so that it no longer conflicts with their interests. For example, agreeing to pay a particular dividend to cash-seek shareholders instead of retaining all the profit to spend on growing the business, or negotiation of an industry code of practice with government to avoid them passing strict laws against aspects of the industry's behavior.

- *Key players*: this is a strategy of significant engagement. Engaging these stakeholders via consultation and participation ensures that the strategy is determined by their interests. For example, major shareholders would expect to be consulted on any changes in the strategy of the business, and business partners such as suppliers or joint venture partners might participate in the formulation of the strategies by offering investment and support where necessary.

3. Stakeholder engagement

Definition
The process by which an organization involves the people who may be affected by its decisions or can influence the implementation of its decisions

Benefits of stakeholder engagement

1. Enables compliance with some systems of sustainability accounting

Frameworks including GRI and IIRC require that firms engage in and report their stakeholder engagement. It is also part of assessing *materiality* in sustainability reporting because deciding whether something is material involves understanding whether the issue is important to any stakeholder.

2. Helps align the business with its ecosystem
   Understanding what customers, investors, partners and regulators want can help drive long term sustainability and shareholder value maximization

3. Sustains competitive advantage
   Stakeholders can be an important source of information about changes in the business ecosystem such as changing tastes, technology changes, or about their own plans which can affect the business

Developing a stakeholder engagement plan

1. Identify key stakeholders and key stakeholder groups
   This requires the stakeholder mapping described earlier

2. Develop a stakeholder engagement plan
   This should specify channels for engagement and the frequency and methods of engagement

3. Assign responsibility to persons to facilitate the engagement plan
   Different managers and teams should be assigned to respective stakeholders

4. Identify the interests and concerns of the stakeholders
   This is a process of listening and clarification conducted by the appointed facilitators

5. Design a mechanism to feed stakeholder concerns into the strategic plan
   The findings from the engagement should be presented to the Board in reports and the senior leadership team should decide which concerns to address and how to address them

6. Provide feedback to the stakeholders
   This feedback could be specifically to the stakeholder groups in debrief or reports, and also via the firm's Integrated Reports

Stakeholder Engagement

The procZess by which an organization involves the people who may be affected by its decisions or can influence the implementation of its decisions

1. Identify key stakeholders and key stakeholder groups
2. Develop a stakeholder engagement plan
3. Assign responsibility to persons to facilitate the engagement plan
4. Identify the interests and concerns of the stakeholders
5. Design a mechanism to feed stakeholder concerns into the strategic plan
6. Provide feedback to the stakeholders

Managing relationship with stakeholders

1. Create teams to engage with each group of stakeholders
   - Investor relations
   - Government relations
   - Public Relations
2. Hold face-to-face meetings with stakeholders
3. Ensure diversity within own organisation
4. Include stakeholders within the management process
   - Consultation committees and staff associations
   - Board representation for staff, CSR etc.
5. Develop an organisation communications strategy

ACTIVITY 3.1

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context
Couchweb is a provider of subscription video on demand services (SvoD). It is similar to Netflix and Disney+. It streams content to over 200 countries, and generally requires a license from the government of each country to stream content there.

Couchweb does not accept advertising but rather relies on subscriptions from viewers for its revenues. Subscribers pay M$12 per month to watch Couchweb. However, it does sell its content to other cable television operators who may take advertising.

Couchweb is following a strategy of reducing its reliance on bought-in programs and movies. It has begun its own production studio to make unique content. It has a strategic preference for series and serials rather than movies because they lead to more repeat subscriptions from viewers.

Task
The Director of Finance has invited you into her office.
"Thanks for your notes on financing our future production and for spotting potentially profitable subjects for new programs.

Investing in new production raises the same worries about retaining our creative talent such as key actors and also our script writers and production staff. We need to work harder to retain these people.

I was surprised at some of the program areas the creative team want to develop. I don't want Couchweb to get a backlash for making programs on unacceptable topics

Who are the stakeholders we should consider before we commission a range of new TV series?"

[sub task (a) = 55%]

Solutions to Activities

3.1 Couchweb deciding in new TV series

Approach: answering the question requires you to identify how new series will affect the interests of a stakeholders and to identify the Key Players and Keep Satisfied group (according to Mendelow's Stakeholder Mapping Matrix). These are the ones who must support the strategy. It is not usual to include competitors in a discussion of stakeholders because it would be unusual for a firm to try to accommodate the interests of competitors in its strategic choices. The best approach is to identify the likely power and interest of each stakeholder group first and then use this to justify needing their support. The question particularly requires a discussion of their interest.

Stakeholders to consider when commissioning a new series

Stakeholders are 'those persons and organizations that have an interest in the strategy of Couchweb'. In this case commissioning new programs.

They have interest in its consequences, and they have power to comply with or to disrupt the strategy (Mendelow's Matrix). This strategy may meet some of the interests of some, but leave other interests frustrated. Couchweb's management must be aware of this before it proceeds with commissioning a new series

<u>Subscribers</u>

The purpose of the new programs is to retain existing subscribers and to attract new ones. The *power* of subscribers as a group is potentially high. They are the main source of Couchweb's revenue and, they are willing to change TV source if content is not good. Because they are not organized to exert the power collectively, their power and influence remains limited.

The *interest* of the subscriber in program decisions is high. They want to receive good entertainment for their M$12 per month

Therefore, a strategy of 'keep informed' is appropriate. Subscribers must be shown trailers and pilots of proposed programs, and notified in advance of the catalogue for the next season. Couchweb needs to listen to their preferences, expressed in focus groups or on social media.

Some subscribers may differ on their willingness to pay a subscription to a broadcaster that produces violent content for example. In India a censored version of Game of Throne is broadcast to remove the violence and nudity that appears in the unedited American version

<u>Governmental and regulatory bodies</u>

Most countries have a government watchdog that screens and censors broadcasting content. These bodies have enormous *power* because they can refuse a broadcasting license and can make the selling of subscriptions a crime

Their *interest* can be kept low providing the content of the programs comes within their rules

Therefore, the strategy for this group is 'keep satisfied' by not providing programs that will lead to a regulatory response to Couchweb

<u>Creative talent providers</u>

The main purpose of the programs is to make profits for Couchweb's shareholders. The creative talent in Couchweb however have limited interest in profits. They are likely to have high *interest* in the quality of the final program and the chance to improve their reputation in the entertainment business.

The *power* of the creative talent will vary. Most will have low power because they can be replaced, but particular scriptwriters and actors will be essential to the success of the program and therefore have high power.
This means that some talent will require minimum effort and keep satisfied strategies over consultation and involvement in production decisions, whilst others may be key players who will need to be offered some participation in script and character development and plot lines.

<u>Other broadcasters</u>
Other broadcasters may stream or sell the Couchweb shows through their set top boxes and bundled with their subscriptions

The *power* of these broadcasters is potentially high. They can reduce the coverage and subscription income of Couchweb by refusing to distribute the new programs and Couchweb's catalogue.

Their *interest* will be the same as Couchweb's. They want to attract subscribers or, in some cases, viewers who will watch paid for advertising. This means they want to have attractive programs that will ensure viewers continue watching night after night and renew their subscriptions.

They will need to be kept satisfied or, in some cases, may be key players. The joint venture between Netflix and Baidu/iQiyi that gives it access to the huge Chinese market but will need to be sensitive to government concerns over content. [*student note: the occasional use of real-world examples learned from research into the industry behind the pre-seen material will gain extra points in the exam].*

<u>Investors</u>
The main investors will be the shareholders of Couchweb. But it is possible that programs may attract other investors under collaborative or syndicated arrangements.

Investors' *interests* are the financial return they will receive and the risks they will take. They will be particularly interested in new and untested programs, or new plot lines that could affect the popularity of existing cash cow programs like Tumbling Down or The Politics Game [*these programs were mentioned in the pre-seen material. Referring to them help show the examiner that you have read the pre-seen and helps to make your argument here more persuasive*]

The *power* of investors will be high due to the possibility that they will be asked to provide further equity or other finance for programs, and their votes on the composition of the Board and their influence on the share price.

Most investors will wish to be 'kept satisfied', in other words they will expect that the investment in new programs will yield a positive financial return and share price improvement and that it will not generate unacceptable risks

Conclusion
Balancing the interests of stakeholders may be difficult for Couchweb. The main problems will be accommodating the interests both of investors and creative staff

The vision of Couchweb includes commitments to innovation and creativity [*mentioned in the pre-seen material*] which means that Couchweb should have attracted investors expecting this rather than cash cow game and talent shows

Student note: Other stakeholders could include suppliers, banks and lenders, TV broadcasters and movie companies that Couchweb provide audiences for as it needs to ensure new series do not overlap copy or compete with their existing programs nor breach copyrights

Chapter 4
Strategy Formulation Process

1. Business strategy

Business strategy refers to the courses of action that management have taken that impact on the sustainability and value created by the organization. There are several other terms associated with this:

- *Strategic plan/planning*: this involves management creating a framework document saying what the firm will try to achieve over the next 3 to 5 years
- *Strategic decision*: a management decision that has an impact on the sustainability and value created by the business over the medium to long term
- *Strategic management*: an alternative to strategic planning that emphasizes the ability of management to respond to events and shape the strategy accordingly
- *Strategic management accounting*: the assistance given to management by the management accounting function through its provision of information, decision making techniques, implementation, and performance measurement systems

Characteristics of strategic decisions

- Impacts on the future prosperity and survival of the organisation
- Concerns the fit the organisation achieves with its environment
  - satisfaction of owners, investors, and other stakeholders
  - the products it makes and the markets it works in
  - the risks the business is exposed to
- Outcome sometimes not obvious at the time and may not be clear for some years
- Subject to high degrees of uncertainty

Contrast with other types of decisions

In managerial theory, and also in the levels of the CIMA/CGMA syllabus, a distinction is made between levels of decisions and issues.

Operational decisions: routine decisions concerned with achieving short term goals and targets, such as for a time span measured in weeks. They concern the best use of existing business processes and do not change the business model of the organization. They can be taken by relatively junior supervisory staff.

Managerial decisions: these are more tactical decisions concerned with implementing plans developed at a higher level, such as for a time span of several months and up to a year. They are generally taken within the existing business model of the organization but they can include decisions to change processes and procedures to find a better way to reach the organisation's goals.

Strategic decisions: generally long term and large decisions with significant resource consequences. These can affect the direction of the business over many years and also its ability to survive. These decisions should be taken by, or approved by, the most senior management such as the corporate board.

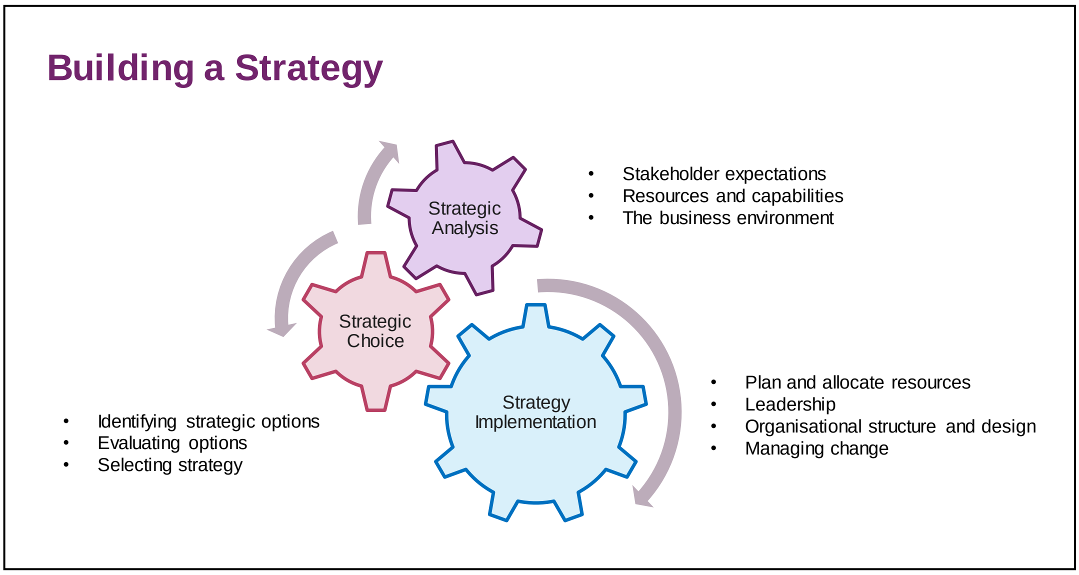
Exam guidance
- In the real examinations of 2021 the Examiner set several tasks which required candidates to assess whether something was a 'strategic matter' or not. If not 'strategic' then it would be 'operational' (i.e. routine and business as usual) or 'managerial' meaning concerned with implementing a decision already taken at a higher level in the organization.
- The examiner has also asked for assessments of the 'strategic logic' of something. By this the Examiner seems to require an assessment on whether it will maintain or improve the long term fit of the organization with its environment
- The Examiner set a task asking whether a fall in the firm's share price was a 'strategic issue' or not. Presumably this means whether the fall indicates a problem with the long term strategy of the firm that may require it to be changed, or whether it is merely a short term market movement that does not require management to re-think its business strategy.

Elements in strategy formulation
The process of formulating strategy requires management to undertake three activities:

1. *Strategic analysis*
   Assessing the present position on the business and understanding the expectations the stakeholders who will judge the success of the strategy

2. *Strategic choice*
   Deciding on ways to take the organization forward

3. *Strategic implementation*
   Making the strategy happen



Approaches to strategy formulation
1. Intended Strategy
   - Also called the 'Rational Approach' or the 'Strategic Planning Approach'
   - Grew from 'long term budgeting'
   - Formal process initiated by senior management

- Follows a set template and arrives at a plan for next 5 to 10 years
- Suffers from problems of uncertainty about the future and the divorce between planning and implementation

2. Emergent Strategy
   - Termed 'strategies in the absence of intentions'
   - Strategies evolve as responses to unexpected events
   - Recognition of the limitations of Intended Strategies
   - Requires divisional management to be alert and dynamic
   - But doesn't work if the industry requires long term strategic investments

3. Freewheeling Opportunism
   - Characterizes an entrepreneurial approach to business strategy
   - Managers/owners invest in ideas that take advantage of opportunities
   - No strong corporate direction or mission
   - Suffers from lack of control and lack of suitable opportunities

Approaches to strategic formulation

1. **Intended Strategy**
   Formal planning process
2. **Emergent Strategy**
   Develops and adapts as circumstances occur
3. **Freewheeling Opportunism**
   Entrepreneurial seizing of opportunities

Strategic Leadership

An intended strategy may develop in one of three ways

1. *Strategic planning*: developed top-down by senior management or *corporate parent*
2. *Strategic management*: developed and implemented bottom-up by divisional management
3. *Strategic leadership*: developed by a senior leadership figure or leadership team that inspire others to follow them

Approaches to strategic decisions

Strategic Planning (Rational approach)
- periodic process using a set template
- usually led by corporate management with limited divisional involvement
- implemented through the performance management system

Strategic Management (Emergent approach)
- divisional management teams manage business within strategic guidelines
- use of strategic analysis tools and business awareness
- framework of controls in performance management system to monitor implementation

Strategic Leadership
- Key individual or small group able to influence others to take decisions to improve the organisation
- Requires a balance of strategic analytical skills and human skills
- Develops a strategic mind-set instead of developing a one-off strategy

Benefits of the top-down approach are:
- It creates a control framework for the whole organization based on one strategy and group wide plans to implement it
- It will be long-term in outlook whereas divisions may be more concerned with short term results and rewards
- Will help ensure group adds value to separate divisions by releasing synergies
- It will reassure significant stakeholders that the group has a plan

Drawbacks of the top-down approach are:
- Erodes managerial ability at lower levels so organization cannot be flexible according to events
- May be based on wrong assumptions about the future business environment
- Slow to create and often needs revision as events unfold
- Detached from managers responsible for its implementation and may be impractical and resisted by them

Strategic leadership can be exercised by the Senior Leadership Team as a whole, or within a particular business function. This includes Freewheeling Opportunism but also a longer-term strategic approach based on a vision shared by the top team such as Amazon, Google/Alphabet, or Alibaba

Finance Leadership
Finance leadership refers to strategic leadership within the finance team and leads to *finance transformation*.
- Aligning financial strategy better with business needs through building partnership between the finance function and the business manager
- Improving the availability and use of financial and non-financial data for decisions and control
- Transform finance processes to gain efficiency – for example though technological investments including process robotics, data visualization, the application of Fin Tec (Financial Technologies of which the most obvious example is cryptocurrencies) and the application of Artificial Intelligence (AI or sometimes termed Cognitive Computing).
- Improving organizational structure and efficiency for example through creating shared service centres and by outsourcing

The CGMA is a strong advocate of the transformation of the finance function although, to date, this has not appeared as the topic of SCS exam tasks and, disappointingly, barely features in the strategic syllabus for CGMA.
https://www.cgma.org/resources/reports/re-inventing-finance-for-a-digital-world.html

ACTIVITY 4.1

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context

Pixlwizz develops and sells video games. It is in the same industry as PlayStation and Xbox. The pre-seen material states that Pixlwizz has a Northland Division that conducts research into Virtual Reality/Augmented Reality technologies (VR/AR).

The Chief Commercial Officer is considering ensuring that all new games made by Pixlwizz will have a VR version sold separately which can be played by people with the right headsets. These VR versions with be immersive and also allow player to play with smart devices, such as guns, swords or games racquets, and to see them in three dimensional space as they play. These smart devices will be sold by other firms and Pixlwizz games will be programmed to recognise and include them.

There is a Board meeting next week and the Chief Operating Officer has demanded that the move to developing VR games should be discussed as a key strategic decision.

Tasks

- Firstly, is the Chief Operating Officer justified in calling this a strategic decision?

[sub task (a) = 60%]

- Secondly what three stakeholders need to be engaged to make this successful and how can Pixlwizz engage with each?

[sub task (b) = 40%]

2. Sources of sustainable competitive advantage

Competitive advantage

Competitive advantage means the ability to create value better than rivals and therefore aids sustainable business

- better returns to investors
- better products and services for customer
- better social impact

Positioning versus Resource-Based Approaches

The origins of strategic business thinking were the 1960’s and coincided with the growth of marketing. So, it emphasized the need to make a better product to fit with customer needs. Later thinking generalized this into the need for the organization to have a strategy that enable it so adapt to fit with all aspects of its environment. This is the *positioning approach*.

Later thinking adopted an economic analysis and attributed profits and survival to the ability of the organization to carve our industries in which they are protected by barriers to entry resulting from their ownership of unique resources and competences. This is the *resource-based approach*.

Sources of competitive advantage

Positioning approach (outside-in)

- Competitive advantage comes from ensuring the organisation fits its environment better than rivals
  - better products
  - better financial returns to owners
- Requires that the organisation is changed as the environment changes

Resource-based approach (inside-out)

- Competitive advantage comes from identifying *distinctive competences* that provide better performance
- Competences should be *levered* to enter or create new industries

This resource-based approach requires management to identify the *core competences* (or sometimes called *distinctive competences* or *unique resources*) of the business which it can use to sustain itself in its present industry and also lever (or apply) to gain entry to new industries.

Features of a Core Competence

- Core competences are resources, relationships, and know-how that enable the firm to deliver high levels of shareholder value
- Conditions for a core competence (distinctive competence) are:
  1. They are *unique* to the firm
  2. They are *non-substitutable*: meaning that other firms can't find alternatives that are as good
  3. They are *valuable*: this means that deliver something that customers will pay for and/or they reduce the costs of the firm
  4. They are *leveragable*: this means they can be used to expand the business into new industries in the long run

ACTIVITY 4.2

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context
Couchweb is a provider of subscription video on demand services (SvoD). It is similar to Netflix and Disney+. It streams content to over 200 countries, and generally requires a license from the government of each country to stream content there.

Couchweb has recently announced an agreement to merge with a main film and media producer, Majik Media to form a new business to be called Pegasus Media

Task
To: Senior Manager
From: Martine Andels: Director of Finance
Topic: Response to investor queries

One of Couchweb's large institutional investors has raised concerns about our merger with Majik Media. They have sent in a list of queries

Each of these matters they raise will need a lot consideration by the Board of Pegasus Media if the merger goes ahead. I don't expect you to answer them in detail yourself.

Please brief me on the things we need to discuss with Majik Media's Directors before responding the issues raised by this investor on the questions, they raise

Note: you are not required to write response to the investor. Your job is to brief Martine Andels *on the things the merger parties will need to discuss in relation to each issue raised by the investor.*

- What will be the strategic financial and non-financial objectives of Pegasus Media after the merger?

  [sub task (a) = 33%]

- What is the business strategy to reach these objectives over the next 5 years?

  [sub task (b) = 33%]

- How will Pegasus Media engage with stakeholders after the merger?

  [sub task (c) = 33%]

Solutions to Activities

4.1 Pixlwizz video games publisher considering developing VR versions of its games

Sub section (a)
Is the Chief Operating Officer justified in calling this a strategic decision? (60%)

Student note: this question is based closely on a question from the August 2021 real exam. Presumably the Examiner intended candidates to debate what is meant by 'strategy' which is a question that has occupied entire textbooks without a uniform or agreed definition having been arrived at. The Examiner's report comments 'Higher-scoring answers demonstrated some thought as to the distinction between strategic and non-strategic matters, which then led on to an evaluation of the question. Candidates thought of several unexpected twists on that initial sub-task and all were marked on their merits. Level three answers generally suggested a logical starting point and then evaluated the strategic nature of the decision accordingly'. The Examiner relegated to Level 1 marks any answers that merely evaluated the business decision because that was not what the question asked.

The solution below constructs the argument for this being strategic or not from elements of definitions of strategy and steps in the strategy formulation process.

<u>Reasons why it may be strategic</u>
There is no agreed definition of 'strategy' or 'strategic'. Generally a strategic decision concerns the fit of the organisation with its environment, and has impacts on the long-term future prosperity and survival of the organisation.

Strategy formulation normally requires systematic analysis and emphasis on the need to assure the decision fits with the strategic position and competences of the business, the expectations of key stakeholders, and is possible within the resource constraints of the business. This is turn requires systematic resource audits and environmental analysis.

<u>Affects the long-term future of Pixlwizz</u>
Developing, or not developing, VR games could affect the survival of Pixlwizz (PW) in the games market. For example failing to develop them could allow PW to be overtaken by rivals and so suffer a decline in the value of its franchises and its profits. Firms like Meta (via its platform Facebook Games) are moving decisively in the direction of VR games. This move by PW needs to be done properly.

<u>Will be affected by the business environment</u>
The acceptance and success of VR games will be influenced by the actions of competitors and the extent of developments in technology. If large competitors like Amazon/Twitch, Meta/Facebook, Google and Microsoft/X-Box also launch VR games it will make them more popular and they will become an industry requirement for survival. VR requires affordable headsets and good bandwidth, especially 5G if they are to be used on the move. PW should assess these trends before making this decision.

<u>Has resource implications that must be planned for</u>
PW does not presently develop VR games. We will need to ensure PW has adequate developer skills, game formats that will be enhanced by VR, and the impact of developing VR on sales of non-VR formats needs to be considered. This decision seems simple but the COO is justified in demanding that its operational implications be properly considered otherwise it may fail.

<u>May take Pixlwizz into new industries</u>

VR may also have roles in education and in the facilitation of business and personal meetings. There is no proposal at the moment that PW may wish to move into these new areas, but given that PW already hosts MMO (Massive Multiplayer Online) games it would be a small step for it to use this expertise for wider purposes. This would have impacts for the development of expertise and the need for information systems to support it. Understanding the potential strategic directions that this could take PW in would be helpful now to maximise the benefit and identify the costs and risks properly.

<u>Has caused concern of COO</u>
The COO is a key stakeholder in the implementation of this decision because he is responsible for game development and programming. He does not consider this to be a routine operational decision, and he should have the chance to argue that point. The principles of good corporate governance emphasise the need for open discussion. If the COO thinks it is a strategic decision it needs to be discussed properly.

Reasons why it may not be strategic

By saying it is not strategic it is necessary to say that it is managerial or operational. That means that it is a routine decision or is the implementation of a strategy than has been agreed already.

<u>Extension of existing strategy to develop VR/AR</u>
This seems to be a continuation of past strategic decisions and not a new direction for PW. The Northland Office has been tasked with research into VR/AR so presumably PW is already keen to harness this technology in its game. Launching VR games may be just the latest step in implementing an agreed strategy.

<u>Consistent with VMV</u>
There is little that is radical in this proposal. The Vision of PW refers to virtual worlds, and the mission refers to ‘places’ to work and play. Moving from a 2D screen to a 3D headset is not revolutionary and would be expected by investors, staff and players.

<u>No specific or new risks</u>
This is not a high-stakes gamble for PW. The proposal is merely to design games that can be used on VR headsets as well and on other consoles and devices. This seems to be an organic development of the existing business model, not a new strategic direction.

<u>Limited resource implications</u>
It seems unlikely that this will require huge new numbers of additional staff to implement. Therefore adjusting additional training programs to include the development of VR skills would be relatively routine. Many games authoring packages already include VR capabilities (e.g. Unity, Amazon Sumerian, Google VR, and Unreal Engine) and developers from college will have been taught them.

<u>Within existing competences</u>
This is not outside of the core competences of PW. It will use PW’s gaming expertise, its research into VR/AR, and will build on the marketing capabilities and brand values of existing franchises such as Jakob Plunge. It is not in discontinuity with the present business model of PW.

Conclusion

This decision is probably strategic, in the sense that if PW doesn’t do it, or do it well, we could be in difficulty as the VR market grows.

The fact that some stakeholder engagement is needed (see answer to sub-task b) also suggests that this decision is more than merely operational or managerial.

For these reasons it is best to consider it properly.

Sub section (b)
What three stakeholders need to be engaged to make this successful and how can Pixlwizz engage with each?(40%)

Stakeholders to engage to make this successful
Stakeholders and persons or groups with an interest in the development of VR/AR games by Pixlwizz.

<u>Players</u>
The *interest* of players/gamers will be in whether VR/AR adds value to the playability of the games. They will also be concerned that it will be adopted by the player community, especially if it is applied to MMO games which require opponents and collaborators to be enjoyable.

The *power* of players/gamers in in their economic power of purchase. If they do not buy VR enhanced games the decision will be a financial failure. Players also have powers of influence over other gamers in the reviews they write and these can influence the commercial success of these VR enhanced games.

<u>Developers</u>
These are the staff (and freelance developers) who will be required to develop VR enhanced games for Pixlwizz.

Their interest will be in the training they may receive to cope with this new development. This will range from developing game concepts that benefit from VR to programing and coding for VR. Many may relish the challenge of developing skills in this area. Some will have concern over increases in workload, especially if there is not extra time allocated for training.

The power of developers lies in the contribution they can make to developing engaging games. This will depend on their acceptance of the new formats and their motivation to learn how to develop them. If they are unhappy they may leave and endanger the ability of PW to develop games.

<u>Headset manufacturers</u>
Some game publishers also produce consoles and headsets (e.g. Meta via Oculus, Sony PlayStation VR) but some headsets are made by electronics firms (e.g. HTC, Valve).

The *interests* of headset makers are the need for attractive games to run in their headsets to encourage players to buy VR headsets. The headset makers that belong to games publishers will see PW as partly a threat but also will recognise that helping PW will expand the number of players seeking the right VR headset. The popularity and VR quality of PWs franchises will be important here.

The *power* of headset manufacturers will be in the functionality they include with their headsets, whether they can link to peripherals (guns, swords and so on). They will need to provide coding information to allow PW to configure games for their platforms. They may also provide collaboration such as selling PW's games by downloads from their websites and involvement in joint marketing initiatives

How to engage with each stakeholder
Given their importance it is necessary for PW to get the required support from each stakeholder:

Players
There should be consultation with leading players and influencers. They could be provided with downloads of betas of the games (prototypes which may be incomplete and still contain errors), or the games could be exhibited at eSports events. PW could pay influencers to play the games in their online sessions to develop interest. PW could also seed social media with videos of the game play to encourage interest.

Developers
There should be an assessment of the training needs of existing staff and programs of training should be provided. PW should explain the opportunities of VR (if developers do not already know them) and the strategy of PW to harness it. It should be made clear that professional development within PW will depend on being able to contribute to these VR games. PW could create a steering group for the VR project to include some developers. It should provide assurances on workloads and resources. Having some element of performance related pay, such as bonuses or options over the shares of PW, would also improve engagement.

Headset manufacturers
There are relatively few of these. It would be possible for PW managers to meet key management in each headset maker to explain the roll-out and marketing of PW's VR games. This could be used to establish points of contact to help with technical collaboration and joint marketing.

4.2 Couchweb answering investors on its intentions after merging to become Pegasus Media

Sub section (a)
What will be the strategic financial and non-financial objectives of Pegasus Media after the merger?
Approach: this task does not require you to suggest objectives. It requires you to describe the types of objectives and the factors that will determine the final selection of objectives. As a listed firm it is inevitable that Pegasus Media will need to set shareholder-related financial objectives. But there will also be strategic business objectives designed to deliver the financial objectives, and perhaps some objectives related to CSR.

This solution is longer than could be produced in the 15 minutes available for this task. This is in order to help prepare you for more detailed questions on specific matters raised in the solution.

Strategic financial and non-financial objectives after flotation
Strategic objectives are long-term objectives that are supposed to form the basis of Pegasus Media's planned strategy. Strategic objectives are set at two levels:
- Fundamental goals and objectives which serve the needs of the persons or groups who sponsor and control the firm
- Secondary or enabling objectives that are steps on the road to the achievement of the fundamental objectives

The fundamental strategic objectives will become related to the goals of Pegasus Media's external investors and based on shareholder value

The considerations that the Board will need to bear in mind when setting strategic objectives include the following:
The goals of the shareholders in Pegasus Media
It can be assumed that the economic goals of the shareholders are the maximization of return to their investment, i.e. the maximization of shareholder wealth. Investors will also have a risk appetite which will influence the levels of risk that they are prepared tolerate arising from the strategic decisions of Pegasus Media's Board. Investors may also have a preference for the balance between dividend income and capital growth. Finally, investors may also require that Pegasus Media limits its scope of activity to entertainment and not to diversify into other areas.

It is important that the Board establish the goals of the shareholders before setting strategic objectives

The business strategy that will best achieve the goals of the shareholders
The present business strategy of Couchweb is to gain more subscribers and (backward integration into program-making. This strategy depends on secondary objectives related to innovation, service reliability and so on.

There are also risk management objectives related to foreign currency exposure, protection of intellectual, avoiding litigation, reputation risks and so on to protect shareholder wealth

The setting of strategic objectives will depend on the strategy that Pegasus Media adopts after the merger (discussed below)

The goals of other powerful stakeholders who may influence strategy
Stakeholders are persons or groups with an interest in what Pegasus Media does (see discussion later) Shareholders are one stakeholder group and we assume their interest is in the maximization of shareholder wealth.

Modern business enterprises cannot merely pursue the goals of investors and expect no reaction from other stakeholder groups such as the governments of the countries in which it sells programs, its customers and its staff. Therefore, secondary strategic objectives may be set which gain and retain the support of other stakeholders as an essential element in securing the primary objective of maximizing shareholder wealth.

The inclusion of CSR/sustainability goals
Pegasus Media needs to decide whether it wishes to include CSR/sustainability in its strategic objectives.

Sustainability refers to the ability of Pegasus Media's business model to endure and be successful through time. Some of the non-financial enabling objectives discussed above are also sustainability objectives, for example cultural improvement, responsible broadcasting, fostering new talent and so on. To these could be added objectives that would help Pegasus Media to develop strategies to align with other developments in its environment, for example reduction in environmental emissions, encouragement of diversity in the workplace.

In addition, Pegasus Media may include some specifically CSR objectives relating to charitable giving, ensuring inclusion in sourcing scripts and actors from both genders and from across ethnic groups, equal pay, and avoiding bullying at work

There is bound to be a discussion about whether these CSR objectives are essential for sustainability, and whether they contribute to the maximization of shareholder wealth.

The factors that affect whether strategic objectives are feasible

It would not be appropriate to set objectives which cannot be achieved. This would mislead investors and would subject the Board to criticism later when the objectives are not achieved. This means that the Board will need to consider the resources and competences of Pegasus Media and set objectives which are feasible.

The Board will also need to consider the environment and the likely demand and competition in its market. This means that some strategic objectives will need to be reviewed as the strategy process develops and it is discovered that some objectives are not realistic

Sub section (b)
What is the business strategy to reach these objectives over the next 5 years?
Approach: this task requires you to provide a summary of the elements in strategy formulation. You should try to apply each to the TV industry and the situation of Pegasus Media. This leads on logically from the previous task on setting strategic objectives and you could make this point. Where you see a link between tasks you should still mention it. It will impress graders.

Considerations in establishing a business strategy
A business strategy is the way that a business assures a fit with its environment (or assures its sustainability) in the longer term. A planned strategy is a 5 to 10-year plan of how this strategy will roll-out.

Pegasus Media needs to decide whether it will adopt a planned approach to strategy and produce plans for 5 to 10 years, or whether it will rely on indicative objectives and leave the management at the time to choose the best courses of action to achieve them.

Given that the purpose of your response is to encourage this investor to support the merger to create Pegasus Media it would probably be better to provide a 5-year plan with some financial forecast resulting from it

The stages in developing this 5-year strategy are discussed below

<u>Setting the strategic objectives for Royals</u>
Couchweb has a mission statement and vision, but these are not yet decided for Pegasus Media. These will need to be established.

Strategic objective setting has been discussed above

Following the corporate appraisal stage of the strategy process it may be necessary to return to the strategic objectives and to revise them in the light of the assessment of environment and competences

<u>Assessing the business environment of Pegasus Media</u>
The business environment contains the factors not within management control, such as the state of the global economy, the behavior of competitors, changing technology, and changing tastes in entertainment.

This environment will need to assessed, and forecast over the life of the plan, to assess its impacts on Pegasus Media's present business and the opportunities it may create for new lines of business. This environmental analysis needs formal information gathering and could involve hearing from consultants and other experts as well and researching the trade and academic media.

<u>Conducting a position audit/internal assessment</u>
This requires an objective assessment of the competences, resources and organizational situation of Pegasus Media, including its financial results and the availability of finance. The

purpose is to identify the main strengths and weaknesses of Pegasus Media as a prelude to developing strategies to leverage them and to address them

This will require detailed and objective information gathering and assessment by the Pegasus Media Board. The Board should be prepared to accept the weaknesses of Pegasus Media as well as the strengths.

<u>Completing a corporate appraisal</u>
A corporate appraisal evaluates the present and potential future position of Pegasus Media against the pressures from its business environment This process sometimes culminates in a SWOT analysis (Strengths, Weaknesses, Opportunities, Threats).

<u>Generating strategic options</u>
These are potential courses of action to help Pegasus Media reach its strategic objectives and to deal with the weaknesses and threats in its present situation. Some of these will involve decisions on expanding sales, through market penetration in existing country markets, but also expansion to new products and new markets and the adoption of new technologies. Other strategies may be internal efficiency strategies to improve returns from existing operations and to future-proof them against threats.

<u>Evaluating and choosing final strategies</u>
The process of evaluation involves assessing the suitability, acceptability, and feasibility of each option and choosing those that are attractive. This can involve financial analysis, risk analysis and the assessment of resource availability and competences.

It will also require negotiation and agreement between the personal positions of the various members of Pegasus Media's Board.

<u>Implementing strategies</u>
A five-year strategy will take five years to implement. This will require policies, programs and plans to carry out the essential elements of the strategies

Sub section (c)
How will Pegasus Media engage with stakeholders after the merger?
Approach. Avoid interpreting this task as a question about stakeholder mapping. It is a communication question. Stakeholder engagement is a requirement of some approaches to corporate governance and is an element in Integrated Reporting and in other model of global business citizenship.

Stakeholder engagement after flotation
Stakeholders are persons and groups with an interest in what Pegasus Media does. Some approaches to strategy formulation emphasize the need to consult with stakeholders to set the goals of the business.

A global citizen perspective of business, in which business is required to take on a duty to society and be a good neighbor, also emphasizes stakeholder engagement.

The principle of stakeholder engagement is to ensure that management are aware of stakeholder requirements, needs, or concerns, and that these are formally monitored and noted. This then impels management to take notice of them and manage the business to satisfy stakeholders

<u>Engagement with shareholders</u>
This requires a statement on how Pegasus Media will maintain its awareness of the perspectives of its investors.

Potential solutions could refer to the annual shareholders meetings, investor communication pages on the company website, periodic meetings or teleconference events with Board members.

Engagement with viewers
This will require Pegasus Media to monitor and listen to the needs and suggestions of customers. Potential solutions could include regular on-line surveys sent to members, monitoring a social media page set up by Pegasus Media for viewers to post comments, focus group meetings with representative viewers in each geographical area, and a formal system of noting viewer comments made during meetings or other interactions

There should also be feedback to the viewers, for example by regular social media posts explaining actions that have been taken, formal responses to complaints and so on.

Engagement with governments and regulators
Governmental bodies issue the license to broadcast via Video on Demand and to show programs in their countries. Therefore it is important to maintain regular contact with the governmental bodies through regular meetings.

Monitoring the pronouncements and decisions of the bodies will also give insight into their thinking

Engagement with staff
Staff can be engaged through workplace meetings with their line managers and regular staff surveys can be conducted using online survey applications.

Specific staff liaison roles could be created within business units and at head office.

Pegasus Media will need to keep staff informed of decisions and of responses to complaints and suggestions via sections on the intranet, a management blog and so on.

Senior management should also have programs of visits to business units during the year to talk with staff

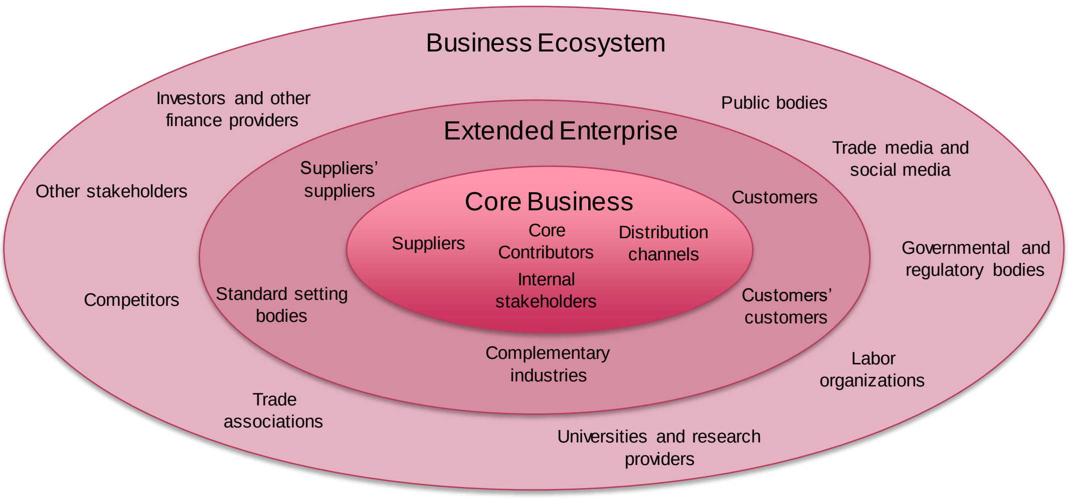
Chapter 5
Analysis of the Business Ecosystem and Environment

1. The Business Ecosystem

The business ecosystem was introduced in Chapter 1.
Earlier chapters reviewed stakeholders. This Chapter focuses on a few elements of the ecosystem that are specifically named in the CGMA syllabus

- The competitive environment
- Competitor analysis
- Customer analysis
- PESTEL analysis

Ecosystem



Because these external factors have the potential to change the earnings and operations of the business, they can also be called *risks*. We shall see in a later Chapter, the term 'risk' can refer to the potential for unexpected upside gains as well as for unanticipated downside losses).

2. Porter's Five Forces Analysis

It is unlikely that you will face a task that refers to Porter's Five Forces, or obliges you to describe or apply it. But you are very likely to be asked to advise management of risks, or to advise them how to assess the market.

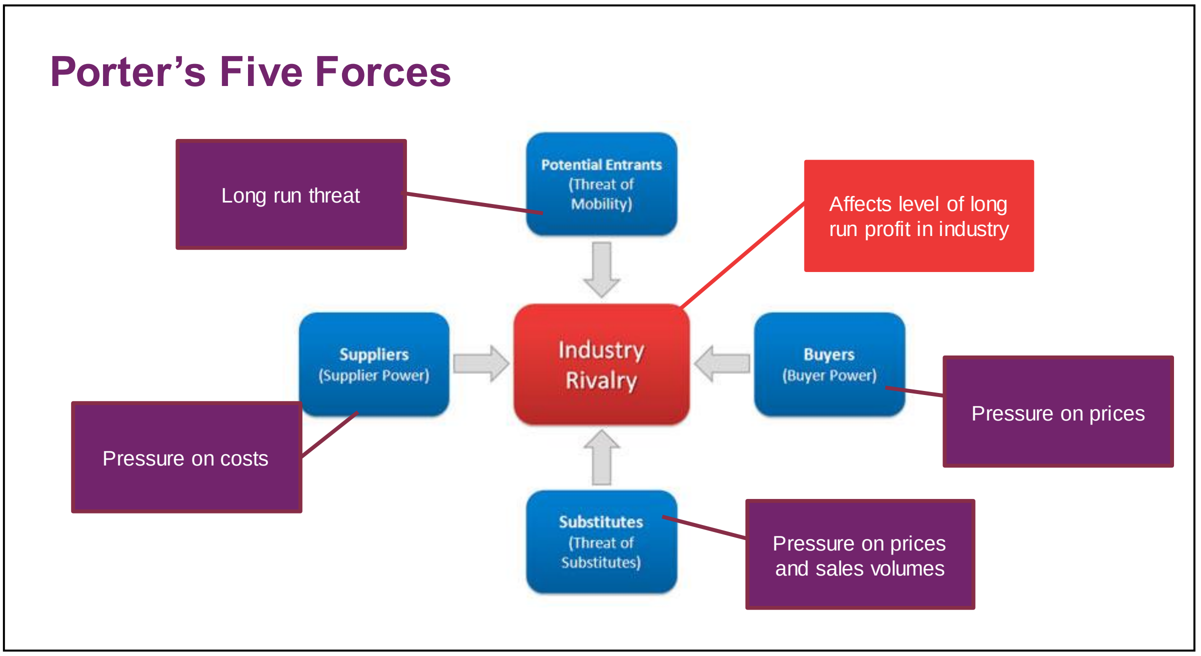
Porters 5 Forces Model attributes present and long run industry profitability to the impact of five competitive forces

1. *Buyer power* leads to pressure on prices but also pressure on service standards and product expectations that will raise costs
2. *Supplier power* affects costs but also some suppliers can cut the industry out from the supply chain by going straight to the buyers via disintermediation (e.g. as Disney did to movie theatres when it launched home subscription video streaming)

3. *Substitutes* are alternative technologies that can rival the industry and therefore put pressure in prices and sales volumes
4. *Potential entrants* are a long-term threat to profits, volumes and may also increase service standards and costs by bringing product/service improvements
5. *Competitive rivalry* is between firms in the industry for market sales and will lead to reduced prices and costly service and product improvements

Use the framework to

- Help analyze the pre-seen material for the case and guide your research by setting you to look for buyers, suppliers, entrants and substitutes.
- Provide a checklist of things to discuss if you are set a task asking for a market or industry assessment



Porter's model overlooks modern thinking that emphasizes the need for firms to partner with suppliers and buyers instead of treating them as opponents. It also suggests that the forces can be identified and predicted over the medium to long term. This may not be possible in a digital age of 'move fast and break things' (the internal motto of Facebook).

Five forces and digital disruption in the ecosystem

The Porter model can be adapted to provide some understanding the threats of some forms of digital disruption on the markets and industry of a business.

This assessment often features digitally disrupted industries e.g. (smartwatches, video streaming, ride hailing) and the Five Forces Model can help in analysis of the impacts on the industry. The digital technologies will be discussed further in later chapters.

1. Digital disruption and buyer power

   Online marketplaces and aggregators provide greater price transparency and wider choice than instore shopping. Digital commerce can also improve information to buyers (e.g. peer ratings systems, samples of sound files/videos). Electronic delivery (e.g. streaming and eBooks) and crowd shipping apps (e.g. UberEats and Deliveroo) give much quicker and more convenient fulfillment of wants.

   However, buyer power is directed through and influenced by a small number of portals. In the West they are known as FANG (Facebook, Amazon, Netflix and Google), and additionally Apple and Microsoft, with a Chinese counterpart of BAIT (Baidu, Alibaba, iQiyi and Tencent). These helps lock in buyers and influence choices by sponsored listings and advertising. They present sellers with a monopoly channel that pays low prices but charges high prices for access to their users, data or advertising. Social media also impacts to manipulate buyer power through peer ratings and recommendations, targeting promotion and memes, and the creation of online role model/influencers

2. Digital disruption and supplier power
   Supplier power may be increased by digital media giving immediate access to buyers (e.g. websites, e-publishing/broadcasting) or break the monopoly of traditional distribution channels (e.g. selling via online aggregators like Bookings.com, online financial services comparison sites, peer-to-peer lending portals). It is also less regulated which enables suppliers from an off-shore country to avoid compliance with laws of another (e.g. banking, medical products)

   However digital technologies can also increase the number of potential suppliers available to an industry and reduce supplier revenues (e.g. purchasing portals, crowdsourcing and crowdfunding of finance, staff, goods, and services). It can also lead to greater scrutiny of suppliers (e.g. buyer ratings, price comparisons, tracking and surveillance) and increase their costs (e.g. investment in IT systems, blockchain platforms and so on).

3. Digital disruption and substitutes
   The predominant trend is for digital upstart firms to be a substitute that threatens the existing industry (e.g. Amazon in retail, Google in advertising, Airbnb in hospitality, Netflix in home entertainment, and Uber in public transportation). Digital disruption has brought in entirely different solutions to consumer needs which has eliminated or harmed many conventional industry players (e.g. retail stores, record and movie firms, package holiday operators, newspaper publishers, video game producers, gambling shops, fixed-line telephone service providers, auditing and accounting firms, and investment markets)

   The threat of the digital substitutes can force industry members to undertake a ‘digital transformation/pivot’ of their business model (e.g. Disney, New York Times, Sony Computer Entertainment, Walmart, and Microsoft Xbox Live)

4. Digital disruption and market entry
   Digital disruption has brought in greater numbers of suppliers due to lower costs of entry (e.g. AirBnB entered the hospitality industry, and Uber the transport industry). It also brings entry from information industries (e.g. Alphabet/Google entering healthcare, Apple into automotive industry, Amazon into groceries and healthcare). End-to-end technologies, such as item tracking and Blockchain, and the use of cryptocurrencies also help new suppliers enter by providing buyers with assurance of provenance, transparency of supply chain, and ease of payment.

   However digital technologies can also act as barriers to entry for existing firms due to their access to better customer profile information, and the cost of investing in digital technologies.

5. Digital disruption and competitive rivalry
   The basis of rivalry amongst may be digitally enhanced either by a digital transformation of the business model, or by harnessing aspects of digital technology to reduce costs and prices (e.g. outsourcing, use of cloud-based hardware and software, process automation, use of machine or artificial intelligence to reduce staff costs, de-skilling of staff, and use of crowdsourcing to provide labor and supplies), or digital enhancements to the customer offering (e.g. affinity sites, faster delivery, online buying channels, app-based interaction, social media marketing)

This means that new digital business models or technologies can be evaluated by assessing how they will disrupt these 5 forces

3. Competitor Analysis

Competitor analysis focuses on rivals, their strategies, and their potential impacts on the strategies of your business.

Analysis can be conducted by:

- Observation
    - Analysis of their product
    - Reading their reports, pronouncements, and articles on them
    - Infiltration of people into their business (unethical)
    - Monitoring social media posts and blogs of their management, staff and customers

- Interview
    - Discussion with industry experts and investment analysts
    - De-briefing former employees of the rival
    - Networking at industry events

- Experiment
    - Small changes to prices or products to gauge response
    - Announcements of new strategies and monitor trade media and investor for their reaction

Competitor Analysis

1. **Identify competitors**
   - Brand competitors: similar appeal to similar customer group
   - Industry competitors: similar products and technologies
   - Form competitors: sell substitute technologies that satisfy same need
   - Generic competitors: compete for the same income

2. **Analyze competitors**
   - Objectives: the and goals of the competitor and how well it is reaching them
   - Strategy: the markets it competes in and the methods it uses
   - Assumptions: the way the rival sees the future of the industry and its place in it
   - Resources and competences: key strengths and weaknesses, its costs and its brands

3. **Develop competitor response profiles**
   - How it may react to our firm's strategies
   - Laid back: does not respond
   - Selective: reacts only in selected markets that it believes are important
   - Tiger: always responds aggressively
   - Stochastic: its responses are random and hard to predict

Exam point: *this assessment often sets tasks asking you to explain how to find something out. The simple model of Observe, Interview, and Experiment provides a checklist that is very useful to develop an answer providing it is customized to the task at hand.*

Gathering, storing and analyzing this information requires specialist staff to co-ordinate it. Investing in a Knowledge Management System (KMS) may be undertaken to enable codification of information (using referencing systems called hypertext tags) and its storage in a searchable database.

Management accounting function can assist in competitor analysis by:

- Developing a cost and pricing model to simulate the competitor

- Providing information and analysis of competitors
- Providing simulations of the impact of competitor actions on our own business

4. Customer Analysis

Customer analysis assesses the value, behavior and dynamics of customers in the industries involved in the firm's business strategy.

Customers are part of the firm's ecosystem and the source of its revenues. Customer analysis examines existing customers and potential customers that the firm may wish to set strategies in place to gain.

Customer Analysis

1. **Assess potential value**
   - Purchasing frequency and price
   - Range of product bought
   - Costs of supply and maintenance
   - Influence on other buyers
   - Forecast growth or decline of segment
2. **Identify buyer behavior**
   - Purchasing characteristics
   - Influencing factors
3. **Understand their potential future value**
   - How satisfied and well served they are at present
   - Potential to acquire or lose them
   - Potential lifecycle value
   - Changes and development in their needs

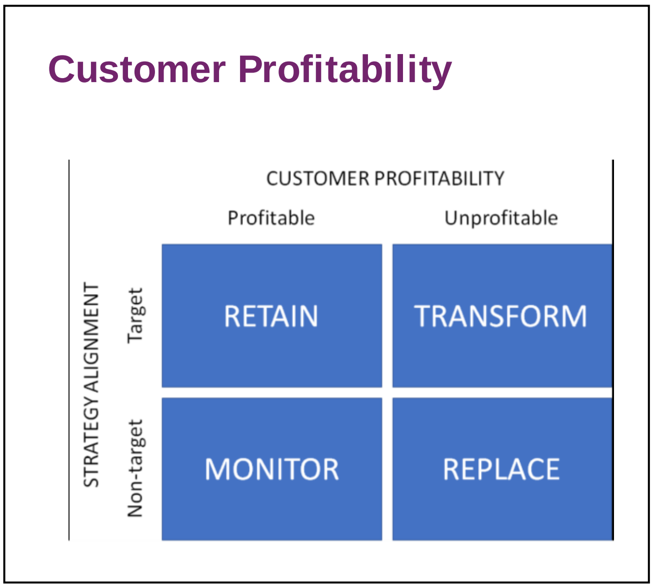
Assessing potential value

This is termed Customer Profitability Analysis (CPA) and involves making a holistic assessment of the profit that is available from the particular customer account (used in business to business marketing and called *Customer Account Profitability*) or to the segment of types of customer

- Revenues include the sales of a particular product, the sales of complementary products, and also some of the revenues from customers influenced to buy by the original customer
- Costs are all costs of supplying the customer, including manufacturing, distribution, advertising, administrative support, technical and service support, and credit costs

This is used to make customer strategy decisions:

- Profitable customer segments
  - Retain existing ones
  - Acquire more
- Unprofitable customer segments
  - Replace them with profitable customers (i.e. cease to serve them)
  - Transform into profitable customers by cost engineering (e.g. service reduction, reduction in product specification) or by increasing revenues (e.g. price, encouraging them to buy more or better)



Identifying customer behavior
The approach to this varies according to whether the customer is an industrial buyer (B2B) or a consumer of the product (B2C).

- Industrial buyers
    - Buyer motivated by impact on their business not on their own taste and needs
    - Larger volume and greater bargaining power
    - Likely to be a complex buying process with multiple stakeholders involved in the decision
    - Industry demand level is *derived* from the level of activity for what it sells

- Consumer behavior
    - Seek to meet 'needs' that can be deeply rooted in their psychology (self-image, wish to belong, care for others)
    - Products 'positioned' to suggest benefits that meet these needs and by brand personality
    - May be influenced by other people (fashion, group pressure, peer-opinions)

Understanding future value of customer
Strategic time frames last for 5 years and longer. Customers may not stay the same throughout this period.

The firm needs to assess:

- Whether they are satisfied or likely to switch in future
- How the customers' needs and requirements may develop and change
- Availability of new customers to replace present ones who will move on (e.g. children's toys, buyers of small apartments, university tuition)
- Lifecycle value of customer: their potential value over the length of time they are a customer which may repay initial costs of attracting them (e.g. rationale for discounted smartphone handsets, free banking for small businesses)

Some basic marketing concepts
Marketing is described as 'the management function that identifies, anticipates and fulfils customer needs efficiently and profitably'

1. Market segmentation/target marketing
   This recognizes that not all potential customers want the same thing and that not all potential customers are of equal potential profit. Segmentation divides customers into groups with similar buying characteristics using *segmentation bases* such as age, gender, location, income level and lifestyle. The attractive segments are identified and a suitable *marketing mix* is developed for each one. The opposite to market segmentation/target marketing is called *mass marketing*.

2. Marketing Mix
   These are the factors that marketers use to *position* a good in the market to justify its price and to gain attention and sales. Initially comprised of 4 Ps (*Product* specification, *Pricing* strategy, *Promotional* strategy, and *Place* (distribution strategy and sales channel selection). This has been extended to include *People* (the service face of the supplier), *Process* (the method by which the customer is handled), and *Physical* (the quality and atmosphere of the sales outlets or the tangible indicators of product positioning such as the quality of promotional materials)

3. Brand strategy
   Brands are more than the product (e.g. 'Apple' is a brand, not a product) and comprise a name, a logo and color scheme, but most importantly a 'personality' that implies special benefits. These together are termed *brand equity* because they require investment and

building but also, they should give long term returns from enabling the firm to charge a premium price and gain more sales from the target market that recognizes the brand. The following brand strategies are used:

- *Line extension*: additional similar products under an existing brand (e.g. Coke Zero)
- *Brand extension*: new products under an existing brand (e.g. Model S, X Y, 3 and Cybertruck under the Tesla brand)
- *Multi brands*: creation of several brands by the same firm (e.g. Estee Lauder also has Clinique, Bobbi Brown, Jo Malone, Lab Series and so on)
- *New brands*: a new product that is distinguished from the range by its new brand (e.g. Toyota launched Lexus)

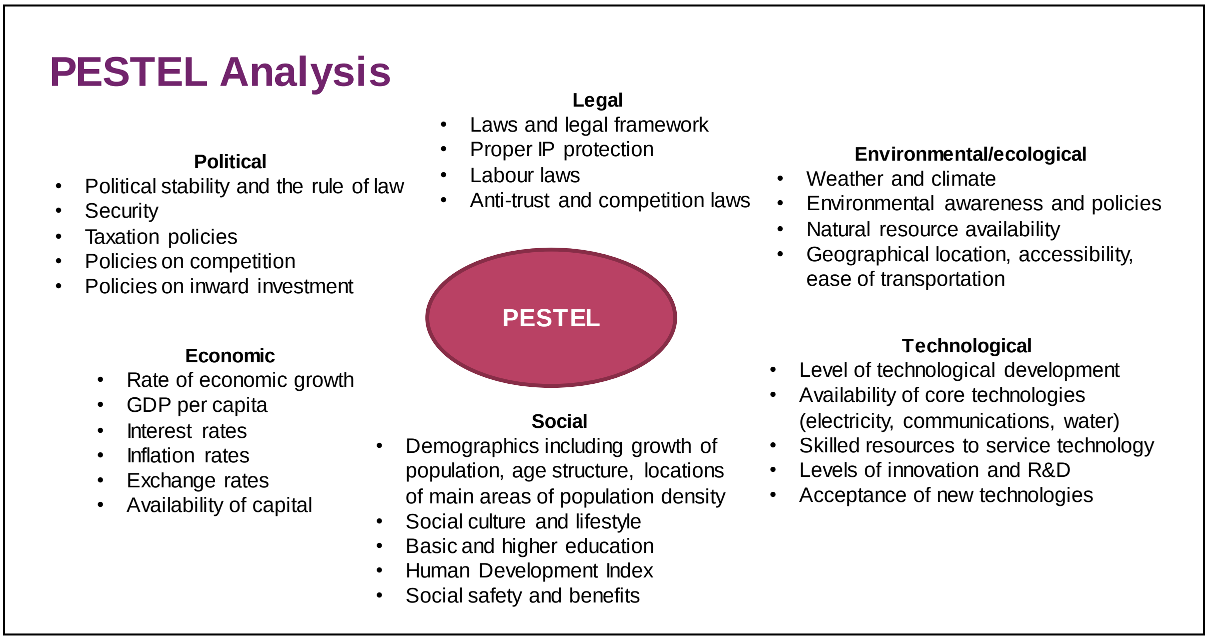
4. Customer Relationship Marketing
   An approach to marketing that moves away from gaining revenues from selling extra units of product (*transactions marketing*) towards maximizing the long term spend of the customer by building a relationship. Based on the principles that staff should be able to identify and know the customer during transactions (e.g. via loyalty scheme account and phone number or facial recognition harnessed to customer profiles on the Customer Relationship Management system), should ensure they receive excellent service to retain them, and helpful information and advice to up-sell them

5. Web 2.0 and marketing
   In addition to e-selling the interactivity of Web 2.0 provides marketing opportunities to enhance *customer engagement*. These include *rich content* on the product (sound and video files and interactive virtual displays), social interaction such as online help chat, peer-to-peer discussions between users, and social media reviews (stars) and endorsements (likes). Web 2.0 also provides the opportunity to track the potential customers online behavior, target online advertisements, and gain access to their social networks

5. PESTEL Analysis

PESTEL analysis refers to the wider forces impacting on the business and the industry
Each country will have different PESTEL factors

Most businesses are part of a global supply chain and so the PESTEL factors a business needs to assess are transnational



Macroeconomic analysis

Macroeconomic analysis is a part of economic, political and currency risk. It is only examined rarely. The following provides a summary of the main macroeconomic factors and their likely effects:

Macroeconomic factor	Impacts	Consequences for business

The business cycle (or trade cycle)	• Mature economies tend to follow a four stage cycle
	  • Boom: very rapid rise in economic activity and prices
	  • Recession: initial slowing down of economic growth
	  • Depression: prolonged low demand and high unemployment
	  • Recovery: initial upturn in economic activity usually originating in the construction and capital goods sectors
		• Boom stage
		  • Higher demand for non-essential products
		  • Shortages of labour and supplies causing increased costs
		  • Government follows deflationary policies such as raising taxation, cutting government spending and raising interest rates
		• Depression stage
		  • Low general demand ('low aggregate demand')
		  • Tendency for prices and wages to spiral downward as spending falls and unemployment rises
		  • Government seek to improve economy by encouraging competition, increasing spending and reducing interest rates

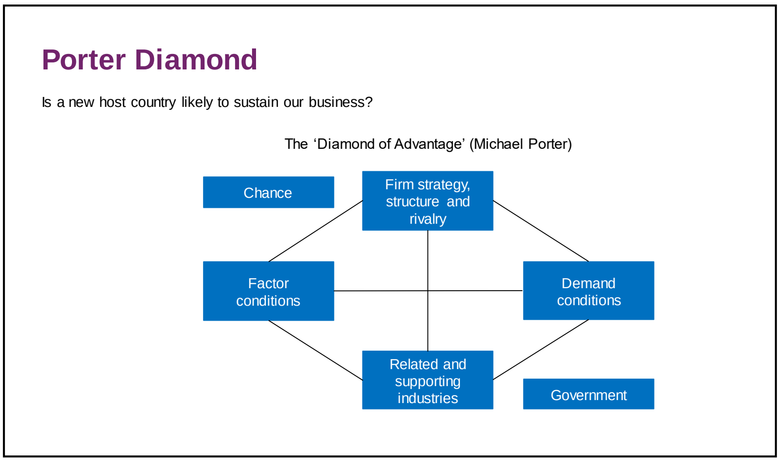
High rate of price inflation	• Reduces the value of money • Upward pressures on costs or labour and supplies • Tends to increase interest rate to maintain real value of returns to capital • Tends to reduce the exchange rate • At extreme levels (e.g. 2000%) can lead to desertion from money and collapse of market (hyperinflation)	• Loss of export sales due to higher costs • Higher costs of imported materials and components • General fall in demand due to lower value of money and counter-inflationary policies of government (see boom stage above) • Uncertainty in pricing and budgeting • Increased foreign exchange risk and need for foreign exchange risk management

Macroeconomic factor	Impacts	Consequences for business
Protectionism in trade	• Tariffs levied by domestic governments on imported goods and materials to discourage their use • Tariffs levied by foreign governments on exports from domestic country to discourage their use • Administrative barriers to trade (e.g. ease and speed of customs clearance, prohibitive changes to regulations on product design) • Total embargos on trade (trade sanctions) to gain political leverage • Tendency for protectionism to degenerate into successive rounds of retaliation between conflicting nations (e.g. between USA, China and EU)	• Loss of foreign sales • Increasing costs of imported components and materials • Pressure to ensure multi-national production to get behind protectionist barriers
Foreign trade imbalances	• Particular countries suffer external *deficit* ('balance of payments deficit) through its population spending more on imports than it earns through exports • Other countries have external surpluses • In the short run the deficit nation needs to exchange its currency for foreign currency and this weakens its foreign exchange rate • Deficit nations may be forced to borrow foreign exchange from international banks (recycling the funds from the surplus nations) and institutions (e.g. IMF) • Longer term may lead to *structural readjustment* of the economy such as austerity programs, sale of state assets, sharp reductions in government welfare programs	• May lead to protectionist policies (see above) • Foreign exchange rate instability in deficit nation • Danger of deliberate creation of recession by government to remove external deficit, usually though deep cuts to government expenditure • Potential for opportunities to buy industries and assets sold by the state and to gain state funding for export business investment

6. Porter's Diamond

Porters Diamond is a framework for assessing the conditions in a country of operation, and whether they are favorable to our business or not. In the SCS Assessment you may find this framework helpful for four sorts of questions:

1. What are the reasons our business is successful in its home country?
2. What information should we gather before investing in taking our business to a new country?
3. What risks are there in going to a new country?
4. How can we assess the attractiveness of investing in the new country?



The main factors to consider are:

- Demand conditions: in effect the Addressable Total Market (ATM) which involves assessment of what product/service specification is needed, likely pricing point, the number of potential buyers and their spending power
- Factor conditions: the resources available to support the business such as land to build on, labour availability and costs, skills, the availability of natural materials
- Related and supporting industries: the supply-chain partners such as transportation, energy, components, technological infrastructure and support
- Firms strategy, structure and rivalry: this means the competition, who owns them, how the compete, and their strengths and weaknesses

Two other factors include:

- *Government*: this means what assistance or obstacles the government might place in the way of inward investment and competition such as grants, low taxes or high taxes and obstacles to staff working in the country
- *Chance*: this is added by Porter to explain why sometimes firms succeed or fail despite the unfavorable or favourable configuration of the other factors

ACTIVITY 5.1

Context

Pixlwizz develops and sells video games. It is in the same industry as PlayStation and Xbox. It relies on skilled and educated young programmers and digital artists to develop its games. It also sells its games predominantly to people under 35 years of age, most of whom view them on mobile devices or by streaming them to home PCs and consoles.

A recent press article reports a growing trend for younger people to seek a better work/life balance and showing a preference for shorter and more flexible working hours, and for working most of the week from their homes.

Task

The CFO of Pixlwizz has emailed you to ask you to evaluate the following:

- The impact of these changes in workforce motivation on Pixlwizz's routine PESTEL analysis

[sub task = 40%]

ACTIVITY 5.2

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context

Couchweb is a provider of subscription video on demand services (SvoD). It is similar to Netflix and Disney+. It streams content to over 200 countries, and generally requires a license from the government of each country to stream content there. It does not accept advertising.

At present Couchweb does not permit its subscribers to record or download the programs it streams.

Task

You have received an email from Martine Andels: Director of Finance of Couchweb

To: Senior Manager
From: Martine Andels: Director of Finance
Topic: Music streaming service
Javier Mercot has asked for my comments on Couchweb launching a music streaming service. He has described his proposal in the attached memo.

What factors should we consider about the industry that may affect the success of this venture?

[sub task (a) = 55%]

What should an assessment of our present movie subscribers and of potential new music subscribers include?

[sub task (b) = 45%]

Extract from memo from Javier Mercot
Firms like Amazon, Apple, Tencent and Google offer music streaming services. We need to do so as well.

We can offer two grades of service: free streaming but paid for by advertisements, or a monthly subscription without advertisements. Unlike our TV product I want subscribers to be able to download and keep some music each month, and have the ability to pay extra if they want more than the monthly allowance.

Solutions to Activities

5.1 PESTEL analysis at Pixlwizz

Sub task
[Evaluate] the impact of these changes in workforce motivation on Pixlwizz's routine PESTEL analysis (40%)

This task is based on a sub task in the August 2021 exam. The Examiner expected candidates to show and understanding of the purpose of a PESTEL analysys, i.e. to spot issues affecting the future success of Pixlwizz's business model, and to recognize that a social change like this would have implications for several of the elements in a PESTEL analysis. Note the solution below speculates on what this change might lead to. This shows the purpose of a PESTEL analysis clearly.

Impact of changes in workforce motivation on routine PESTEL analysis

The trend toward seeking different work/life balances and also more remote working, if continued, will affect several areas of PW's existing and proposed new business. Some of these impacts can be monitored using the PESTEL framework.

Social
This trend is mainly social in origin, coming from demographic changes and also from changes in tastes and preferences. PW will need to assess how to retain and attract key staff, perhaps by offering more flexible working arrangements. It also needs to consider the effects of home working on isolation of its staff, which could mean some staff would leave. But similarly such remote working, flexible work times and so on may increase the demand for games which would benefit PW's business.

Political
This social change may impact on political forces, such as lawmakers seeking to respond favorably to these demands or, alternatively, trying to support management and the office property sector by discouraging them. PW will need to monitor how policies change in response to this, for example lawmakers may encourage this trend as a way to reduce the emissions caused by mass transit systems and office complexes. Alternatively they may put in place policies to rescue cities, and commercial areas, for example by improving mass transit systems or setting trends by requiring state employees to continue travelling to their place of work. PW has employees and it has 4 centres around the world, so it is a affected by this trend too.

Legal
If the social trend becomes stronger the lawmakers may respond, for example by imposing an obligation on employers to offer home working, and to introduce protection of employees from harassment or discrimination against them for asking for home working or time off work. PW will need to ensure it complies with these laws but also that it takes them into account in its manpower planning, recruitment and development activities.

Economic
Increased home working will reduce the jobs and incomes of those serving office workers, such as those working in transport, lunch stands, stores and so on. This could impact on the demand for games. The more flexible working may also affect incomes if people decide to work less hours. The government might use tax incentives to encourage or discourage staff and employers to embrace home working and flexible working.

Technological
If remote working becomes more widespread it may mean that employers, communications providers and government may work to improve the bandwidth of telecommunications. Employers may also provide equipment such as better PCs and screens to help home working. These may be useful in promoting gaming and making more complex games playable by more customers. PW should monitor this as part of its game development strategy.

5.2 Music streaming at Couchweb

Sub task (a)
What factors should we consider about the industry that may affect the success of this venture? (55%)

Approach: the question does not mention business ecosystems, nor does it ask for Porters Five Forces or PESTEL Analysis. But recalling these frameworks can help generate sufficient ideas to answer this. Avoid discussing customers in your answer to sub section (a) otherwise your answer will be repetitive in the sub section (b)

Industry factors to consider

Competitors

The proposal from Javier Mercot makes clear that there are already rivals in the music streaming market. They will have already attracted most of the subscriptions, and will retaliate against market entry by Couchweb.

It would be important to assess the quality of the service the competitors presently offer, the pricing points of both the subscriptions and also the advertising rates changed to firms. Couchweb will need this information to identify a market space for them, such as an overlooked segment or an offering that will be better than the established competitors' services.

It would also be important to understand the strategic value of the music streaming business to each rival. For example, is the service a major income source for them, and whether it provides other benefits such as enhancing sales of their smartphones, or additional data that can be used to sell or target advertisement to. This information would help Couchweb to forecast the likely strength of their competitive response.

Suppliers of recorded music

As Couchweb has found with television programs and movies, suppliers of content can exercise supplier power and may withhold content if they intend to launch their own music streaming service.

Couchweb needs to assess who owns the copyright to the most popular artists, and whether they are willing to provide the music and at what price.

Couchweb should also assess the strategies of these suppliers, for example do they plan to launch a streaming service, or are they willing to engage with Couchweb's platform. Their ownership should be examined to assess if there is a danger that they may be purchased by another rival platform.

In some cases, the attitudes of major artists might be sought. For example, Taylor Swift withheld her songs from Apple following a fee dispute, whilst other artists have shown a willingness to provide exclusive content to particular streaming services.

Regulatory considerations

Recorded music is a new industry for Couchweb. It will be governed by regulations on intellectual property rights and acceptable content that Couchweb needs to understand.
There will also be regulations around the advertising within the streaming service. This will include permitted content and product promises, regulations on collection of listener data and over targeting listeners.

Failure to understand and accommodate these regulations could lead to fines or a failure of the business model, as the original Napster found when its file-shared songs and breached copyrights.

Quality of information networks and receiving devices

Couchweb must ensure that target customers can receive the music without interruption. This will depend on the available bandwidth in each target market, the coverage of the networks, and the quality of the listening devices.

The streaming service will need to be made appropriate to the technological limitations in each market, or otherwise Couchweb will need to focus only on those markets with the ability to receive the music streaming.

<u>Willingness of advertiser to pay for advertising</u>
The business model of the free streaming service depends on funding from advertising. It is important to assess whether this funding will appear.

Couchweb presently has not experience of attracting advertising.

Potential advertisers, or media buying agencies, should be approached and their attitude to the new service assessed. They will want to understand who the streaming will attract, and in what numbers.

Sub task (b)
What should an assessment of our present movie subscribers and of potential new music subscribers include? (45%)
Approach: note that this sub task implies two questions: present movie subscribers AND potential music subscribers. The marking guide will have separate mark allocations for each, of about 8 marks each. Structure your answer to keep them separate.

Assessment of present movie subscribers
An obvious, and identifiable, market for the music streaming service is Couchweb's existing television subscribers. These need to be assessed as critical to up-take and revenue from the services. This will affect subscription revenues and, for the free service, advertising revenues.

<u>Potential to purchase music streaming subscription</u>
Assessment should be made, in each country, of the likelihood that existing subscribers will also purchase a subscription to the music service.

This may involve experiments such as limited access to the service for free, and market research to establish the optimum pricing point.

<u>Likely tastes in music based on analysis of viewing behavior</u>
Couchweb already has considerable data on its subscribers and their tastes based on their choices of viewing and amount or viewing. This could be used, via data analytics, to reveal potential tastes in music such as preference for older or classic songs, orchestral music, or more modern songs.

The crossover of movies into having soundtracks of particular genres of music, and the crossover of actors into music stars, such as Ariana Grande, or the reverse, such as Lady Gaga, will help identify the sorts of music to feature in the initial service and promotions to gain adoptions.

<u>Attitude to inclusion of advertising and downloads in Couchweb service</u>
Several aspects of the music service are different from the television offering and could have unintended consequences.

The inclusion of advertising in a free service may make some Couchweb subscribers call for a similar free television service, whilst others may fear that Couchweb will start to interrupt their viewing with advertisements.

The availability of music downloads will build pressure for the same to available from the television streaming service. This would be unfortunate and should be assessed before launching the service.

<u>Preferred platforms and functionality of service</u>
This information would also apply to potential music subscribers who are not presently subscribers to Couchweb's television services.

The platforms relate to how listeners access the service and the devices they use to listen to it and to download it. For example, will it be on laptops, smartphones, televisions, or tablets? Factors such as download speed, music file size, and the appearance and navigation of the site will be influenced by this.

The functionality of the service refers to what the listens want it to do. For example, will the songs be accompanied by videos of the artist, and how can the songs be searched for and whether there are automated playlist functions.

<u>Attitude to present methods of listening to music</u>
Presumably Couchweb's television subscribers already have a source of music. The new streaming service would aim to improve on these existing methods (e.g. rival streaming services, private CD collections, file sharing sites and so on)

Subscribers should be assessed on the strength and weaknesses of their existing solutions, and their attitude to the features and benefits of Couchweb as a streaming service should be assessed.

This would provide insights into gaps in the market and the necessary characteristics for the new service to gain users.

Assessment of potential music subscribers
In addition to Couchweb's television subscribers, it needs to attract subscriptions from other too.

<u>Circumstances in which they listen to music</u>
Couchweb understands television viewing behavior but not music listening behavior. For example, music can be listened to while engaged in other activities such as driving, working, and travelling.

Knowing how music is listened to will help design the interface for the service, the promotional messages, the features of the service such as playlists, and the devices on which it will be heard (earphones, home smartphone docking stations, via home hi-fi systems, car audio systems and so on) which will affect the desired audio quality.

Getting these matters right will help improve adoptions. This information is relevant to adoption by Couchweb subscribers too.

<u>Potential subscriptions to music streaming service</u>
The level of adoptions of either service will affect revenues from subscriptions or from advertising.

Some assessment should be made of initial and long-term adoptions at each price point to establish an optimal price and entry strategy for the subscription service.

<u>Distribution channels to reach new subscribers</u>
Couchweb needs to consider whether to offer its service direct to non-subscribers, or whether to use alternative channels such as offering the service in a bundle with television streaming, offering through mobile network providers or in conjunction with handset makers, or possibly other consumer outlets and brands such as Starbucks or fashion brands.

<u>Promotional methods to attract new subscribers</u>
Couchweb can use its television service to promote its streaming service to its subscribers. It cannot use this to reach people who are not its subscribers.

Couchweb needs to identify promotional channels to reach these potential customers, for example advertisements on social media and conventional press and music magazines and websites. There also needs to be a promotional message and attractive pricing.

<u>Potential for new subscribers to also buy television subscriptions</u>
An attraction of this strategy is the possibility that offering both music and television, perhaps in a single bundle, will enable Couchweb to rival services such as Amazon Prime.

This means it could lead to a rise in subscriptions for television streaming.

The potential customers should be assessed for the reasons they don't presently view Couchweb, for example lack of popular series, inability to download, poor bandwidth and interrupted programs.

These can be addressed to ease the potential for picking-up additional television subscribers on the back of the streaming services.

Chapter 6
Drivers of Change in the Business Ecosystem

1. Overview

The purpose of this Skills Workbook is to prepare you for the CGMA final assessment, the Strategic Case Study.

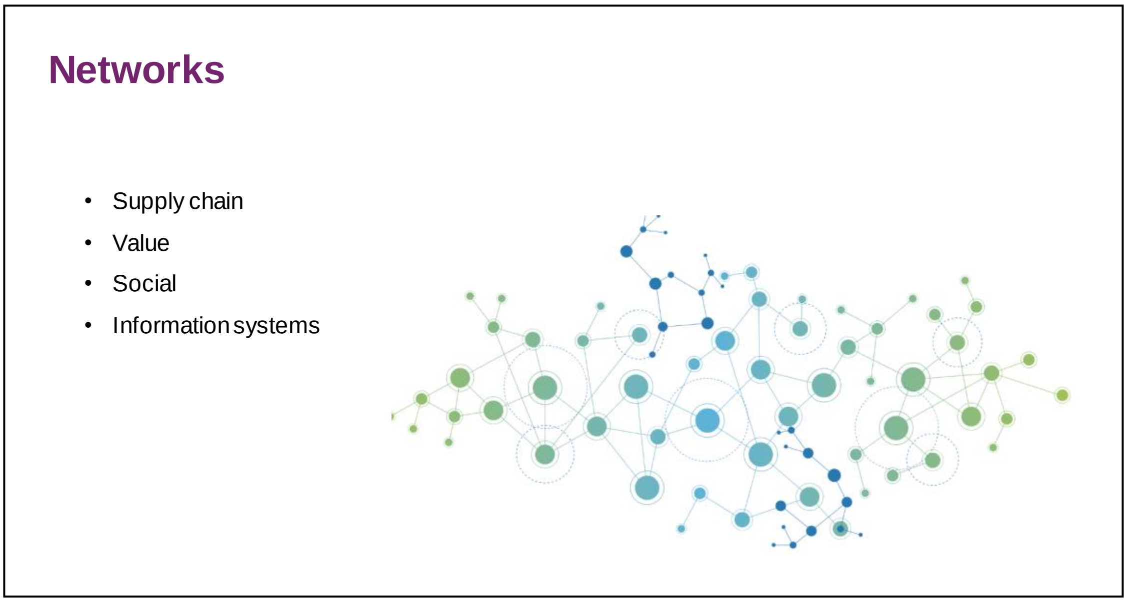
CGMA recognizes a wide range of institutional, social, market, technological and ecological drivers for change in the business environment. But for exam purposes you need to recognize these in the context of the pre-seen case for your exam. Therefore, this chapter provides an overview of each, but not how it relates to your specific case study.

The program materials for your CGMA/CIMA course will provide specific coverage related to your specific case study. But also, the topics in this chapter will help you conduct your own analysis and investigation. *Student tip: enter a technical term in this Chapter in a search engine, followed by the name of the industry in your case to understand the ecosystem for the specific industry.*

2. Networks

A network describes a set of linkages which can change and be added to, but in which all parts are affected by a breakage in another part.

Businesses are often described as being part of networks and are affected by changes in the networks



Disruption to the ecosystem can be understood in terms of its impact on these networks. Disruption may come from changes in these networks or in their ability to function.

Supply chain networks
This describes the total sum of organizations involved in supplying a customer with an output. It stretches *upstream* of the firms back through the sources of its materials and components, and *downstream* through its delivery channels.

The supply chain networks can be disturbed by:
- Business failure of a member of the chain
- Logistical difficulties resulting from climate, congestion, or problems in cross-border trade (e.g. protectionism)
- Changes to ordering, tracking, or payments systems (e.g. blockchain)
- Investment or lack of investment by members of the network

Value networks
Businesses rely on value networks. It is the collaboration between the parties in the network, and the number of actors in the network, that enable it to create value, and therefore enable the participants to capture some of that value for themselves

<u>*Examples:*</u>

Network	Actors	Outcomes created (value)
Medical services	Universities, hospitals, governmental bodies, pharmaceutical firms, consultants, public health organizations, investment funds, schools, social care organizations, sports organizations	• Better health outcomes • Research • Profits and shareholder wealth • Fitter labor force • Export earnings • Employment • Happier and longer lives • Government revenues
Automotive industry	Dealerships, repair shops, insurance firms, government licensing bodies, universities, credit providers, car manufacturers, car wash centres, driving schools, , component makers, GPS service providers, drive –through restaurants, training providers, filling stations, automotive design houses, motoring media and television, race events, team sponsors, recycling industry, metal foundries, gasoline providers	• Better communication between people and regions • Research • Profits and shareholder wealth • Enhanced commerce and supply chains • Enjoyment of hobbies • Aesthetic designs • Greater personal safety • Employment and careers • Government revenues

Recognizing value *networks* takes us away from earlier notions of firms as a part of a *pipeline* that takes inputs from suppliers and sells outputs to customers. For example, what is the relationship between a car manufacturer and a firm that runs filling stations? Mutual dependence: if one did not exist then neither could the other, and both gain if the total size of the industry grows (as car makers wishing to produce electric and hydrogen cars are trying to do by extending charging points and finding filling stations willing to sell liquid hydrogen).
This leads to the notions of *collaboration* between actors to increase the value in the network, instead of *competition* between them to get the largest share of a fixed amount of available value.

Social networks
These are networks of people and institutions. They are important for understanding, and affecting:

- the dissemination and adoption of new ideas and discoveries
- influence over beliefs, fashion, or government policy
- the passing across of business opportunities
- access to and exercise of power in an industry or society

These social networks are disrupted by changes in methods of personal communication (e.g. social media technologies), and the arrival of new actors (e.g. tech entrepreneurs into established industries).

Commercial applications include:

- promotion of product awareness by hiring/creating 'influencers' with large followings on social media
- creation of games, group deals, instant messaging and celebrity live-streaming to support e-commerce sites and channels
- viral marketing through encouraging posting of short-form videos and memes that support the product
- monitoring of social media to understand customers tastes and interests

Information systems networks
CGMA will focus mainly on disruption to the ecosystem resulting from changes in information systems networks. But changes in information system networks have no impact of their own. They underpin many of the supply chain networks, value networks and social networks descried above. This means that changes in information systems networks will impact on these other networks.

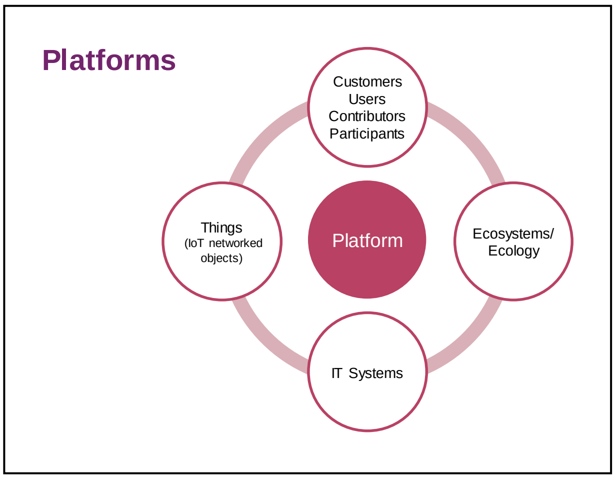
Examples include:

- Enterprise Systems (Enterprise Resources Management, Knowledge Management)
- Internet and 4/5G telecommunications networks
- Satellite based GPS systems
- Cloud based computing and web services providing hardware as a service (storage and processing) and software as a service (applications technologies) and analytics
- Video conferencing and instant messaging services
- File sharing and chat forums (e.g. Facebook, Dropbox, LinkedIn, Reddit)
- Crowdsourcing sites for goods, services, labor, finance, ideas (MTurk, Uber, Upwork, PiggyBee, HeyParcel)
- Electronic marketplaces (eBay, Alibaba, TotalJobs.com)
- Handheld, desktop and wearable communication devices (e.g. smartwatches, smart eyewear, smart clothing)
- Networked devices: the Internet of Things (Radio Frequency Identification RFID, smart appliances, vehicles, handhelds and wearables)

3. Platforms

The terms 'platform' and 'platform strategy' have become very common. The term is derived from IT but has become more generalized.

Technology platforms
A group of technologies that are used as the base on which other applications, technologies, or processes are developed
Examples
- Operating system platforms: Linux/Android, Windows
- Computing platforms: Google Cloud Platform, Amazon Web Services
- Database platforms: Oracle, MySQL, PostgreSQL
- Game platforms: Xbox Live
- Content management system platforms



Platform strategy
An approach to entering a market, or creating a new industry, by allowing participants to benefit from the presence of others, usually mediated by an information technology.

The principles of a platform are
- Platform owner creates the platform and takes value from it (e.g. membership fees, license fees, commissions, introduction fees, extracts data on the participant to use, gains ideas and intellectual property)
- The value of the platform is maintained and increased by the contributions of the participants, the communities, and the networks that use them
- The greater the number of participants, the greater the value of the platform to participants and to the platform owner (e.g. more riders and drivers using Uber, more Netflix subscribers and more movie-makers licensing content to Netflix)
- The platform is moderated by the platform owner (e.g. removing posts and participants, ensuring quality of apps and content, ensuring personal safety and security of participants)
- This creates a *platform ecology* that sustains the participants and gives then value (or not, if the ecology is poor)

A platform strategy involves:
- Developing a platform and expanding the participants
- Maintaining the integrity of the platform
- Using the platform as basis for further platforms to utilize the technology, brand, or subscriber base (e.g. LinkedIn Premium, UberEats/UberFreight)

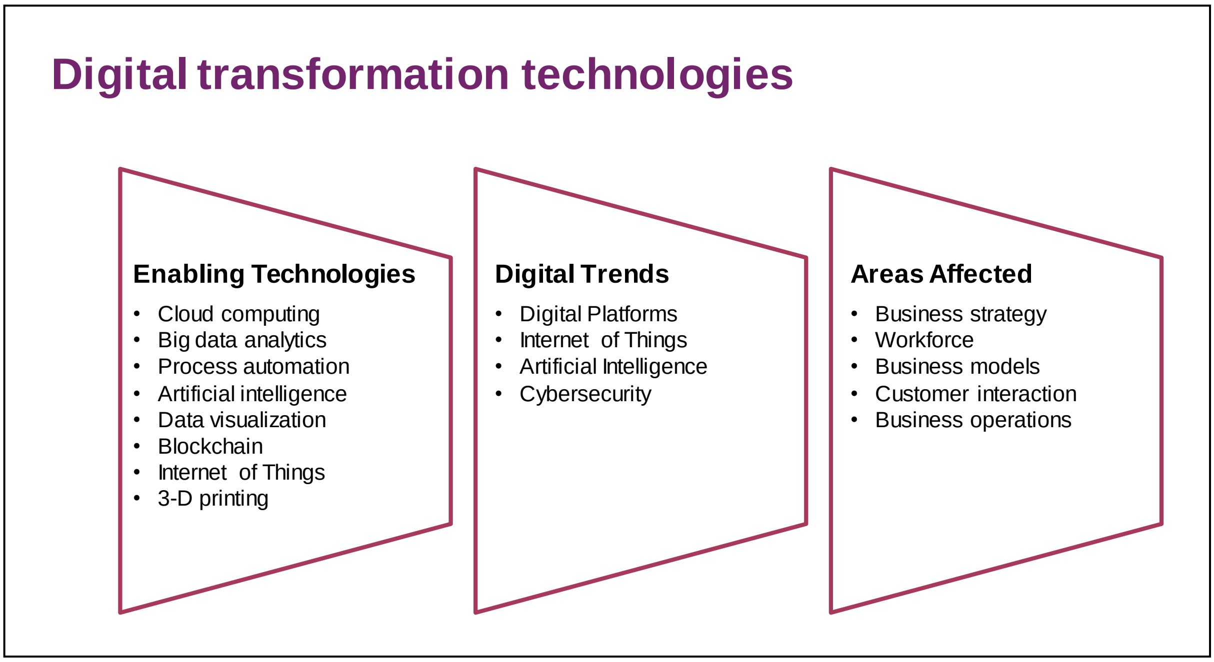
Examples:

Classification of platform	Secondary classification	Examples
Intermediation Markets	• Social network sites • Social network software • Aggregators and price comparison sites	• Facebook, Linkedin, Tencent Weibo • Snapchat, TikTok, WhatsApp, WeChat • Skyscanner, Uswitch, Pronto, Pandacheck
Audience-making markets	• Portals • Video and digital content	• Amazon, QQ.Com, QQ Game.Com • Tencent Video, Netflix, New York Times, Tencent Comic, Tencent Literature, Kindle, Google Books

Shared-input markets	• Game platform • App Store	• Steam, QQ game platform • MyApp, GoogleStore, App Store
Transactions based markets	• Crowdsourcing • Third party payment	• Uber, Lyft, Didi, Airbnb • Google Pay, Alipay, Bitcoin

Digital platforms have the potential of extracting value by putting themselves at the center of the networks discussed earlier.

4. Digital transformation technologies



The CGMA has specified a number of digital technologies relevant to your examination. Several of these rely on technology and associated platforms of the types discussed above.

Each of the technologies reviewed are themselves IT Platforms. To access them a business must contract with a platform provider, inevitably a cloud provider, to use their technology.

Cloud computing

- Internet-based data centers operated by third parties including hardware as a service (HaaS) and software as a service (SaaS)
- May be limited to a single organization (enterprise/private/government cloud) or available to many organizations (public cloud)
- Enables users the benefits of high-power computing on-demand but without the need to invest in or maintain a large data center of their own
- Charged on a pay-as-you-go basis which reduces fixed costs
- Increasingly used due to the growth in digital applications (see digital technologies discussed below)
- Cloud operators claim that centralization of IT services in a specialist operator gives greater expertise and security, transfers operator risk and yields the benefits of low-cost locations
- Examples include Amazon Web Services, Microsoft Azure, and Google Cloud

Big Data and Analytics

This is discussed in detail in the next chapter

Process Automation

- Also called Business Process Automation (BPA)
- Technology-enabled automation of *complex* business processes with minimal human intervention
- BPA tends to work to set procedures and processes
- Robotic Process Automation (RPA) is an advanced from of BPA that uses software robots (bots) to observe humans performing less structured tasks on an IT system in order to learn from them and to copy them (e.g. it learns how a human receiving an email, rather than a formal purchase order, can extract relevant data, check it, enter it into a sales order system, and then respond to the email)
- Frequently applied to workflow (invoicing, order processing, ERP entries), payroll and on-boarding of staff, accounting reconciliations, business contract document creation)
- Benefits include greater accuracy/consistency of process, contains costs, and streamlines and speeds tasks

Artificial Intelligence

- Also referred to as Machine Intelligence (MI) or Cognitive Computing
- Development of information systems with much greater ‘human-like’ potential
- Utlises human language recognition/processing, information gathering and searching, techniques of problem solving, and an ability to learn. There can be paired with an ability to manipulate objects (robotic ability)
- Already evident in applications such as autonomous vehicles, route guidance, medical diagnosis and prescribing, and IT interfaces such as search engines, handwriting and voice recognition, chatbots (Siri, Alexa, Xiaoice), and predictive texting and through the replacement of a growing number of routine jobs by AI
- AI is forecast to reach a take-off point of explosive growth (referred to as a *singularity*) by 2025
- Benefits of AI include:
    - Improved quality of decisions through elimnation of human error and emotions
    - Reduced processessing costs through elimination of labour costs
    - Greater effectivness, innovation and learning from AI’s ability to be multi-disciplinary and use knowledge and data from wide sources
    - Liberation of people’s time from performing routine and trivial functions
    - Increased competitiveness for firms and economies who are able to harness AI
- AI raises a numer of issues:
    - Unemployment and social exclusion of persons who traditionally performed the tasks performed by AI (e.g. truck drivers, clerical workers)
    - Greater vulnerabilty to errors in decisions
    - Changed social control – who is now in charge, humanity? the owners of the AI? The AI itself?

Data Visualization

- Graphic representation of data to make it more intelligible to users
- Increasingly important as businesses pivot their businesses to greater use of data as result of digital transformation of their business model (e.g. big data, internet of things and so on)
- Reconfigures data into a form that provides information for users who need to make decisions
- Benefits of Data Visualization
  - Faster decisions and action
  - Better communication between different team members and facilitates discussion
  - Quicker identification of major issues or trends
  - Encourages interaction with data, such as via data analytics
- A related technology is Data Virtualization which is an approach to data storage and extraction that permits a user to gather data from all across the organization, regardless of who initially obtained it

Augmented Reality/ Virtual Reality

These both involve the use of electronic devices to change a user's perception of the world around them.

<u>Augmented Reality (AR)</u>

Adds to the user's information about the real world around them, often by the use of screens or eyewear, that provide information in addition to what can be seen.

Examples include:

- Video games which superimpose characters and games on the real world in front of the player's handheld device e.g. Pokemon GO, Amon, Knightfall
- Marketing/selling technologies that show the customer and their home with the make-up, clothes, spectacles, furniture etc that they are considering buying e.g. See-my-fit, IKEA Studio App, YouCam Makeup app
- Promotional devices which use AR to draw attention e.g. Pepsi MAX AR enabled bus shelters showing alien attacks in the street
- Inventory location applications that allow warehouse pickers, shoppers and others to scan an area and have the relevant items, and information about them, highlighted on the screen
- Tactical AR for military applications e.g. showing troops what is behind walls, superimposing virtual battlegrounds for training
- Technical/instructional such as allowing mechanics to point the device at the real world object and receiving instructions and illustrations of the approach to removing an engine part or on the right settings for technical equipment

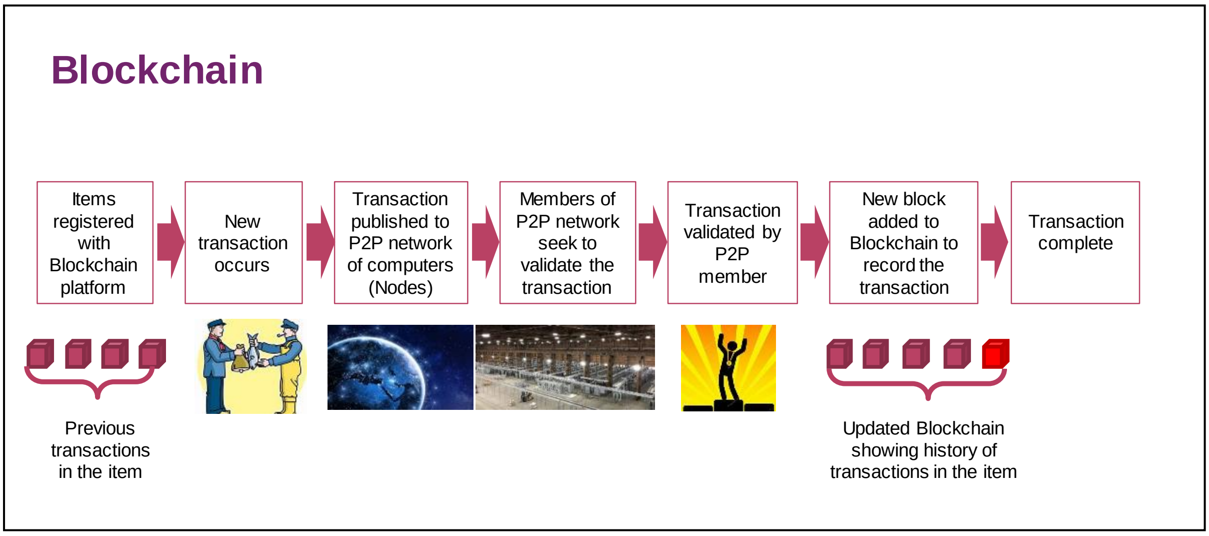
<u>Virtual Reality (AR)</u>

Dispenses with the real world by offering an online virtual world, often through the use of VR headsets.

Examples include:

- Healthcare: used to simulate complex operations for training or to control robots to conduct actual operations at long distance, to provide therapies for persons with mental health issues
- Gaming: providing 3D games environments with avatars of the other players.
- Metaverse: a network of 3D online worlds where each provider creates a platform/virtual containing to allow subscribers create avatars that can meet other persons/participants for the purpose of social interaction & dating, education, business meetings, game playing, and trading activities. The movie franchise ‘The Matrix’ approaches a Metaverse concept. There is a market in ‘crypto real estate’ in which real world businesses and individuals buy the rights to locations within a potentially-popular Metaverse in order to sell from or inhabit there online, and the fashion industry seeks to sell items to dress the avatars.

Blockchain
An electronic *public distributed notarised ledger* that has replaced the role of traditional *trust service providers* including currency issuing authorities, conveyancing lawyers, and land and property registers.



- Blockchain solutions are offered by platform operators e.g. Azure, Ethereum, IBM Blockchain and businesses pay to use their platforms
- Data items, termed *blocks* are stored electronically on the platform in a cryptographically secured form
- The data item can relates to an underlying asset, such as a consignment of coffee or a ticket to a sports match
- Each block has a unique identification number. Blocks cannot be changed without creating a new block and a new identifier
- Steps in the process
  - When the underlying asset undergoes a transaction the giver and the reciever each notifies the Blockchain platform
  - The Blockchain platform relies on third-party ‘miners’ to match the two halves of the transaction by undertaking complex mathematical analysis using supercomputers to verify the past chain, the validity of the transaction, and to publish their result
  - The first Blockchain miner to validate the chain and transaction recieves a payment from the platform. The payment varies with the type of asset and the strength of the encryption. The remaining miners receive nothing for their work
  - The valid transaction is recorded as a new block added to the end of the chain with a new number but which can be tracked back to the previous transactions of the item

This creates the *chain*, in effect a trail of blocks linking one to another, with each link in the chain denoting some change to the block or transfer of its ownership, for example the steps in which a medical file was updated, the transfer of custody as a parcel travels through a delivery process, or most famously the transfer of blocks of cryptocurrency from a buyer to a seller

- The security of the system is assured by the huge potential number of miners that scrutinise every transaction and are incetivised to spot fake transactions, or bad matches by their rival miners
- However criminals with sufficient mining power could potentially ‘corner the system’ and set up false blockchains by being the first to validate the fradulent transaction, and so create a fake block and divert the underlying asset to themselves. This reveals two flaws in blockchain:
    - Some chains are not mined by the majority of miners because the commission for matching is very low
    - Miners are profit-motivated and so do not check the entire length of a blockchain when validating a new transaction and, also cease trying to match a transaction once a winner declares they have found a match

The implications of Blockchain include:

- Potential for much faster and efficient transactions processing
- Improved security of transactions through better trails of provenance and transfer
- Elimination of many in-house transactions processing functions, and the jobs they sustain
- Greater opportunities for selling (*monetisation*) of information and data

However, at this time, many actors in supply-chains do not yet use blockchain and this means that adopters of blockchain depend on an incomplete chain of transactions and verification

Cryptocurrencies and Non-Fungible Tokens
These are specific applications of Blockchain technologies.

Cryptocurrencies: specific Blockchain currencies such as Bitcoin, Etherium, Litecoin or Tether which can be used to buy things within an online transaction or sold for an amount of a conventional currency, such as US$. They have become speculative assets which are traded on exchanges such as Binance or Hotcoin Global and are characterised by large swings in their values against conventional currencies with holders making large capital gains or losses on the swings in value.

Cryptocurrencies have been a large part of the FinTech revolution in borrowing, lending and exchange and have the potential to undermine conventional banking and also the use of the national currencies issued by central banks, not least because the owners are anonymous and so invisible to regulators. For this reason several countries have sought to regulate them but also to introduce sovereign/government cryptocurrencies tied to the value of the nation’s hard currencies to regulate the market and to benefit from the transactions advantages of cryptocurrencies.

Non-Fungible Tokens: a Blockchain asset that is ‘minted’ (i.e. created) to represent a single and unique physical or digital asset, such as a digital artwork, song, winning goal, or video clip. They are called non-fungible because they are unique and their value is unique, whereas a unit of crypto currency is identical to another unit and its price is set on the exchange. i.e. a Bitcoin is identical to and always worth the value of another Bitcoin.

The owner of the NFT may be recognised as the sole owner of the underlying asset and obtain rights to how the asset is used or displayed. Alternatively the NFT confers no rights over the underlying but rather its value is somehow enhanced by the innate popularity of the underlying unique thing to which it relates. Therefore they are speculative assets and buying them is similar to betting on a racehorse where one can own the betting ticket but still not own the racehorse. Frequently the actual asset is already in wide circulation, such as downloaded songs, artwork or video clips, but the person who buys the NFT relies on its being unique and the thing it represents being known-about, popular and conferring some prestige on its owner. The value of NFTs is entirely determined by market forces of demand for them and their scarcity. Celebrity involvement and endorsement has been influential in raising the value of NFTs, particularly if the Blockchain record shows a celebrity has owed it previously. Many of the characters and locations in the metaverse (see above) are sold and traded as NFTs.

It is hard to resist the suspicion that a major attraction of cryptocurrency and NFTs is that the anonymity of ownership and trading can permit money laundering and tax evasion, and the hype around them enables expolitation of the foolish.

Internet of Things (IoT)
This is a system of interrelated computing devices, digital machines, digitally enabled objects and animals or people. Each item has a unique identifier (UID) which enables it to be tracked and operated remotely through an information systems platform.
Applications of IoT include:

- Smart homes: lights, alarms, climate control, security and entertainment systems form an ecosystem that can be controlled remotely by smartphone app, or by occupants of the home using voice commands
- Medical and healthcare: called the Internet of Medical Things (IoMT) it provides remote monitoring of patients (exercise, heart rate, alcohol and sugar in bloodstream) via wearable monitors, voice recognition systems and intelligent monitors in the home listening for falls, lack of movement, labored breathing and so on, and monitoring whether medicines have been opened and used. Some regular consumer devices are being repurposed to become, in addition, smart devices to monitor the health of the user, for example contact lenses, toilets and electric toothbrushes.
- Transportation through vehicle-to-vehicle/-everything communications to avoid accidents, adjust speeds and route to conditions on the smart highway, permit ride hailing, autonomous vehicles and so on
- Agriculture through sensors monitoring rainfall and soil condition, pest infestation and crop readiness to help farmers decide on watering and fertilization, harvesting and so on.

IoT provides enormous amount of information. For example Google division Sidewalk Labs is using an area of Toronto waterfront as a ‘sandbox’ to develop a ‘smart city’ including optimal layout and design of accommodation and services, self-driving delivery vehicles, automated snow-clearing pavements, and optimized traffic and parking systems for autonomous passenger vehicles.

3D Printing
Also called additive manufacturing it takes Computer Assisted Designs (CAD) and creates three-dimensional objects by successive adding of layers, or sometimes by cutting away unwanted material to leave a 3D object. The materials used include synthetic plastics, food proteins, fabric fibers and polymers, and active pharmaceutical compounds.

3D printing is presently mainly used for:

- Rapid creation of product prototypes
- High value urgent replacement components for manufacturing processes, mining and aviation

Potentially it may be developed to provide:

- Household goods and furniture
- Food
- Clothing
- Prescription pharmaceuticals

The implications of 3D printing are:
- Potential to create unique customization of products (medicines, clothes, food)
- Simplification of supply chains and reduction in emissions
- Disintermediation of manufacturing and logistic industries
- Increased importance of maintaining control over cybersecurity to protect Intellectual Property and to ensure reliable supply chains and safe end-products (e.g. imagine someone hacking a drug prescription)

5. Drivers of change

The CGMA has specified a number of specific drivers of change

Globalization
This refers to increased interaction and integration between people, industries and governments worldwide. Its effects can be subdivided into
- Economic globalization:
- Cultural globalization
- Political globalization

Its initial drivers were:
- Improved information communications: enabling co-ordination of global subsidiaries, supply chains and partners and disseminating cultural artifacts including movies, social media, and news media
- Improved transport infrastructure: including faster and larger ships and aircraft, better docks and dock-handling
- Political realignments and opening-up: including reductions in protectionism via the World Trade Organization, and the opening-up of economies in Eastern Europe, India, and China

It has four main aspects:
1. Increased trade and transactions
2. Improved capital and investment movements
3. Greater migration and movement of people
4. Greater sharing and dissemination of knowledge

Factors resisting and reversing globalization include:
- Increased nationalism and ‘country-first’ policies
- Hostility and resistance to migrants
- Criticism of poor terms of trade that allegedly enable rich countries to exploit producers and migrant labor from weaker countries
- Concerns over the emission impacts of the global supply chains and travel and the concern that poorly regulated countries are using social dumping to harm better countries and reduce standards generally (social dumping refers to gaining competitive advantage by adhering to poorer worker and environmental protection than other client countries)

Geopolitics
Technically-speaking this refers to the study of the impact of geographical factors on international political policies and relations. These geographical factors include location, land topography, population size, water and food resources, and so on.

A major geopolitical trend affecting the business ecosystem is global climate change. This threatens to lead to

- Increased incidents of extreme weather (typhoons, wildfires, flooding)
- Competition for water supplies
- Migration of populations
- Loss of productivity and output in some industries (e.g. lower fish stocks, loss of arable land, migration from the villages, plant blight and disease)
- Imposition of regulations to halt emissions and the introduction of carbon taxes
- Civil and national wars

It can also mean 'international political relations'.

Therefore, consider also:

- New political alignments
  - G4 countries seeking bigger role at UN (Brazil, Germany, India, Japan)
  - Belt and Road countries within China's strategic trade initiative

- The development or erosion of political and economic groupings
  - G20/G7 countries
  - The World Economic Forum
  - European Union
  - BRICS (Brazil, Russia, India, China, South Africa)
  - United Nations

Demography
This means the size, age composition and location of population.
Some key trends:

- Aging populations in many developed countries and in some developing countries due to lower birth rates and migration of young people to countries with better job prospects
- Migration from village to city in developing countries leaving older people left behind
- Increased life expectancy in most developed and developing countries
- Increase in living alone
- Increased diversity in populations (due to migration, individualism in lifestyle, different domestic circumstances, different levels of income and wealth)

Customer empowerment
Four stages in customer empowerment due to digital technologies:

1. Demand-based power
   Development of websites and search engines increased consumer access and choice and put a downward pressure on prices. It also led to an increase in the range of choices firms made available to customers to ensure they obtained a sale, with consequent increases in costs

2. Information-based power
   The development of Web 2.0 (faster and more interactive) allowed customers to demand more information about products to help them choose (e.g. sample chapters/songs, visualization of products, reviews and 'likes') and it also enabled them to influence the market through uploading reviews and ratings, to bid on items, to express their demands for producers to note and respond to (e.g. uploading pictures of things they like and don't like, comments, blogs). Some markets feature online opinion leaders

3. Network-based power
   Products and content were enhanced or changed online by consumers who disseminate it though their network groups. Examples include re-mixed and hacked music and video

content, posting pictures and image memes, publishing reviews of items, and engagement in forums. These all enable consumers to express themselves and also influence others in positive or negative ways towards the producer and their products

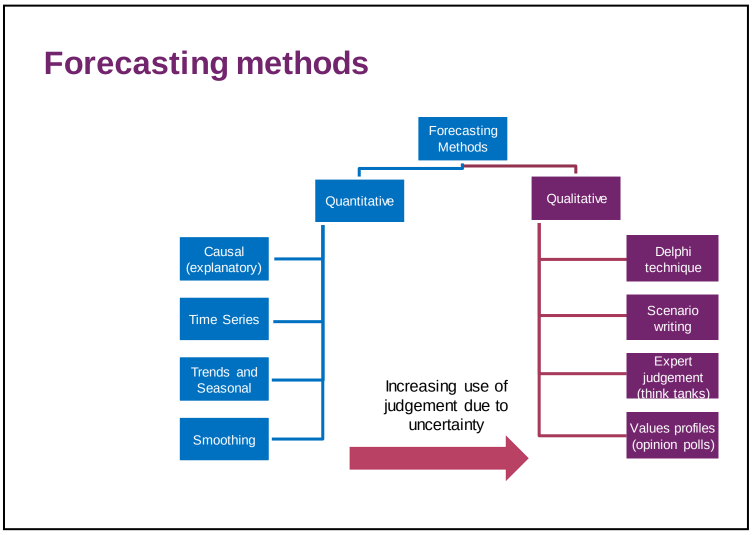
4. Crowd-based power
   This involves mobilizing large numbers of consumers through the use of digital media. Sometimes called 'crowdsourcing' examples, crowd creation and sharing (such as Soundcloud and Wikipedia), crowdfunding, crowd support in peer-to-peer problem solving (e.g. user forums)

Strategies to deal with consumer empowerment include:
- Facilitation of the interactions: firms like Microsoft establish and monitor peer-to-peer problem solving to replace proving online help themselves
- Close monitoring of the interactions and tastes of consumers: for example, some song-writers will upload a small part of a song through a chat group and monitor responses before writing the rest
- Engagement with influencers in the design and marketing of products
- Customization of products for the individual consumer

6. Forecasting

Uncertainty
This is defined as 'the problems in making reliable assumptions about the future'
Uncertainty is increased by two factors:
1. Greater complexity: this means the difficulty in understanding the factors affecting the future such as caused by having a greater number of competitors, or new and unfamiliar government or technology disturbing the ecosystem
2. Greater dynamism: this refers to the rate at which things change, for example quicker changes in customer tastes, or more rapid development in digital technologies

Forecasting the business environment
Forecasting tries to reduce *uncertainty* to allow management to make decisions.
You are not required to have a detailed knowledge of the *Quantitative* (i.e. statistical) techniques because these have been assessed at earlier levels



- *Causal/explanatory*: build a mathematical model that shows the impact of different variables on the factor under investigation (e.g. impact of product price, rival price, and consumer incomes on demand for furniture), or a regression analysis (e.g. impact on change in price on quantity demanded)
- *Time series*: extrapolating from past data using techniques such as method of least squares
- *Trends and seasonal*: using past data to model changes in demand or prices, particular where there is a seasonal element

- *Smoothing:* using past data to project but establishing underlying trend through exponential smoothing

A drawback of Quantitative techniques is that they are only valid if the past behavior on which the data is based is likely to continue into the future.

The use of *Big Data and Analytics* (discussed later) has increased the reliability of quantitative techniques of forecasting.

Qualitative techniques are part of the CGMA syllabus at Strategic level and therefore knowledge is needed:
- *Delphi Technique*: panel of experts assign probabilities to events happening on a scale of 0 to 1 and the medium average is used for the forecast. Sometimes experts can be asked to justify very high or low assessments to the group to help improve group understanding and modify the average
- *Scenario writing/planning*: experts write narrative pictures of the future by analyzing how present trends may play out
- *Expert judgement/think tanks*: management are advised by people who believe they know the future of something (e.g. technology, global politics, consumer tastes and so on)
- *Values profiles:* research by interview with people, notably young people, about their values and what they want. These will drive demand, and regulation, in the medium-term future.

Please see the solutions to the activities in this chapter to gain more insight into how these forecasting techniques work in practice.

ACTIVITY 6.1

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context
AutoAuto is a listed company that makes intelligent electronics to sell to automobile makers. It is based in the country of Upland. Much of its research and development expenditure is into developing technologies for driverless cars. These are cars that do not need to have any human intervention in their operation.

The share price of AutoAuto has been falling and investors are concerned at the amount of money being spent on developing these technologies when there is no clear evidence of future sales demand for the driverless cars.

Task
The following email is waiting for you when you arrive at the office
From: Andrew Jarvis, Director of Finance
To: Senior Financial Manager
Subject: Future strategy

Hi.
I'm sure you have heard about yesterday's shareholders' meeting. They are not happy. If you haven't heard I attach an article from the website of Upland Business Times.

We have an Executive Committee meeting in an hour to try to formulate a response.

Please write me a paper that I can show to the Executive Committee that explains the techniques we can use to forecast the likely date for commercializing driverless cars and forecasting the likely profits in the first five years

[sub task (a) = 55%]

ACTIVITY 6.2

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context

Knowledge City Technology Park (KCTP) is a publically listed provider of buildings for lease on a Science and Technology Park (STP). KCTP has links to a local university and also to large corporations and providers of investment capital. This means KCTP is a good location for start-ups that wish to exploit a new innovation. KCTP seems to specialize helping tenants engaged in biotechnology and in materials development.

Task

The following email is waiting for you when you arrive at the office

From: Shereen Peros: FD
To: Senior Finance Manager
Topic: Cancellation of 3D Printing project

The Board has decided to cancel the development of the 3D printing centre within the new building that is under construction. The main reason is that the increased popularity of 3D printing has attracted several large corporations into the industry and this will reduce the demand for the 3D facility on KCTP, and probably also the prices.

Changing competition is something we need to be aware of. But there are always changes in our business environment. We do need to take these environmental changes into account in our strategic decisions for KCTP.

- Please explain some of the main sources of the change that KCTP will face in its business environment and provide ideas on how forecasting may help KCTP's Board to incorporate environmental change into its strategic thinking

[sub task (b) = 60%]

ACTIVITY 6.3

Context

Shinepodd (S/D) manufactures and sells coffee pods. These are small plastic containers with ground roast coffee inside that can be put into a special machine to make fresh coffee at home or in the office. S/D's real world rivals would be Nespresso and Keurig. S/D sources coffee beans from across the world and has a mission that includes paying fair prices to struggling coffee growers. S/D is aware that its business model has adverse consequences that could impact on its sustainability including the pollution caused by poor disposal of its plastic pods and the impact of its buying operations on the incomes, water availability and agriculture of the developing countries that supply its beans.

Task

You receive the following email from Liela Dorcas: CFO of Shinepodd

From: Liela Dorcas: CFO
To: Senior Finance Manager
Topic: Transformational technologies

Please see the attached report. The Board has all been given a copy of it.

Anders Lunt, our CEO, is concerned that Shinepodd lacks a digital strategy. He wants to hold a focus meeting on how Shinepodd can undertake a digital transformation of its business.

I admit it. I don't know much about this stuff. I hope you will know something.

I would appreciate your advice on the following matters before I discuss this with Anders:

- I need a briefing on how blockchain and other transformational technologies may affect Shinepodd's business

[sub task (a) = 50%]

- Secondly, discuss how Shinepodd might go about implementing a formal system of scenario planning. Please identify and discuss two examples of scenarios in order to support your discussion

[sub task (b) = 50%]

Coffee: a good place to use blockchain

Blockchain could transform the coffee industry with new levels of transparency. For the first time, farmers, traders, roasters, and consumers can be fully aware of their coffee's journey. Blockchain can be used to store information about the coffee supply chain. This could include where the coffee comes from, its price, who bought it, and even how long it took to get from farm to cup.

With a public blockchain anyone, from bean to cup, can be given access to the blockchain record to see where the coffee beans come from, where they end up, and how much is paid at each stage.

But a private blockchain would be controlled by the main actor that established the blockchain and be used to simplify tracking and transactions.

In a public blockchain farmers could see the price of their coffee at every stage and potentially connect with buyers through blockchain-based platforms. This allows farmers to be better informed in their negotiations and more knowledgeable about the market. It also would cut out the wholesalers who presently take a big slice of the profit.

Ivonne Herrera is a producer. "It is an opportunity for us farmers to provide a better cup of coffee, to be known, and for you to be able to recognize our coffee," she says.

Other players in the coffee industry would also benefit from increased transparency. Roasters could have accurate data about green beans, which could influence how they choose to roast.

If a problem was found with a coffee lot, it would be possible to search the blockchain records and pinpoint where fault lies.

There is also a benefit for consumers. Are you unsure of what fair trade really means or skeptical of where your coffee comes from? Using blockchain can allow a café customer to truly know the origin of their beans and be confident that they are paying a premium that benefits the producer.

Solutions to Activities

6.1 AutoAuto forecasting the future of driverless cars
Explain the techniques we can use to forecast the likely date for commercializing driverless cars and forecasting the likely profits in the first five years
(55%)
Approach: the task requires you to discuss forecasting firstly a date for commercialization and secondly likely profits in the first five years. The first element is principally reliant on unknown external factors plus forecasting the readiness of AutoAuto's own technologies. The second element allows a greater use of quantitative forecasting based on AutoAuto's cost models and its past experience of launching new products.

Forecasting Methods
Forecasting refers to techniques to help management make reliable assumptions about the future. Forecasting techniques are divided into

- Quantitative: numerical and reliant on past data about similar events/instances
- Qualitative: requiring judgement about future unfamiliar events/instances

Quantitative techniques include causal models and projections. Qualitative techniques include think tanks, Delphi technique and scenario planning.

Forecasting a date for commercializing driverless cars
<u>Potential passing of enabling legislation</u>
Driverless cars would not be legal at the moment. Gaining legislative approval for use on the roads of Upland and other countries will be crucial.

The likelihood and timing of gaining approval will depend on the outcomes of tests, the political atmosphere and attitude to passing the legislation, including the proximity of elections, and the progress being made, and pressure being brought, by rivals.

A Think Tank (or panel of experts) would provide answers to these questions. These should include experts on the political trends in Upland and other target countries, on the policies of the political leaders and so on.

The Think Tank might use the Delphi Technique to assign subjective probabilities to the likelihood of favorable legislation being passed over the coming, 12 or 24 months. Big Data may be used to gain insight into the political environment around driverless cars (blog posts, comments by politicians, social media posts by influence leaders). Statistical techniques such as opinion polls, may be consulted by this Think Tank.

Alternative scenarios may be explored using scenario writing/planning techniques. These scenarios could include

- Widespread public concern over impact of driverless cars leading to a ban
- Public willingness to allow the cars in rural areas, or limit it to certain types of vehicles

- Public enthusiasm for cars
- The impacts of successful trials or disasters such as the July 2016 death of a Tesla test driver in the Florida, USA

Rate of gaining consumer acceptance
AutoAuto should not commercialize the car until there is a prospect of demand. Therefore, the date when a critical mass of demand appears in each market will affect the commercialization date of driverless cars.

Qualitative approaches to forecasting this have been discussed above. Some quantitative analysis could be conducted using Big Data by modelling the adoption rates of the past auto innovations made by AutoAuto. This could identify the markets, automakers, and environmental circumstances that contributed to the adoption, or non-adoption, of past innovations

Big Data could also be used to analyze customer attitudes to the driverless car (e.g. social media comments, blog posts, new articles, media reviews)

Preparing a technologically safe driverless car
This should be developed qualitatively from the estimates of AutoAuto's own experts, perhaps using the Delphi technique. It will need to be updated regularly in the light of the trials of the technology

Profit forecasting for first five years
The forecasting of possible adoption levels each year can be modelled on past adoption and diffusion sequences using Big Data (discussed above)

The impact of competitors should be forecast
Competitors will affect price and volumes sold by AutoAuto. Given the relatively small number of major manufacturers of cars and commercial vehicles there is probably insufficient data for a Big Data forecast of volumes brought and prices paid for past innovation.
The better way to forecast this would be by using expert opinion of AutoAuto's manager based on the past behavior of rivals and any competitor information that can be gained. Scenarios of market penetration or market skimming strategies by rivals could be developed.

Costs of production
AutoAuto's costs will determine profitability of the new product. Quantitative analysis can be used to forecast costs of production, such as a cost analysis (causal forecasting)

Forecasts of the level of costs as sales volumes rise can be derived from data analysis of past cost bookkeeping records and/or Big Data from the AutoAuto's management accounting systems. This would identify any learning curve effects, experience effects, and economies of scale that may help reduce the costs of the driverless car systems over time

Profits would be affected by the level of amortization of the R&D costs, which would need to be written off to the statement of profit and loss over the useful life of this new technology

6.2 KCTP the Science and Technology Park forecasting change
Explain some of the main sources of the change that KCTP will face in its business environment and provide ideas on how forecasting may help KCTP's Board to incorporate environmental change into its strategic thinking (60%)

Approach: the task requires two answers: first a description of main sources of change and then, secondly, a discussion of how forecasting may help it meet these challenges. The *specific change examples are based on research around some of the themes in the pre-seen material.*

Main sources of change in business environment of KCTP
The environment of KCTP impacts on the needs of its tenants and on the operation and financing of its business.

<u>Technological</u>
KCTP is a science and technology park (STP) which offers space to firms wishing to develop and commercialise new technologies. The supply and nature of new technologies will therefore affect its business. For example the developments in quantum computing will impact how materials and medicines are tested, genome research developments create new avenues of biotech research and so on.

Technology also affect operations. KCTP has its own data centre but increasing amounts of data and better communications are encouraging firms like KCTP to migrate to a cloud-based service provider. The use of on-line collaboration, especially after the impetus given to it by Covid 19, means that the localised academic network that supports KCTP may now become global or, potentially, make KCTP less attractive and collaboration moves online. Technological developments also impact on the cyber risks of KCTP

<u>Government policies</u>
Government policies affect the wages and conditions of KCTP's staff, the safety and design requirements for the buildings, and whether funding for transport infrastructure may improve communications for KCTP.

Government policy also affects the funding of research and the incentives given to start-ups on KCTP's campus. It will affect commercialisation, for example by impacting on regulation and permissions for new materials and organisms.

<u>Legal changes</u>
Legal regulation increases the issues facing, and operating costs of, KCTP. These include minimum wage or equality laws and the costs of assuring compliance with them. It can impact on the costs of maintain buildings, such as in the UK an upsurge in the replacement of cladding on buildings in the wake of several lethal fires in high-rise apartments and the impact of statutory regulations designed to reduce carbon emissions and increase energy efficiency.

Legal changes will impact on tenants and rents too.

<u>State of the economy</u>
The Advland and the world economy will undergo booms and recessions. Potentially some economies will decline whilst trade and wealth move to others.

This will change the amount of funds available for investing in KCTP and its tenants, the markets for what the tenants produce and so on.

The state of the economy will also affect factors such as the interest rate on loans, the foreign exchange rate of the A$ and so on. These can change the profitability of KCTP and also the value of its shares.

<u>Geopolitical</u>
This refers to the political and economic alignment of nation-states. The decisions to enter or to leave common trade areas, or collaborations to share technology and knowledge, are part of geopolitics. So too are the encouragement or restrictions placed on the movement of people between countries, such as academic experts and business people, the restrictions on the

export of technologies or products, and protectionism that can increase the price of materials and technologies imported from outside A$.

Use of forecasting in strategic thinking

Forecasting is the prediction or estimating of future trends and development.

Quantitative forecasting relies on past data from which can be extracted causal models or times-series forecasts that can enable so degree of prediction of costs, sales, economic variables or social trends.

Qualitative forecasting is used where there is an absence of reliable past data, generally where there is uncertainty and a break with the past situation and trends. Most strategic thinking involves dealing with longer term and unknown futures, so it is served better by qualitative forecasting methods.

<u>Expert judgement</u>
This requires engagement with experts in their chosen fields, such as academics, technologists, futurologists (student note: people who specialise in writing about the future of a technology or society). These will not give predictions that are bound to come true.

By the Board hearing from them and questioning them the Board will be more aware and sensitive to the potential for changes and how to exploit them or prevent harm to KCTP from them.

<u>Think tanks</u>
These are collections of experts, often in discussion and conference with each other. Examples include the World Economic Forum, the Group of 20 world economic leaders (G20) and other institutes with particular interests in areas of science, social planning, legal reform and so on. They can be influential in shaping public debate, legislation and political policy.

KCTP should join relevant think tanks, either as a corporate body or enroll individual managers, to keep aware of the developing issues and trends

<u>Delphi technique</u>
A technique to clarify and quantify diverse opinions about the future into central tendencies and outliers.

Participants, such as experts, provide subjective assessments of the probability of events or tipping points that will affect the future such as the outcome of crucial legal decisions or political elections, the speed at which a technology will be available for commercial use, or the timing of interest rate changes by the central bank. These will be grouped to identify the median average central assessment and the extreme outliers and either accepted or rendered more consistent by asking the outliers to justify their assessments to the others in the expectation of a cleared consensus being established.

<u>Scenario planning</u>
A technique of weaving various trends and forecasts into coherent pictures of potential future situations in which KCTP may operate. These are developed by discussion and collaboration. The experience of these discussions are to help senior managers appreciate the potential forces driving the changes and their effects. The managers can then develop

6.3 Shinepodd: blockchain, digital disruption and scenario planning

<u>I need a briefing on how blockchain and other transformational technologies may affect Shinepodd's business (50%)</u>

Note: this is a very comprehensive solution. Given that Blockchain and other digital technologies are new to the syllabus in 2020 there will be some tasks set on it, and/or the other technologies, in the real exams.

Impact of blockchain on Shinepodd

Blockchain is a distributed notarised ledger system. This means that it records transactions in a form that is available to many viewers and hence provides greater transparency and validation that would a single logging system such as a ledger or enterprise system.

Blockchains run on cloud platforms, such as Microsoft Azure, and the extent to which it is private (owned by one firm and only visible to the people the owner chooses to show it to), public (visible to anyone), or co-operative (run by a group for the benefit of each other) varies. For example the Starbucks blockchain will be private but the Bext360 blockchain will be either co-operative or public. The validation of the transactions usually involves multiple miners using their supercomputers to crack the mathematical algorithms and earn money by validating the transactions.

Blockchain has three obvious uses for Shinepodd:

1. The most obvious use of blockchain for S/D is in tracking the buying, sorting, drying and supply of coffee berries and coffee beans from grower to cup.
2. A lesser use would be in tracking inventory and sales of coffee pods from manufacturer through warehouse and logistics partners, to consumer
3. There is also the possibility that blockchain may be used for payments by managing a crypto currency that customers, logistics providers, farmers and so on will accept

There is likely to be a high cost in setting up blockchains for Shinepodd.

Coffee-growing areas are not well-served by the telecommunications systems needed for blockchain, although firms like Google are seeking to address that with Project Loon which will use high altitude balloons to create an ariel wireless network. S/D may need to help invest in building the infrastructure.

Shinepodd's partners, such as farmers, shipping firms, couriers and so on will not have blockchain platforms that S/D can communicate with. To have an unbroken chain they need to all be on the same or compatible systems. S/D will probably have to build the platform and train them in how to use it

S/D may need to provide farmers and others with the devices needed to access and utilise its blockchain (smartphones, PCs, automated weighing and milling machinery and so on).

There are some potential benefits to S/D

<u>Reduction in costs of transactions</u>
The use of a distributed ledger means that less staff will be needed to keep records, note the arrival and despatch of items. There would need to be less inspections of the green coffee on arrival in Middland because it would have been validated at the point of purchase.

This will reduce wage costs and other overheads (office space, stationery, travel and so on)

Improved traceability of beans
This is promoted as the main benefit of blockchain. It is argued that socially conscious millennial consumers want to know the origins of their coffee and that fair prices have been paid to farmers.

Coffee connoisseurs will want to trace their coffee back to the particular country, region and even farm they came from.

If the blockchain is public, or relevant parts are made public, the customer can track the beans in their consignment back down the chain to see where they came from, how fresh they are, where they were roasted and so on.

Given that S/D has Values that attract socially conscious consumers, and it has published social responsibility statements that refer to its ethical purchasing policy, this could be an important step.

It may be a big improvement over the assurance systems used by S/D at present. The present system comprises sporadic visits by the Director of Overseas Operations and his team to the farms and sorting stations. Many of the coffee growing regions have a reputation for corruption. A blockchain solution will provide greater assurance that farmers received the money paid for the beans, that the beans that arrive in Middland are the ones that were paid for, and also where they travelled and how long they remained at a particular place.

Possibility of buying direct from the grower
The blockchain will provide S/D with a platform for coffee trading. At present S/D will probably buy from the sorting/milling stations (wholesalers) to whom small growers will have sold their crops. This muddles the berries together in a way that makes it impossible to check provenance later (say if some turn out to be diseased, or some have a particularly good flavour). There is also no guarantee on the price that the wholesaler gives the grower.

A blockchain platform could enable growers to sell consignments of beans direct to S/B and to receive the entire value of their crop without the wholesaler taking a cut. This is the intention of Bext360 which offers, in effect, a reverse vending machine in coffee growing areas into which growers can pour berries, have them automatically inspected and sorted by the machine, and a price offered. If accepted the fee is credited to the growers crypto currency account automatically, the berries are parcelled and assigned a block which then can be traced along the supply chain right to the cup.

Potential to pay in crypto currency
There is not automatic need for blockchain to pay in a cryptocurrency, but it is a possibility.

If S/D sets up a blockchain it will need to pay in cryptocurrency to the miners who use the supercomputers to validate the transactions in the chain. But it could extend this to paying farmers, couriers and even accept cryptocurrency for the sales of its pods.

This could potentially reduce transaction costs such as bank fees. But also it could reduce exchange rate risk. At present S/D earns its income in M$ and other currencies but 100% of the costs of beans are paid for in US$. Using a crypto currency for purchases and sales could reduce this element of its currency risk.

However S/D should be wary of unstable cryptocurrency values. Many cryptocurrencies have become speculative assets rather than mediums of exchange and this has made them unreliable as a store of value and as a unit of account. This would deter stakeholders from using the system and could impose another exchange risk on S/D.

<u>Risk of consortium blockchain increasing supplier power</u>
This is a treat to S/D from blockchain.

Blockchain solutions lie behind consortium platforms like iFinca and FarmerConnect which have the stated intent to ‘democratize’ the coffee supply chain using blockchain. This will empower farmers to cut out wholesalers and to see the prices being paid by firm like S/D and to ensure they get the best price for their coffee.

This holds the wider threat of disintermediating the coffee firms altogether as customers buy the beans directly from their favourite farmers and have them roasted locally or in home roasteries attached to their coffee makers. This will impact mainly on the coffee connoisseur segment, which probably doesn’t use S/D pods, but it will also interest commercial organisations like cafes and hotels who might wish to do this.

Additional impacts that could have been mentioned include:
- *Need for higher level of information skills*
- *Shift of business systems into cloud with cost and security implications*

Other transformational technologies

Internet of Things
Coffee makers are becoming smart machines. This provides an opportunity for S/D to sell new machines with enhanced functionality and also to gain more information from customers

Smart machines enable remote control by the user, and can have brewing timed to alarm setting on the customer’s smartphone or home digital assistant, or be triggered by the proximity of their smartphone indicating they are arriving home.

Smart machines may order beans or pods automatically based on use, notify users of low water of the need to descale or clean the machine. They can store or set up playlists of coffee where several people want different brews. They can also send data back to S/D on preferred combinations of pods (suggesting new blends)

RFID chips (Radio Frequency Identification) in pods enable machines to distinguish original pods from third-party pods and refuse to process pods that are not from S/D. The same RFID could be configured to track recycling of the pods.

The addition of voice recognition software could turn the machines into a digital assistant that can process customer information requests, control household lights and other appliances either as a the main hub or as part of an intelligent home or part of ubiquitous computing (intelligent home is one that has smart controls for heating, lights, music/entertainment, security and so on. Ubiquitous computing, also called ‘everyware’ or pervasive computing, is interlinked IoT in the home, car and on the body that enables the individual to access computing power and information anywhere are not just by using their smartphone or PC. It also allows closer tracking and surveillance of the individual)

IoT poses a threat as it can improve the functionality of connoisseur machines that use beans and green coffee rather than S/D’s pods. These machines can use programs, user instructions, tips from coffee blogs and so in to determine how they soak, roasting, grind, steep and brew coffee to give a superior and customised taste beyond that which S/D can achieve using 50 pods

Data visualization
This converts complex and large amounts of data into visual displays.

This could assist S/D with planning based on crop supply and readiness. It could also help production planning based on trends in buying.

Process automation
This is a form of Machine Intelligence (MI) that learns to perform routine clerical operations from monitoring and learning from real operatives. Eventually it replaces them.

This can reduce costs of processing orders and payments and also speed up the processes and reduce errors.

Big Data and Analytics
This is the use of massive real time and historic data sets to identify patterns and trends.

S/D already has data on coffee crops, prices, harvest times, types of bean, pod sales, customer location and buying preferences and so on. This data can be used to extract trends in tastes for coffee.

It would also be used to choose which brands to put in store (e.g. supermarket lines) if S/D decided to broaden its availability to become a convenience product.

It could improve inventory management by spotting fast and slow moving items, or forecasting when demand is about to rise or fall.

It would help tracking crops and predict yields and prices based on data on climate and on the amount of bushes that had been sown.

Artificial Intelligence
This is a technology that enables machines, or information systems, to think and to learn. It is accelerating in its capabilities and use and is set to change many industries in the coming 10 years.

Applications include:

- Providing advice to farmers on factors such as irrigation, soil fertility, pests, and when to harvest via apps on their smartphones that can photograph and monitor the crops using photographs and the results from simple soil testing kits
- Machine vision: cameras linked to AI that can automatically inspection the berries being offered for sale and accept or reject them, price them, sort them and so on.
- Monitoring customers to provide coffee when needed, based on their eye pupil size or readings on their metabolism and activity from wearable trackers and smart watches. It can also take in diary appointments and the people at the meeting to decide on when to recommend coffee and take account of their known preferences
- Self-driving delivery vehicles. S/D needs to deliver pods to customers and it uses external couriers. AI gives the potential for self-driving delivery vehicles, such as vans or drones.

Secondly, discuss how Shinepodd might go about implementing a formal system of scenario planning. Please identify and discuss two examples of scenarios in order to support your discussion (50%)

Student note: questions like this are either easy or hard for you depending on your imagination. They are also hard to structure where you are trying to illustrate a process with two examples. Scenario planning is a forecasting technique, so the things that 'ought to be planned for' are events in the future that impact on the business. Look upstream and downstream in the supply chain for ideas: political unrest in growing regions, transportation or trade problems, fire in the roastery, breakdown of the website and ordering system, changes in tastes away from caffeine in drinks, new intelligent coffee machines and so on.

Implementing a formal system of scenario planning

Overview

Scenario planning, or scenario writing, is a technique of foresight to help management to understand the future under conditions of uncertainty.

One type of scenario planning can be used to visualize a foreseeable event, such as a rise in raw material prices or a problem in the online sales systems or operational problems at the roastery. Contingency plans can be developed to deal with each of these (student note: the examiner's solutions don't discuss this type of 'disaster scenario' approach which perhaps suggests the examiner doesn't recognize it. The solution below does not use this as an example)

Another type of scenario planning can also gain foresight into the future of the business ecosystem. This could include changing tastes in beverages, political or economic changes in the countries which supply coffee beans, technological changes and their impacts on the working lifestyles of coffee drinkers, and so on. Strategic plans should be made in advance to deal with these.

Scenario planning requires a planning team. This can include management but might use expert facilitators and others to guide the process and to stimulate thought and bring research ideas into the process.

1. Identify drivers of the future

Drivers of the future can be identified using PESTEL analysis or other ecosystem frameworks. Two factors driving change in Shinepodd's ecosystem are:

1. Economic development of Asian economies. The rapid rise of economies such as China and India will be the main economic opportunities in the next 20 years or so as they overtake, and perhaps supplant, the western economies of USA and Europe. They have a history of being closed to trade, which may have been abandoned or which may re-assert itself in a wave of nationalist thinking, and also each has a complex set of political arrangements that affect policy and relations with other countries.

2. Growing environmental awareness and concern. This includes concerns over climate change, blamed on pollution from fossil fuels and destruction of forests for meat production, and the disposal of plastic and other non-degradable waste into the oceans, land dumps and its ultimate infiltration into the food chain. There is a rising belief that resolving these existential threats will require fundamental abandonment of globalized, consumption-based models of economic activity.

2. Postulate outcomes of drivers for the firm and its ecosystem

This is a stage of visioning where the drivers might lead.

Economic development of Asian economies could lead to the emergence of large new markets for products associated with luxury consumption. There is huge potential for rising average incomes given that China and India are still 47th and 68th in the list of annual incomes per head with incomes at 15% and 3.2% of USA respectively. But in both cases the enjoyment of incomes is and will remain very unequal with metropolitan knowledge workers and entrepreneurs taking a disproportionate share of the national income. There may be a swing towards western lifestyles and tastes, or possibly a rejection of these as the newly rich countries seek to restore traditions and maintain an independent cultural identity.

Growing environmental awareness and concern may lead to more general criticism of impact of consumerism on environment, especially from younger 'millennial' age groups in developed countries, spilling over into direct action against factories and stores that sell 'irresponsible' products. There may be a backlash against 'trivial and selfish' promotional messages at a time of crisis and a preference for brands and products that appear more sustainable. There may be specific areas of concern touching on S/D's business such as concern over water resources, carbon emissions, and recyclability of packaging. Investment funds move towards industries that have more sustainable business models.

<u>3. Resolve outcomes into series of potential scenarios</u>
This is developing a narrative of a likely future world based on the drivers.

Another Western Market Scenario would be of China and India increasingly adopting the secular culture of the market economy and abandoning older faiths and political creeds. The lifestyles in the countries would resemble closely the lifestyles of Western Europe and USA with high mass consumption, long working hours in office environments within a lengthy commute from home, high automobile ownership, and search for companionship and community around coffee shops, several of which feature in popular television serials as meeting places for the cast.

Green World Scenario would be countries, or population groups, in which firms are judged and rated on their green credentials and the expectation of recyclable and biodegradable packaging, clothing, and products becomes normal. In Green World people carry own cups and take-away boxes which they require to be filled by vendors. There will be a preference for brands that emphasize being natural, caring for the world, and a superficial collectivism around a 'save the planet' ethical code.

<u>4. Write/fill out scenario</u>
Both scenarios above would be elaborated. At this stage there might be an attempt to identify tipping points which are events, or thresholds, that make particular scenarios more likely. These could include threshold rises in income, appointment of particular parties or political figures, judgements in legal cases and so on.

<u>5. Identify actions for each scenario</u>
This adopts an evolutionary analogy and seeks to identify the characteristics of businesses that would survive better in the new ecosystem.

Another Western Market. S/D should identify beverage drinking habits in the countries, for example the preference for tea as a beverage, and consider developing pods to brew tea and supply chains to support this. It should seek to find business partners in the more promising countries able to make machines, distribute pods. It would need to decide an entry strategy for each market, for example targeting Western corporations and shop brands with machines, or adopting a local brand name.

Green World. S/D might develop coffee machines to replace its reliance pods. It could launch brands under a fair trade regime and promoted as good for the planet and providing incomes and opportunities to impoverished farmers in low income areas. The marketing message could change to showing S/D as part of community and its drinks as an allowable pleasure and respite in a troubled world.

6. Monitor and update scenarios

Scenarios are mere guides and are invariably wrong about the precise way in which the future will play out. Therefore the assumption on which they are based, the drivers and tipping points, need to be monitored to assess whether the scenario is developing as anticipated. New developments in the ecosystem need to be included.

Chapter 7
Big Data and Analytics

1. Overview

Big Data refers to data sets that are so large or complex that traditional data processing applications are inadequate to deal with them.

The SCS Assessment sets tasks which require candidates to assess the business value of Big Data to the organization in the case material. Your course materials will cover the specific applications relevant to your case study. This chapter discussed the use of Big Data and Analytics in general.

Applications of Big Data

Big Data has applications in many areas:

- Provision of government and public services (efficiency, public attitudes, security and crime)
- Targeting international development activity (health, education, natural disaster prevention, security)
- Manufacturing (component performance, reducing inefficiency, preventative maintenance)
- Healthcare (disease trends, treatment efficacy, risk assessment, epidemiology)
- Entertainment and media (targeting content, customized advertising content)
- Sport to achieve improvements in player selection and coaching, player performance, team selection, and team fitness
- Retail (inventory management, customer relationship management, targeting)

2. Big Data

Characteristics of Big Data

When Big Data was an emerging field may IT experts argued it was only another grand name for database analysis. To distinguish it from database analysis the consultancy Gartner developed the 5 Vs model.

Characteristics of Big Data

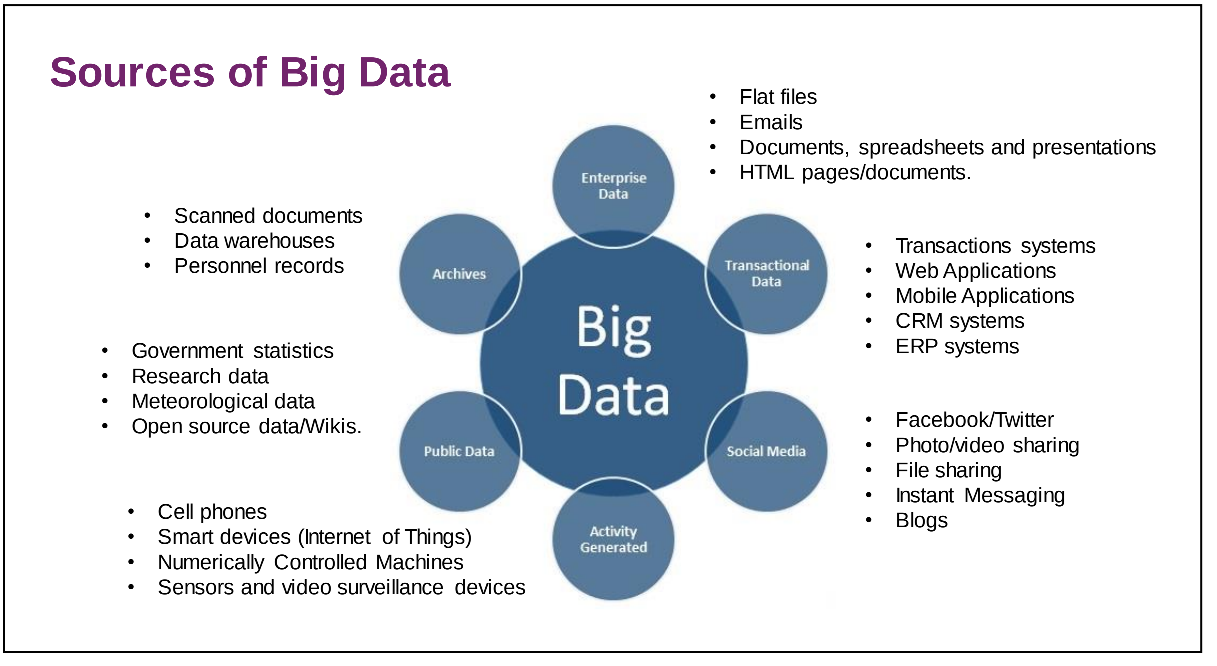
Characteristics of Big Data
Volume: Vast amounts too large to be stored and analysed by conventional data bases. Need for distributed processing (eg cloud)
Variety: Unstructured eg messages, social media postings, photos, sensor data, video and voice recordings
Velocity: Constantly generated and rapidly changing. Requires real-time capture and analytics
Veracity: may be more true due to large numbers
Value: can provide huge business benefits

More recently the original 5Vs have been augmented by two additional ones

- Variablity: care should be taken because the quality and conventions used to gather the data are not always within the firm's control and so there will be problems in combining data to extract findings
- Visualization: the huge data sets must be subjected to Data Vizualization to make the useful and inteligable

Sources of Big Data
Big Data is gathered from systems the firm operates, but also make take in external and additional data sold by data brokers like LiveRamp, Datalogix and Experian.

The supply of this data is set to increase massively with the greater use of social media and the wave from adoption of IoT.



3. Data Analytics

Big Data is analysed using statistical techniques, or algorithms, to reveal three classes of business information:

1. Descriptive
2. Predictive
3. Prescriptive

Analytics can be of two types:
<u>Real time analytics</u>: this is data arising at the current time and is used to mae decisions on operations day to day. For example you would look at todays temperature and atmosperic pressure when deciding whether to take a raincoat

<u>Data mining:</u> this is looking at the layers of past data in order to see patterns and sequences to help make decisions. For example you would look at the weather in past years at a resort when you are planning whether to take a holiday there and when in the year you will visit the resort.

Descriptive Analytics	Examples
• **Associations** between data items	• Items commonly bought together • Weather/season and store footfall • Personal characteristics and illness/mortality • Components/suppliers and product failures
• **Clusters** of data items	• Customer segments/types • Hot spots of illnesses • Slow moving inventory items/locations • Peer networks in the organisation

Predictive Analytics	Examples
• **Sequences** in data items	• Symptoms that are lead indicators for an illness • Customer actions that may be followed by loss of the customer • Route followed giving guide to final destination
• **Trends** in data items	• Increasing/ decreasing popular interest in something • Changes in prices of materials or foreign currency • Warranty claims by cause rising indicating a manufacturing defect

Prescriptive Analytics	Examples
• Identifies the best course of action or decision given the known parameters	• Suggesting interventions to deal with illness • Likely future prices of commodities • Automated share dealing • Fault finding in a piece of equipment • Best tactics in a competitive game • Where to drill for oil • Optimizing layout and product range in a store • Optimizing pricing of a product or service (dynamic pricing) • Spotting bogus transactions or fake products in a supply chain

4. Concerns over Big Data and Analytics

The following concerns have been raised about the role of Big Data and you should be willing to mention them in your assessment

1. Concerns over personal privacy
   This technology, when applied to people, is effectively surveillance. Many consumers and staff do not realise this when they sign agreement to be monitored. There is also the real danger that this data may be sold to third parties or leaked to harm the person

2. Exploitation of the individual
   Data firms like Google and Facebook, and their owners, have grown very rich by using the data given to them free, or almost free, by people. Yet the value of what is given 'free' in return such as a search engine or instant messaging system, is not an adequate reward.

3. Manipulation of the individual
   Data firms initially used data analytics to target advertisment and messages to potentially responsive consumers. More recently it has been suggested that this targeting has manipulated social and political behaviour and the outcome of elections using messages that the individual cannot always filter out or ignore.

4. Diminishing returns to data
   The volume of data is growing exponentially but the value from its findings may not be. The data firms have a vested interest in encouraging firms to buy more data and capture and analyse more data. But whether having data from an IoT that tells you what temperature they like their houses at, and what time they turn off the lights and go to bed will help reveal a secret that allows the firm to improve its revenues is less clear.

The SCS examination has featured tasks on Big Data including:

- What can be learned from Big Data to help forecast future oil prices
- How can Big Data (derived from sensors in player's boots) improve the performance of a soccer team?
- What can be learned by Analytics from analysis of the viewing and voting behaviour of people wathing televison talent shows?

The SCS examination solutions have also referred to Big Data as potential sources of:

- Investors feeling about the quality of management (e.g. from analysts publication, social media posting and blogs)
- Forecasting data for understanding future sales (e.g. from social media comments, pictures posted, trending comments, and causal models of income/climate on demand)

ACTIVITY 7.1

This activity is indicative of the sorts of assessment tasks in the SCS assessment. Each sub-section has a weighting that denotes the percentage of 60 minutes that should be devoted to answering that subsection.

Context

Runn Group is a global Courier Express Parcels firm, like Fedex and DHL.
Its business operates a network of 1,600 depots across the world and its market is growing at 15% per year. It uses 208 cargo aircraft to transport parcels long journeys, and 104,000 road vehicles, some are large trucks to move bulk amounts between depots and a fleet of vans to manage collections and deliveries in the immediate locality of the depot, the 'final mile'. Its vehicles are equipped with GPS and they are tracked. The parcels are tracked using scanning technology, and each van driver carries a handheld tablet to scan parcels in and out and to record customers' signatures when they accept delivery.

Task

You have received an email from Chen Yanyun: Director of Finance for Runn

To: Senior Finance Manager
From: Chen Yanyun: Director of Finance
Topic: Project Insight

Please see the attached from Shashi Raheja, our Director of Information and Engineering.

The Board realizes that the use of Big Data is transforming many industries. But it is also aware that develop the 'data lakes' and 'real time data collection' and 'data analytics' will require a lot of investment in data collection technologies, data storage, and data analysis.

We have a discussion of Big Data and Analytics tomorrow.

What I need from you is a briefing on the following things:

What is meant by 'Big Data and Analytics'?

[sub task (a) = 55%]

How might identifying the four things Shashi lists bring us any business benefits?

[sub task (b) = 45%]

To: Board: Runn Group
From: Shashi Raheja: Director of Information and Engineering
Topic: Project Insight

The Background
There has been an explosion in the amount of data potentially available to courier firms
The main sources of this data are:
- Our GPS and tracking systems
- The information systems we offer to corporate clients, including Runn Now, Runn Web and Runn Back Office
- The records we collect from low volume businesses through our parcel logging systems at service desks, Drop and Send, and through our online booking system, Click and Send
- External data sources such as from road monitoring, weather reports, government and business publications, and internet search records

This data can be stored as 'data lakes' and analyzed to reveal patterns in the data whilst real time analytics can spot patterns as they arise.

The Proposal
Runn sets up Project Insight to gather and use Big Data more effectively. This should help us in the following ways:
1. Improve route optimization for collections and deliveries
2. Assistance in deciding where to build new depots
3. Improvements to customer value management by helping us to identify the value users and the less valuable ones
4. Planning operational capacity to meet anticipated demand

Knowing these will be of enormous value to developing our business.

The Cost
We need expertise to develop apps and wearable technologies to gather data on individuals. We need to write algorithms and we need a lot of processing power. I propose the following:
- Recruitment of a Project Manager with responsibility to oversee this project
- Recruit 30 full-time designers and developers to write the algorithms and to manage the outsourcing of parts of the task to contractors in a developing Asian country
- Contracting world leading cloud services provider to upload our data to its systems for our analysis

I assess the annual expenditure on this to be in the region of G$1bn per year initially
Shashi

Solutions to Activities

7.1 Big Data in Runn the courier firm
What is meant by 'Big Data and Analytics'? (55%)

What is meant by Big Data?
<u>Sources of Big Data</u>
The growth in connectivity and information-based technologies has greatly increased the range and types of data available to courier businesses like Runn

External data also comes from social media (blogs, postings, instant messaging), surveillance systems (video cameras, GPS, tracking technologies and increasingly from intelligent devices such as smartphones and digital assistants that listen to human conversations and translated them into machine code), public media (news and weather broadcasts, websites) and from voice communications and messages

Internal data can come from Runn's own Customer Relationship Management systems (CRM), its logistics systems and so on

The data is captured as code and archived in a Data Warehouse for later analysis and use
It can also be used in real time to improve operations immediately

<u>Characteristics of Big Data</u>
Gartner Group refers to the following five characteristics of Big Data

- ***Volume***: the amount of data is massive (measured in Terabytes, Zettabytes and Brontobytes) and is too large to be stored or processed by the data centres of firms that have been scaled only to cope with the firm's routine transactions. Therefore, they state that Big Data inevitably involves using cloud computing (and 'the cloud' is likely to the main source of external data)
- ***Variety***: the sources and types of data is very diverse. It can include keyboard strokes, pictures, videos, voice, and machine data
- ***Velocity***: the data constantly being generated and moves around ('viral') and is changing rapidly through time
- ***Veracity***: the truthfulness of much of the data is in question (e.g. 'fake news') but the size of the data and the self-policing aspects of social media may make it better than relying on small size samples or the guesswork of individual managers
- ***Value***: it is difficult to predict the value to be gained from capturing, storing, and analysing Big Data but there are many cases where it has delivered information and insights of enormous value to a business

<u>The uses of Big Data</u>
Internal uses of Big Data refer to using it to improve business *efficiency* by spotting or predicting problems and dealing with them

External uses and aimed at improving business *effectiveness* through understanding clients, suppliers, competitors, business partners and stakeholders better and adapting the firm accordingly

Big Data is not information, it is data, and by itself it cannot tell management anything

Data Analytics (or Data Mining) is an information extraction process that uses intelligent agents to search through the data to look for patterns using advanced statistical techniques and algorithms (sometimes referred to as and application of Machine Intelligence/Learning or Artificial Intelligence)

This has led to three generations of outputs

Descriptive analytics**:** this is the discovery of ***associations*** and ***clusters*** in the data. For example, linking particular weather conditions to patterns of traffic can help in route planning, or attaching orders to clients can identify clusters of high value and low value customers

Predictive analytics: this is the discovery of ***sequences*** and ***trends*** in the data. For example, the lead indicators for the emergence of a large number of orders, or identification of locations or industries with increasing use of courier services

Prescriptive analytics*:* this goes beyond predictive analytics and assesses one or more possible outcomes and their effects in order to identify the optimal decision. This can be used to instruct drivers on the routes to take, and the provide parcels to couriers in correct sequence in which to load their vehicle (the things that need to be delivered first should be nearer the doors)

[Sub task (b)]
How might identifying the four things Shashi lists bring us any business benefits? (45%)
Note: to help deepen your understanding of how Big Data and Analytics works the solutions below contains explanations of the techniques that provide the benefits. These were not required by the task.

Business benefits from Big Data and Analytics
<u>Improve route optimization for collections and deliveries</u>
This is known as 'the travelling salesman problem' and is dealt with by an operational research technique called 'branch and bound'. Each morning at each depot Runn must allocate the parcels that have arrived overnight to the available vans. Some of these parcels will have guaranteed delivery times, whilst others will be 'anytime' deliveries. Also, it needs to allocate any advance notifications of collections to the vans. It will do this using prescriptive analytics to choose the optimal route to minimize travel distance and also to hit the guaranteed delivery times. It will use geolocation algorithms combined with real time data on traffic conditions. During the day more parcels may arrive at the depot and more collections will be booked. To a large extent predictive analytics based on past data will have anticipated both things and will have been used to set routes for the vans at the start of the day to put them in the right zones for the predicted demands later in that day. The information system will dynamically update the van's routing to incorporate collection of the new parcels from the depot and also making the deliveries and collections required

The main benefit of this will be that it will ensure that Runn hits the times required for collections and deliveries and therefore retains its customers

It will also help to reduce costs of fuel and vehicle use (such as brakes, tires and so on). The reduced distances will also reduce the carbon emissions and potential for road accidents

It should also enable reduction in staffing costs through minimizing number of vehicles and drivers in each depot through greater logistical efficiency

<u>Assistance in deciding where to build new depots</u>
This will use descriptive and predictive analytics initially. It will identify clusters of delivery locations, or collection points, where parcel volumes are growing. It may combine this data with data on late collections/deliveries, delivery distances, value of clients, and external information on road traffic and on plans for new businesses, homes or roads. Predictive analytics will be used to pinpoint best locations for the new depot given the above factors

Depots are a significant investment of Runn's capital and will also lead to operating costs each year. It is vital that they are built only when needed, in other words where the extra costs can be recovered from extra revenues from clients and/or from greater operational efficiency

Having the new depot and in the right place may reduce costs by reducing journey distances or by requiring less vehicles and drivers due to avoiding congested areas. It may help avoid breaking service level agreements with clients due to late delivery/collection and so avoid penalty payments to client. It might also win new clients by expanding capacity and reducing delivery times

Having new depots may also help to reduce some of the adverse social impacts of Runn such as emissions from vehicles, contributing to road congestion, and emissions

<u>Improvements to customer value management by helping us to identify the value users and the less valuable ones</u>
Data analytics could identify clusters of high value clients. These would be ones that pay higher prices, and others whose consignments are not expensive to move and to deliver or who do not place orders last-minute and who also are able to complete documentation accurately enough to avoid misdirection or deliveries being held up at customs and so on. The ordering, tracking, management accounting and marketing systems of Runn could provide this information. Low value clients could also be identified. The systems would help identify the core reasons for the high value and low value clusters

Attracting and retaining valuable customers is the basis of Runn's profitability more than the volume of parcels shipped. Identifying which customers are valuable should drive the customer relationship and marketing activities of Runn to retain the existing high value customers and to attract new ones with similar characteristics

Data analytics should identify the characteristics of the valuable customers such as their industry, the way they place orders, the quality of their packaging, the location of their customers and so on. This information can be used to spot non-clients who could potentially be valuable

It can also be used to improve the profitability of low-value customers by encouraging them to be better, say by using better ordering or packaging and not demanding urgent delivery, or by charging them more.

Knowing who and where the valuable customers are is essential to profitable capacity planning and depot location decisions.

<u>Planning operational capacity to meet anticipated demand</u>
This would use predictive analytics to spot factors that together indicate a wave of business, or a shortage of business, over the coming few days. Some of this will be internal data, for example a large amount of collections on one day will predict a large number of deliveries for the next day. This can be made more precise by analysis of what type of client the collections are from and therefore whether they are likely to be timed deliveries. For example, if the client belongs to the building supplies cluster it is likely that the items will be required early morning the following day to enable building work to continue on site. Also, past data will show patterns in the collections and deliveries such as seasonality and days of the week when there are likely to be more parcels

External data may also be used within a causal predictive model such as noting the impact of weather on demand for clothing, building or heating/cooling equipment and also the effect on road congestion and delivery speeds. For example, a rainy few days may lead to a big rise in shipments of waterproof clothing, roofing materials, and dehumidifiers, but also to a slow-down in delivery speeds as the roads become congested by nervous drivers and minor collisions

Volume of business will change from day to day. Runn needs to match its capacity to the volume of deliveries at each depot

This will avoid missing delivery deadlines, or incurring excessive costs by paying drivers to stand in the depot unneeded

It will also help in deciding whether to add non-urgent parcels to the vans for delivery, or whether to leave space for the anticipated wave of extra collections during the day.

Chapter 8
Business Ethics

1. Overview
Ethics means the proper application of moral principles in our decisions and behavior.

The behavior of the CGMA member and the advice they give must be of the highest ethical standard to ensure trust.

Discussions of ethics focuses on two things:

1. The *rights* of individuals. If someone else breaches these rights they will have acted unethically.

2. The *duties* of individuals. If someone does not perform their duty they will have acted unethically.

In either case the unethical action will have cause some degree of *harm* to another person.

Discussions of the ‘rights’ of individuals is problematic and there is unlikely to be agreement. The better approach in this exam is to focus on ‘duties’ e.g.
- The duties of a management accountant because they have joined a professional body like CIMA/CGMA
- The duties of Board directors and managers that arise because they have taken on those positions
- The duties of a public administrator or political leader because they have been appointed to the office

2. The Ethical Code for Accountants
The *Ethical Code for Accountants* (sometimes called the CIMA or CGMA ethical code) is a valuable source of guidance on the following issues:
- Your personal ethical duties as a CGMA member and a character in the case
- To provide terminology for the personal ethical duties of other managers, or political figures, in tasks, whether or not they are accountants
- The terminology to use in the case of ethical issues facing the entire business

Ethical problems are rarely clear-cut. There are always arguments back and forth on serious issues. Your SCS Assessment response should show that you are aware of this, rather than merely arguing one point of view.

Ethical frameworks

Professional & Accounting Ethics

IESBA/CIMA/CGMA Code of Ethics

- **Professional behavior**: comply with laws/regulations and not discredit profession
- **Integrity**: straightforward and honest
- **Professional competence and due care**: maintain professional knowledge and be diligent
- **Confidentiality**: don't disclose business information without authorisation nor use it to benefit yourself
- **Objectivity**: no bias, conflict of interest, or undue influence

The Ethical Code for Accountants defines the ethical duties and obligations of a professional accountant. It is not intended as a general code of business ethics. Therefore, is necessary to adapt it to the circumstances

Ethical principle	Exam examples of unethical behaviour
Professional conduct	• Unfair discrimination between staff • Exploitative pay and work conditions • Bullying and oppressive behaviour • Breaking the laws or regulations
Integrity	• Deliberate false claims about product or CSR • Lying when asked for account of something • Not warning others of something
Professional competence and due care	• Not doing job to best of ability • Failure to supervise situation properly • Not responding to a clear risk • Dangerous products
Confidentiality	• Intrusion or surveillance of someone • Leaking or loss of confidential data • Using inside knowledge to make a profit for yourself
Objectivity	• Offering or accepting bribes • Discrimination against staff member or customer without evidence

3. Approach to tasks

It is vital that you use the precise terminology of the CIMA Ethical Code when you answer questions concerning ethics. You need to memorise the 5 terms above and their exact

Approach to ethics tasks

State main ethical issues
- use key concepts/principles
- show how action or proposal of management conforms to/conflicts with the principles

Describe the consequences of complying/not complying
- business or brand reputation
- trust and commitment of stakeholders
- legal challenges
- attitudes of regulators
- organizational culture or motivation

State any counter arguments
- ethical duties to others that may be harmed by treating one group ethically
- business issues/impact on competitiveness and profits
- Action or harm is not exceptional or extreme

definitions.

4. Resolving ethical problems

From time to time the examiner sets tasks that require you to suggest ways that management should respond to ethical dilemmas

Responses to ethical dilemmas

- Transparency and disclosure
- Express empathy and awareness of the harm
- Communicate with victims and regulators
- Compensating actions
- Remedial action

ACTIVITY 8.1

Context

Zoom is a Transport Network Company (TNC) offering a ride-hailing service in the country of Jayland (like Uber, Lyft or DiDi). These services have revolutionized the taxi industry by using a technological platform to match self-employed drivers with riders. Zoom wants to expand into a new country, Kayland. But to do this it will need an operating license from the Kayland government.

Task

You have been called into the office of Kevin Sato: Chief Finance Officer for Zoom

Dev Khan has made a few contacts in Kayland and the government there is willing to consider granting Zoom a TNC license to operate there.

But since his visit Dev has received a letter from a member of the government offering us assistance in establishing ourselves in Kayland. Here is a copy of the letter (Kevin passes you a letter).

My concerns are:
What are the ethical issues raised if Zoom takes up the offer in the letter?

[sub task (a) = 55%]

Palladia House
Capital City
Kayland
Mr D. Khan: Chief Operating Officer
Zoom

Dear Mr Khan.

I am a Special Advisor to the Minister of Transport and I enjoy his support.

I am in a position to facilitate your business in Kayland. In particular in helping you to obtain the necessary operating licenses from the Ministry of Transport and in ensuring that the repeal of the laws that at present forbid civilians from providing taxi services using their private cars.

There would of course be considerable work involved in facilitating a favourable legal climate for Zoom. I believe that the Minister would welcome a contribution from Zoom to the funds of the political party he is a member of. This donation would help to ensure that the present government remains in office past the next election, and is able to complete the necessary reforms to our public transport systems.

I look forward to working with your company here in Kayland

Yours sincerely
P J Jason

ACTIVITY 8.2

Context
KCTP is a listed operator of a science and technology park. Its management has been discussing adapting its business model to provide virtual collaboration between firms as well and face-to-face collaboration within its park. A recent news article reports confidential information taken from a personal computer that was lost by one of the staff of KCTP, possibly from his car whilst it was parked at a shopping mall.

Task
You have been called into the office of Kevin Sato: Chief Finance Officer for Zoom

You receive the following email

From: Shereen Peros: FD
To: Senior Finance Manager
Topic: News article

I have forwarded you an article that went online this morning. It has caused a great deal of embarrassment for KCTP. The article distorts the facts. As you know the Board is still talking about options to develop a hub and some of the data here is obsolete.

The tablet PC belonged to Kester Michaels, a member of the operations team. The tablet was Kester's personal property. He is very keen on encouraging KCTP to consider launching a virtual collaboration hub so he downloaded some of the documentation around the proposal to take on holiday with him. The tablet was missing from his briefcase when he got home, having stopped on the way to collect shopping for his trip. He is not sure whether it was stolen from inside our offices, or from his car whilst it was parked near the shopping mall.

Sanjay Gupta, our CEO, has reminded me that we have a strict 'zero tolerance' rule concerning data security and has warned that Kester Michaels may face dismissal for this.

- Please advise, stating reasons, whether it would be unethical for Kester to defend himself against dismissal by arguing that he is a busy and hard-working executive?

[sub task (b) = 50%]

Solutions to Activities

8.1 Entry of Zoom into Kayland

What are the ethical issues raised if Zoom takes up the offer in the letter? (55%)

Approach: Ethics is about the moral rightness or wrongness of human actions. The best approach to ethics questions is: state the issue, state the ethical arguments against it, state the practical consequences of behaving unethically, state any counter-arguments, and then conclude. Try to use terminology from the CIMA/CGMA Ethical Code if possible and underline them.

Ethical issues raised by letter

Ethics concerns the moral rightness or wrongness of decisions and actions. 'Good' behaviour involves doing things that are ethical and avoiding actions that are not. 'Bad behaviour is doing unethical things, or not stopping others from doing them.

Issues

Zoom has been asked for a political donation in return for assistance from the Ministry of Transport in Kayland to assist its entry into the market.

Ethical aspects

Breach of integrity

Gaining access by making political payments looks like bribery. This is not a straightforward or honest way to do business

Breach of objectivity

Making this payment will encourage the Ministry to be biased in favor of Zoom. This is putting political interests ahead of the interests of the people of Kayland. Paying this donation creates a conflict of interest for the Ministry

Breach of professional behaviour

Most countries will have laws against bribing politicians, for example the US Foreign Corrupt Practices Act. Making this payment would probably breaks laws in Jayland or Kayland. It would also bring Zoom into disrepute as a firm that relies on bribes to win competitive battles.

Potential breach of professional conduct and due diligence
Zoom could choose to make the donation to the political party but Mr Jason may not act with professional behavior. He may not assist Zoom by obtaining it a TNC license. It would also mean that Zoom would have wasted its shareholders' money.

Consequences
Zoom may be prosecuted
Zoom could be prosecuted and fined under anti-corruption laws or laws on political funding. It could also have restrictions placed on its ability to operate in Kayland. This could harm the success of the venture in Kayland.

Zoom would lose the goodwill of other political parties in Kayland
By showing support for the ruling party alone, Zoom would raise the animosity of other political groups. This could lead them to block legislation that was intended to help Zoom's business. They could also reverse support for Zoom if they were to become a government in the future.

Suspicion over ethical standards of Zoom generally
If the Board of Zoom is discovered to have bought political influence it will raise doubts in the minds of stakeholders about whether Zoom acts ethically in other areas. Investors may be concerned that Zoom's managers do not report profits or their payments accurately, or act in the interests of shareholders. Other governments may be suspicious of working with Zoom in case it is assumed that they have taken bribes too

Counter arguments
Duty to make a profit
The Board of Zoom has a duty of professional competence and due care to maximize the shareholder wealth of investors in Zoom. Paying this political donation may be only way to do business in Kayland

Political funding very common in business
It is not very usual for businesses to pay political donations to one or more political party in the country. These donations are normally transparent and recorded and the public can see that they have been paid and therefore to guard against abuses of influence by the political parties and firms involved

Having Zoom will be good for Kayland
Zoom will improve transport and provide jobs for drivers who might otherwise be excluded from work by the high costs of gaining a taxi license. These benefits could offset any harm caused by the payment of the donation.

8.2 Ethics of dismissing a staff member for losing his personal computer
Please advise, stating reasons, whether it would be unethical for Kester to defend himself against dismissal by arguing that he is a busy and hard-working executive?

Ethical defence of Kester against dismissal
Ethics is the application of moral conduct. Kester owes various duties to his employer, KCTP.

Kester has breached several principles by allowing this data to be stolen.

The principle of professional competence and due care has been breached. This is a duty to show diligence in the conduct of his job. Leaving a tablet computer in a car at a shopping mall is not diligent behaviour.

Kester has also breached the duty of confidentiality. This means not to disclose information belonging to KCTP without authorisation. Downloading data to a personal device, and then proposing to use it on holiday is inconsistent with this. Others could see the information on the screen and the chance of theft is always present.

Trying to argue that he should not be disciplined due his hard work as an executive brings two further breaches of the ethical code.

There is a breach of professional behaviour. This means the duty to comply with regulations and not to bring the organisation into disrepute. Kester knew the rules on zero tolerance for data loss, but he has broken them. The result has been an article that harms the reputation of KCTP.

Finally it is a breach of objectivity for Kester to try to claim mitigating circumstances of being hard working. This is the duty to avoid self-interest and bias in actions. Kester should reflect that he is arguing to be spared discipline for entirely personal reasons. If someone else had lost the information Kester is unlikely to have applied the same lenient standards to them as he is now asking for himself

Chapter 9
Developing New Strategic Options

1. Frameworks for business analysis

The following techniques can be used by management to understand the present business model of the firm

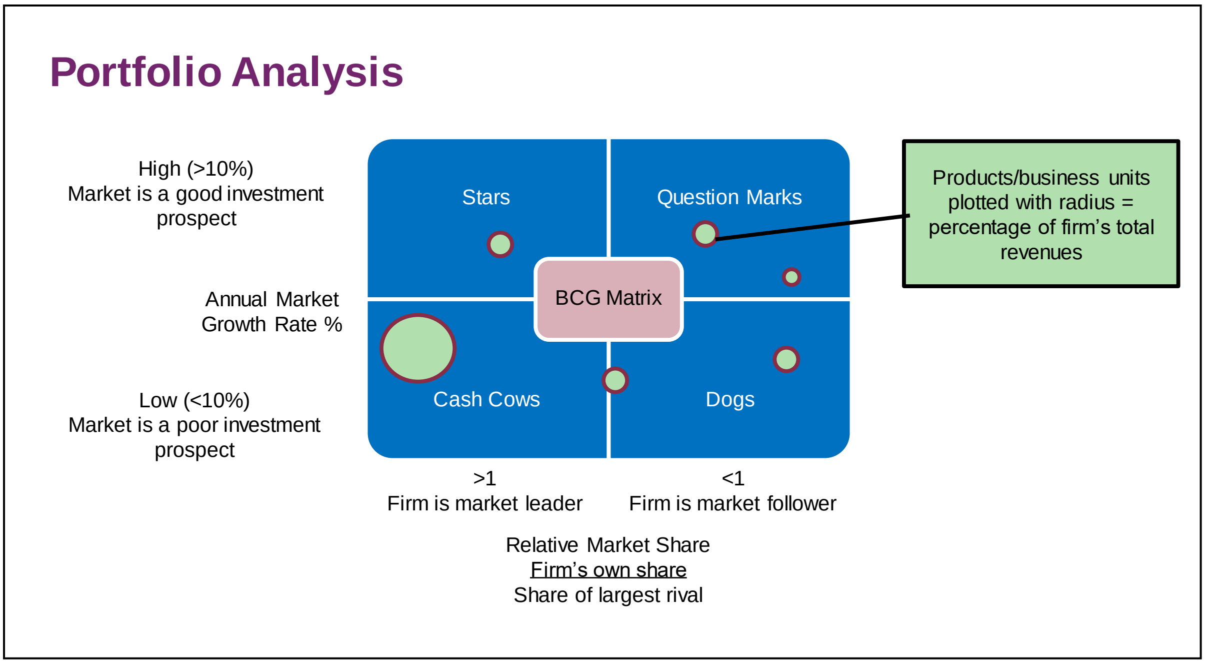
Resource audit

This was examined in November 2020. It is where management, using suitable experts including consultants, assess the available resources and competences of the organization. Sometimes these are classified into *strengths* and *weaknesses* that will form the goals of the strategy, to overcome weaknesses and to build on strengths.

The relevant resources have been classified into a series of ‘M’s to make them easier to remember:

- **Members/Manpower**: human resources, their cost, skills and morale
- **Management**: quality and expertise of the top team, their vision and experience
- **Money**: financial health such as available cash, and ability to raise finance
- **Make-up**: the organizational structure and culture of the business
- **Machinery**: the non-current physical assets of the firm and the flexibility, costs and quality of them
- **Methods**: the processes used by the firm, including whether any processes have been outsourced to partners
- **Markets**: the firm’s products and the markets it presently serves and how well its products are reputed in the market
- **Materials**: supplier relations, inventory management including use of just-in-time purchasing and manufacture, and the extent of reliance on a small number of suppliers
- **Management information**: the quality and timeliness of information available on which to base decisions and respond to market signals

Boston Consulting Group Matrix (BCG)
This is a graph which views a firm's products, or business units, as a group of investment assets. It is used to analyze cash flows and to recommend investment priorities

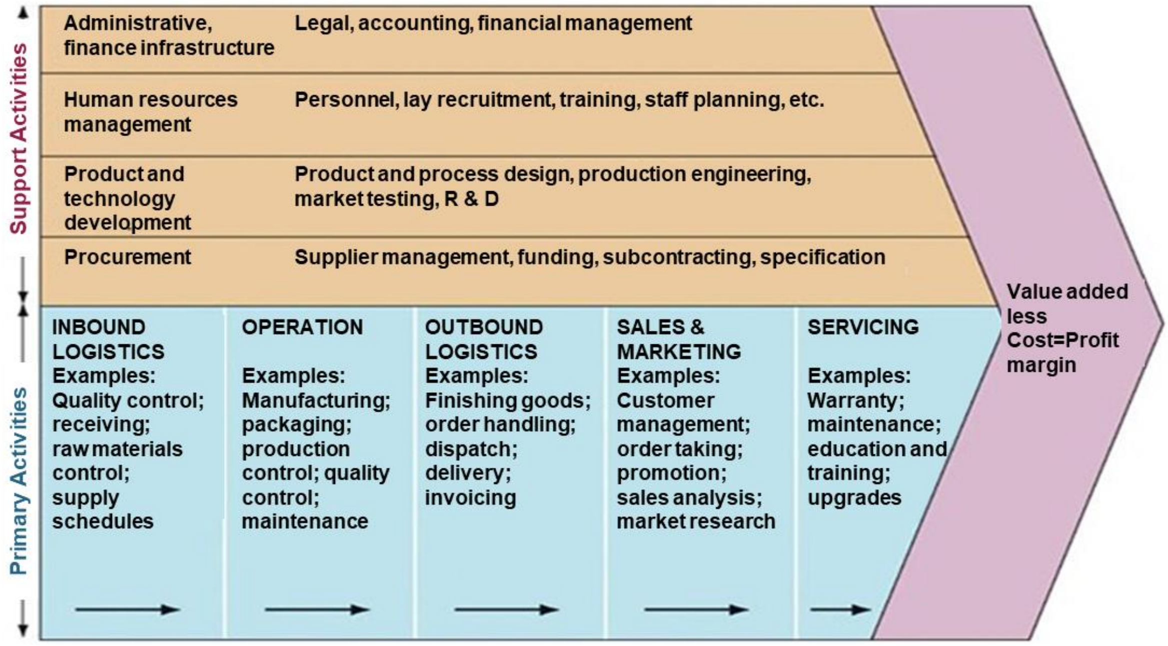


- *Question Marks*: invest free cash into these to build long-term market dominance or alternatively cease to support them and try to sell them
- *Stars*: continue to invest cash into these to retain dominant market position
- *Cash Cows*: limited future growth prospects so avoid investment and instead utilize the free cash they create to support investment in Question Marks and Stars
- *Dogs*: consider selling or closing these 'money traps' unless a sustainable niche market can be found for them

- The main contributions of the matrix are:
  - Helps management to visualize the investment/divestment decisions
  - Emphasizes the cash yielding/cash using relationship between its portfolio of products
  - Stresses the need to have development products to replace established products in the later stages of their product life cycle

- The limitations of the matrix are:
  - Rapid life cycles and innovation mean that few products are cash cows for long and so profit needs to made sooner in the life cycle
  - Overlooks the reasons other than cash flow that a firm may take into account when deciding when to develop and retain products. These include:
    - Risk reduction (for example having a substitute product available in case the main one hits problems)
    - Potential to use a product to cross-sell other products
    - Maintaining the reputation of the business (such as a Dog product that is important to significant groups of customers)
    - Filling supply channels: some products are made to stop rivals get a foothold in a market area which could later be used to attack the firm

The Value Chain
Developed by Michael Porter to assist management to identify the *value adding activities* of their firms



Application of the Porter Value Chain requires management to

- *Examine the linkages between the Primary Activities*
  - Can they be used to provide greater *added value*? (such as swifter order fulfillment, fresher/better product, superior customer service)
  - Can they be improved to reduce *non-value adding activities*? (such as inventory building up due to poor links between activities, excessive costs in sales and servicing due to poor quality operations)

- *Understand the true costs generated by the business model*
  - Recognize that doing things a particular way generates costs that may not be related to sales volumes (such as retaining functions in-house, offering multiple channels and product lines)
  - Appreciate that many overhead costs are caused by the Support Activities needed to facilitate the Primary Activities (such as the procurement costs generated by having multiple components and suppliers due to offering multiple variations of a product)

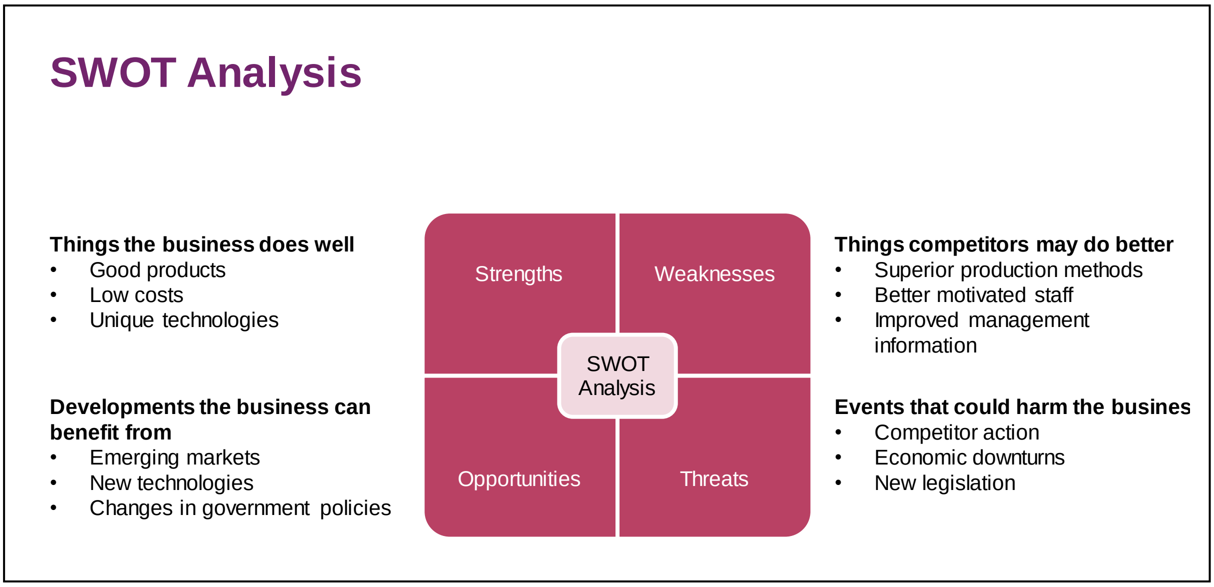
- *Undertake Business Process Improvement*
  - Use the value chain to support *strategic cost management* by changing the business activities to cut costs radically
  - Drive improvement to the performance of activities by *benchmarking* them across the firm, or against other firms
  - Use activity performance to set *Key Performance Indicators* to control value creation

2. Frameworks for strategic option generation

The assessment tasks generally focus on asking candidates to *evaluate* options that are under consideration. Candidates are not asked to *generate* ideas on what the firm could do. These tasks sometimes refer to the *strategic logic* of doing somethings, as well and to the *benefits* and *drawbacks*.

SWOT Analysis
The initial framework here is *SWOT analysis*. This strictly speaking belongs as part of strategic analysis.
The Strengths and Weaknesses are under management control and therefore internal. The Opportunities and Threats are external.
The basis of strategic choice requires management to improve the fit between the organization and its environment by

- Reducing the weaknesses that leave the firm exposed to threats (or unable to grasp opportunities)
- Building and developing strengths (or competences) to grasp the opportunities



Porter's Generic Strategies Model
This model suggests that superior competitive advantage is only sustainable by choosing to utilize one of three competitive strategies
Sustainable competitive advantage creates superior value in the long run. It can result from either:

- Being the lowest cost firm in the market
  - Higher margins
  - Able to withstand price wars better than rivals
  - Acts as a barrier to entry
  - But won't grow as incomes grow and people trade up to a better product

- Being differentiated as the premium provider in the market
  - Not bought on price and so unlikely to get into price competition
  - Differentiated status acts as a barrier to entry
  - But may lose sales in bad economic times

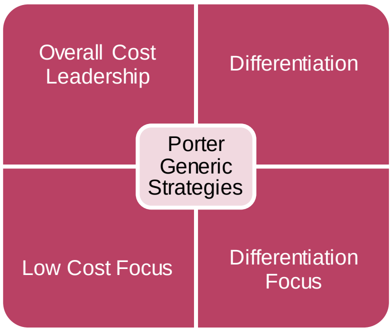
1. *Overall cost leadership*: lowest cost operator in the main industry
2. *Differentiation*: seen as a premium provider in the industry (note: multiple differentiators can co-exist in the industry providing the distinguish themselves in different ways)

3. *Focus*: target a small segment of the industry and develop either a super-premium product or a sub-budget product for the segment

Porters Three Generic Strategies

Lowest cost in industry
- Survive by undercutting competition
- Low cost production locations
- Mid-market product/service

Focus on very budget-conscious customers/clients
- Limited distribution
- Stripped-down product/service



Superior product positioning
- Survive by charging high price
- High brand value
- Superior product/service

Focus on exclusive premium customers/clients
- Niche distribution
- Low volume/customised product

Ansoff Product-Market Matrix

The *Ansoff Matrix* classifies directions for the strategic development of the business. Each quadrant has particular benefits and risks.

Ansoff Matrix

Increase sales and market share
- Greater promotion
- Expand capacity
- More distribution channels

Expand the firm into new markets
- New customer segments
- New uses for product
- New countries

Launch new products
- Complements for existing produc
- Something the customer buys fro another business
- A breakthrough technology

Enter new industries
- Vertical integration
- Technology extension
- Risk avoidance

- *Market Penetration* staying in the same industries as at present and increasing sales and output
  - Relatively low investment costs, and low risk because firm knows the industry
  - Not an attractive investment strategy if the industries are mature or declining or firm is suffering intense competition

- *Market Development* selling existing products to additional customer groups
  - Enables firm to capitalize on core production competences, brands and so on and use them to gain foothold and returns from new market. Also reduces exposure to decline of competition in present market

    - Investment risks from the problems of dealing with unknown markets, different customer needs, operating abroad

- *Product development* is using existing market distribution channels and pipelines to sell additional sorts of products
    - Reduces reliance on present products and can enable harnessing of customer relationship and brands to gain value
    - Investment at risk from unfamiliar products and production technology. Also, danger that firm will lack credibility in making the new product or that brands won't stretch to encompass wider product range

- *Diversification* requires investment in a new industry which involves unknown markets and new technologies
    - May help to diversify the risks of the company away from its present industries. Also, may enable leverage of competence from existing industry to revolutionize the new industry
    - Management competences may not stretch to new industry and investors may be suspicious of conglomerate/diversified firms

ACTIVITY 9.1

Context
Zoom is a Transport Network Company (TNC) offering a ride-hailing service in the country of Jayland (like Uber, Lyft or DiDi). These services have revolutionized the taxi industry by using a technological platform to match self-employed drivers with riders. Zoom wants to expand its business.

Task
You have been called into the office of Kevin Sato: Chief Finance Officer for Zoom
'As you know Zoom doesn't make a profit and yet we have 30% of the taxi business in our home country. Investors are becoming restless and want to know how we can extend our platform and make more sales and, eventually, more profits.

The Board has a meeting tomorrow to generate some new ideas on how to leverage Zooms platform. I need some ideas to present

Please suggest some ways in which Zoom could grow its business

[sub task (a) = 55%]

Solutions to Activities

9.1 Growing Zoom's business
Please suggest some ways in which Zoom could grow its business (55%)
Approach: generating ideas can be improved by considering in turn each of the quadrants of the Ansoff Matrix. The solution below suggests many ideas, but in the assessment, you would only need to suggest five or so. Also, this question does not require you to classify the ideas into the quadrants, and neither does it require an idea for each quadrant. The task is testing whether you know how to develop strategic options, not whether you can illustrate the Ansoff Matrix.

[Market Penetration]
<u>Increase and retain subscribers by loyalty discounts on next rides</u>
This could be offered through the app. It would help to convince riders to use Zoom rather than rivals, and also to take cars rather than buses or trains. This would help improve market share and sales revenue.

<u>Extend services to additional locations in Jayland</u>
Zoom may not cover all locations in its home country. For example, it may not serve smaller towns and settlements. Rival taxi firms may not serve these areas either. By recruiting drivers in these areas, perhaps offering better rates, Zoom could gain sales.

<u>Greater promotion of service to attract non-users</u>
Zoom may have attracted as many riders as it can using its existing promotional strategy. It could change the strategy to attract non-users, for example by inviting price comparisons with rivals, or stressing its ecological credentials.

<u>Service improvements</u>
This could include dedicated pick-up points to enhance convenience and rider security, or cars that are more suitable for people with limited mobility. The service could also be improved by ensuring the cars are warm, and that passengers are offered help with luggage and offered bottles of water.

[Market Development]
<u>Create premium and discount versions of Zoom</u>
The present service may be pitched at the mass market with a good service. It might be able to attract riders from the limousine market by offering a premium service for executive travel. Alternatively, it could penetrate more price-sensitive segments by offering discount for household use, or discounts for less attractive cars.

<u>Expand operations to new countries</u>
Zoom presently operates only in its home country. Its platform could be modified to attract riders and drivers in new countries. This would help spread the fixed costs of maintaining and improving the platform and spread risk from reliance on just its home market.

<u>Offer specialist versions of Zoom</u>
New customer segments could be attracted by varying promotion and making small changes to Zoom's service. These new markets could include medical transportation (doctors, supplies/samples), assisting persons with special needs (e.g. wheelchair access), or Eco-Zoom: enable riders to specify electric cars

[Product Development]
<u>Sell in-app services to users</u>

Zoom has its apps on the smartphones of people who travel. Earnings could be increased by offering related services to riders such as accommodation or tourism guides and narration for tourists.

<u>Extend model to other transport needs</u>
Zoom's platform could appeal to specific transport needs such as food collection and delivery from restaurants to homes, collection and delivery of children to and from schools or friends' homes

[Diversification]
<u>Use software to match other suppliers and buyers</u>
Zoom has a powerful matching platform. It could be used to match provider and customers of other services including hotel accommodation, holiday accommodation, restaurant bookings and sporting or musical tickets. Commissions could be earned from this matching.

<u>Sell client data to advertisers and analysis firms</u>
Zoom's app is able to track the user, it records their home address and other details, and it could be modified to access other data from their smartphone such as their social network, their browsing history, and their spending. This data could be sold to data brokers and provide an additional source of income.

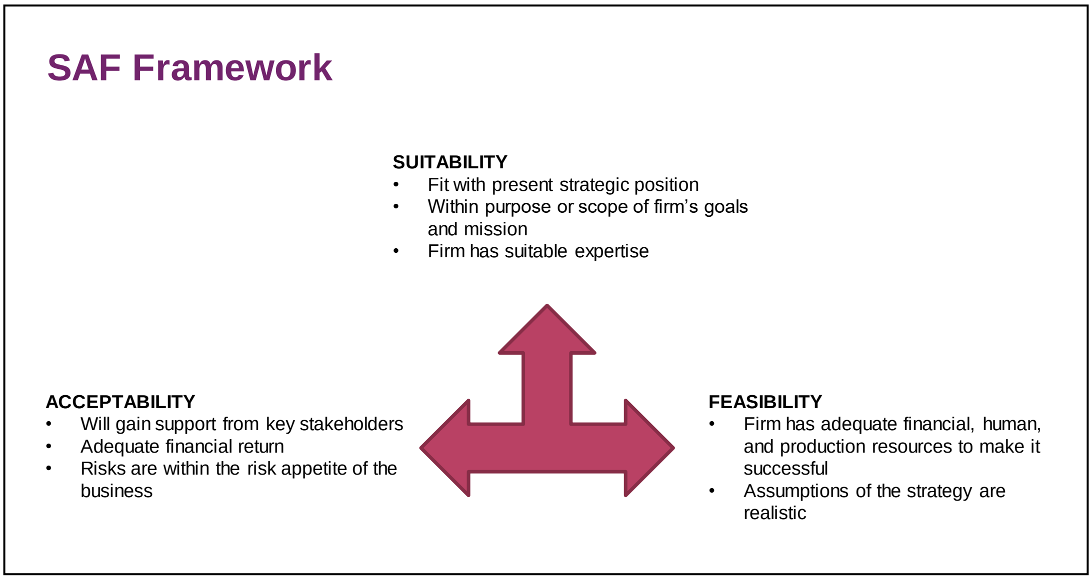
Chapter 10
Evaluating Strategic Options

1. The SAF Framework

The SAF Framework
This framework (by Johnson, Scholes and Whittington) is useful to evaluate a strategic option. It encourages management to evaluate each option using three criteria:

- Suitability
- Acceptability
- Feasibility

Although no strategic option us likely to meet all three criteria completely the model helps management to see areas where the project might need to be modified to gain support from all stakeholders.



2. Digital strategies

The SCS Assessment Outcomes includes the requirement to evaluate and to recommend strategic decisions ‘digital or otherwise’. This recognizes that there is a separate type of strategy called a ‘digital strategy’.

Digital technologies

Artificial Intelligence
- Technology that problem solves and learns

Cloud and mobile computing
- Flexible and collaborative data at low cost

Internet of Things
- Networked objects that report their position, status, and behavior

Big Data
- Large volumes of data that provide basis for real-time analytics and data mining analytics

Blockchain
Public, distributed, notarized ledger system that can track and validate transfers and transactions

Data visualization
Visual representation of data such as DNA molecules, or hotspots of crime and accidents

3D printing
Additive manufacturing of prototypes, components medicines, clothes and food

Process automation
Robotics in administration and manufacturing

These will be based on the main digital technologies described earlier

The following is a useful summary of the areas of impact of a digital strategy. Examples are based on Disney’s Digital Transformation under Bob Iger after 2018

Impacts of a digital strategy

- Business model
- Structure
- People
- Processes
- Platforms
- IT capability
- Offerings
- Engagement

- **Business model**
  It can impact on how the business makes money
  For example, Disney moved away from being a film studio that operated leisure parks, and using digital technology to create movies to a new business model that uses digital streaming, online games and engagement, and big data at the heart of its business

- **Structure**
  It can impact on how the business in organized and the extent to which it uses partnerships and value networks to operate its business
  Disney consolidated into two divisions: ‘Direct-to-customer’, and ‘Parks, Experiences and Consumer Products’, bought Fox movies, and created streaming services Disney+, Hulu (movies) and ESPN (sports)

- **People**
  The increased use of digital technology will change the sort of skills and people that will be required by the business. There is a need for digital skills (IT skills and so on) but also for people with a ‘digital mindset’ which is a set of attitudes and behaviors that understand the capabilities of technologies, actively enjoy disruption, grasp interconnections between things, and respond quickly to changes and opportunities.
  Disney gained digital skills by recruitment of a new generation of ‘digital and social media creators and influencers’

- **Processes**
  This is how things are done in the business. It can include use of process automation and AI to streamline workflows, the acceleration of product development, and agile working and project management
  Instead of buying-in characters from children's books and other franchises Disney shifted to the deliberate creation of characters to maximize cross-platform potential (films, games, toys) and trialed them online to assess their potential before adopting them for films

- **Platforms**
  Many digital strategies involve creating a digital platform that will attract users and suppliers to engage with the firm and share content and opportunities

  Disney has regularly used its movie as platforms for collaboration with toy makers, hoteliers, restaurants and so on. It now uses its digital streaming platform to attract subscribers and program makers, and its live online games platform and social media platforms to engage with its customers

- **IT Capability**
  The ways information is used and how it is managed will become central to the business.

  Disney utilizes Big Data collection from 'listening posts' including smart wristbands, apps, monitoring use of streaming services, analysis of correspondence and website posts to fine tune its offering, and to develop new products and services

- **Offerings**
  These are the products and services that the firm offers to its customers. Digital strategies tend to shift towards a more personalized offering including providing you with your own version of the home page with recommendations for you, or custom-made clothes, furniture, and financial products

  Disney from 'same to all' cinema films, shops and parks to personalization via streaming, creation of digital worlds and online 'clubs', apps, and personalized engagement

- **Engagement**
  How stakeholders are engaged with. The principle is that stakeholders are no longer the passive recipients of the things the firm decides to make and tell them. They are actively engaged and their comments and needs influence the way the firm behaves. Obvious examples are the use of social media comments and rankings to improve product and service, and the close monitoring of comments and usage using analytics to create response strategies

  Disney claims that it is now entirely customer-centric

More on digital strategy and digital thinking

The term 'digital strategy' is very poorly defined in literature and it can take several meanings in this exam.

The approach above suggests that 'digital strategy' means the harnessing of new digital technologies to revolutionise the business.

In practice the use of terms like 'digital strategy' and 'digital thinking' seems to have widened beyond merely the applications of digital technologies.

Writers have noted the success of digital industries like Amazon, Alibaba, Tesla, Facebook/Meta, Tencent and so on (and their founders: Jeff Bezos, Jack Ma, Elon Musk, Mark Zuckerberg and Ma Huateng), and attributed their success to strong leadership and vision to undermine established industries by the use of revolutionary digital technologies. This is now being promoted and taught as a better alternative to conventional strategic thinking (see Chapter 3).

Slogans like ‘move fast and break things’ (Facebook), approaches that emphasise setting clear simple goals and swift experimentation in ways to achieve them, and approaches that prefer revolutionary rather than incremental improvements to the business model have all been described as ‘digital’.

Be prepared for the Examiner to use ‘digital strategy’ to mean this sort of strategic leadership too.

However, in May 2021, the Examiner used the term ‘digital strategy’ in yet another way, in effect to mean ‘a strategy for information systems’. This is the approach in Activity 10.2 below.

ACTIVITY 10.1

Context
Fizz is a manufacturer of soft drinks, including cola, lemonade and energy drinks. Many of these drinks have a high sugar content and a high caffeine content, both of which are believed to lead to bad health outcomes due to strain on the consumer’s heart, and the promotion of obesity that will lead the consumer to develop chronic illnesses such as diabetes and cancers.

The government of its country has announced that it is considering imposing a tax on soft drinks based on the sugar content that will raise the prices by up to 25%

The Director of Marketing has proposed that Fizz creates a smartphone app that allows users to scan the barcodes of the Fizz drinks, and other foods, that they purchase. It will also prompt them to upload pictures of them drinking Fizz drinks to a social media site. He calls it a ‘digital strategy’.

The benefits of the app to the wearer will be the opportunity to obtain discounts on future purchases of Fizz products, some enhanced social engagement, and messages from Fizz that suggest exercise or changes to consumption based on what they are buying.

The benefits to Fizz will be that they can be seen to be trying to address the poor health effects of their products, but also that they will be collecting data that can be used to help shops analyze their sales, and that will help Fizz to understand its customers better.

Task
You have received an email from the Director of Finance of Fizz
Please evaluate this proposal as a digital strategy

[sub task = 55%]

ACTIVITY 10.2

Context
Pixlwizz designs and publishes video games, in the same industry as PlayStation and Xbox.

It has decided to launch its own metaverse and also to experiment with 4D games. In consequence it will need to invest in building its own data centre to take its games to the next stage. This will require it to pay cancellation fees to Cloudbor, its present cloud services provider. It will also need to develop its own consoles rather than continue to rely on third-party consoles because they are not fast enough to run metaverse and 4D applications.
This is likely to cost nearly W$100m to set up.

Task
You have been called into the office of Maria Tektonidou: CFO of Pixlwizz

I need a report explaining:
- The need for Pixlwizz to develop an explicit digital strategy to address issues such as raised by these new developments

[sub task = 50%]

3. Mergers and Acquisitions
SCS Assessment tasks frequently ask candidates to provide an analysis of the strategic logic of a merger or acquisition between their company, from the pre-seen, and another firm, often one featured in the pre-seen material too.

Technically these methods of *business combination* are different.
- *Acquisition*: the share capital of the target company is bought by the predator and its assets become the property of the predator
- *Merger*: the share capital of both companies is exchanged for share capital of a new company which will become the merged entity

Benefits of acquisitions	Drawbacks of acquisitions
Gain access to new market quickly	Pay too much for a bad firm
Synergies • Higher joint revenues • Lower joint costs • Better management	High costs of integration • Closures/dismissals • Integration of systems
Reduce risks • Greater control of market/industry • Spread across business sectors	Financial strain on business
Defence against market entry	Lack of management competences

Synergies
One firm buying another should expect to pay a price which is at least as high as the discounted value of the target's future earnings. Therefore, to be worthwhile it must be possible for the combined business to have higher earnings than they would be if the two firms

had remained independent of each other. This is why synergy is often summarized as the *2+2=5 effect*.

Sources of synergy	Examples
Operational	• Economies of scale from combining operations • Lower costs of inputs from larger volumes • Fixed costs spread over larger product range • Rationalisation of common operations • Disposal of surplus assets (eg one head office, duplicated stores)
Financial	• Cash flows from one support finance needs of the other • Lower borrowing costs due to larger asset base • Larger firm able to access cheaper finance (eg becoming listed and issuing shares) • Shared Research and Development expenditure
Managerial	• Weak management replaced by strong management • Transfer of technical skills and know-how
Market	• Reduction of competition • Transfer of brands to add value to each other's products • Sharing of common distribution and sales channels
Risk spreading	• Wider range of businesses gives more stability of income • Greater market power

Synergies have four beneficial effects for profitability
1. Increase revenues
2. Reduce costs
3. Economize on assets used
4. Reduce risk
5.

Reducing the risks of an acquisition
1. Conduct due diligence (e.g. by instructing consultants and professional advisors)
2. Conduct forecasting on the market or industry in which the target company operates
3. Use a risk adjusted cost of capital to evaluate the acquisition
4. Assign probabilities to the likely outcomes and assess best-case and worst-case scenarios
5. Defer payments to the seller, or use equity, to limit expenditure
6. Identify potential exit routes from the investment if it is not successful (e.g. dates of break clauses in leases/contracts, assets that could be sold, valuable divisions or operations that could be retained)

ACTIVITY 10.3

Context
Couchweb provides television programs and movies as a subscription video on demand service (like Netflix and Disney+) business. Its strategy is to increase global subscribers but also to develop its own series, movies and programs to reduce dependence on movie makers and also to differentiate itself from rivals. HomeVideo is a rival to Couchweb

Task
You have received an email from Martine Andels: Director of Finance of Couchweb

Confidential
The Board is considering entering negotiations to buy HomeVideo.

HomeVideo produces some successful series that are very popular in Mayland. It also has a contract with a broadcaster in Upland, a country we have not been able to sell to yet.
But it seems to be in financial problems since it started making its own programs and its profits fell last year.

Upland is a very large market and subscriptions are paid in U$. But both HomeVideo and Couchweb rely on programs which we pay for mainly in M$.
Before we go too far with this, I need to have a briefing from you to show the Board

What are the potential benefits and drawbacks to Couchweb of the proposed acquisition of HomeVideo? *[sub task (a) = 55%]*

4. Other methods of business development

SCS Assessment tasks frequently ask candidates to provide an analysis of the strategic logic of a merger or acquisition between their company, from the pre-seen, and another firm, often one

Collaborative growth

These are sometimes used as an alternative to organic growth or acquisitions.

1. *Joint Ventures*: this is a separate firm, or vehicle, set up by two or more business partners. The partners are shareholders in the joint venture firm and use it pool resources and expertise to jointly realize their business goals

2. *Strategic alliance:* similar to a Joint Venture but without the use of a separate business vehicle. The partners agree to work together to achieve mutual benefit. There is usually a legal agreement between the parties which may also be backed up by some exchange of equity between the parties to ensure each benefits from the success of the other.

3. *Franchise:* the leasing of the firm's business model to outside firms. This requires the *franchisee* to be trained in the business model and to buy supplies from the franchisor. There will also be a capital sum paid by the franchisee for the exclusive right to use the model in a given territory.

4. *Licensing agreement:* the firm licenses its intellectual property to other firms, such as recipes, brands, mineral rights, technologies. The *licensee* exploits these commercially in their area and pays a royalty, often based on sales turnover, to the licensor.

5. *Outsourcing/contracting out:* instead of using its own resources and management to carry out the whole strategy a firm may sub-contract it to other organisations to carry out for a set fee.

Benefits	Drawbacks
Reduces the amount of capital needed for growth	Potential for partner to break partnership unexpectedly
Gains the expertise and knowledge of the partner	Danger from sharing intellectual property rights
Shares the financial risks	Difference in strategic objectives may reduce effectiveness
Avoids competition	Inability of management to work together
May overcome legal obstacles to entering a market	Loss of control over operations, quality and brand

5. Divesting business units

The opposite of a merger or acquisition

Divesting Business Units (demerging)

Reasons:

- To escape loss-making units
- To gain cash in a liquidity crisis
- To re-focus on core business
- To improve P/E ratio by removing low growth business
- Due to change in business strategy
- To satisfy competition regulations

Methods:

- Sell to existing management (Management Buy-Out)
- Sell to another business (trade sale)
- Float division separately (equity carve-out)

6. Real Options Theory

An approach to making strategic decisions that can assist under conditions of uncertainty.

Once the decision has been taken to pursue a strategy there are still choices to me made about the timing of the investment and the amount of investment to commit.

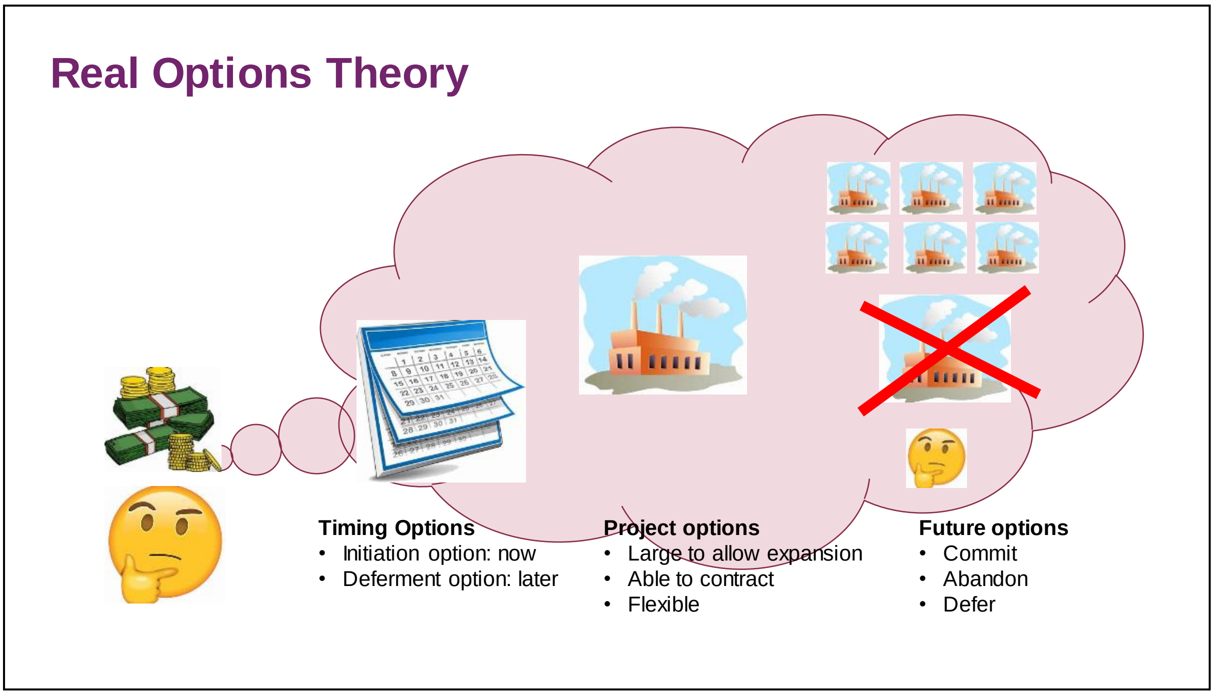
Example: a firm is deciding whether to build a new factory to make driverless cars. The precise features that will be demanded are unclear, and the level of demand is also unclear. A lot depends on how regulation, customer tastes, and digital technologies develop over the next 5 years.

Real options relating to investment in the project

Having decided to make a driverless car management needs to decide how to invest in such an uncertain product.

- *Initiation and deferment options*: this is the choice on when to invest in the project. If things are favorable, they may invest now, but if they are awaiting something important, say a decision on the legality of driverless cars, they may defer the investment until that is certain
- *Later stage options*: are available in the future and can be taken in the light of how things have developed, and perhaps when things have become less uncertain
    - *Option to commit*: this means to make the larger investment and scale up production from the initial pilot investment earlier
    - *Option to abandon*: this means to give up on the project, spend no more money, and try to sell it or to close it down

- *Option to defer*: this means to delay further investment because things are still not clear



Real options relating to project size

- *Option to expand*: the factory could be built with capacity in excess of the initial production. For example, it is built on a larger piece of land than necessary, and suppliers are chosen on their ability to supply more than will initially be needed. If the driverless car is successful in the market it can scale up capacity further

- *Option to contract*: the factory is designed to operate at lower output if the car fails to sell. For example, the factory is flexible and can be switched to another product, and more of the functions have been outsourced to reduce fixed costs

- *Option to expand or contract*: the factory is able to change output as demand changes, for example by having flexible manufacturing systems or by relying on outsourcing for components and focusing on assembly

ACTIVITY 10.4

Context
Runn is a large global Courier Express Parcel (CEP) firm based in Kayland. It wishes to grow its business. It has no experience of food distribution. The margins in its parcel delivery service are falling due to the trend towards costly urgent deliveries for which customers are unwilling to pay more.

Runn has a Vision to be the ‘World’s favorite delivery company’. Its Mission is to ‘link businesses and individuals in a responsible and sustainable manner’. Its Values are that it ‘cares deeply about its customers and stakeholders and it honors its commitments’.

Task
You have received an email from Chen Yanyun: Director of Finance for Runn
‘We are bidding for a supply-chain contract from Down Home Chicken. It’s Runn’s first venture into food delivery.

Down Home Chicken operates about 900 restaurants in Geeland and it specializes in serving chicken meals, and some other foods, to the fast-food market.

Its present delivery contractor uses 8 depots to serve Down Home Chicken and this makes it very costly. We can win this contract by basing our operation at a single distribution center on our site at Semon. This is a central location and any truck dispatched from Semon can reach the furthest Down Home Chicken store within 6 hours.

Chicken and other foods must be temperature controlled for health reasons, so Runn will need to build a refrigerated distribution center at Semon and also invest in refrigerated trucks to make the collections from chicken farms and the deliveries to the 900+ Down Home Chicken restaurants. We will build in extra capacity to the network because we expect to win additional similar contracts to come from food stores and restaurants once we show we can handle this Down Home Chicken contract.

The budgeted capital expenditure is G$300m for the specialized refrigerated distribution center and the trucks.

Reference Material An article from
A national newspaper that states:

- Fast food sales are growing at 15% per year
- Fast food is produced in a healthier way than home-cooked food
- Fast food is very poor for peoples' health if consumed in large and regular amounts

- **Please give me your strategic evaluation of this contract proposal**

[sub task (a) = 55%]

Solutions to Activities

10.1 Fizz soft drinks launching smartphone app

Please evaluate this proposal as a digital strategy (55%)

Approach: this task was provided by CGMA in a specimen exam for the exams from February 2020. It is very unclear. It seems to be asking 'to what extent does this proposal conform to the principles of a good digital strategy? However, the examiners own solution seems to answer the simpler question 'to what extent is this proposal a good idea?' and it ignores the digital element entirely. The solution below uses SAF to analyze the proposal as a strategy but in the conclusion, it assesses whether it is a digital strategy or not

<u>Evaluation of proposed digital strategy</u>

This strategic proposal can be evaluated using SAF analysis

Suitability

This assesses the fit between the proposal and the strategic position and mission of Fizz. The positive aspects of the proposal are as follows:

<u>Provides a way to increase sales of Fizz drinks</u>

The app will help to improve customer loyalty towards Fizz drinks by providing a value-added engagement through giving diet and exercise advice, discounts, and a social media experience (it is assumed that Fizz intends to create an affinity site that can be accesses by this app).

<u>May offset risks from sugar taxes</u>
The recent announcement of possible taxes on sugar in soft drinks leaves Fizz exposed to potential sales reductions. This app engages with that threat by nudging customers towards healthier Fizz drinks such as sugar-free, fruit juices or bottled water. This may protect revenues from the effects of the tax

The negative aspects of the proposal are as follows:
<u>Fizz lacks expertise in creating Apps</u>
This is a new technology for Fizz and we lack the staff expertise to develop this app. It may not be suitable for Fizz to embark on using apps to drive a digital strategy unless it can acquire this expertise.

<u>Fizz lacks expertise in management of social media communities or Big Data</u>
The main rationale behind this digital strategy is the engagement and data it may yield. These are both complex areas and Fizz has not operated social media sites before and will need new social media savvy staff to develop and monitor the posts of users as some could be derogatory to Fizz. Data analytics of the sort suggested are also sophisticated activities that Fizz does not presently conduct.

Acceptability
This means the acceptability of the proposal, and its risks, to stakeholders

The positive aspects of the proposal are as follows:
<u>Customers may like the app</u>
Customers such as millennials will be very accustomed to using apps on their smartphones. They may like this app if it provides discounts and other engagement and it allows Fizz to position its brand for the digital age

<u>Shops will welcome increased sales of products</u>
Shops are important stakeholders in this. Providing it leads to more sales of Fizz, and perhaps protects their sales from the sugar tax, they will welcome it.

<u>Government may welcome app</u>
The Minister for Health wishes to reduce consumption of unhealthy drinks. If this app advises exercise and reduced consumption of sugar it will help the government in its aims. It may agree to the license of the app, and perhaps to delay the implementation of its proposed tax on sugar.

The negative aspects of the proposal are as follows:
<u>Some customers will dislike the intrusion of the app</u>
This app collects data on the user and their consumption. It also tries to give advice. Some customers may cease to use the app, or publicly criticize it, due to its intrusion into their purchasing and consumption choices and implied criticism of them

<u>Shops may resent time taken</u>
There seems nothing in the app to stop the user from scanning products on the shelves that they haven't bought just to claim discounts. Shops will need to validate purchases and they will need to administer discounts. They will also have customers standing by counters dealing with the app and deterring shoppers

<u>May not be a financial success</u>
This would concern shareholders because it will be expensive to develop and maintain and the Director of Marketing has not provided any clear financial benefits

<u>Fizz could face risks from regulation on data collection and use</u>
The trend is for stricter regulation over the collection, storage, and use of digital data. If Fizz is not compliant with these regulations it could face fines and this would annoy the regulators and the shareholders

<u>Other food makers may rival it and reduce its usefulness</u>
Fizz intends to use the data to affect the presentation of snack foods and perhaps to develop its own range. This will alarm the snack food makers and may lead them to create a rival app that will make the Fizz app less successful.

Feasibility
This means the feasibility of the proposal, including the cost of resources, to Fizz

The positive aspects of the proposal are as follows:
<u>Fizz has barcodes on its drinks</u>
Shops presently use bar codes at the electronic point of sale (EPOS). Therefore, the cans will have barcodes that can be scanned. Presumably snack foods made by other firms will also have bar codes although Fizz will need to develop a reference table to identify what these products are if it wants to give advice on their sugar content or produce rival versions.

The negative aspects of the proposal are as follows:

<u>Fizz has no experience in app development</u>
This was discussed above under suitability. It may prove to be a significant cost to Fizz and also may affect the functioning of the app.

Conclusion
This is a very basic app and not a ‘digital strategy’ or ‘new digital business model’ as the Marketing Director claims. It is very little more than a digital version of a coffee bar loyalty card. The limitations of the app mean that is unlikely to have major influence on sales or behavior without more support and features. This app may be a worthwhile first step towards a digital presence providing it doesn’t cost too much. However, Fizz needs to be more imaginative in its understanding of ‘digital strategy’.

For example, creating a wristband that monitored blood sugar levels and engaged the wearer in a program to improve their health would provide a platform on which to base a digital strategy and involve government health authorities, medical organizations, fitness centres, insurance providers and more. It would help pivot Fizz away from its present unsustainable business model based on selling sugar and caffeine enriched drinks and towards positioning itself as an organization providing assistance to leading a healthy lifestyle.

10.2 Digital strategy for Pixlwizz – video games publisher
Please explain the need for Pixlwizz to develop an explicit digital strategy to address issues such as raised by these new developments (50%)

Need for an explicit digital strategy

Digital strategies focus on using technology to improve business performance. As an information-based business Pixlwizz needs to implement the various parts of its digital investment and development within a clear strategy.

Developing a strategy encourages management to take a considered approach to the use of digital developments and to plan their introduction systematically to achieve clear commercial objectives and with a business case to back it up. This is better than a series of ad hoc responses to digital developments urged by developers, technologists or as a response to the urging of technology firms wishing to sell them.

To harness digital opportunities
Digital technology is a driver in the business ecosystem (see above) and creates opportunities for Pixlwizz. The Board needs to deal with these in the same way as it might deal with other opportunities, such as changing laws or potential acquisitions. It needs to retain awareness of them, and have plans in place to exploit digital opportunities.

To help set investment priorities
There are many new digital technologies available to Pixlwizz, but it only has limited investment funds. Investing in OtherLife may reduce the funds available for investing in VR/AR, mobile gaming, or game development.

A digital strategy will help prioritise investment cases in terms of which are essential to realisation of the digital strategy. For example if Pixlwizz has prioritised developing its place in the VR/AR gaming world it would mean that investing in OtherLife should be passed over to keep the focus and funding on the main strategy.

To gain support from stakeholders
Having a clear digital strategy may excite shareholders and investors, raise the share price, and enable Pixlwizz to get the funding it will need. Similarly it may attract and motivate expert staff who wish to be part of the developments that PW is working on.

As the present situation shows, the needs of Pixlwizz that have emerged have moved faster and further then Cloudbor or the console manufacturers were able to support. It will cost Pixlwizz money to move on. Had a digital strategy been developed it should have noted these limitations and either encouraged the partners to move forward more quickly with investments of their own, or at least warned Pixlwizz that there could be delays and costs.

To plan staffing
One outcome of a digital strategy will be changes to the staff and organisational structure of Pixlwizz. For example the development of OtherLife may require forum moderators, whereas the development of 4D games will need new programing skills and also new game designer skills. These cannot always be recruited quickly externally. Instead a digital strategy would highlight these needs and part of the strategy would be the recruitment and development of suitable people ahead of time.

To manage cyber security
Digital technologies often promise rapid enhancement to earnings and/or deep reductions to costs from the use of novel technical innovations. Many of these are not secure or carry risks that have not been fully considered.

Having a digital strategy would normally include a strategy for managing the cyber risks associated with the new developments.

To enable monitoring and control
A digital strategy would employ digital innovations to achieve clear business goals. The investment cases should be based on these goals. This makes it possible to monitor the deployment and development of new digital innovations against the time scales, operational efficiencies, and revenue goals in the strategy.

This can help avoid wasting time and money on failing implementations.

10.3 Couchweb acquiring HomeVideo

What are the potential benefits and drawbacks to Couchweb of the proposed acquisition of HomeVideo? (55%)

Approach: evaluation of a strategic option features in many real exams. There are a number of approaches that can be used, including using the SAF framework to discuss the Suitability of the option (i.e. how well it satisfies the aims of the business and its present strategic position), the Acceptability of the option (i.e. its financial returns and risks and the likelihood that key stakeholders may support or oppose it), and the Feasibility of the option (i.e. the availability of resources to carry it out such as finance, and management capacity and know-how).

This task requires an answer structured into benefits and drawbacks, but you can still use the SAF framework to help you generate ideas under each.

Proposed acquisition of HomeVideo

This acquisition should be judged solely on its likely impact on the value of Couchweb's share price. This impact will be determined by the future earnings impact of the acquisition and the impact on shareholder risk.

Benefits of acquiring HomeVideo

<u>Gain access to new country markets</u>

HomeVideo has access to at least one country not presently served by Couchweb, Upland. There may be other countries also that this acquisition would give Couchweb access to. This helps to diversify the revenue sources of Couchweb to help insulate it further against a downturn in its home economy

<u>Gain access to programs and content</u>

HomeVideo has produced successful series. These could be added to Couchweb's catalogue This may gain extra subscribers. It may also convince more cable and broadcast firms to package the Couchweb catalogue with their offers because it would be more popular than HomeVideo or Couchweb alone.

<u>Resist amalgamation of media firms</u>

The program making and the broadcasting firms are amalgamating and converging. For example, Fox sold its catalogue and production business to Disney, and cable firm Comcast has launched a takeover bid for Europe's Sky broadcaster.

Having a more substantial catalogue and access to a greater number of countries can provide subscribers with an alternative to these amalgamated rivals. It can also force the program makers to continue supplying Couchweb because after buying HomeVideo Couchweb could offer program makers a greater subscriber and country portfolio to sell their content to.

<u>Gain synergies</u>

Synergies are financial benefits arising from the combination of two previously separate businesses (the 2+2=5 effect). Acquiring HomeVideo may yield a number of synergies including economies of scale in administration, technology, and the buying of program. The acquisition would give access to popular TV program produced by HomeVideo that Couchweb could exploit or develop spin off TV program from.

This means that the eventual value of Couchweb may rise substantially more than the cost of buying HomeVideo (*student note: technically speaking an acquisition can only be justified by reference to synergies. This is because the price paid for the target will be at least the present value of its future earnings. Unless the acquisition can increase these earnings, or increase the earnings of the predator, it will not increase shareholder value)*

<u>Reduce competition in Video on Demand market</u>
At present HomeVideo competes against Couchweb for subscribers. This means that both Couchweb and HomeVideo must charge lower subscription prices, and offer better service, than they might do without the competition between them.

Being able to raise subscription prices would boost the cash flows of the combined firms. This cash would increase Couchweb's share price and would provide funds to finance even greater production and global expansion

Drawbacks of acquiring HomeVideo
<u>May be a poor purchase</u>
HomeVideo profits are falling. This indicates that HomeVideo has a deteriorating return on investment due to poor investment decisions and rising expenses.

Couchweb needs to understand fully the reasons for the fall in the operating profit last year.

<u>Increased risk of higher debt finance</u>
It is not clear how Couchweb intends to pay for the shares in HomeVideo. If some of this involves extra debt this will increase the capital leverage of Couchweb.

HomeVideo has M$5,130.0m of debt [*data in pre-seen material*] which Couchweb will need to take liability for or refinance. Increased capital leverage will increase the risk of Couchweb

This risk comes in part from the need to make higher interest payments to service the debts. Also, the higher capital leverage will make the returns to equity more volatile. It will also strain cash and capital availability of Couchweb further.

<u>May not be supported by Couchweb shareholders</u>
It is not clear whether Couchweb's shareholders support this acquisition, not whether they will provide extra equity capital to pay for HomeVideo's shares. If they do not support the acquisition it could lead to a decline in the share price of Couchweb in the run-up to the acquisition

This could make raising the capital more difficult and could make Couchweb subject to.
a takeover itself

<u>May distract management from developing Couchweb</u>
Acquiring HomeVideo requires management to oversee the process of acquisition, take over managing HomeVideo, and to get to grips with the differing market and operating conditions in Upland. These extra responsibilities may be undertaken at the expense of maintaining proper control over the existing business of Couchweb.

Conclusions
This acquisition poses a number of risks for Couchweb and the benefits are not clear and need investigation.

It is recommended that a full due diligence investigation be undertaken into this proposal before it is considered further. The impact of the financing and operation of HomeVideo on the future growth of Couchweb also needs to be undertaken.

If you have any further questions please contact me.

10.4 Runn taking on fast food supply contract

Please give me your strategic evaluation of this contract proposal (55%)

Approach: The email and reference materials contain several hints that should be used in the answer. This is Runn's first venture into this area, which suggests possible conflict with mission and also suggests risk. The decision to centralize distribution is also a potential risk. There are clear pointers in the email and the article to the risks of transporting chilled foods and it is widely known that chicken must be kept cool to avoid illness. The article does present a conflict with Runn's mission of 'responsibility' and also 'caring deeply about stakeholders'. The article refers to 15% growth and the email expresses the hope that winning this contract will bring additional contracts in future. The capital expenditure is for non-current assets and ignores the potential extra that will be needed for working capital (wages, energy, fuel, and so on).

A strong answer would draw attention to the pros and cons of this proposal under each of the headings of SAF.

Strategic evaluation of contract

This proposal represents a diversification for Runn according to the Ansoff Matrix because it is a new product to a new market: i.e. fast-food supply chain to national restaurant chain. This needs careful evaluation and SAF framework of Johnson, Scholes and Whittington is a good framework for doing this.

Suitability

This assesses the extent to which the option fits with the strategic position, the mission, and the competences of Runn

<u>Still a delivery service</u>

The Vision of Runn is to be 'the World's favorite delivery company'. Delivering chicken and other food supplies fits within this definition. This means that investors and other stakeholders should not be surprised by this contract or resistant to it

<u>Helps escape falling margins in urgent parcels</u>

In 2019 Runn failed to meet its budgeted profit due to the increase in low margin urgent deliveries at the expense of higher margin non-urgent deliveries [*data in pre-seen*]. This is a trend in the CEP industry as clients seek to enhance their online offer at the same prices but with urgent, or timed, delivery to the customer.

Providing food delivery under a contract allows Runn to differentiate itself by integrating with the client. This enable Runn to control its costs better, because they are more predictable, and to move away from price-driven competition

The effect will be to raise Runn's profits and share price

<u>Builds on the core competences of Runn</u>

Core competences are abilities and assets that Runn has that other firms do not

Runn has excellent tracking and GPS technologies that allow it to be more profitable than Geeland rivals, such as Fronnt. It also has a strong brand. These will be leveraged further by moving into fast food delivery

However, there are aspects of this contract that are not consistent with Runn's Vision, Mission, and Values

Fast food may not be 'responsible and sustainable'
The Mission of Runn includes a commitment to responsibility and sustainability in its business The article makes clear that fast-food may not be a responsible industry because of the adverse impacts it has on the health of the diner. This may also run counter to Runn's Values of 'caring deeply about its stakeholders

If fast-food has such negative consequences then it is not 'sustainable' because, at some point, public tastes or regulation will adversely impact on the industry.

This lack of responsibility, sustainability and care may deter investors and staff from engaging with Runn, leading to a fall in its share price

Lack of competences in refrigerated transport
Runn needs to buy new refrigerated trucks and to build a new depot. This suggests a lack of competence in this area.

Refrigerated foods have special handling conditions and perishability that Runn is not familiar with
This may mean that Runn will not be able to offer this service easily and that it will devote extensive resources and management time in trying to succeed in an industry that is not suitable for it

Acceptability
This assesses whether the strategy will gain acceptance from crucial stakeholders and what its impact on risks will be.

The following aspects are in the strategies favor:
May boost share price of Runn
Fast-food market is growing at 15% per year according to the article. This will enhance the growth prospects for Runn and raise market expectations of its future free cash flows.

This will help to improve Runn's share price and increase shareholder value. This will satisfy shareholders and make them welcome this new fast-food delivery business

May lead other clients may place contracts
When Runn demonstrates competence in fast food delivery it is likely that other fast food and food retail businesses may start to use Runn. This will introduce Runn to a completely new industry where it can lever its core competences in order tracking, deliveries and handling.

This will improve Runn's profitability and share price further

Reduces risk of reliance on CEP industry
Runn's earnings are entirely dependent on the CEP industry. This sector is subject to risks from the entry of new rivals and the emergence of technologies such as 3D printing and drone delivery networks. Fast food is not subject to these same risks.

This means that Runn's earnings are, to some extent, shielded from changes in the CEP business by having a food delivery business. This may help make Runn's share price to rise However, there are some aspects of this contract that may not be acceptable

May reduce investment in main CEP business

The G$300m does not include working capital, such as wages and fuel, that will need to be paid monthly whilst waiting for Down Home Chicken to pay its account to Runn. This means the capital investment is likely to exceed the G$300m forecast.

This will reduce the amount of capital needed to support the global growth of Runn's CEP business

This could harm Runn's competitive position in its core business and reduce its share price

Chicken and chilled food subject to bacteriological infection
Foods are subject to the risk of becoming infected during their journey from farm to restaurant Chicken is particularly subject to this and if it occurs it can lead to very serious and sometimes fatal illnesses. This means the contract brings the risk of litigation from Down Home Chicken if it is sued or loses its reputation and the possibility that Runn may have to destroy infected consignments and replace them at its own cost

This is a new risk for Runn and may discourage investors

Fast food and animal welfare under political attack
The fast food industry is under attack for its poor impact on health. Also, some the intensive farming that provides it has been criticized, particularly by animal welfare activists.
This could lead to reputation risk for Runn as a part of this business.

There is also the risk of disruption to the operations of Runn, such as blockading of vans or the depot

This will impose extra risk on shareholders and may lead to a fall in Runn's share price

Feasibility
This criterion is used to assesses the availability of resources to carry out the strategic option

Runn has a site and drivers
This new depot is an extension to Runn's existing operations at Semon. This means it has the site for extension and the necessary operating licenses and road connections. Runn is already employing driver around the Semon area and it is assumed that more will be available

Lack of finance
However, Runn may not have the G$300m to carry invest in this strategy, even before allowing for the extra working capital that it will need in addition to this G$300m

Run has a high dividend policy and this means it lacks free cash to support initiatives like this without cutting-back on the planned investment in its CEP business

Conclusion
This contract is an important strategic initiative for Runn. It is not possible to recommend investing in it without more detailed information and financial projections.

Runn should undertake these calculations as a matter of urgency. I would be keen to help with this.

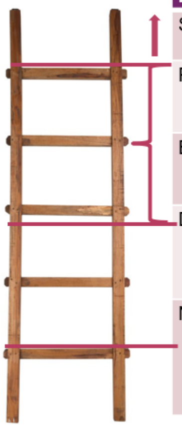
Chapter 11
Business Valuation

1. Business valuations

Tasks involving business valuations are amongst the most common in the SCS Assessment. These tasks have involved

- Flotation of the company (Initial Public Offer)
- Acquisition of another company
- Negotiations for a merger
- Sale of new shares to external investors to raise finance

Business/share valuation methods



Method	Comments
Synergies	• Factors that will improve the earnings of predator and target above that reflected in their present share prices and financial performance
PV of cash flows	• Closest to efficient market valuation • Difficulty in making accurate cash flow forecasts over long period • Difficulty in obtaining accurate discount rate/cost of capital
Earnings method	• Post tax earnings x an appropriate P/E ratio • Valuation distorted by differences between target and firm whose PE is used • Distorted by 'window-dressed' earnings that can't be sustained
Dividend valuation	• Assesses the amount to compensate shareholder for loss of future dividends (i.e. PV of future dividends) • Relies on a consistent and sustainable dividend policy to predict dividends • Overlooks capital growth potential of shares to shareholders
Net assets	• Based on alternative value to seller = liquidating its assets, and buyer's alternative to buy similar assets from the market • Using book value of assets leads to distortions from basis of valuation and are not market values • Overlooks 'goodwill' (brands, know-how, people, business relationships)

As the diagram suggests, it is possible to arrive at high or low values for a business depending on the approach used and on the assumptions made. Ultimately it depends a lot on argument and negotiation.

Quoted and unquoted businesses

In valuing an unquoted business the following should be remembered:

- *The sellers may be motivated by more than the economic value of their business*: they may feel loyalty to it and its staff, see it as a family asset that provides jobs to family members, or be concerned about whether they will have a role in the business after they have sold it
- I*ts financial record may not be representative of its true value*: it may have been poorly managed, or have sought non-economic goals. Its financial systems may not be accurate
- *There may be very few similar firms to compare it to*: it may be small with unique technologies in a start-up industry

In valuing quoted business the following should be remembered:

- *There will already be a market price for the shares of the target firm*: sellers will expect at least this price plus a *bid premium*

- *The buyer's shareholders will need to hear a good justification for paying more than the market price for the shares:* this might be reference to synergies, or suggestions that the target has been undervalued by the stock market
- *Showing any interest in the target will cause its share price to rise*: this is due to speculative buying by people wanting to sell at a profit

In addition to valuation on the basis of the current earnings of the business, the following should also be considered:
- Potential synergies that might increase the earnings of both firms once they are combined
- The degree of enthusiasm the buyer has to buy (or the seller has to sell)
- Potential contingent liabilities that may need to be taken on by the predator
- The trend of earnings and the general outlook for the industry

Negotiation and business valuation
Past exams have featured questions about the process of arriving at a final price in negotiations between management teams over the acquisition or merger of a business. Although the technical approaches above form a basis for discussion, they are not bound to give an answer acceptable to both sides.

Setting and agreeing an acquisition price involves the following factors:

1. Understanding how keen the seller is to come to a deal: the more keen they are, the less they will accept
2. Understanding the seller's desired outcome: in addition to price this might include the timing of the sale, whether it must all be paid in cash, and whether they seek safeguards for their own jobs, or the jobs of staff
3. Clarifying what is being offered for sale: the actual assets and intellectual property but also the willingness of the seller to agree to a non-competition clause
4. Offer and counter offer: there will be bargaining between the parties and the seller may claim they have a better alternative if the price and terms are not right for them
5. Negotiation: a balancing act in which the offer made by your firm must be sufficient to keep the seller engaged in the process, but not so high as to result in your firm paying too much for the purchase

2. Process of business acquisition

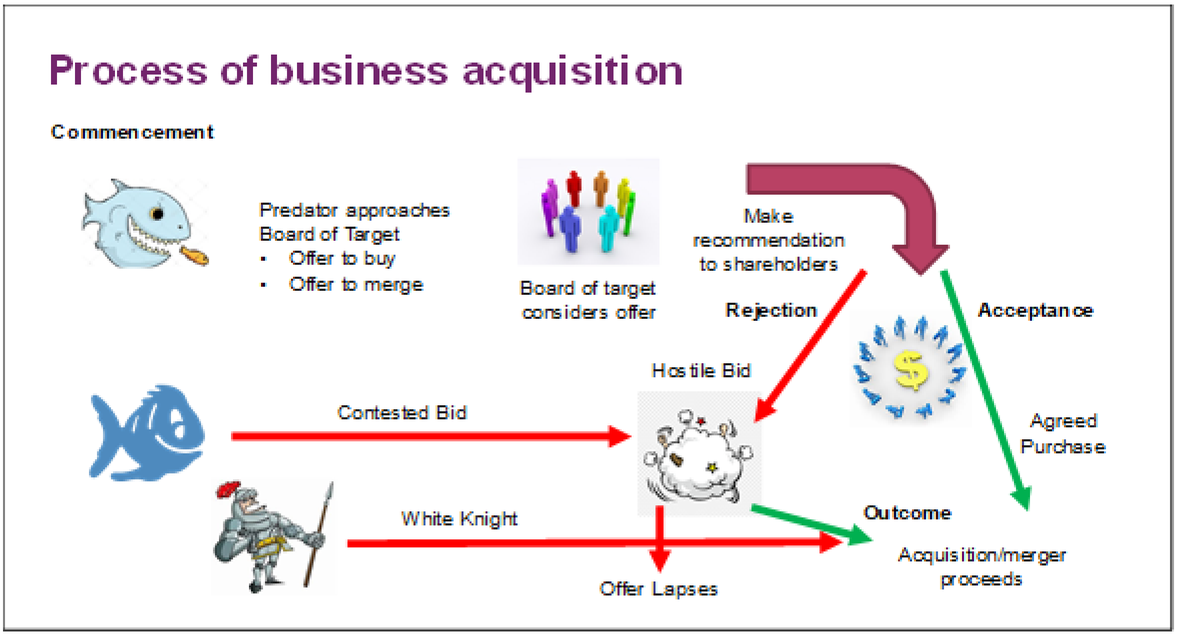
Process is as follows:
1. <u>The buyer (or *predator, bidder,* or *offeror*) approaches the Board of the seller (or *target*)</u>
   This is to express interest in combining the businesses. Sometimes a price may be suggested. If the target is known to be resistant to acquisition, and a struggle is expected, the predator may make this offer public knowledge at the same time to avoid the Board of the target from trying to conceal the bid or mount some other defense. Most bids are *conditional*, meaning that the predator is only obliged to actually buy shares in the target if it has received *acceptances* from shareholders totaling more than 50% of the target's issued share capital.
2. <u>The Board of the target will consider the approach/offer</u>
   They may seek to open discussion with the predator about improving the offer. Or they may reject the approach. At some point the Board of the target must communicate the offer to all its shareholders together with a recommendation on whether to accept it or not.

3. <u>Agreed and Hostile takeovers</u>
   If the advice of the target's Board is to accept the offer it is called an *agreed takeover* but if the advice is to reject the offer the bid becomes a *hostile takeover*

- Agreed takeovers are followed by the predator undertaking *due diligence* into the target and they may reduce or withdraw their offer if they find bad things
- Hostile takeovers are characterized by the target seeking to defend against takeover by explaining that the target has a better outlook on its own, criticizing the business record of the predator, seeking to get another bidder to rescue it (a *white knight defense*), appeal to stakeholders to show resistance to the bid (e.g. staff, customers, regulators). The predator will make a *conditional offer* to the market to buy shares in the target at a specified price. The predator will make clear that they will only exchange cash for the shares in the event that they receive *acceptances* (pledges by shareholders to sell a quantity of shares to the predator) equal to 50%+ of the issued share capital of the target. Generally, the shareholders are not being asked to sell 100% of their shares to the predator, although stock exchange rules usually insist that the predator should be willing to buy all the shares. Many shareholders will decide to sell sufficient of their shares to give control to the predator, or to take the predator's shares as payment for their own, because they expect the predator to manage the target better than its existing management can. Therefore, many of the shareholders are interested in what success the predator will make of the acquisition. The predator will also criticize the existing management of the target, increase its bid price, and make promises and undertakings to stakeholders.

4. <u>Conclusion of the bid process</u>
   The process ends in one of two ways:
   1) The acquisition is successful because the predator obtains acceptances of 50%+ of the issued shares. At this point the bid becomes *unconditional* which means that the predator must be willing buy all the shares that are offered to it, up to 100% of the issued shares of the target, and at the highest price it has offered. However, there is no obligation on the target's shareholders to sell. This means that the predator may find itself having a majority interest in a business in which there remains a *minority interest* of the target's original shareholders. Minority interests have legal protection to compensate them for their lack of voting strength and they can be problematic, sometimes being difficult in order to force the parent company to buy them out. This practice is called *greenmail*.

   2) The bid 'lapses' which means it has not been resolved within the maximum time laid down by the stock exchange. Stock exchanges require that bids only last a maximum time because of the uncertainty it causes to businesses, their management, and to the share prices of the predator and target. Sometimes the bid will be 'withdrawn' early by the predator if it believes its bid will not succeed.



Note the following things about an acquisition that have featured in past exam questions:

- The duty of the Board of the target is to act in the best interests of its shareholders. The board should not seek to preserve their jobs by concealing the bid, making untrue statements or issuing unrealistic forecasts, nor should board members seek to gain assurances from the predator that their jobs will be safe in return for their recommending the acceptance of the bid. These would be *unethical conduct*.

- Speculators will take positions in the shares of the target in the hope of making a profit. These positions are contracts to buy or sell the shares at current market price in the future. Some will make contracts to buy the shares (go *long*) in the expectation of a successful bid at a higher share price. Others will contract to sell the shares (go *short*) in the expectation that the bid will fail and the share price will fall. This behavior, based on rumors, can sometimes distort share prices in advance of a bid being made.

- Some boards of potential targets will have designed *poison pills* to deter takeovers. These include very high pay-offs to board members in the event of a takeover (a *golden parachute*), the release of equity at discounted prices in the event of a takeover such as employee and executive share options, or discounted share offers to existing shareholders, but excluding the predator (a *flip-in* or *flip-over*)

- A Board that is resisting a takeover can be defended ethically if they believe the target is unrealistically undervalued (e.g. due to short term market sentiment), if they believe the predator is misrepresenting its intentions for the target, or if they believe that the predator is an unsuitable owner for the business.

ACTIVITY 11.1

Context

Runnabout is a listed firm that hires electric hoverboards for use by commuters and tourists to travel short distances between transport hubs, such as railway stations and coach stations, and their places of work or tourist attractions. Their competitors are bicycle and scooter hire firms. Their hoverboards need to be returned to racks, or base stations, to be recharged.

Task
You receive the following email from Geo Pataros: CFO of Runnabout

From: Geo Pataros: CFO
To: Senior Finance Manager
Topic: Acquisition of MobiCharge

The Board is considering the acquisition of MobiCharge in order to unchain our hoverboards from their docking stations.

At the moment our hoverboards must return to docking stations to be recharged regularly. MobiCharge offers a team of technicians with equipment to charge our hoverboards away from the docking stations.

MobiCharge offers this service to other electric vehicle firms, mainly to help people with electric cars to get power when they have accidently run down their batteries. But MobiCharge also offers this service to Dokbyke in its six cities of operation.

We will need to upgrade our hoverboards to network them so that customers and MobiCharge can locate them away from docking stations. And the hoverboards will also need to signal when they are running low on charge so that MobiCharge can go to service them.

I have attached details of MobiCharge to this email.

I need your views on a number of issues:

- Evaluate the benefits of this proposal and explain the factors we should consider before we acquire it

[sub task (a) = 55%]

- What methods are there for putting a value on MobiCharge and how should we decide the final price?

[sub task (b) = 45%]

Reference material
MobiCharge

Background
Unquoted business founded in 2015 to provide a service to the emerging electric vehicle market

Owned by two brothers Pino and Georgio Agula, and members of their families are involved in driving the charging trucks and in administering the business

Offers charging service across Capital City and 9 other cities in Geeland

Users join the scheme and pay a monthly fee of G$10 plus a variable cost based on the number of times they use the service each month plus the amount of electric charge they consume in the month

MobiCharge has 8 million registered members

	G$ million
Total Revenue	1,530
Operating profit	229
Finance charges	183
Profit for the year	45
Net assets	25

3. Process of floating a firm (IPO)

Unquoted firms do not have access to the equity markets, or stock exchanges. This reduces the amount of investment funds they can attract to grow their business. Also, the fact they are not quoted means that their existing shareholders will not get a good price for their shares if they try to sell them, assuming they can even find a buyer. Therefore, they may float their firm through an Initial Public Offer (IPO).

Benefits and drawbacks of flotation

Benefits	Drawbacks
Raises capital to help expand the business	Forces the management to prioritise share price and profits in managerial decisions (short-termism)
Gives access to additional capital in the future	Increases the regulatory burden on management
Provides an exit route for the original investors	Increases scrutiny of management actions by shareholders and analysts
Increases the reputation of the firm in the eyes of customers, suppliers, and regulators	May enable a hostile takeover bid to be made for the firm
Enables firm to use share options and equity-based schemes for incentivising the management and staff	Forces management to respond to share price movements caused by market sentiment and speculation

The steps in floating a firm are:

1. *Identify which stock exchange to float the firm on*. This may be a choice between the main stock market (‘full listing’) or a second-tier market. It can also involve the choice of location of the market and the currency in which the shares will be paid for. The firm will want to float on a market with plenty of liquidity and where investors recognize the potential in its business model.

2. *Appoint financial advisors to assist with the flotation.* These advisors will be familiar with the rules governing admission of a firm to its chosen market and will be able to advise on making the firm compliant with these. The advisors will also advise on how to market the shares to get the greatest amount of capital, and may act as an introducer to the market

in order for the stock exchange regulator to agree to have the firm traded on its market. One aspect of this is to ensure at least two market professionals are willing to trade in the shares of the firm in order to establish a secondary market for its shares (a secondary market means the buying and selling of shares between investors after the IPO)

3. *Prepare a prospectus:* this is a data pack of past financial results and assets, but also the plans the business has for the future and its financial projections. It should detail the backgrounds of the members of the corporate board. The prospectus should make clear what the equity will be used for. It should also indicate the number of shares being issued and any special rights that the shares carry, such as voting rights

4. *Decide the offer price for the shares:* this is done shortly before the day of the IPO and it will be based on the amount of interest in the shares that has been generated by the prospectus and by the meetings the Board has had with influential investors and analysts. The setting of the price seeks to balance the desire to get as much equity as possible with the wish not to see the issue *undersubscribed* and therefore see the unsold shares fall back into the underwriters' hands and drag the value down in the longer term (an underwriter is a bank or large investor which agrees to buy unsold shares at a set low price if the IPO is undersubscribed in return for being guaranteed some of the shares at the same low price even if the IPO is *oversubscribed* and the shares are a bargain.

5. *Conduct the issue on the day of the IPO*

ACTIVITY 11.2

Context
Couchweb is a global provider of subscription video on demand (SVoD) like Netflix and Disney+. It streams movies, series and other television programs. It is dependent on program and movie production firms for its content and has taken the strategic decision to invest in producing its own content. It has to date created two successful series: The Politics Game and Tumbling Down.

Majik Media is an unlisted program maker that presently provides 26% of Couchweb's content.

Task
You have received an email from Martine Andels: Director of Finance of Couchweb
To: Senior Manager
From: Martine Andels: Director of Finance
Topic: Proposed merger with Majik Media

Confidential
We have been approached by the Board of Majik Media with the proposal that we merge Couchweb with Majik Media.

There are no precise details yet but I assume we will create a new company that will take over running the businesses of both Couchweb and Majik Media. Our shareholders will be asked to exchange their Couchweb shares for shares in this new merged company.

Majik Media is not listed. It's owned privately by a group of wealthy individuals.

As you know Majik Media provides 38% of our movies and 26% of our TV programs on a license agreement. It is the largest production company in Mayland but it doesn't broadcast its own content. It licenses its content to us and to several other broadcasters around the world.

I need your thoughts on two things:
What will we need to tell our investors in order to encourage them to support this merger with Majik Media?

How should we value the two businesses to come to an agreement on a fair rate of exchange for the shares of Couchweb investors?

Solutions to Activities

11.1 Runnabout acquiring MobiCharge

Evaluate the benefits of this proposal and explain the factors we should consider before we acquire it (55%)

What methods are there for putting a value on MobiCharge and how should we decide the final price?(45%)

Benefits of this proposal
This is an example of diversification in the Ansoff Matrix. Charging services is a new product/technology and car owners, Dokbyke and supplying ourselves is a new market. (student note: some markers like to see theoretical structures in answers if appropriate . This illustrates how far to go in explaining theory).

<u>Greater convenience to users</u>
At the moment riders must locate a convenient docking bay with an available hoverboard before they can travel. At the end of their journey they must find a docking station near their destination that has a spare docking bay to return their hoverboard to. Having a remote charging service means that hoverboards can be left where the rider ends their journey.

This also enables riders to use them to travel home, at a distance from the docking bays, which may be more convenient that needing to transfer to traditional public transport.

This approach of untethered micromobility vehicles has been used for bikes and scooters, for example Dokbyke used MobiCharge. It can lead to some inconvenience when vehicles are deliberately left by riders in places where they cannot be located easily by the next rider.

<u>Less fees for docking bays</u>
Runnabout pays fees for the spaces to put its docking bays. Presumably it has three or four docking spaces per hoverboard in order that the rider can leave it where they want to. These costs could be eliminated if the hoverboards could be untethered from the need to return to a charging bay.

This could increase profits and it would reduce the additional capital needed for Runnabout to expand its business.

<u>Diversification of income for Runnabout</u>
Runnabout is solely dependent on income from the hire of hoverboards. This could be severely reduced if the regulations on the use of hoverboards changed against it, or competition were allowed in Geeland. The purchase of MobiCharge would give other sources of income including the income from electric car owners and Dokbyke. There would also be the monthly membership fee from registered MobiCharge users.

However the present revenues of MobiCharge are much less than Runnabout's annual revenues and therefore the protection from this diversification is likely to be very little.

<u>More data from hoverboard usage</u>
This proposal requires that hoverboards become part of the Internet of Things by being networked in order to report their location and maintenance status. This could provide large amounts of additional data to Runnabout. This enhanced data will help it plan its operations and investments better and will also be useful in providing consultancy to, or deepening partnerships with, local and national governments.

<u>Increased membership data</u>
Runnabout has over 30 million app users, and MobiCharge has a further 9 million app users. These apps already access location of the user, but they may also access other aspects of the smartphone used, for example social media, web browsing, payments made, personal contacts and so on. This data is useful for profiling the app user and can either be developed by Runnabout or sold to data brokers for use by analytics firms. For example in the Covid-19 pandemic many 'Big Data' firms revealed their tracking ability by mapping the social interactions of spreaders to forecast the spread of the illness across the globe and alert medical services and governments.

Factors to consider before acquiring MobiCharge
The attractiveness of MobiCharge depends on how well it benefits Runnabout's business model and how good a business MobiCharge will be.

<u>Ability to convert hoverboards to make them trackable</u>
This is essential for the hoverboards to become untethered and locatable by users and by technicians. The supplier, Minnering Robotics, will need to be consulted on the feasibility and cost of including network capability and on-board diagnostics in new hoverboards, and on the feasibility and cost of converting the hoverboards already in service.

<u>Ability to provide prompt service to all Runnabout's hoverboards</u>
Will these chargers reach hoverboards quickly? Micromobility seeks to resolve congestion on roads and this congestion may stop MobiCharge's trucks from reaching the hoverboards quickly. This could leave riders stranded and would reduce the capacity of our micromobility network leading to a loss of revenues each day and potentially a loss of users in the long run.

MobiCharge will also need to scale up if it is to serve Runnabout in addition to its service to Dokbyke.

Also MobiCharge does not serve all 15 cities served by Runnabout. A decision will need to be made on whether to restrict the service to the 8 cities it does serve, or whether to seek to extend its service to the remaining 7 cities.

<u>Financing the acquisition and development of MobiCharge</u>
Capital will be needed to buy MobiCharge, to expand it, and to convert hoverboards. Runnabout will need to decide how that capital can be raised, say from debt, new equity, retained profits and so on.

This may mean that Runnabout will need to consider reducing its capex budget in other areas.

<u>Impact on licenses of providing untethered hoverboards</u>
Untethered vehicles have been criticised as litter and obstructions in other cities. Runnabout depends on licenses from the 15 local authorities but these have been granted for hoverboards that are returned to docking stations to end the hire session.

Before buying MobiCharge the Board should approach local governments for an indication of their attitude to continuing the licenses if hoverboards were untethered.

<u>Retaining Angula family to work for MobiCharge</u>
If Runnabout decides to buy MobiCharge it should ensure that the Angula family remain with MobiCharge to assist in the transition and to help management get to grips with the new business.

There is a danger that once they have sold their business, and obtained a significant amount of money, the family may decide to cease work or to seek for new business challenges. This could endanger the operations and management of MobiCharge in its initial years of ownership under Runnabout.

Other considerations could include:

- *Quality of cyber security at MobiCharge*
- *Environmental impact of charging trucks*
- *Attitude of Runnabout's shareholders to acquisition*
- *Potential to lose Dokbyke's business and reduce revenues*
- *Why are the Angula family selling MobiCharge?*

Methods of valuing MobiCharge
There are various theoretical methods for valuing firms. They are not all equally useful in valuing MobiCharge.

<u>Net assets method</u>
This is the break-up value of MobiCharge and is based on what its assets might be worth if sold individually and not as a going concern. The data suggests this is G$25million but this is clearly not likely to be acceptable to the Agula brothers because it is less than the profit they enjoyed in 2019.

This is because the net assets method ignores the contacts and human capital of the business, and these give it the ability to make the trucks and so on achieve an income. However Runnabout should note the figure because an alternative to buying MobiCharge is to set up their own charging services for which the investment required in physical assets may be close to this G$25 million, even allowing for differences between the depreciated assets owned by MobiCharge and the costs of new equipment.

<u>Dividend method</u>
This values a firm at the present value of the cash dividends it has paid. Shareholders will require the purchase price to be high enough to compensate them for the dividends they will miss if they no longer own the business.

MobiCharge does not appear to have paid a dividend. But even if it had it will not be a reliable guide to an acceptable value. The Agula family may have foregone dividend over the past 5 and more years since it was established to build the business with the intention of selling it at a capital gain. It is not a mature 'cash cow' investment for them.

<u>Earnings method</u>
This approach seeks to assess what the stock market would pay for MobiCharge. It uses the PE ratio of a similar firm that is listed and applies that to the profits of MobiCharge to engineer a potential market value.

This approach is appealing because on alternative the Agula family has to selling to Runnabout would be to float MobiCharge in an IPO (Initial Public Offer). Runnabout could offer this potential market value, less some allowance for the savings the Agula's will make by not having to go through the process of floation, and the Agula's may accept it.

The difficulty is that there is unlikely to be a company listed on the Geeland stock market that is similar to MobiCharge. This will open the potential for inaccuracy and dispute as the Board of Runnabout choses firms with low PEs to value MobiCharge and the Agula's seek to compare the prospects of MobiCharge with the firms that have high PE ratios.

<u>PV of future cash flows method</u>
This is the efficient market valuation approach because it reflects the theories of how stock markets set share prices. It would require Runnabout to forecast the free cash flows of MobiCharge into infinity (or more likely 10 years) and discount them at an appropriate cost of equity to come to a present value, the value of MobiCharge.

The problems will be in arriving at a forecast of MobiCharge's future cash flows. The efficient market does this by pooling the wisdom of millions of investors whereas Runnabout only has its own wisdom and best guesses. Also the cost of equity will be difficult to assess because this would require an assessment of the risks of MobiCharge which, like Runnabout, are very dependent on the legislation affecting the future of new forms of transport

<u>Synergies and strategic value</u>
The approaches above are all derived from approaches to valuing shares on the stock market to help investors identify when they are undervalued in order to buy them. Runnabout wants to gain commercial advantage, not just income, from buying MobiCharge. The Agula family will also realise that and hold out for a price that is higher than the price that could be justified by just MobiCharge's future earnings.

Therefore Runnabout needs to assess how much more Runnabout could earn with MobiCharge as a subsidiary compared to its likely earnings without MobiCharge. This would be the maximum Runnabout should pay, but obviously it should aim to pay less in order to reserve some of the extra value for its own shareholders.

Deciding the final price
Ultimately this will be based on negotiation between the Agula family and its advisors, and Runnabout and its advisors. This negotiation should take into account:

<u>The value of MobiCharge to Runnabout</u>
This was discussed above as its strategic value and potential synergies that might increase the earnings of both firms above their 2019 level, such as shared administration costs, shared IT development costs and so on.

<u>The willingness of Angula family to sell</u>
It is not clear whether the brothers want to sell their business or not. If they do want to sell then this would help Runnabout to gain the firm at a lower price, although care should be taken to ensure the sale is not due to some problems that Runnabout might inherit.

This willingness to sell will be affected by the other aspects of the offer that Runnabout makes such as promises of management positions, safeguards to staff and so on, or threats such as setting up in competition if they cannot buy MobiCharge (possibly an unethical approach)

<u>Potential alternative buyers for MobiCharge</u>
The purchase of MobiCharge may threaten its existing customers, such as DokByke, and they may make a counter-offer or offer other support to the management of MobiCharge.

Alternative buyers lead to a contested bid, and this has the effect of pushing the price up to the level of the value to the most desperate buyer.

<u>Alternatives to buying MobiCharge</u>
As mentioned, Runnabout should not lose sight of the possibility of setting up its own charging network, as Tesla did, or of buying other mobile charging services or technologies.

The price agreed for MobiCharge should be below the price of setting up an alternative to it.

<u>The structure of the payment for Angula family's interest in MobiCharge</u>
The price will depend on the structure of the deal. The initial price could be reduced by an agreement to pay more in future years depending on the retention of the Agula family and the performance of the combined business.

11.2 Couchweb merging with content provider Majik Media

What will we need to tell our investors in order to encourage them to support this merger with Majik Media? (55%)

Approach: evaluation of a strategic option features in many real exams. There are a number of approaches that can be used, including using the SAF framework to discuss the Suitability of the option (i.e. how well it satisfies the aims of the business and its present strategic position), the Acceptability of the option (i.e. its financial returns and risks and the likelihood that key stakeholders may support or oppose it), and the Feasibility of the option (i.e. the availability of resources to carry it out such as finance, and management capacity and know-how).

This task requires an answer structured into benefits and drawbacks, but you can still use the SAF framework to help you generate ideas under each.

Convincing Couchweb investors to support the merger with Majik Media
Couchweb is a listed company and so investors will be mainly institutional funds. These shareholders will be principally concerned with the impact of the merger on the value of their shares, the dividend they will receive, and the risks of the merged business.

There are legal and ethical limits on what the Board of Couchweb can say to convince investors to support the merger. For example, they should not be misleading.

Investors will wish to have reassurance on the following things:
<u>The strategic logic for the merger</u>
Investors will need to be convinced that the merger is necessary and that it will put Couchweb in a better place that it would be if it remained independent. They will also need to be convinced that Majik Media and Couchweb are both strong companies, or otherwise that Majik Media is undervalued and a bargain for Couchweb

Reference could be made to the convergence in the media industry between studios and between broadcasters and the treats these pose to Couchweb if it cannot obtain programs

They would need to understand the benefits of the merger (below)

<u>The benefits of the merger to shareholders</u>
The benefits of the merger should be expressed in terms of revenue increases, or cost reductions, that will make the earnings of Couchweb higher than they would be without the merger. Benefits could also be expressed in terms of reductions to the potential volatility of Couchweb's earnings, in other words in terms of reduced risk.

The merger could enhance earnings by enabling Couchweb to get unique content from Majik Media and participation in the earnings of Majik Media from other channels such as cinemas and other broadcasters. Costs could be reduced by economics of scale in administration, paying less for programs from Majik Media, reduced finance costs by paying back loans from the sales of duplicated assets (studios, administration buildings and so on)

Risks would be reduced by Couchweb having assured access to Majik Media's programs, the diversification of revenue sources away from just reliance on Couchweb's subscriptions to include additionally Majik Medias sales to rival broadcasters and other channels. There might also be gains such as the experience of Majik Media's production team, improved quality of Majik Media productions by access to Big Data collected by Couchweb and so on

Assurance that they will get fair value for their shares
Couchweb investors will want a generous percentage of shares in the new merged business. For example, if Majik Media and Couchweb are firms of equal value, Couchweb's shareholders will want at least 50% of the shares in the merged firm to maintain the value of their investment.

Investors will need to have the share exchange ratio explained them but this value will need to include a fair share in the enhanced value of the merged group.

Future dividend policy
Couchweb operates a generous dividend policy at present. In 2017 it paid 43.7% of profits out as dividend (*data in pre-seen material*). This high dividend will have a clientele effect: it will have attracted investors to its shares with the prospect of reliable high cash flows.

These investors will want assurance that the dividend payout from the merged firm will not reduce dividends below that promised by Couchweb when it was independent.

The future strategy of the merged business
The future strategy will impact on the earnings and risks of their investment. Investors will wish to know what the merged firm will do that is different from what Couchweb was intending when it was independent.

This would include understanding the operations and business model of Majik Media because many investors will not also be investors in program making firms. They would also want to understand the motives of Majik Media's management in seeking the merger.

This strategy will need to be explained in terms of markets and products and any factors that may demand more capital, or release capital, in the future.

How should we value the two businesses to come to an agreement on a fair rate of exchange for the shares of Couchweb investors? (45%)

Valuing Couchweb and Majik Media for the merger
The setting of an exchange ratio between shares in the merged company and shares in Couchweb depends on obtaining values for each firm.

Couchweb shareholders will require their shareholding percentage to pass two tests:
1. Be at least as valuable as the shareholding they held in Couchweb
2. Give them the same proportion of the value in the merged firm and the percentage value of the new firm brought by Couchweb

The second value should be higher but it depends on a lot of speculation about potential synergies and which of the two firms contributes more to the synergies and growth of the merged business

The valuation is complicated by the fact that there is already a market value available for Couchweb that reflect the value that its investors put on it. There is not similar market value for Majik Media because its unlisted.

In a merger the shareholders of Couchweb are likely to argue that the market value of Couchweb understates the firms' real value, and may seek to enhance this using some of the methods below

There are a number of technical methods that could be used to value the two firms
- The net assets methods
- The dividend method
- The PE method
- The future cash flow method

In addition, there are some other practical considerations that need to be born in mind

<u>The net assets method</u>
This values the book value of net assets of each firm at the time of the merger. This would represent the disposal value of each firms' physical, intangible, and financial assets or, put another way, it represents what the assets might cost if bought separately on the free market.

This approach will struggle with the fact that in both cases the biggest value of assets is the content of the firms. This represents the licenses each holds or the value of programs it has made. These are hard to value, especially if the rights and licenses expire in the event of an acquisition.

It would overlook the 'goodwill' element that represent the unique skills of staff in each firm, the program formats and franchise possibilities, and the respective brands and value networks each firm can bring to the merger

Net assets are likely to be a very conservative valuation for each firm and would be distorted by differences in accounting treatments between the firms and differences in their business models.

<u>The dividend method</u>
Based on the annual dividend that each firm pays to shareholders.

The principle is that the shares are worth at least the present value of the dividends that they will receive over time. This approach ignores the potential for capital growth as each firm becomes larger and more successful.

This approach might be distorted by differences in dividend policy. For example, if Couchweb pays a very high dividend each year but Majik Media pays no dividend this would means that Couchweb was very valuable but that Majik Media was without value.

This is approach is extremely unlikely to give a value for Couchweb that would be acceptable to the current investors

<u>The PE method (or *earnings method)*</u>
Takes the post-tax profits of each firm and multiplies them by an appropriate P/E ratio.

This is not necessary for Couchweb because as a listed company it already has a share value and market value. For Majik Media this P/E ratio could be the market sector P/E for listed production companies generally or, perhaps, specific production companies that are similar to Majik Media, perhaps Pinto Studios of Ronsteel Productions. Another approach would be to use Coachweb's P/E ratio.

This approach is less reliable if there are significant differences between Majik Media and the firm whose P/E is being used. If Couchweb's P/E is used it might overvalue Majik Media because Couchweb has enjoyed a rapidly rising share price and it has a different business model from Majik Media.

Also, the share price of firms can differ due to differences in non-systematic risk between the two firms. The most significant element here might be differences in capital leverage. This approach is also compromised by short-term volatility in earnings. For example, if Majik Media has reported a high profit in the most recent year due to, say, a particularly successful movie, this will boost its value despite this being a one-off success and make the value of Majik Media exceptionally high and unrealistic.

Also, for the purpose of valuation this P/E should be applied to prospective earnings and this will depend on the business plans of Majik Media.

<u>Future free cash flows</u>
This is technically the superior approach because it reflects the approach of the efficient stock market. These cash flows should be discounted to present value terms at a cost of capital which reflects the risks of investing in the industry

Majik Media should provide a cash flow forecast as part of the merger discussions. The calculation will be complex due to the need to have clear financial forecasts for at least 10 years and also a cost of capital to discount these. The cost of capital would need to be engineered from an existing proxy firm, such as a listed production company, and therefore will, like P/E approach, suffer from any lack of comparability between Majik Media and the firm used as the basis for calculating the cost of capital

<u>Other considerations that will influence the respective values</u>
The negotiating powers of both management teams should be considered. Inevitably each will be trying to argue up the value of their firm in order to get more shares for their present investors.

The extent to which each side needs or can benefit from a merger. For example, if the merger doesn't get agreed will either penalize the other by withholding programs or by refusing to broadcast them?

The level of liabilities of each firm. Couchweb has significant borrowing and possibly long-term commitments to pay program suppliers. Majik Media will have similar loans and commitments. These will become the responsibilities of the merged firm. They need to be discounted from the eventual merger values

Conclusions
The merger can only go ahead if a majority of Couchweb's shareholders approve it. They will need to see that the shares they will receive will be of at least equal value to their present holdings and in a firm that has a better future than Couchweb on its own. Negotiation will set the final shares allocation.

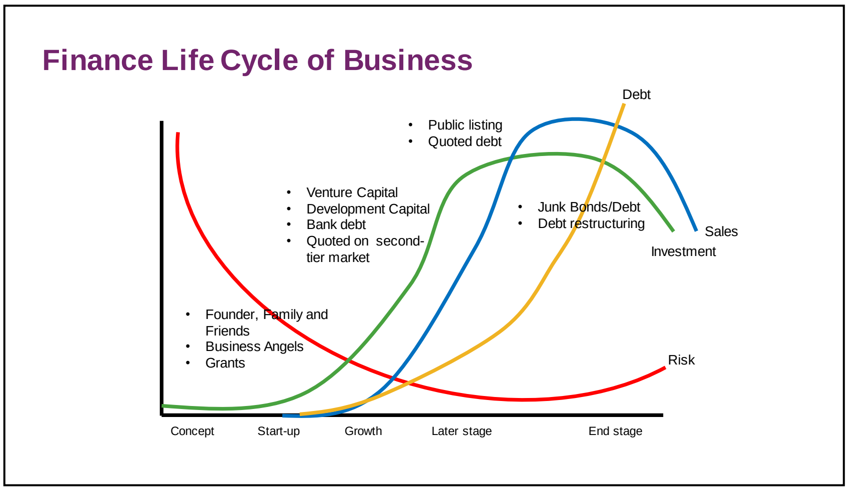
If you have any questions please ask me

Chapter 12
Business Finance

1. Capital structure – financing the organization

The SCS assessment very frequently features tasks involving the capital structure of the firm

- Suggesting ways to improve cash flows
- Suggesting ways to finance an investment of acquisition
- Evaluating proposals to raise finance in a particular way



Sources of Equity Finance:
Provided by shareholders in several points during the firm's development

- *Unlisted/unquoted firms*
  - Personal equity: founders, family and friends (FFF) will invest money: sometimes from savings or by borrowing in their personal capacity
  - Grants: these come from state and other sources with a non-financial interest in developing this business or its founders
  - Business angels: individual rich persons who use their personal wealth to invest in business ideas
  - Venture Capitalists (VC): investment funds that buy into a start-up business with the express intention of selling their holding at a gain within 3 to 5 years
  - Development capital: similar to venture capital but with a preference for less risky investment in already established and profitable businesses
  - Business partners: such as suppliers, major clients, research partners/universities and so on

- *Listed/quoted firms*
  - *Initial Public Offer of shares (IPO)*: shares admitted to a stock market at flotation
  - *Secondary issues/offering*: a listed firm issues additional shares to the market

- *Retained earnings:* undistributed profits that are invested in growing the business

Sources of Finance - equity

Source	Advantages	Disadvantages
Equity – retained earnings	• No special process or costs of acquiring • No fixed interest commitments • Can be flexible - repay it later as dividend from reserves	• Depends on firm having good positive profits and cash flows • May disappoint investors who want dividend • Failure to pay dividend can appear to be cash flow problem and over-investment

Equity – new issues	• No cash flow costs to pay interest
	• Capitalises on market sentiment and support for the business
	• No requirement to repay later
		• Dilutes existing shareholders
		   • May reduce share price if capital not going to high profit project
		   • May change votes and control
		• Hard to repay if capital not needed later
		• Need to borrow large amount at once – leaves firm with surplus cash until it is invested

Equity financing
Put simply, equity means ‘ownership’. That means the a shareholder with 10% of the issued shares in a firm owns

- 10% of the annual profits
- 10% of the annual dividend
- 10% of the net assets of the business (total assets minus total liabilities)
- 10% of control, the votes to appoint board members and to pass resolutions affecting the firm

When shares are issued the firm receives the value that the shares were first sold for on the *primary market* (i.e. the ‘new share market’). That amount is recorded as part of equity in the Statement of Financial Position, and the cash goes into the firm’s bank account to be spent on assets and expenses in the years to come. The subsequent rise and fall in the share price on the *secondary market* (i.e. the ‘used share market’) does not affect the Statement of Financial Position nor the cash flows of the firm. The winners and losers on the secondary market are investors and not the firm itself.

Equity finance is increased in two ways:

1. *Retained earnings (retentions)*: annual profits are earned but some percentage of these are not distributed to the shareholders as dividend, but instead are kept within the firm to buy additional assets and to build a bigger business

2. *New issues of shares*: these are additional shares sold to investors. If it is an Initial Public Offer (the first time the firm sells its shares on the stock market) it is termed a *primary offering*. If the firm has already sold shares to the market and now wants to raise extra capital by selling more new shares it is called a *secondary offering*. NOTE: don’t confuse primary and secondary offerings with primary and secondary markets. Either type of offering goes to the primary market.

Before issuing new equity management must consider:

- The higher the price is on the secondary market, the more the capital that can be raised by selling each new share
- Issuing extra shares in a secondary offering will have an impact on the share price on the secondary market, usually it reduces it, because after issue each share, both existing ones

and new ones, will now have a smaller percentage of the profits, dividend, net assets, and control than before. This is called *dilution*.

- In fact most secondary offerings are priced at a discount to the market price of the share to encourage investors to buy them, and this supply of cheaper shares is bound to reduce the price of all the firm's shares on the secondary market once the shares are issued
- The announcement of a secondary offering may be enough to make the share price fall because existing shareholders realise that their existing shares may lose value and they may rush to sell them
- To avoid this fall in share price management must convince investors that the capital raised by the issue will be used to increase the profits of the firm sufficiently to eventually increase the earnings per share (EPS = Total profit/number of shares in issue)

Rights issues
This was a question in the November 2020 exam

It is a convention of secondary offerings to give the existing shareholders the first chance to buy all the new shares. This is termed a *rights issue* and requires that management sends a *rights letter* to each existing shareholder offering to sell them, say, 1 new share at a discounted price, for every five shares they already own.

This forces the existing shareholder to choose between a number of options:

- buy the extra share to keep their % holding the same
- don't buy the share and suffer a fall in the value of their existing shares, but try to sell the rights letter to someone else for cash so that the new owner of the letter gets cheap shares and the existing shareholder gets some cash to compensate them for the fall in value of their shares
- sell their existing shares as soon as possible to avoid the fall in their value

If support for management and its strategy are high existing shareholders usually buy the extra shares, or, if they can't afford to invest more, sell the rights letter and keep hold of their existing shares.

But if support for management is weak, and shareholders don't support the strategy, it can lead to the disaster of a collapse of the share price to such a low level that no-one will buy the new shares because even at a discount they are now more expensive that the share price on the secondary market. In this event management has failed to raise extra capital but has alienated its shareholders who have lost value, and ruined its reputation on the stock market

Sources of Debt Financing
This is money borrowed from a source that will need to be a paid back at some point.

- *Suppliers* that extend credit
- *Customers* who advance money on account and deposits
- *Banks* providing overdrafts, revolving credit facilities, and underwriting for the debts of the firm
- *Investment markets* if the firm issues debt instruments that can be traded on the *bill and bond markets*.
- *Finance houses and leasing firms*: branches of banks or the finance or credit services branch of supplier

Debt and shareholder risk

- It is commonly stated that firms with higher capital leverage are more risky for shareholders to invest in. This is for two reasons:

1. Debt ranks higher than equity in the event of a liquidation of the firm. This means that if a highly debt levered firm goes out of business the equity holder is much less likely to get any of their investment back.
2. High levels of debt make the profits to shareholders, measured by Earnings per Share (EPS), more volatile which in turn makes the share price rise and fall more. Buying a share at the peak could lead to a huge loss if it falls.

The following numerical example will illustrate the point (taxation on profits has been ignored for simplicity):

	Unlevered firm $0 interest	**Levered firm $20m interest**
2019 Operating profit	$500m	$500m
Less interest	$0	$20m
Profit to shareholders	$500m	$480m
2020 Operating profit	$600m	$600m
Less interest	$0	$20m
Profit to shareholders	$600m	$580m
% rise in profit	20.0%	20.8%

- illustrates that the same 20% rise in operating profit leads to a greater than 20% rise in the shareholder profit of the firm with capital leverage
- if the operating profit for 2021 were to fall back by 16.7% to $500m again the percentage fall in shareholder profit for the levered firm would be 17.2% ($100m/$580m) compared to only 16.7% for the unlevered firm ($100m/$600m)
- assuming that the share price of a firm is EPS x Price/Earnings ratio, the share prices of the two firms would rise and fall by the percentage change in shareholder profit
- this means that the share price of the levered firm will have greater volatility then the unlevered one for the same changes in operating profit
- greater share price volatility means greater risk for the investor buying the shares
- later we will see that this effect leads to different *beta coefficients* between levered and unlevered firms and the requirement to *un-gear* the *equity beta* mathematically to reach a proper comparison of business risks based on the *asset beta*

Sources of Finance – working capital

Source	Advantages	Disadvantages
Supplier credit	• No interest costs • Rises and falls with business activity	• May lose discounts for settlement • May lock firm into unsatisfactory supplier • May jeopardise supplier's survival • Potentially unethical

Invoice financing
• Discounting = bridge loan
• Factoring = sale of AR
	• Improves working capital management
	• Factoring reduces costs of invoice collection
		• Costly
		• Factoring loses control of customer relationship
		  • Collection methods
		  • Rising bad debts

Improve working capital management
• Increase supplier credit
• Reduce inventory
• Reduce customer credit
	• Reduces finance costs
	• Reduces risk
	  • From client bad debts
	  • From obsolete inventory
	• Encourages streamlined operations
		• Reducing inventory
		  • Potential for stock-outs
		  • May reduce choices available to client
		  • High dependence on supply chain
		• Reducing customer credit
		  • Loss of some customers
		  • Disaffection of sales teams
		  • Rise in costs of credit control

Sources of Finance – debt for working capital

Source	Advantages	Disadvantages
Bank overdraft	• Flexible – rises and falls with working capital needs • Quick to arrange • Interest is tax deductible	• Expensive • Can be withdrawn quickly • High overdraft signals cash flow problems
Commercial Paper	• Flexible – can be issued and redeemed in line with working capital needs • Quick to arrange if bank has agreed an underwriter • Shifts cash outflows to the maturity date of the bill/bond	• Variable discount/interest rate depending on when they are issued • Depends on good credit rating of firm and may dry up

Sources of Finance – debt for asset purchasing

Source	Advantages	Disadvantages
Term loans 1 – 15 years	• Can be borrowed as it is needed • Match loan and repayments to economic life of the asset • Lower cost if secured on asset • May be fixed rate or floating rate of interest • Interest roll-up possible (balloon loans)	• Restricts asset use and sale if secured on them • Restrictive covenants from lender (maximum borrowing, minimum profit, lines of business) • High levels of borrowing increase capital leverage and can reduce share price • Firm needs to have good credit rating to borrow
Bond markets (debt capital market)	• Competitive rates • Can be secured on tangible or intangible assets (securitised assets) • Possible to phase maturities to suit long term cash flows	• Requires initial admission to the markets • Needs good credit rating and trading record (otherwise junk bond status)
Convertible debt (converts to equity at election of lender)	• Interest rate reduced by expectation of capital gain on equity • May raise finance for firms with limited credit history	• Costly • Dilution of shareholders if exercised • May ratchet up if share price falls

Sources of Finance – additional types

Source	Advantages	Disadvantages
Mezzanine finance • Convertibles • Warrants • Preferred stock • Subordinated debt	• Cheaper than common equity due to preferred status in winding-up	• More expensive due to lack of security compared to loan stock
Leasing • Of new assets • Of existing assets (sale and leaseback)	• Payments may be lower because ownership of asset gives security to lender • Potential to return asset if no-longer required • May benefit from incentives from asset seller	• Lease payments are a compulsory payment • Cancellation costs for breaking lease • Constraints on use of asset • Costs of restoring condition of asset at end of lease
Rent-to-own Also called • Hire Purchase • Closed-end leasing	• Payments may be lower because ownership of asset gives security to lender • May benefit from incentives from asset seller	• Rent payments are a compulsory payment • Cancellation costs for breaking agreement • Constraints on use of asset

Debt versus equity – a summary

Source	Advantages	Disadvantages
Debt	• Quick to arrange – assuming firm has good credit rating and relationship with banks • Funds can be raised as they are needed to reduce cost and surplus liquidity • Loan length can be matched to economic life of the asset to ensure it is repaid from earnings of the asset • Provides boost to shareholders providing return on the assets exceeds interest cost on loan • Loan interest qualifies for tax relief and this boosts profits and cash flow • Avoids shareholder dilution or changes in control • Possible to borrow in foreign currency to match revenues in foreign currency and so reduce currency risk	• High levels of borrowing increase capital leverage and can reduce share price because it makes the firm a more risky investment • Restrictions imposed by lender over assets use/sale and other restrictive covenants • Cash flow risk from interest payments and repayment schedule of loans • Lenders may refuse to continue lending after original loan matures • Could be affected by rising interest rates
Equity	• Permanent capital that does not need to be repaid • No compulsory interest or repayments to make • Easy to borrow if support for firm is high • Keeps control with shareholders rather than with banks	• Hard to repay so not suitable for short-run financial needs • New issues threaten dilution of shareholder which will reduce share price • Dividend reduction unpopular with shareholders • Difficult to do if firm lacks profits or is in financial problems • May led to arrival of hostile shareholders

Roles of corporate treasury

Role	Description
Cash management	• payment transactions • cash management, account organisation and cash flow monitoring • managing bank accounts, electronic banking • pooling and netting group cash as well as the functions of in-house bank
Liquidity planning and control	• cash flow forecasting • managing liquid investments to deposit and redeem cash as required
Management of interest, currency and commodity risks	• control of risks through use of derivatives and other policies • handling and monitoring , terms and conditions, hedging periods and contractual dates • maintaining documentation of hedging transactions and risk exposure
Procurement of finance and financial instruments	• money market dealings • working capital finance through corporate bonds • factoring/invoice discounting • financial investment and active asset management
Contacts with banks and credit rating agencies	• cultivate points of contact and maintain successful business relationships
Corporate finance	• medium and long term financing • issue of capital market instruments, Asset Backed Securities • use of group financing, credit, leasing, shareholders' loans

ACTIVITY 12.1

Context

Couchweb is a listed global provider of subscription video on demand (SVoD) like Netflix and Disney+. It streams movies, series and other television programs. It is dependent on program and movie production firms for its content and has taken the strategic decision to invest in producing its own content. It has to date created two successful series: The Politics Game and Tumbling Down.

Task

Martine Andels, Director of Finance of Couchweb, calls you into her office one morning.

"The Board has decided to significantly increase the amount of content that we produce. The success of The Politics Game has shown us what can be achieved.

The aim of the Board is to increase the percentage of TV programs made by Couchweb from 16% of our catalogue to 35% before the end of 2019. This means us having to finance the production of another eight series each year.

I have summarized some of the financial implications of this in a three-year cash flow forecast.

I need you to consider something for me:

Is it better for us to finance these new programs by debt or by equity? **(sub section = 55%)**

Financial summary prepared by Martine

	2018 forecast M$m	**2019 forecast M$m**	**2020 forecast M$m**
Revenue	16,532.1	20,665.2	35,831.4
Cash from operations (before expenditure on program production)	1,983.9	2,479.9	5,733.1
Expenditure on program production	2,130.2	5,343.2	6,411.9
Cash flow after expenditure on programs	(146.3)	(2,863.3)	(678.8)
Tax and finance costs	421.9	492.1	584.5
Dividend	320.5	352.6	387.8
Net addition to cash for year	(888.7)	(1,131.0)	(1,651.1)
Cash position at end of year	1,053.9	(77.1)	(1,728.2)

ACTIVITY 12.2

Context

Knowledge City Technology Park (KCTP) is a listed operator of a single campus science and technology park (STP) at Capital City in Advland. It specializes in the niche market of leasing buildings to firms engaged in technological innovation. KCTP is 20% owned by Capital City University which founded KCTP 35 years ago to support innovation in Advland. The university sold KCTP in a flotation over 20 years ago but retains close links with it. KCTP's share price is very stable at A$14, shows no growth, and its shareholders seem to prefer high dividends instead of bearing the risks that may be associated with new strategies and growth. This preference is reflected in its low capital leverage of 13.9% represented by loans of A$960m despite having property plant and equipment (PPE) valued at A$6,600m.

KCTP is considering issuing additional shares, equivalent to 13% of total shares, to raise A$9.3bn to buy a struggling campus called Exland Technology Park (ETP) in the developing foreign country of Exland. ETP's profits were the equivalent of A$15m last year. The profits of KCTP were A$146m in the same year.

Task

<u>You have met Shereen Peros, FD of KCTP, in her office</u>

'I've just left a Board meeting that was called to discuss the acquisition of Exland Technology Park.

Here are some notes I took from the meeting (*these have been summarized in the description of context above*).

I need your thoughts on some things that occurred to me in the meeting:

- What will be the difficulties in raising the capital through a rights issue and what other sources of finance can be used?

[sub task = 50%]

2. Share prices and shareholder risk

The SCS Assessment frequently sets questions that require candidates to describe the potential impact of a business event on the share price of the firm.

Level of the share price

According to financial economics the total market value of a firm is equivalent to the discounted present value of the expected future free cash flows from its operations and strategy.

This theory, and the factors impacting on expectations of future free cash flows, should be the starting point for any discussion of the factors influencing the price of a firm.

Factors influencing share price

Factor	Description
Future earnings (forecast free cash flows)	• Market value assumed to be related to PV of future earnings (free cash flows) • Past earnings may be used as a basis or indicator for this forecast
Risks of business (impacts the rate of discount applied to cash flows)	• Potential for unanticipated increases (hope) or falls (problems) in earnings or value • Capital structure of the business • Exposure to external events that impact on its business model
Dividends (impacts when cash will be received by investor)	• Share price will rise when shares are 'cum div' • Changes in dividend policy tend to move share price in same direction (contrary to dividend irrelevance hypothesis)
Rumours and events affecting the business	• Changes of management or strategy • Major client wins or product discoveries • Speculation about acquisition activity • Potential for favourable or adverse regulations coming into force
Industry growth and outlook	• Growth rate of the industry as a whole • External impacts affecting the industry (costs, exchange rates, competition)
Investment market sentiment	• Poor economic outlook will reduce all share prices • High market liquidity (cash) and low interest rates will boost all share prices

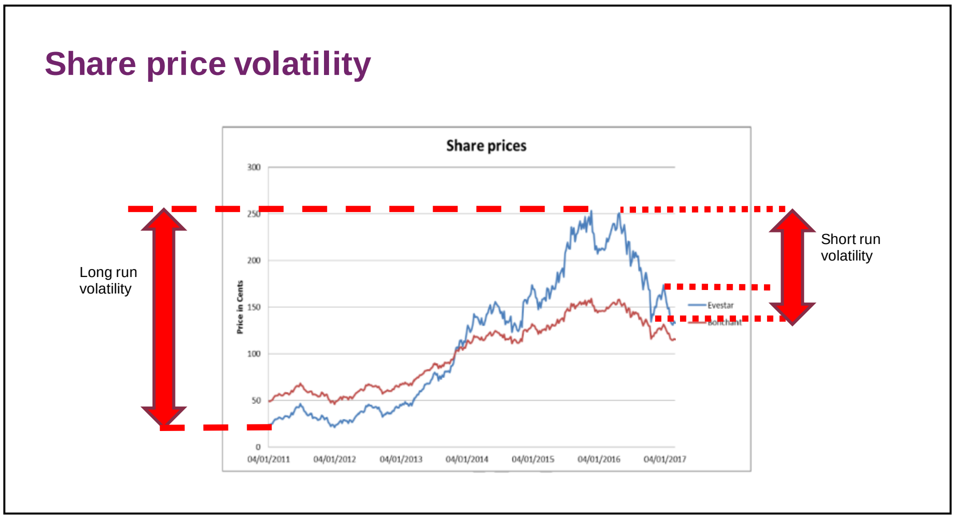
Share price volatility

Volatility means the tendency for a company's share price to rise and fall sharply over time In a recent pre-seen case, the examiner provided the following chart of the share prices of two firms from the same industry over a 6-year period.

The following can be noted

- The rise in value of Evestar between 2011 and its peak in mid-2016 was from 25 cents to 250 cents, 1000%

- Over the same period Bonchant only rose from 50 cents to 150 cents, 300%
- Since the peak in mid-2016 Evestar has fallen from 250 cents to 130 cents, 48%
- Over the same period Bonchant has fallen from 150 cents to 120 cents, 32%
- This shows that Evestar has a more volatile share price than Bonchant
- It can also be noted that the short run volatility of Evestar seems greater too



The volatility is caused by two sets of factors:

1. *Industry-wide factors (systematic risks)*
   These are termed *systematic risks* because they are present for all firms in the market (or *sector* – meaning the industrial sector in which the business operates). It is clear that both Evestar and Bonchant enjoyed the same helpful industry factors until mid-2016, and suffered similar unhelpful factors after. Examples could include
   - outlook for the industry
   - state of the economy
   - level of yields on alternative investments (e.g. interest rates on bonds)

2. *Factors unique to the firm itself (un-systematic risks)*
   These are the *specific risks* of the particular firm and they arise from the way the firm is structured and operated. Evestar seems to have greater specific risks than Bonchant. There are several potential causes for this:
   - Higher capital gearing/leverage: this increases the volatility of shareholder returns because interest on the debt imposes a fixed charge on operating profit. So small % changes in operating profits create significant % changes in profits after interest
   - Higher operational gearing/leverage: this is where the firm has a high fixed cost base so that if revenue falls slightly the operating profits fall significantly because the firm is unable to cut its fixed costs
   - Market perception of the management of the firm. This can change due to rumors or because investors doubt the ability of management to cope with a change in the business environment
   - Concerns over the sustainability of the firm's business model

The Beta Coefficient

The *Beta coefficient* is a measure of the volatility in the price of a share, or the value of all the shares in a sector, in comparison to the movement of share prices in the market as a whole.

- $\beta = 1$: the share moves in the same proportion as other shares

- β > 1: more risky - the share moves in a greater proportion than other shares
- β < 1: less risky - the share moves less than proportionately compared to other shares
- β is negative: *negatively co-variant*: the share moves in the opposite direction to the price of other shares

There are two beta coefficients to consider:
1. The *Geared Beta*: this is the volatility of the share price on the market. Sometimes called the *equity beta.* It shows the risk from buying the shares of the firm.
2. The *Ungeared Beta*: this is a measure of share price volatility with the volatility caused by the level of capital leverage removed mathematically. It shows the risk of the business model of the firm, called the *asset beta*. This indicates the potential risk of buying the company if the capital structure could be revised.

Shareholders and risk
Shareholders are assumed to be *well diversified*. This means they are assumed to hold *portfolios* of shares that include the shares of firms in different industrial sectors.

This means that they have reduced their exposure to the unsystematic risks of relying on one single sector, or worse, one single firm. Sometimes investors may deliberately seek shares that are *negatively co-variant* to neutralize their risks further i.e. that when the value of one falls the value of the other is expected to rise.
- Systematic risk cannot be reduced by diversification in a share portfolio. Systematic risks would affect the value of all shares in their portfolio
- Non-systematic risk can be diversified

This concept of diversification of risk through holding portfolios has several implications for the practice of financial strategy
- The management of a firm does not need to seek to diversify business risks through assembling a portfolio of unrelated businesses
- The actual financial returns to a share are important in establishing its price, but so are its risk characteristics
- The cost of capital of a firm (i.e. the minimum level of return acceptable to shareholders) will be influenced by how the risks of the firm compare to other firms that an investor may hold in their portfolio. This is the basis of the *Capital Asset Pricing Model (CAPM)*

The SCS examination has featured questions that expected candidates to recognize the lack of diversification of risk where shares are held in the following circumstances:
- As part of an employee share scheme: if the firm fails the staff lose both their incomes and their investment
- Where the shares are being granted as share options: this again means the recipients are employees of the firm but their risk is *not symmetrical*. There is no *downside risk* because if the firm’s shares fall in value the option would become worthless but also it cost them nothing to receive it. But there is *upside risk* because if the firm’s shares rise in price above the exercise price of the option, they can exercise the option and get a capital gain
- Where the shares are in a family firm: this will be the main wealth of the family members and they will not be able or willing to reduce their risk by selling some shares to buy others because it would mean relinquishing control of the family firm
- Whether changes in the share price of a firm due to adverse revelations about its CEO are an example of unsystematic risk and therefore should be ignored by a well-diversified investor. (It’s nonsense. Because it is unsystematic risk it is unlikely to have been offset by gains elsewhere in the diversified portfolio and no damage to share price by management misconduct should be ignored)

Significance of the share price to business decisions
The Examiner has asked several questions on this in the real examination.

Ultimately the share price is the wealth of the shareholder: if it rises they will be richer but if it falls they will be poorer. Therefore a duty of management is to protect and increase the share price. But that is not the same thing as saying that the share price should determine management decisions.

How important is the share price to management?

Reasons for importance	Factors reducing its importance
• High share price is indicator of investor support for management and its strategy	• Much share price movement is speculative and short term falls will be followed by recoveries
• High share price makes it easier and cheaper to raise equity capital for investment	• The stock market does not understand the business and its issues or potential as well as its management does
• Low share price means shareholders unhappy and they may put pressure to change Board and business strategy	• Management should lead the company in the way it thinks right and not become fixated by fluctuations in the share price and market sentiment
• Low share price could make firm a potential takeover target because it is *undervalued*	• Share prices can be affected by factors outside management control (interest rates, economic and political outlook) which management will not be blamed for or need to respond to

ACTIVITY 12.3

Context
Couchweb is a listed global provider of subscription video on demand (SVoD) like Netflix and Disney+. It streams movies, series and other television programs. It is dependent on program and movie production firms for its content and has taken the strategic decision to invest in producing its own content. It has to date created two successful series: The Politics Game and Tumbling Down.

Task
You have received an email from Martine Andels: Director of Finance of Couchweb
To: Senior Manager
From: Martine Andels: Director of Finance
Topic: Kyle Johnson
I have attached an article from today's Mayland Independent News.
I can confirm that the Board has received a petition from staff on Tumbling Down calling for Kyle Johnson to be removed from the series. The allegations are serious.
What will be the likely impact on Couchweb's share price of this publicity? (sub section = 55%)

Reference Material

Mayland Independent News

Tumbling Down star accused of inappropriate conduct

TV star Kyle Johnson is accused on living-up to the antics of his buccaneering on-screen character, Josh Carter, in real life.

We have learned that staff on the Couchweb series 'Tumbling Down' have complained to the management of Couchweb about the behavior of the star of the series, Kyle Johnson. According to reports Mr Johnson creates 'a bullying and toxic atmosphere' on set. He shouts at co-stars and physically abuses production staff. He is also accused of unwanted attentions towards female staff.

One female makeup assistant is reported to have said 'He demands that I attend to his makeup in his private trailer but it's not safe for a woman to be left alone with that man'.

An analysis of the credits for the past eight seasons of Tumbling Down reveals a very high turnover in actors and production staff. Mayland's Guild of Creative Staff has promised to look into these allegations.

The Efficient Market Hypothesis (EMH)
The examiner has called this the 'rational economic view' of the financial markets. It's probably more accurate to say that it provides a basis for such a view (or perhaps gave a reprieve to the economic view at a crucial time in its history).
The EMH makes two assumptions:
1. The stock market will respond to new information about the firm and this will affect its share price
2. The stock market is efficient in its processing of information and so in a perfect market the share price of a firm will reflect all known information about the firm and the circumstances affecting its cash flows and risks.

The EMH concludes that an individual investor cannot out-perform the stock market in the long-run because they cannot have information that the market as a whole doesn't have, and neither can they possess a secret system for spotting potential share price movements, such as by predicting patterns in the share price charts.

Later writers recognized that information may not be universally shared and proposed the following versions of the theory:
- Weak form: the market knows the past share price behavior
- Semi-strong form: the market knows the past share price behavior plus extra publically available information about the earnings fundamentals of the business (note: this version allows that some private or insider knowledge may exist and that therefore some shareholders can gain from this)
- Strong form: the market knows past share price behavior and all relevant knowledge of things affecting the share price (i.e. there is no 'private or insider knowledge')

Non-rational views of share price behavior
In real-world financial markets there are additional forces at work to influence share prices. *Speculative forces*: investors and fund manager buy shares (or traders increase the price of the shares they are offering for sale) if they expect the price of the share to rise. This enables them to sell fairly quickly and make a capital gain. One example of this is *High Frequency Trading* (HFT) which uses supercomputers to detect minor movements in share prices and then automatically deal in the same direction and thereby exaggerate the price movement

Behavioral forces: a developing field that recognizes that the reasoning processes, biases and enthusiasms of market participants can influence share prices. This has been used to explain the history of speculative waves such as the dot.com boom of the late 1990's and the enthusiasm for shares in loss-making platform business models in the present decade. The existence of these forces calls into question the economic theory of share prices and, by extension, many economic explanations of business behavior.

More can be found in the following article, but this would be well-beyond the requirement of the CGMA assessment

https://www.researchgate.net/publication/331210273_Theories_Explaining_Stock_Price_Behavior_A_Review_of_the_Literature

ACTIVITY 12.4

Context
Runnabout is a listed micromobility provider. Its main business is offering electrically-powered hoverboards for hourly hire to help tourists and commuters to travel from rail and coach hubs to their destinations. Its shares have an historic beta of 1.33 and its shares have traded at between G$6 and G$14 in the past two years. Three weeks ago Runnabout received an offer of G$18 per share from a rail operator, Railcorp to acquire it.

Task

Yesterday, August 25 2020, the Board of Runnabout announced that it was going to recommend to shareholders that they accept an offer from Railcorp of G$18 per share

From: Geo Pataros: CFO
To: Senior Finance Manager
Topic: Share Price Behaviour

Extremely Urgent

We have received a formal Notice of Enquiry from the Supervision Section of the Geeland Stock Market.

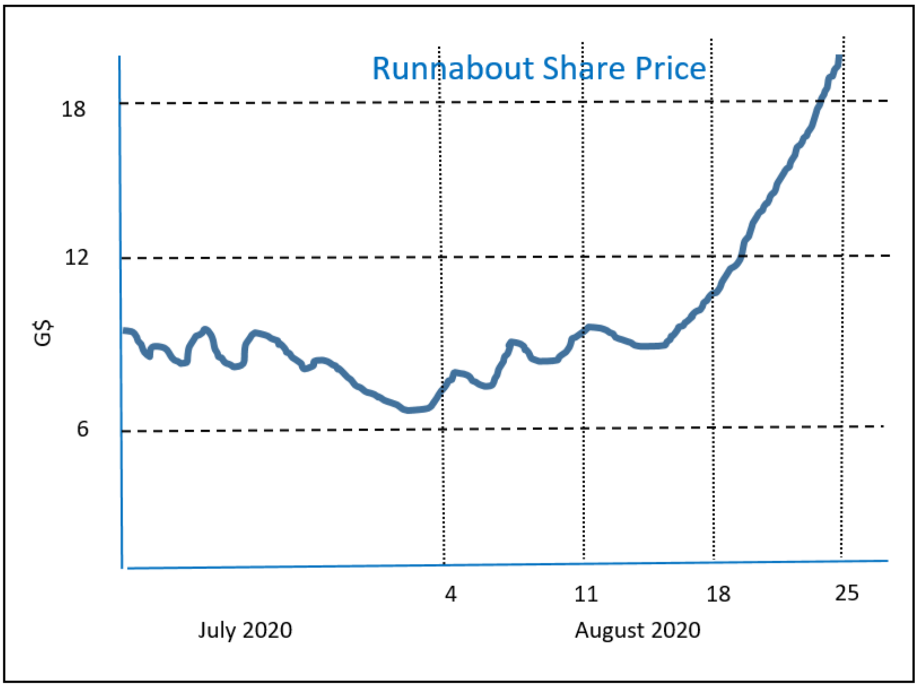
The Stock Exchange wishes to investigate the movements in Runnabout's share price in the weeks before yesterday's announcement of a recommendation to accept Railcorp's offer.

The interest from Railcorp was treated confidentially and yesterday was supposed to be the first time that anyone outside the Boards of the two firms knew anything about the potential acquisition of Runnabout.

I have attached a share price chart showing the share price of Runnabout over the past two months.

What factors normally influence Runnabout's share price and do you think the share movements in the chart prove insider dealing by someone?

[sub task = 60%]



3. The Cost of Capital

Exam note: the assessment will not require you to calculate the Cost of Capital in the exam, and the pre-seen case to date has never contained sufficient information to calculate it. But the examiner likes to set questions involving elements of the determinants of the Cost of Capital so you do need to understand it.

Cost of capital and the maximization of shareholder wealth

Recall the following from an earlier chapter on the goals of a profit-seeking business

- The management of firms seek to maximize shareholder wealth.
- Management does this by ensuring a positive return on investments after discounting them by the required rate/discount rate of the business in an NPV calculation
- This required rate represents the best alternative return that the investor could have obtained by investing in the next best business

The cost of capital of a firm is the minimum rate of return it must make on the funds that various types of investors have tied up in the firm. If its rate of return falls below this cost of capital it will cause the share price to fall as disappointed investors sell their shares to buy better-performing shares elsewhere.

The cost of capital is used in two ways in making management decisions:

1. It is used to evaluate new investment projects: the cost of capital is used as the *required rate* or *discount rate* in calculations such as Net Present Value. Providing the project returns a positive NPV it will increase shareholder value and boost the share price
2. It is used as a target for return on existing assets owned by the firm. Simply put the return on capital employed of the firm should be managed to exceed its cost of capital

The Weighted Average Cost of Capital (WACC)

The *weighted average cost of capital* is an approach used to calculate this cost of capital

- The percentage of the capital structure financed by debt is multiplied by the interest rates on the debts of the business (which are determined by the loan agreements). It shows the minimum return to debt holders.
- A second percentage is calculated by using the cost of equity to multiply the market capitalization of the business. Note, it uses the market value of equity and not the balance sheet value. This is because the market value represents the actual money the shareholder has tied up in the business. This is the minimum return to equity holders
- The two percentages are added together to give the WACC

This means the WACC is influenced by four things:

1. The Cost of Debt (Kd)
2. The amount of debt in the firm's capital structure
3. The Cost of Equity (Ke)
4. The amount of equity in the firm's capital structure

The Cost of Equity also must be calculated. It represents the minimum return acceptable to the equity holder.
There are two approaches to calculating the Cost of Equity:

1. Dividend method:
   Divides the dividend by the market value of the firm's shares.

   This is the dividend divided by the share price. So if the share price is stable at $10 and the firm has been paying $1 a year dividend it implies a Ke of 10% (i.e. $1/$10). Shareholders are content because the share price is constant, and they are getting a 10% return,

   If Firm A and Firm B both have the same market value but Firm A pays more dividend it follows that Firm A is riskier for investors and therefore must pay more dividend to compensate them for the risk of holding its shares. It has a higher Cost of Equity

2. The Capital Asset Pricing Model (CAPM)
   This uses three things to obtain the cost of equity:
   1. *The risk-free rate*: this is the level of return available on a risk-free asset such as a government bill. If governments raise their interest rates it will have the effect of raising the Cost of Equity and therefore reducing share prices. To this is added an equity market risk premium.

   2. *The equity market risk premium*: a general percentage which reflects the extra that investors require to make them invest in equities rather than invest in risk-free assets. In times of stock market pessimism this premium may increase and therefore it will have the effect of raising the Cost of Equity causing share prices as a whole to fall. The equity market risk premium is increased or reduced according to whether the Beta coefficient is greater or less than 1.

   3. *The Beta coefficient for the firm or sector:* this a numerical measure of the volatility of the share prices of the firm or of its industry compared to share prices as a whole. If this volatility has a value of greater than 1 it will raise the Cost of Equity above the rate for equity investments generally. If it is less than 1 it will reduce the Cost of Equity below the rate for equities generally. A higher Cost of Equity leads to lower share prices.

In the exam the examiner asks questions on parts of the theory such as:

- Should a fall in the firm's share price affect the cost of capital used to evaluate investments? (Yes: technically it should lead the firm to adopt a higher cost of capital in future because the investors perceive greater risk in holding the shares and/or management have underestimated the cost of capital and the share price fell due to them undertaking investments with too small a return)
- What is a Beta and why does it affect our investment decisions? (It's a measure of volatility of the shares in our sector. If it rises it will raise the WACC and so we should apply higher discount rates to our investments)
- How would downgrading our debt to junk status affect the share price of the firm? (junk bonds have higher interest rates and so this will increase our WACC. This will cause the share price to fall until a lower market capitalization means our present return are satisfactory)

Considerations in use of Cost of Capital

In her solutions the examiner expresses reservations over the reliability of these technical economic approaches to cost of capital and business decisions.

- Share prices are always volatile in the short run, especially where a firm is announcing new business investments, and this makes calculating a cost of equity very difficult
- The WACC is based on the existing funding and risks of the business and assume a 'pool of funds' that can be dipped into to finance the project. It is not reliable for evaluating investments that are
    - In a different industry from the firm's present industry
    - Financed in a different way from the present funding of the firm (e.g. if the project is 100% debt funded the firm should evaluate it using its *Marginal Cost of Capital*: meaning the cost of the new debt plus the rise in the firm's cost of equity in consequence of the greater risk the higher leverage present to shareholders)
    - Very large in comparison to the firm

4. Shareholder Value Management

These are approaches to performance evaluation of a business based on the same economic reasoning as net present value: that *shareholder value maximization requires that the return to invested cash be at least equal to the cost of capital of the firm*. However, these approaches are applied to assess how existing capital is being used by the use of special performance measures such as Economic Value Added (EVA) or Economic Profit (EP).

Taking EVA as an example. The formula is
Economic Value Added = Net Operating Profit after Tax minus (Invested Capital x Required Rate)

Essentially the formula asks whether the cash actually generated by the business (NOPAT) is greater than the minimum satisfactory return in the cash invested in the business (Invested Capital x Required Rate). If the cash generated is less than the minimum return the business is destroying shareholder value and should change its ways.

Some points can be made about shareholder value approaches

- Requires management to replace traditional performance measures for business units and divisions, such as return on capital employed, with shareholder value metrics such as EVA above
- May focus managers on maximizing cash flows but also on economizing on the cash tied up in their business and on reducing the required rate (WACC) by reducing the risks created by their capital structures and business strategies
- Requires that traditional management accounts be supplemented with additional accounts and forecasts that use a cash-based approach to valuing capital

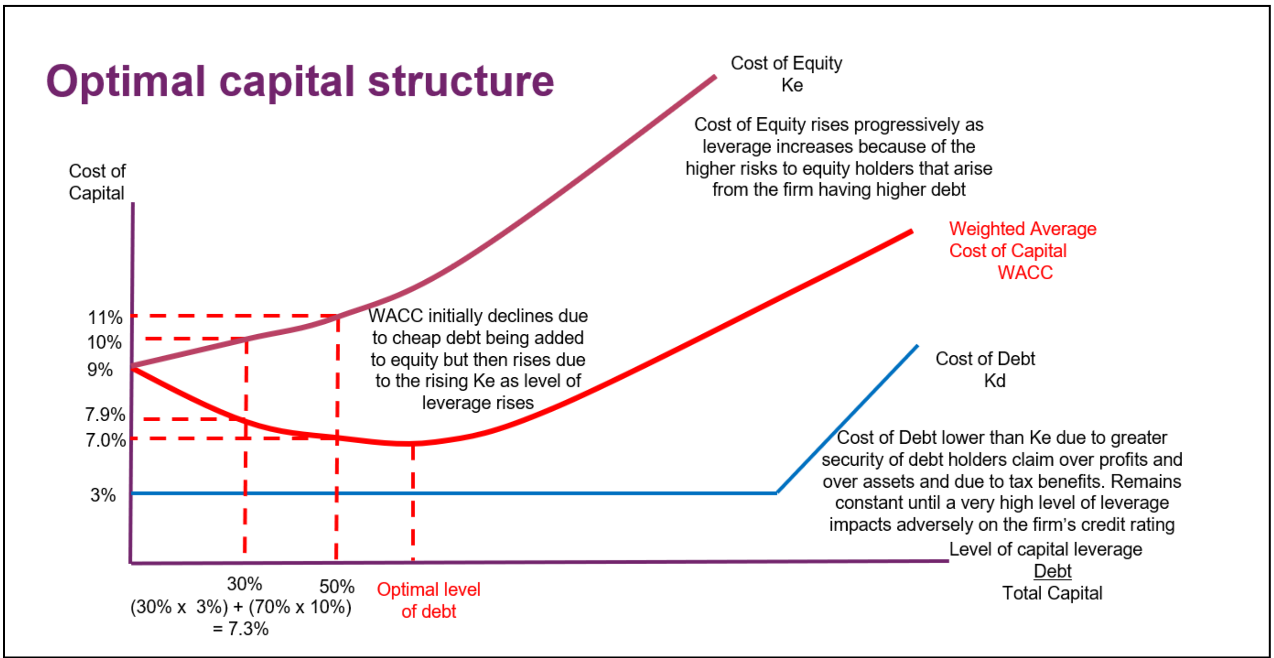
- There may be a short-term bias in some shareholder value measures because they seem to favor maximizing cash flows in each year at the expense of investing in the business

5. Optimal Capital Structure

Conventional economic theory focuses on identifying an *optimal capital structure* that will give the lowest cost of capital for the firm by striking the right balance between debt and equity.

Capital Structure of a business refers to the ratio of debt to equity in the firm's financing sources.

Conventional economic theory focuses on identifying an *optimal capital structure* that will give the lowest cost of capital for the firm by striking the right balance between debt and equity.



Solutions to Activities

12.1 Couchweb financing program development

Is it better for us to finance these new programs by debt or by equity? (55%)

Approach: The sources of finance, and in particular the relative merits of debt finance versus equity finance, is often requested in this exam. Most candidates focus on the cost of capital. But notice that answers on capital structure also need to consider cash flow implications, issues of control and risk, and also the length of time the finance is needed for and the flexibility of the borrowing to be extended or paid back according to peaks and flows in the project.

Note that this task requires advice on whether debt is likely to be better than equity. To get high marks candidates should be willing to give advice.

Raising finance for new programs

The cash flow forecast shows that during 2019 Couchweb will require additional finance which rises by 2020 to over M$1.7bn. The accuracy of the forecast is subject to errors in projections of revenues, operating costs, and the costs of producing a large number of programs and this needs to be born in mind in assessing financial structure.

In deciding on capital structure Couchweb needs to consider the length of time that the capital is likely to be needed for, the risks it will bring such as changes in interest rate and variations in liquidity affecting our ability to service the debt, and the willingness of shareholders and banks to lend to Couchweb

It seems likely that this borrowing will be long term because after 2020 Couchweb will still be producing a lot of programs and may wish to expand its catalogue further

Debt borrowing
Couchweb may be able to issue financial instruments, such as long-dated bonds, to the financial markets. It already has over M$5bn of loans [*data in pre-seen*] and so will have an established credit rating. This could make the debt less expensive. It should also be remembered that the assets of Couchweb are comprised mainly of content, i.e. programs and licenses, which provide poor collateral for any lender.

Advantages of debt
- May reduce the cost of finance for Couchweb due to enabling us to benefit from tax credits against the interest on the debt
- Can be repaid from positive cash flows in later years if the need for extra finance reduces. However, these are not forecast to occur until after 2020 and may not occur for many years after 2020 if Couchweb continues to expand program production further, for example by creating extra programs for different international audiences
- Cost of the debt can be reduced if convertible debt is issued, this mean the debt holder has the right to exchange the repayment for shares if the share price of Couchweb rises above the exercise price granted in the convertible
- It may be possible to adopt project financing as is the common practice of many movie producers. Here the program or movie is set up as an investment vehicle and external finance is raised to find it and to share in the proceeds of its success. Couchweb would hold an investment stake but not 100% of the investment. This transfers some risk to the other investors and away from Couchweb, although it also reduces Couchweb's share of the profits.

Disadvantages of debt
- Increased strain on cash flow because in addition to the forecast shortfall Couchweb would also need to pay increased finance costs on its increased debt borrowings
- Is not matched to length of borrowing needed. Debt is usually limited to 5 years or so and needs to be renewed periodically if it cannot be repaid. The investment in program production requires a long-term tranche of extra funds that is unlikely to be repaid because once the new series is earning cash flows from subscriptions, these funds will be needed to finance the next season. There is a risk to Couchweb's sustainability if one day the lenders refuse to renew or refinance the loans
- Couchweb is relatively highly geared/levered at the moment at around 59.4% [data in pre-seen material], and an additional M$1.7bn or so of debt would take this leverage to 66.0% which may appear high. This could mean a higher interest rate and a negative impact on Couchweb's share price because high leverage increases the specific risk (or unsystematic risk) of Couchweb's
- Greater volatility of free cash flows and profits due to the need to continue paying high interest costs even if operating profits fall
- Risk of Couchweb being liquidated if it cannot pay interest/capital due to lower than expected profits (e.g. because of competition or making unsuccessful new series)
- Loss of managerial autonomy if lenders impose conditions (covenants) or demand board places
- Potential interest rate risk if loans have floating rate or need regular refinancing

Equity issue of shares
Couchweb could issue new shares to raise capital. This may require some of the shares to be made available to the stock exchange as well as selling shares to existing shareholders.

The ability of Couchweb to attract buyers for these shares, and the price it can charge, will depend on the extent of investor support for Couchweb and for its strategy of expanding program making

Some equity investors may feel that program making is riskier that providing SVoD [*Subscription Video on Demand] as* a distribution service because a bad program not only reduces subscriptions, it also costs money to make

Advantages of equity issue
- Less risk than debt because there is no compulsory interest to pay
- Capitalizes on the high share price of Couchweb

Disadvantages of equity issue
- May disrupt existing shareholders due to them being diluted.
- Will potentially lead to the admission of a new external shareholders who may have different objectives from Couchweb's Board

Other forms of raising finance
Couchweb could use the project financing approach described above

Joint Venture with a TV channel for a new TV series, perhaps with advanced transmission rights for Couchweb

Another approach is to sell advance rights to the program to third parties. For example, sell the serialization rights to other broadcasters, and the merchandise rights to consumer goods firms in advance

Conclusions
The main methods described are probably available to Couchweb

The decision could be based on minimizing the weighted average cost of capital (WACC) by finding an optimal balance of debt and equity in the capital structure.

It is recommended that greater equity should be used to finance this permanent capital requirement

The actual amount raised could be reduced by selling advance rights and inviting other broadcasters or Subscription Video on Demand (SVoD) firms to provide some financed. This would also reduce the risk to Couchweb from cost overruns and producing an unpopular series

12.2 KCTP raising finance by a rights issue
What will be the difficulties in raising the capital through a rights issue and what other sources of finance can be used? (50%)

Raising capital to purchase Exland Technology Park
A rights issue is a new issue of shares to existing shareholders (or more accurately a new issue of shares in which the rights of existing shareholders to pre-empt the sale are acknowledged by giving them first refusal on the new shares). Shares are normally offer pro-rata to existing shares (say 1 new share for each 8 existing shares) at a price below the present market price of the shares (student note: the reason for the discount is because shareholders

could buy more shares at the existing market price by just buying them on the stock market and so it is assumed they have as many as they want. It is necessary to offer a discount to get them to buy more shares). The effect of having cheaper shares issued to the market will be to depress the market share price once they are issued (called the ex-rights price). This threat can lead shareholders to waive their rights to the new shares and sell the rights letter to others but also, sometimes, to sell their existing shares before the price falls.

Difficulties in rights issue

Potential dilution of shareholders

This rights issue increases issued shares by 13% but it will increase profits by only 10.3% (A$15m/A$146m). This means the earnings per share of KCTP will fall following the acquisition. This is a dilution of the EPS and may lead to fall in the share price of KCTP.

In fact the fear of dilution may lead the shareholders to sell their shares now, in advance of the rights issue. This will mean that even more shares will need to be sold to raise the A$3.3bn required, leading to even greater dilution, and so on.

The way to avoid this is to reassure the market that the profits of ETP will increase sharply in the coming years and that this will reverse the dilution and increase the EPS.

Existing shareholders prefer dividend

In 2019 KCTP paid over 50% of its profits out as dividend. KCTP is a cash cow share, giving high incomes each year with low risk (it has a beta of 0.9 which is below the risk of shares in general). This will have had a clientele effect of attracting investors who want cash and low risk. These same investors are very unlikely to change their investment objectives to accept lower total dividend and the higher risk caused by exposure to Exland and the problems of turning ETP around.

KCTP will need to accept that many of its existing shareholders will try to sell their shares. The Board needs to create a market sentiment that means different investors are willing to buy these shares, and the rights issue letters, without requiring a significant fall in the share price to entice them. KCTP needs to communicate the potential for growth and higher future earnings that may come about from buying ETP.

Resistance from Capital City University (CCU) due to loss of influence

CCU has 20% of the shares of KCTP and appoints its own NED to the Board of KCTP to look after CCU's interests in stimulating innovation in Advland, providing opportunities for commercialisation of its research and so on. If CCU is not willing or able to subscribe for its allocation of new shares it would see its control eroded from 20% to 17.7% (20%/113%).

It would also fear a change in the characteristics and objectives of the majority shareholders in the future, from the present risk averse and dividend seeking shareholders, to shareholders who are more risk seeking and desire capital growth in their share price from undertaking this risky acquisition in Exland.

Adverse impact on share price

The financing of this acquisition by a rights issue depends on the existing share price remaining at A$14 or so. If it falls it will undermine the issue by making the new shares unattractive and by requiring more than 650m to be issued.

The stock market values KCTP at the value of the markets expectations of its future free cash flows, discounted by a risk factor based on its present business and capital structure. This will all be disrupted by the news that it is considering buying ETP.

The attitude the market takes to the likely future free cash flows of KCTP with ETP as part of it, and the impact on the risks of KCTP of this foreign subsidiary, will be significant in determining the share price of KCTP now.

To overcome this problem KCTP should focus on explaining the financial benefits of acquiring ETP and describe how it will minimise the risks from it. The success of this will depend a lot on market sentiment towards the opportunities and risks of Exland generally.

Other sources of finance

Loan finance

Debt finance could be used by raising further loans secured on the assets of KCTP, or possibly partly on the assets of ETP that it will acquire.

At December 31 2019 KCTP had loans of A$960m against PPE of A$6,600m and therefore there is spare collateral of A$5,640bn which is adequate to secure a loan of A$3.3bn. Better some of the funding could be raised by an E$ loan could be secured against the assets of ETP.

The difficulty with using loans is the higher capital leverage would increase the risk of KCTP shares and cause a discount in their price. This would be a response to the cash flow costs of meeting the interest payments on the extra loans which will strain KCTP's cash flows and will reduce its profits.

Sale and leaseback

KCTP could sell some of its buildings to a finance house and lease them back over the coming years. This would provide a lump sum with which to buy ETP.

Joint venture

KCTP could seek a partner to help finance the purchase of ETP and therefore reduce the amount that needs to be raised by loans and/or equity.

Grants from governmental bodies

There may be financial assistance available from the Exland government, such as a loan to KCTP at low rates to help it to buy ETP, in effect a scheme of deferred payment.

There may be transnational bodies with an interest in the development of Exland that may be able to arrange or provide funding for firms undertaking inward investment into Exland.

12.3 Couchweb share price: impact of allegations about cast member

What will be the likely impact on Couchweb's share price of this publicity? (55%)

Approach: A very common question in this examination involves assessing the impact of unexpected events on the share price of the business. You are not required to predict the share price movement but rather to illustrate your understanding of share price determination.

Impact on share price of allegations about Kyle Johnson

According to financial economics and the Efficient Markets Hypothesis (EMH), the market value of Couchweb will be equivalent to the discounted present value of its future free cash flows. Therefore, news about Kyle Johnson will affect the share value of Couchweb according to how it affects expectations of future cash flows and the rate of discount, risk, applied to Couchweb's earnings.

However, the EMH has been criticized as excessively theoretical and also as lacking a timescale. In the real world, share values are also affected by rumors and speculative behavior that are not connected with future earnings.

The allegations against Kyle Johnson are not yet proven, and Couchweb has not made any response to them. However, the market will respond.

The share price of Couchweb will probably fall for the following reasons:
<u>Loss of Kyle Johnson from the series</u>
Tumbling Down is a very successful series and has provided the basis of movies. This means it attracts and retains subscribers which provides cash flows for Couchweb. These cash flows help to fund future program-making and also pay dividends to shareholders

The program also provides the basis of potential spin-off such as movies, specials, sub-series and merchandise and licensing of images and intellectual property. These are valued by investors as potential courses of future cash flows.

If Kyle Johnson disappears from the series the value of the media franchise may fall as subscribers drift away. This would reduce expected cash flows and so reduce the value of Couchweb shares.

<u>Litigation against Couchweb by staff</u>
The allegations against Kyle Johnson are potentially serious. Staff could take legal action against Kyle Johnson, but also against Couchweb for allowing this to happen or not dealing with it properly. This legal action would be expensive to defend and, if successful, the staff could recover significant financial damages from Couchweb.

The threat of such litigation, and its impact on future cash flows, will cause investors to sell Couchweb shares to avoid the potential fall in value

<u>Costs of write-downs and re-makes</u>
Kyle Johnson may already appear in episodes of Tumbling Down that have yet to be aired.
It is possible that Couchweb will need to edit Johnson out of these programs, or arrange for a replacement actor to take his role. This will demand re-shooting of the affected parts of the program

Other programs may need to be destroyed as unusable.

This will impose additional costs on Couchweb and so reduce cash flows in the present year. This will reduce the share price due to impact on expected earnings and on dividends.

<u>Further allegations coming to light</u>
Once Kyle Johnson is criticized in the press other staff may feel safe to come forward with more serious allegations against Johnson or against other Couchweb staff or actors. The investment markets will fear this because the bullying actions of one person often go unchallenged due to a culture of poor behavior in the company or industry as a whole.

During recent years the bullying and discriminatory nature of the film and entertainment industry has been widely exposed and it suggests an endemic culture of such behavior.

Investors will sell their shares now rather than risk holding such shares when more serious and widespread allegations are made. This will depress the share price of Couchweb.

<u>Fear of subscriber backlash against Kyle Johnson</u>
The adverse behavior of Kyle Johnson may lead viewers, and broadcasters, to turn against his program. It may also lead to a fall in the sale of merchandise featuring Kyle Johnson's image which has been licensed by Couchweb. This would lead to the cancellation of subscriptions and contracts.

This will also reduce cash flows and therefore the share price.

Speculation against Couchweb shares
Speculators will have already considered the points made above and acted on the assumption that the share price of Couchweb is likely to fall. Speculators will sell their holdings of Couchweb shares and so depress the price. Some will 'short' Couchweb stock by entering into a contract to sell Couchweb shares in the future at a price below present market price. This is based on their expectation that the share price will fall in the meantime and that the contracts will not be taken up (options contract) or that shares will be cheaper to buy in the future (futures contract).

Conclusion
The above factors mean that the share price will have already fallen as a consequence of the new article. The share price will probably have over-reacted and will gradually improve and the speculative fervor around these allegations are replaced by facts about the full position.

12.4 Runnabout share price: insider dealing?

What factors normally influence Runnabout's share price and do you think the share movements in the chart prove insider dealing by someone? (60%)
Approach: Note there are two questions here: describe normal influences on share price and provide assessment of likelihood insider dealing.

Factors normally influencing share price of Runnabout

According to financial theory the total market value of R/T is equivalent to the present value of the markets expectations of its free cash flows in future. This is sometimes termed the Efficient Market Hypothesis (EMH).

Therefore the factors affecting share prices include the following:

Market expectations of future free cash flows
This will be based on investors judgement of the quality of R/T's past performance, its strategy, and the factors in the market that will affect its success in future. The financial forecasts shared with the market by R/T will also be influential.

Market assessments of risk
This affects the discount rate applied to the future cash flows in which the recognition of a higher than expected level of risk will raise the discount rate and so reduce the present value of free cash flows. The past ability of R/T to deliver its financial forecasts, potential for changes to its legislative and competitive environment, and the rise in accidents and injuries, all contribute to this assessment.

Dividend policy
Dividends are immediate cash to the shareholder. A rise in dividend payments from a more generous dividend policy might be expected to raise the share price. In the short run share prices are affected by the time to the next dividend payment because the future cash is added to the long-term share price by buyers and sellers (so-called 'cum-div' prices)

Rumours and events
Rumours of changes to strategy, problems, higher or lower profits, and of takeovers will affect share price. Economists might argue that this is because they affect market assessment of risk, cash flow and so on as stated above. However the speed of reaction does suggest that the market makes an initial guess and then, later, evaluates the exact effect of the rumoured event on earnings and risks.

General market sentiment and values
The equity market as a whole may rise and fall, and R/T with it, if the Geeland economy is expected to boom or to slump, or if the interest rates on bonds is expected to change. Political changes that are good for, or bad for, business will also affect share prices.

Do share price movements prove insider dealing?

Indications of insider dealing
Exceptional rise since RailCorp showed interest
There is a clear rise in R/T's share price over the past 3 weeks, the time frame since RailCorp approached the Board of R/T with its initial offer. Price had risen from about G$6.50 to about $19.00 (192%) in 4 weeks which seems exceptional.

This does suggest that some in the market had reason to expect R/T's shares to be worth more in the future.

Final value very close to actual agreed price
The price at the announcement of the acceptance was very close to the agreed price. It might have overshot because some buyers expected the final offer to be higher. It does seem to be based on a belief that the share value could rise to about G$18.00 or so.

Alternative explanations for the rise
Share price has shown similar volatility before
R/T has a beta of 1.33 which denotes a particularly volatile share price. Inspection of the historic share price chart (not included here) shows a rise on price from G$6.00 to nearly G$14.00 (133%) over 8 months and a rising trend over the last 6 months.

Could be market rumour
Rumours can lead share prices to move without specific insider information. For example if senior management of RailCorp were seen visiting R/T's offices, or administrative staff noticed R/T's board members were cancelling meetings and having rushed teleconferences they might conclude something was happening and mention to people in the investment markets

The rumour might not be about RailCorp but perhaps about better trading figures or the potential of new licenses in new cities.

Copycat trading on a trend
Many market participants might rely on the trend in share prices to decide where to buy. They assume that 'someone knows something' and copy the market.

This is not insider dealing.

Need to know general share price behaviour

Share prices as a whole may have risen due to other factors, such as a fall in money market interest rates or a favourable political development affecting business as a whole.

We would need to see if other shares, and particularly ones in micromobility sector, rose at the same time as R/T's share price.

Conclusion

This rise is suspicious but not conclusive. This is why there needs to be an investigation of the circumstances around it.

Chapter 13
Dividend Policy

1. Methods of dividend payment

A dividend is a payment to its shareholders as a reward to their investment in the business
There are several types of dividend payment

Dividend payment methods

1. Cash Dividend
2. Scrip Dividend
3. Property Dividend
4. Liquidating Dividend

1. Cash Dividend
   Each shareholder receives a cash credit on the payment date

2. Scrip Dividend (bonus issue, equity split)
   A payment in stock instead of cash. It enables the firm to capitalize its existing cash reserves. The shareholder can sell the bonus share and receive money as a capital gain rather than as income which may have tax benefits. However, this would leave them owning proportionately less of the firm.

   Other definitions refer to a script dividend a promissory note for dividend in the future at a time when the firm lacks the cash to pay the dividend at the moment.

3. Property Dividend (dividend in kind, dividend in specie)
   These are relatively rare. The firm distributes assets at fair market value to shareholders. It enables the firm to record the gain or loss on the difference between book value and maker value as a profit or loss in its income statement.

4. Liquidating Dividend
   This is returning the capital that the shareholders originally invested, usually as part of a down-sizing or closing of the business. This dividend may receive special beneficial tax treatment for the shareholder.

2. Dividend policies

The set of guidelines a company uses to decide how much of its earnings it will pay out to shareholders

Dividend policy approaches

Policy	Advantages	Disadvantages
Stable dividend • Constant amount each year • Constantly growing amount each year	• offers investors a predictable cash flow • reduces management opportunities to divert funds to non-profitable activities • works well for mature firms with stable cash flows	• reduced earnings would force a dividend cut • this would disappoint investors • it would highlight the failure of management • may cause management to squander reinvested profits if there is no good use for them
Constant payout ratio A constant proportion of earnings attributable to ordinary shareholders	• retains a clear link between payout and reinvestment rate	• investors discouraged by uncertainty of cash flows • invites greater shareholder pressure for short-term profits • does not signal management confidence about future profits • may cause management to squander reinvested profits if there is no good use for them

Dividend policy approaches

Policy	Advantages	Disadvantages
Zero dividend policy All earnings reinvested in the business to fund its growth	• improves the growth of the firm • reduces the need to source additional external finance • signals investor confidence in growth prospects for business	• focuses investors on demanding very high growth which may not be sustainable • permits management to squander funds in low quality projects • could be interpreted as a cash flow crisis and/or poor cash management
Residual method Dividend is what remains after all investment in growth projects has been funded	• improves the growth of the firm • reduces the need to source additional external finance	• investors discouraged by uncertainty of cash flows • does not signal management confidence about future profits

Practical considerations of dividend policy

Consideration	Description
Legal position	• Laws on maximum dividend pay out ratios designed to encourage investment in growth • Shareholders using past dividend policy as basis for legal action against a future management that decides to change it
Stability of profit	• Unwise to commit to fixed or rising dividend if profits are likely to be volatile
Inflation	• Profits may be inflated by higher prices later in the year but the cash available won't have risen
Growth plans of the business	• Fast growing firms pay less dividend to retain earnings • Fast growing firms can compensate investors for lack of dividend by having significant rises in share price
Control	• Retaining profits reduces need to issue new equity and/or debt • This helps keep control in the same hands
Liquidity	• Profit is not cash • Sufficient liquid (cash) funds must be available to pay the dividend
Taxation	• The personal tax position of shareholders may make them want or not want extra dividend
Signalling effect	• Changes in dividend payouts are taken as signals of managements expectations for the business (eg rise in dividend signals rising earnings)
Clientele effect	• A dividend policy will lead to the firm getting the sort of shareholders who demand that policy (eg paying a high dividend will attract cash-seeking shareholders)

3. Share buybacks

A share buyback is an offer from the management of the firm to purchase some of the firm's shares back from the market using the firm's own funds.

Considerations in a share buyback

Consideration	Description
Impact on share price	• Announcement of buyback will boost share price due to imminent cash pay out • Less shares but same amount of earnings will increase Earnings Per Share and so should increase share price • Returning funds from low yield financial investments could improve group ROCE/ROI and so share price • Market may anticipate further buybacks in future and put premium value on firm • Market may fear that management has run out of profitable uses for equity and downgrade growth expectations and share price • Increases capital leverage of firm which increases risk/cost of capital and may depress share price
Impact on shareholders	• Shareholders can chose whether to cash in shares or wait for share price to rise • Sellers can realise capital gains/losses and pay appropriate taxes • Non-sellers should enjoy a capital gain due to share price rise which can be realised at a later date
Impact on control	• Concentrates control into the hands of the non-sellers • May provide opportunity to engineer exit of unhappy shareholders
Motives for buyback	• May be genuine change in business model away from reliance on capital assets • Could be manipulation of share price to boost value of executive share options • Could be desperate response to falling share price – i.e. to avoid acquisition
Impact on future of the firm	• Will reduce funds available to finance future growth • Will be hard to raise new equity in future
Market perception	• Market may see it as legitimate business decision • Market may see it as share price manipulation • Debt holders may see it as allowing equity to exit and shifting risk to debt holders

ACTIVITY 13.1

Context

Runn is a large global listed CEP firm (Courier Express Parcels), like DHL or Fedex. It has 400,000 staff and 104,000 vehicles.

Runn has decided to move towards becoming a network or platform organization by disbanding its 'final mile' delivery service. It intends to sell its vehicles to its drivers who will become self-employed contractors.

Task

You have been called into the office of Chen Yanyun: Director of Finance for Runn

We believe that selling the vans to couriers will mean that Runn will be able to sell a lot of its depot space and maintenance facilities too. This could lead to about G$3,000m of surplus cash once all the sales are complete.

If we return this cash to shareholders should we pay it as a dividend or use it to finance a share buyback? (sub section = 55%)

Solutions to Activities

13.1 Cash distribution at Runn courier company

If we return this cash to shareholders should we pay it as a dividend or use it to finance a share buyback? (55%)

Approach: AICPA-CIMA students study basic tax in the qualification and will be aware that dividend income is taxed differently in the hands of the recipient compared to capital gains. This is an important consideration to understand in approaching this task element. They will also be aware that double-taxation of dividends is avoided by granting shares a tax credit in *respect of the tax that the company has already paid on its profits. There are also a number of other non-tax considerations that would have obtained good marks.*

Share buy backs
Runn would use the surplus funds to purchase some equity back from existing shareholders. This can be a *market purchase* in which case it buys them from the stock market. It can also be an *off-market purchase* in which a private offer is communicated to shareholders to which they can subscribe

Benefits of a share buy back
Gives cash to investors who sell their shares to Runn
This cash will represent a realized capital gain on their investment for which there may be special tax treatments and allowances that are favorable to them

Will increase the value of the remaining Runn shares on the market
This is because it increases the earnings per share and dividend per share of the remaining shares. This rising share price might lead to further gains as the market seeks to buy a rising stock. This is a capital gain which is not taxed until these remaining shares are sold.

Can be used on a discretionary basis
Share buy backs can be undertaken when funds are available. It does not create an expectation of a similar pay-out next year (unlike a high dividend would).

Can provide an exit route for dissatisfied investors
An off-market purchase can enable Runn to direct funds towards shareholders who are unhappy with company policy. This can release the investors and avoid disruption in the future

(*note: in many jurisdictions there is a principle that all shareholders should be treated equally and therefore all be offered the same buy back terms. This means Runn cannot target particular unhappy investors)*

Drawbacks of a share buy back
May create a tax liability on the recipient
The investors who buy the shares may need to pay tax on the capital gain they make on their share

Indirect effect on shareholder wealth
There is no guarantee of the extent to which the remaining shares will rise in price following the buy back. Some shareholders prefer dividend therefore because it is definite cash to them

Concentrates ownership of the business and changes control
If Runn conducts a share buyback it will increase the control of the remaining shareholders who do not sell. This may discourage other shareholders due to loss of independent control, and reduce the share price.

The share price for share buyback would need to include a price premium (so that buy back price is higher than current share price to entice shareholders to sell)and this may not be an efficient way to return surplus funds to shareholders as the shareholders who sell get a price premium and those that do not sell get nothing.

This is not as fair a way to return funds to shareholders as a one-off dividend.

May look like share price manipulation
Using company money to buy its shares can look like a share price support operation. This may lead the market to conclude that the Board is worried about market sentiment and lead

to a fall in the share price. It may also be seen as a way to improve the value of executive share holdings or options. This will be unpopular with other investors too

Dividend payments
Runn may pay these funds out as a one-off large dividend, or as an enhanced dividend payment over a number of years. These will be an income to the shareholder

Benefits of dividend payments
<u>Returns cash to the shareholder</u>
Avoids Runn having large unused bank balances which will gain low income and draw down group profitability (buy-backs have the same benefit here)

<u>More favourable tax treatment of dividend income</u>
This could be better than they would receive on a capital gain. Dividends may have a tax credit which a buy-back would not.

Drawbacks of dividend payments
<u>Forces shareholders to receive cash</u>
Shareholders may not want cash at the moment. This can affect their tax planning

<u>Obliges funds to use the cash to buy shares in other businesses</u>
Funds do not have a use for the unexpected extra cash and so they will need to reinvest it. This will lead them to incur transaction charges that they will resent

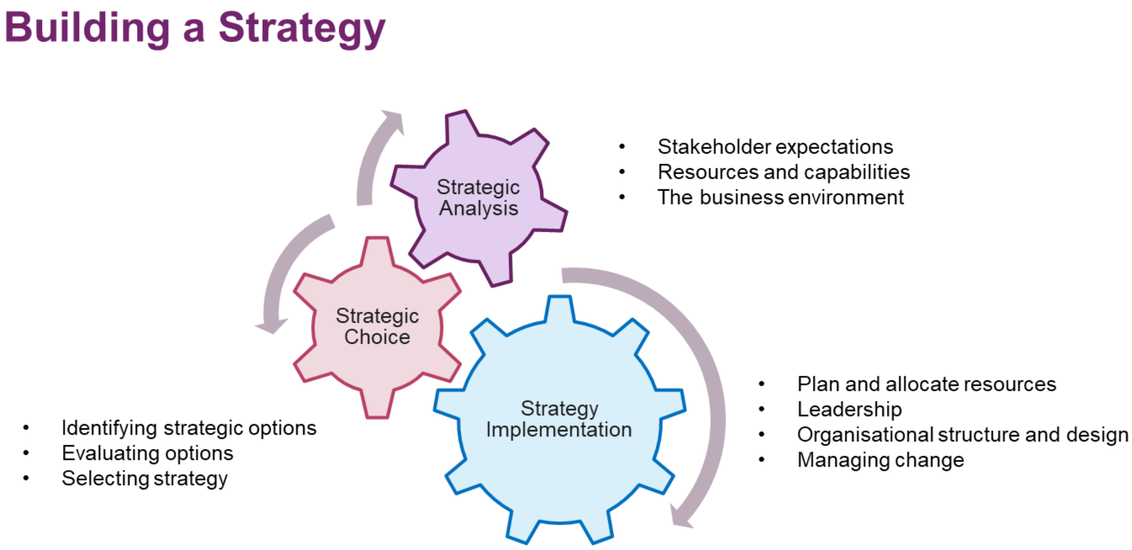
<u>May cause wrong sort of shareholders to buy Runn shares</u>
The high dividend may attract investors who want high cash flows rather than capital growth (the *clientele effect*).

Given that this is a one-off cash windfall Runn will not be able to maintain high pay outs in the future. This will lead to pressure on management to abandon growth and investment of retained earnings in favor of high dividends. This will jeopardize the strategy of Runn

Recommendation
A share buy-back may be better. The circumstances and reasons for the buy-back should be communicated to the market

Chapter 14
Strategic Control and Change Management

1. Overview
This chapter takes us into the syllabus content that deals with strategy implementation.



2. Performance Management
This is crucial part of implementation. It involves setting *metrics* (or *Key Performance Indicators*) to ensure the vision and strategy of senior management is carried out.

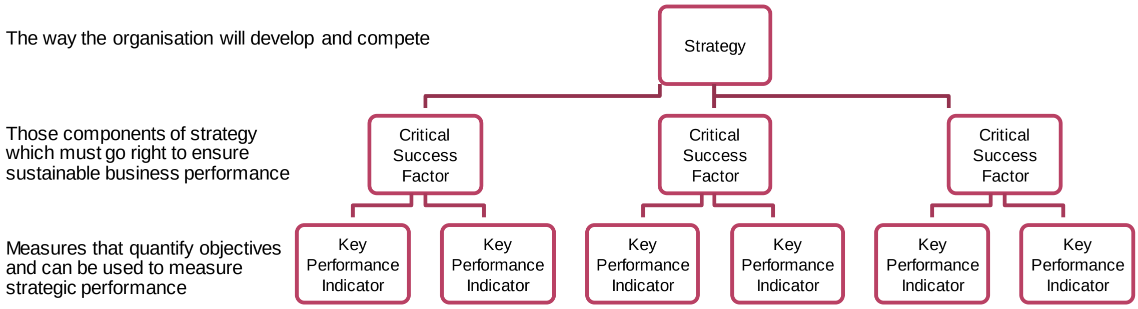
Key Performance Indicators (KPIs)
The purposes of KPIs are the four CPs:
1. *Check Position*: they enable control by allowing management to see where the organization is relative to its plans and to its rivals
2. *Communicate Position*: they are used to report performance internally and to stakeholders
3. *Confirm Priorities*: they are the essential things that managers must focus on
4. *Compel Progress*: because performance against them is used to evaluate managers and staff it ensures that staff will want to achieve them.

KPIs should be based on ensuring that management delivers the Critical Success Factors (CSFs) of the firm's strategy. The purposes of CSFs are to identify the things that will make the organization successful in the eyes of its key stakeholders. Usually these are customers and investors. But in public-service organizations the CSFs may relate to the requirements of regulators, governments or society at large.

In the SCS Assessment you should identify the CSFs from
- Your understanding of the business model of the firm and its present strategic position
- From its Vision, Mission and Values
- From information provided in the task

CSFs and KPIs

An approach to the formulation of performance management that bases performance measures, Key Performance Indicators, on the strategic aims of the business via identifying Critical Success Factors where performance must be adequate to deliver the strategy.



The SMART approach to KPIs
The examiner will expect you to ensure that any KPIs you suggest are SMART.

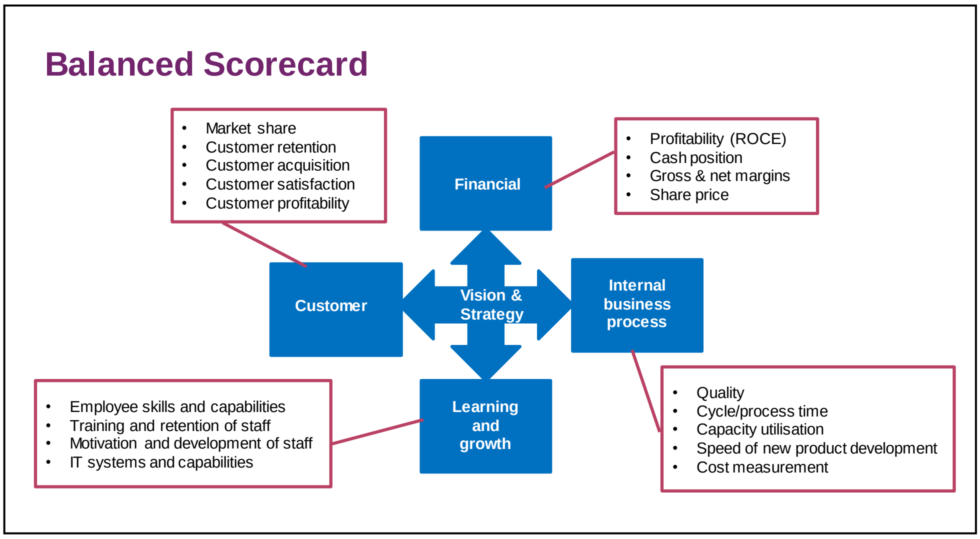
SMART KPIs

Goals:
A. Ensure customer satisfaction
B. Ensure high staff skills

Characteristic	Example
Specific	A Customer repeat purchase rate B Number of staff achieving training standards
Measurable	A % of customers returning to the store to buy again B Number of staff passing final assessment test
Attainable	A Cannot exceed 100% B Cannot exceed 100%
Relevant	A Higher repeat % indicates satisfaction and raises sales revenue B Higher % pass rate means more staff successfully trained
Time bound	A Measured monthly B Measured annually

Multidimensional performance measurement systems such as Kaplan and Norton's Balanced Scorecard can be seen as a development of KPIs.

Senior management can set KPIs across the four perspectives of this framework to monitor all aspects of the operations needed to deliver the business strategy.



Challenges in using KPIs

The SCS examination has involved some situations that are subject to the following challenges:

1. Lack of suitable comparisons: KPIs to monitor performance of a division or a business rely on having similar businesses to compare performance to
2. Distorted by stage of business life cycle: early stage processes or businesses will not return such good performance as mature ones
3. Wrong KPIs will drive managers to deliver wrong performance (e.g. KPIs on profit will cause shirt-termism in decisions of management)
4. Setting the right targets: if KPIs are made into targets it is essential to set them at a level that *stretches* performance but which is also *fair* and unlikely to *demotivate* managers
5. Too many KPIs can confuse and frustrate managers who will spend too long in ensuring they achieve their performance measures and not actually run their businesses

3. Measuring the performance on NPOs

A traditional summary measure of the success of a business is whether it made a profit or not. If a profit has been made it can be seen to have 'added value' because it was able to charge more for its outputs that it needed to pay for its inputs.

For an NPO (not for profit organization) this does not apply. In its place the UK National Audit Office applies three sets of measures to assess whether the public is getting value from the branches of public sector it audits, such as health services, public administration, the Police and armed forces, education services and so on. These are called the *Value for Money* principles (VFM)

1. *Effectiveness*: the extent to which the NPO is achieving its goals and objectives. This is commonly summarized as 'doing the right things'

2. *Efficiency*: how swiftly and reliably it conducts its activities and the amount of resources it uses to do so. Summarized as 'doing things in the right way'

3. *Economy:* the financial costs of the service it provides (or possibly the financial surplus it generates)

Performance Management in an NPO

Effectiveness
- Achieve the goals of the NPO
- Achieves its purpose

Efficiency
- Reliability of performance
- Speed of performance
- Resources used

Economy
- Costs of performance
- Earnings generated

Past exam question – the VFM principles
The 3 E's were the subject of a question in February 2018 in which a profit-seeking gym was entering a partnership with government to improve the fitness and health of people who have been diagnosed as overweight and at risk. The questions asked 'to what extent focusing on the VFM principles, as expected by the Ministry of Health, may cause [the private gym firm] problems when taking part in this program'

One way to approach this would be to make the point that profit seeking firms also seek to be effective in meeting market needs, and efficient in their use of resources and therefore they may be no problem. However, the level of economic return expected by the gym would mean that there may have to be some sacrifice of effectiveness (such as not providing personal coaches to the programs or allowing people unrestricted access to the gyms) to provide the profit. Another aspect would be that the effectiveness of the program is not entirely within the control of the gym because it depends on the effort and attendance of the participants.

4. Change Management
There have been several SCS Assessment tasks on this topic in the past.

Resistance to change
Resistance to change refers to the forces in the organization that operate to prevent or to distort the success of new business initiatives. This is due to the impacts thar change has on people in the organization that disturbs their status and ways of working and may be percieved by them and as a threat.

Impacts of Change

Changes to job roles
- Less jobs/more jobs
- New duties and skills
- Different hours of work
- Less/greater responsibility

Changes to teams
- Loss of workmates
- New leaders
- New team members

New skills
- Old skills may be unwanted
- New skills needed

Change to personal status
- Loss of previous authority
- Resentment at not being consulted

Disagreement with change
- Not necessary
- Wrong reasons
- Will not work

Conflict
- Fight for available jobs
- Fight for status and power
- Attempts to disrupt the change

Kotter and Schlesinger (1979) wrote a key article that summarised 4 sources of resistance to change and suggested 6 strategies to deal with resistance to change.

4 sources of resitance to change

- *Parochial self interest*: the change is resisted because managers and staff fear they will lose something of value, such as money, power, or status at work. They put themselves ahead of loyalty to the firm. This leads to scheming and political manouvering at work.
- *Misunderstanding and lack of trust*: due to misscommunication, rumours, or past experience managers and staff are suspicious of the nature and impact of the change and perceive it will cost them more than they will gain. They hold out on the change.
- *Low tolerance of change*: managers suffer inertia, fear they cannot learn the new skills needed, and a wish to keep things as they are. They may understand that a change is needed but they do not want the change to interrupt their customary work habits and threaten their routines and security. Some are trying to save face by avoiding admitting that things were done badly in the past.
- *Different assessments of the situation*: managers are loyal to the firm but believe that the change is unnecessary, harmful, or just the wrong thing to do.

Resistance to Change

- Parochial self interest
- Misunderstanding and lack of trust
- Low tolerance of change
- Different assessments of the situation

6 strategies to deal with resistance to change

Change Management Strategies

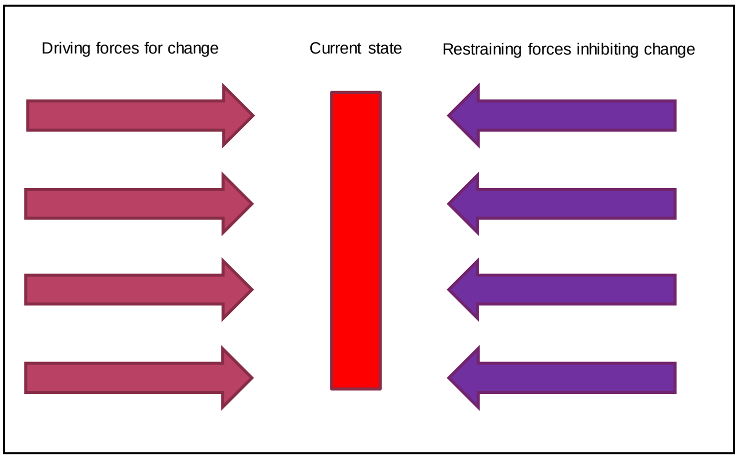
Ethical change strategies

- Education and communication
  - Management explains reasons for changes
  - Management explains impacts of change
  - Presentations, group discussions and one-to-one briefings
- Participation and involvement
  - Create teams and working parties amongst staff
  - Allow suggestions and involvement
  - Project teams, staff delegates join change team
- Facilitation and support
  - Provide training and re-skilling
  - Provide help and counselling to affected staff
  - Use professional change agents
- Negotiation and agreement
  - Provide incentives to individual resisters
  - Be willing to compromise with staff and unions

Unethical change strategies

- Manipulation and co-potation
  - Mislead individual staff into supporting the changes
  - Involve leaders of resistance in the change process but ignore their contribution
- Explicit and implicit coercion
  - Make threats to compel change
  - Aggressive and autocratic management

Force field Analysis – Lewin
Behavioral psychologist Kurt Lewin describes the impact of change on the individual as trapping them between two opposing sets of forces. His original research was into the problems faced by an individual adolescent in making the transition from child to adult. However, his work has been appropriated to describe the impacts of change on an organization



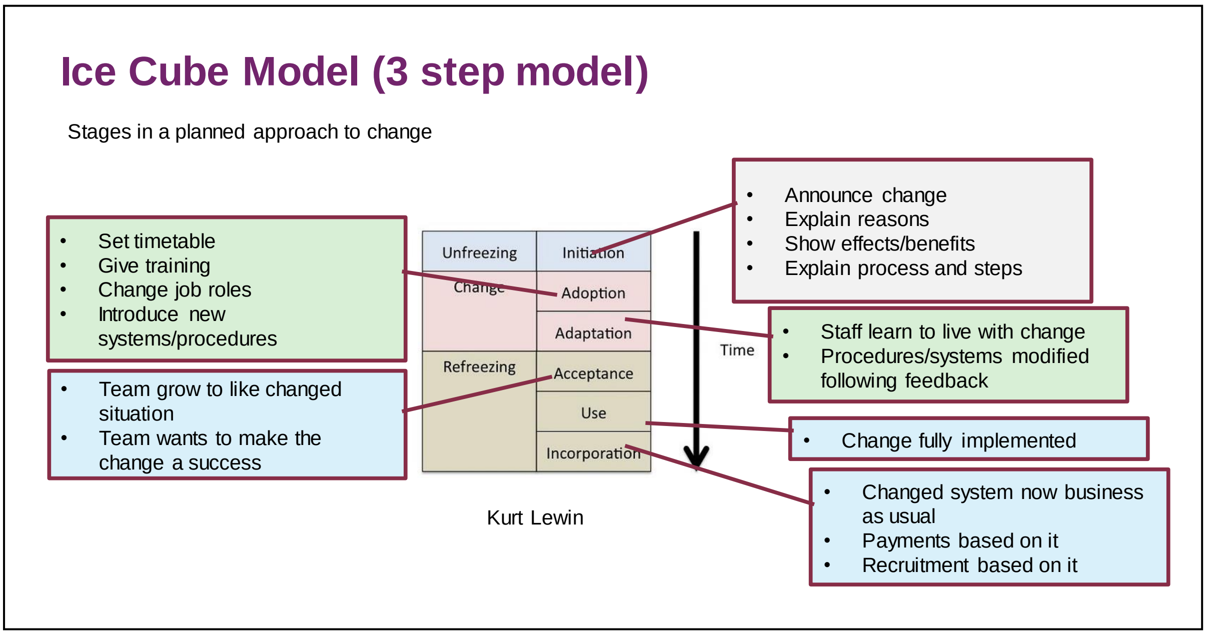
To make change effective management must

Weaken the restraining forces

Strengthen the driving forces

Kotter and Schlesinger – management strategies for change management

Three step Approach to Change (Ice Cube Model)
A successful model for conceptualizing and managing change is the Ice Cube (three step) model developed by Kurt Lewin.



ACTIVITY 14.1

Context
Couchweb is a global provider of Subscription Video on Demand (SVoD) like Netflix and Disney+

It has begun to produce its own programs and movies in order to reduce its reliance on movie studios and to gain some differentiation of its service.

Task

You have received an email from Martine Andels: Director of Finance of Couchweb

To: Senior Manager
From: Martine Andels: Director of Finance
Topic: Controlling extension to production

Confidential

The Board has authorized the scaling-up of our production. This dramatically changes the nature of our business and I need to ensure our internal controls are adequate.

One thing we need is better reporting of the progress of production and the success of each new series. Also, some of these new programs are being filmed in hired studios while others are being filmed in outside in locations all over the world. This seems to expose us to risks

Please suggest two Key Performance Indicators (KPIs) we could use to monitor the production phase of each program and two KPIs to monitor its long-term success once it is broadcast.

[***sub task = 55%]***

ACTIVITY 14.2

Context

Steelcast is a long-stablished maker of premium watches, like Rolex or Patek Philippe. The luxury watch market is subject to digital disruption from smart watches and from less expensive watches that use digital technology instead of traditional clockwork movements.

Steelcast is considering merging with Laine Watches to form a new company 'Premier Watches'

Task

The Finance Director has invited you into her office.

"Thanks for the work you have done on the Laine merger. Its Board is considering our counter proposal.

It seems that Laine is run as a family-owned business, with the Laine family holding about 30% of its shares. This has ensured that the CEO and the Business Development Director are both members of the Laine family.

The non-family investors in Laine have become frustrated at the lack of leadership in Laine and it is these non-family investors that have pushed Laine to request this merger.

We are considering creating one Board to run the merged firm, and to move to having production in one factory and one central marketing and sales team.

Our Board is concerned about two matters:
How can we manage the changes that will result from the merger? [sub section (a) 55%]

Secondly, it seems that Steelcast's Chairman, Viktor Kijec, is one of the shareholders in Laine. Are there any ethical issues raised by this?" [sub section (b) 45%]

Solutions to Activities

14.1 KPIs for Couchweb's new programs

Please suggest two Key Performance Indicators (KPIs) we could use to monitor the production phase of each program and two KPIs to monitor its long-term success once it is broadcast (55%)

Approach. Several past exams have included questions requiring candidates to propose KPIs. In approaching questions like this you should identify critical success factors (CSFs) for the business and then explain how the KPIs you suggest helps identify whether management is reaching this. Try to ensure that the KPIs are precise and that you indicate how it might be measured and what it indicates.

In this task you need to decide for yourself what these CSFs might me, and then develop KPIs for them. Notice that the task requires KPIs for the production phase and the broadcast phase.

KPIs for program production

KPIs serve several functions:

- Communicate priorities to management. If it is important enough to be a KPI it is clearly something that is important to the senior leadership team
- Compels progress because managers realize they will be personally evaluated against their ability to achieve the KPIs
- Check position: the KPIs provide information to management on whether the business is running correctly
- Communicate position: the KPIs can be discussed with staff and managers when improvements are needed. They can also be published for stakeholders such as members, regulators and investors

KPIs for production phase of programs

<u>Expenditure variance</u>

This is the variance between actual and planned expenditure per episode

This can be expressed as a percentage, to show extent of any loss of control over spending, or as the actual financial amount, to show financial impact of any loss of control. Programmes should individually have either a favourable or a zero variance

Programs are expensive to produce. This should be conducted on an episode by episode basis, rather than for the season as a whole, to enable recognition of adverse variances from earlier episodes to feed forward to creating favorable variances in later episodes

Alternatively, separate variances for particular costs could be extracted (e.g. actors pay, transport, hire of locations and so on)

<u>Viewer panel rating of pilot/sample</u>

This is the net approval rating from each pilot/sample (total enthusiasts fewer total detractors)

Couchweb should not commit to filming a complete series before it has assessed preliminary viewer response

Showing a panel of target viewers a pilot episode, or even showing story boards and preliminary characterizations and cast, help Couchweb to decide whether to continue filming the whole season or not. Using the net approval rating, rather than the total approval rating incorporates the loss of viewers that might happen due to the program (*note: net approval ratings convey information better. E.g. saying 70% like the program makes it look very popular,* *but expressed as a net figure it is only 40% approval because it has recognized that 30% did not like it)*

It should be a condition of continued investment that the program obtains a high net approval rating (*student note: this is an example of setting 'gateways' during the life of a project to provide a control on whether to stop the project or to continue it. Other gateways can include peer reviews or making a fresh business case to apply for the next stage of funding. Gateways are very common in research and development projects and was the subject of a past exam task)*

Other KPIs could include cost per broadcast hour for each series

KPIs for long term success of program
<u>Percentage of subscribers viewing program</u>
A high percentage of subscribers suggests that the program is popular enough to be a major contributor to attracting and retaining subscribers

This is a better measure than simply number of viewers because viewer numbers will increase as subscriptions increase. A rising percentage of subscribers shows that the show is actively attracting the new subscribers and/or is becoming generally more popular. This should be an average figure across the season rather than for a single episode to avoid distortion from a particular plot twist

<u>Net positive comments about program on social media</u>
The KPIs will be used to decide whether to invest in the next season or not. Therefore, they need to show the trend of popularity

Positive comments show continued viewer engagement with the program which supports the case for investing in another series. Negative comments suggest disengagement.

Intelligent agents can scan the social media posts, especially if the series has its own social media page, and classify the comments into positive or negative. Adverse KPIs (such as net negative comments) could trigger analysis of the actual comments to see why the program is losing support and allow adjustments to be made (remove unpopular characters, change locations, add plot devices and so on)

Other KPIs for long term success and decisions on continuing the series include 'series cost per subscriber'

14.2 Steelcast watches merging with Laine Watches to form 'Premier Watches'

Sub section (a)
How can we manage the changes that will result from the merger? [55%]
Approach: questions on change management appear occasionally in this exam. The best approach to questions is to set the context and explain why the changes may cause resistance. Theories to support your answer include Lewin Ice Cube (three stage) approach, Lewin's Force Field Analysis, or Kotter's Change Leadership and Kotter's Six Strategies models. But generally, it is easier to state five good ideas that may help manage the change process.

Managing changes resulting from the merger
There are a number of significant changes likely to occur
- Reduction in the power of the Laine family and its two Directors
- Creation of a single board for Premier Watches
- Moving all production to a single factory

- Creating a single team for marketing and sales

These changes will unsettle staff from the Board level to the factory level. This will lead to resistance to change which may lead the merger to fail to achieve its potential. It can also lead to conflict and uncontrolled loss of staff.

Methods of managing the change include the following:

<u>Create a special project group to oversee the change</u>
These changes should be managed as a single project over several years rather than allowed to occur independently and in an uncoordinated way. A project team should be established reporting to the CEO of Premier Watches.

The project team should contain managers from both Steelcast and Laine and also some consultants with expertise in change management on this scale. This project team should oversee all the individual change initiatives.

<u>Set a timetable and project plan for the change</u>
The Board of Premier Watches should set a time frame for each of the changes in consultation with the project team. Change may need to be slower than the Board might wish.

The project team should create a project plan for change with each element having a set of deliverables and a clear set of deadlines for completion. A named individual should be made responsible for each change as project manager.

<u>Communicate the changes to management and staff</u>
Following the merger there will be rumors and uncertainty about its implications for jobs and some concern about the ability of management to make the merger successful.

Initially communication should be reassuring to avoid loss of motivation and staff turnover. However, management should avoid making false promises that they may break in the future because this will erode trust in the long term. Where people are due to be affected in the coming months, they should be told what it happening and what its likely implications are for them

<u>Negotiate with affected individuals to get agreement</u>
Relocating production and changing the Board and marketing teams will disrupt staff There should be negotiation with affected persons to obtain their agreement to new roles and locations, or to agree the terms of their exit from Premier Watches.

These negotiations should involve management at the appropriate level, for example the Chairman would negotiate with Directors whilst the CEO would negotiate with the marketing and sales teams

It is better to obtain a fair agreement than to impose change because this will raise the morale and motivation of the remaining staff.

<u>Employ change agents to facilitate the change process</u>
Change agents are external consultancies with experience in managing change. Change agents will facilitate things like IT systems changeovers, factory relocations and so on.

They are also able to assist in re-aligning organizational culture by involving staff in rethinking the purposes of the firm and how they may adapt their behavior to fit with it

Sub section (b)
Secondly, it seems that Steelcast's Chairman, Viktor Kijec, is one of the shareholders in Laine. Are there any ethical issues raised by this?" [45%]
Approach: Ethics is about the moral rightness or wrongness of human actions. The best approach to ethics questions is: state the issue, state the ethical arguments against it, state the practical consequences of behaving unethically, state any counter-arguments, and then conclude. Try to use terminology from the CIMA/CGMA Ethical Code if possible and <u>underline</u> them. Other ethical terminology can be added, but do try to use the Code where possible.

Ethical issues raised by Steelcast Chairman owing shares in Laine Watches

Ethics concerns the moral rightness or wrongness of decisions and actions. 'Good' behavior involves doing things that are ethical and avoiding actions that are not. 'Bad behavior is doing unethical things, or not stopping others from doing them.

The Issues
The Chairman of Steelcast is involved in negotiating merging with a firm in which he holds shares

This means he will benefit financially if the terms of the merger favor the other firm, Laine.

As Chairman he is a key player (Mendelow's stakeholder mapping matrix: high power and high interest) in deciding the terms of the merger and whether it goes ahead

The ethical aspects
<u>Potential breach of objectivity</u>
This refers to the *conflict of interest* that the Chairman faces between getting a good deal for the shareholders of Steelcast and getting a good deal for himself from assuring he exchanges his Laine shares at a good rate.

<u>Potential breach of integrity</u>
It is not clear whether the Chairman disclosed his ownership of shares in Laine. If he did not then others would be right to say that it should have occurred to him to mention it at that point and that failing to do so displays concealment and a lack of integrity.

<u>Breach of professional competence and due care to shareholders of Steelcast</u>
As Chairman of Steelcast Viktor Kijec is an agent of the owners and responsible to them. He has a fiduciary duty to act in their interests. This is the origins of the conflict of interest described above.

<u>Lack of professional behavior.</u>
It seems unacceptable that a Chairman, or any other Director, should own shares in a rival. This creates a disincentive to work to maximize the success of Steelcast because in doing so they might reduce the value of their shareholdings in competitors. It also does not reflect well in their assessment of the prospects of Steelcast

Consequences
<u>Steelcast investors may get too little for their shares</u>
The Chairman may use his position to promote the merger, and gaining shareholder support for it. This could lead to Steelcast merging with a firm with a poor strategic fit and also getting a bad deal

<u>Anger from investors if it discovered</u>
Discovery of his shareholding and benefitting from the merger will break shareholder trust for the Chairman. This will lead to instability and potentially him facing replacement

Counter arguments

Chairman may have declared this interest

If the Chairman disclosed his shareholdings voluntarily and without prompting or risk of discovery he has acted properly and does not lack integrity. This puts the rest of the Board on notice that his interest and advice might be biased.

Ideally the Chairman should recuse himself from the discussion on the merger and ask another non-executive director to chair the Board during these discussions.

Chairman may believe that merger settlement is fair and in the interests of Steelcast shareholders

This means that he has not deliberately promoted this merger to gain a private interest.

This is essentially a defense and does not change the potential conflict of interest in the situation, nor lack of integrity if he did not disclose at the start.

Chairman is not able to take this decision on his own

There are divisions of responsibility on the Board and other directors had full opportunity to scrutinize this decision.

If they failed to do this, but rather merely agreed with the Chairman, they have failed to act independently and have not shown professional competence and due diligence

Chapter 15
Corporate Governance

1. Sources of Corporate Governance
Governance means 'control'. Corporate Governance means 'control of the corporation as a whole'

There are two main sources for the controls that try to 'make the management of the firm do what it is supposed to do':
1. External Controls
   - Statute and other laws governing companies and their directors
   - Financial reporting regulations, other reporting regulations (e.g. <IR>), codes of corporate governance
   - The ecosystem of the business (e.g. customers, suppliers, competitors and business partners)
   - The investment markets that its shares are traded on (e.g. rules of the exchange, market pressure on price of shares, investment media, and credit scores)
   - The investors (e.g. votes of shareholders, decisions to advance finance, questions at shareholder meetings, threats to sell shares)

2. Internal Controls
   - The quality of the Board and its processes
   - The budgeting and financial control systems of the firm
   - The methods used to manage people in the firm
   - Codes of conduct and policies (e.g. behavior in the workplace, health and safety, anti-corruption, business ethics)
   - The strategy formulation process of the business

In principle the internal controls should communicate and mirror the external controls to ensure the firm continues to fit with its environment.

Failure to have adequate internal controls for corporate governance exposes the firm to risk.

2. Codes of Corporate Governance
These are laid down by the investment markets and/or the finance community to encourage investors to trust the shares on its market and to invest in them.

One motive of the Code is to overcome the *principal-agent problem* in which the management of the firm (agents) can follow their own goals and not the goals of shareholders/owners (principals) due to a lack of effective control by the principal and a lack of transparency and accountability by the agents.

A motive of recent Codes has also been to try to assure the *sustainability* of the business. This has led to an increase in the requirements for stakeholder engagement beyond engagement solely with the demands of shareholders.

Student note: the firms in this assessment are usually fictitious firms based in fictitious countries (Northland, Geeland, Kayland and so on) and listed on the investment exchanges of those fictitious countries. The assessment convention is to assume that the Code of Corporate Governance that applies to the firm is identical to the UK Corporate Governance Code 2018 and supplemented by the UK's Guidance on Board Effectiveness 2018 and its

Guidance on Risk Management (2014). These are part of the requirements of a listing on the London Stock Exchange (LSE).

Corporate Governance Code

Responsibility of Board of Directors
- appointed by shareholders to run the business
- responsible to shareholders for how they run the business (transparency and accountability)

1. Must ensure effective systems of control
    - financial controls (transactions systems and budgetary control)
    - adequate management in each position
    - controls to reduce risks
2. Provide direction and strategy for the business

Issues
- Remuneration of senior management (pay should be related to performance and set independently by a Remuneration Committee)
- Ensuring Board selection (should be independent process such as Nomination Committee)
- Effective controls are in place (Audit Committees should oversee this and Board should report on its management of risk)
- Stakeholder representation: encouragement of participation of smaller investors and methods of responding to staff concerns

Effective corporate board
- Balance of Executive Directors (do) and Non-Executive Directors (advise and monitor)
- NEDs should be independent: i.e. no commercial or financial connection with the firm other then their fee for attendance
- Division of responsibility between Chairman (manages board) and CEO (manages the firm)
- Open participation and discussion and 'strong independent element'

Some details about the Code
- Adoption of the Code is a condition of having access to the full investment market
- For the London Stock Exchange this is called a 'premium listing'. This is the highest of the four segments of the London Stock Exchange and denotes the highest listing standard and requires compliance with the most stringent listing rules. This, LSE claims, will build investor confidence and improves transparency which can reduce its cost of capital by giving it a higher share price
- The Code is not a set of laws. It is a set of principles with firms should adapt to suit their own circumstances
- The Code is based on a 'comply or explain' principle. Boards are expected to comply with the Code and where they do not comply, they must explain the reasons in the Annual Report to shareholders
- The Code depends for its effectiveness on *stewardship* by the shareholders. If shareholders don't check the arrangements, ask questions, cast votes, and expect the firm to apply the Code it will not be effective. Two factors weaken this:
    - The lack of power of small shareholders makes most of them give up on exercising stewardship
    - The dominance of investment funds who are mainly motivated by short-term gains in the price of the stocks they hold rather than exercising stewardship over the long-term behavior of the business and its Board

- The Board is supposed to set the ethics of the business and to build an appropriate culture through:
    - It statements of Vision, Mission and Values
    - The behavior of Board members

- Directors should be appointed for 1 year and subjected to re-election each year by the shareholders
- NEDs should be 50% of the Board and one of these, other than the Chairman, should be designated the 'Senior Independent Director' and responsible for challenging excessive

'closeness' between Chair and CEO or raising concerns of Board members or other stakeholders that are not being addressed by the Chairman or CEO
- Directors should be nominated to shareholders by the *Nomination Committee* comprised of mainly NEDs but with input from the Human Resources function and charged with establishing 'diversity' on the Board
- *Audit Committees* are required and should comprise 100% NEDs and be convened/chaired by a NED with recent senior financial experience. The Chair of the Board should not a member of the Audit Committee
- *Risk Committees* are not required by the Code but the value of them as a means to scrutinize non-accounting risks is recognized. They oversee the risk management systems and are comprised of NEDs
- *Remuneration Committee* sets remuneration for the Chair, Executive Directors and the next level of management and proposing this to the Annual Meeting for approval by shareholders. Other NEDs have their pay set by the Board. The Chair of the Board should not chair the Remuneration Committee and should not be a member of the committee if they were not independent on appointment as Chair of the company

Role of Non-Executive Directors (NEDs)
NEDs are called 'independent directors' in some countries (e.g. USA). However the UK Code classifies NEDs into two types:
1 Independent Non-Executive Directors: these are preferred due to their independence and should be reported as such in the annual report
2 Non-Executive Directors: the Board takes the view that they are not independent due to things such as
– Having served for more than 9 years
– Having been an employee of the firm in the past 5 years
– Having had a business relationship in the past 3 years (shareholder, supplier, customer, advisor and so on) or has cross directorships or significant friendships with other relevant firms
– Receives additional income from the firm (say for consultancy contracts)
– Represents a significant shareholder or is themselves a shareholder in the firm
– Has close family ties with advisors, directors or other senior management

Based on the UK Code of Corporate Governance the examiner views the roles of the NED as follows:
- Primary duty is to the stock exchange on which the firm is listed (not to the shareholders of the firm).
- NEDs are independent scrutineers who try to ensure that firms listed on the exchange are well-managed. This gives investors more confidence in all firms listed on the exchange
- NEDs therefore should have broad business experience in running listed firms and of helping them to develop
- NEDs should not have investment, or options, in the shares of the firm. Otherwise they may be tempted to support irresponsible management actions to artificially boost the share price and damage the reputation of the stock exchange.

3. Executive remuneration
The purposes of executive remuneration are:
- To attract and retain good managerial talent
- To motivate management to take decisions and work in the interests of the investors
- To avoid excessive costs of executive remuneration that is not compensated for by improved business value

Remuneration may include:
- Basic salary

- Benefits package such as pension, personal insurances, accommodation, and so on
- Bonuses based on divisional or group profits
- Executive share options or free shares

Poorly designed remuneration packages lead to risks:
- Profit-based remuneration can lead to short-term cuts in expenditure or tricks to raise revenues that harm the long-term performance of the business
- Divisional profit-based systems may encourage managers to focus on the needs of one division to the detriment of the group
- Profit and equity-based remuneration may encourage aggressive accounting approaches to boost profits artificially
- Equity-based remuneration may encourage executives to mislead the stock exchange (e.g. bad news to depress share price before options granting, then good news to boost the share price at options exercise time
- Equity-based systems can become valueless if market sentiment is poor and share prices as a whole fall. This can mean that executives leave or demand to have share options re-based.
- Excessively high severance payment terms can prolong the retention of poor managers because the firm doesn't want the cost of paying them off

Exam guidance

The examiner sets many questions to test understanding of the principles of corporate governance. These fall into a number two types:

- *Recognising the respective roles of Executive Directors and Non-Executive Directors*: remember that the EDs are the ones principally responsible for the control of the firm and so are mainly responsible for failures of strategy or risk control. The NEDs are also technically responsible because they oversee the EDs' work but they are not ever *mainly* to blame
- *Adding new Directors to the Board:* the only reason to add a new ED is because there is an area of responsibility, or a team, that the present EDs do not have the expertise or capacity to manage, and that a senior level appointment is essential to attracting the right candidate and giving the role the necessary authority. Otherwise it just makes the Board slower and more expensive to operate. Remember that at least 50% of the Board should be NEDs and so an extra ED may lead to the need for an extra NED

ACTIVITY 15.1

Context
Couchweb is a global provider of subscription video on demand (SVoD) like Netflix and Disney+. It streams movies, series and other television programs. It is dependent on program and movie production firms for its content and has taken the strategic decision to invest in producing its own content. It has to date created two successful series: The Politics Game and Tumbling Down.

One of its key actors in Tumbling Down, Kyle Johnson, has been suspended following new article allegations of bullying and other inappropriate behavior towards actors and other staff.

Task
The CFO has invited you into her office.
"I need you to keep these comments to yourself. It's about the Kyle Johnson situation.

It seems that Mr Johnson's alleged behavior is just the tip of the iceberg. Since the news article came out the Board has been deluged with emails and letters from staff and contractors alleging bullying conduct elsewhere by Couchweb staff.

The alleged conduct includes verbal abuse and intimidation by managers, generally demeaning conduct towards woman by male staff, and several accusations of inappropriate sexual conduct by program directors and producers.

None of these complaints involves members of the Board, but we are all horrified.

I need your views on two things please:
What are the ethical aspects of this situation? **[sub section (a) = 45%]**
What roles did the Board fail to perform in allowing this to develop and what should the Board do to prevent it happening in future? **[sub section (b) = 55%]**

ACTIVITY 15.2

Context
Knowledge City Technology Park is listed on the Advland Stock Exchange. It operates a science park of 20+ units that it leases to firms engaged in technological innovation. It was established by Capital City University (CCU) over 30 years ago and floated 20 years ago. CCU still holds 20% of the share capital in KCTP and also works with KCTP to exchange staff and promote innovation. Professor Kress is an NED (but not classed as independent) who is on the Board to act as liaison with CCU and to 'safeguard CCU's interests'. KCTP's mission is to provide a 'platform for innovation' by leasing space and, to date, has not invested in any business start-ups.

The Board has been discussing buying a start-up business, NuFoods,that is researching plant-based alternatives to meat.

Task

You have been called into the office of Shereen Peros: FD of KCTP

'Discussions about acquiring NuFoods are continuing. But we had some upsets during the recent Board teleconference (she hands you some notes).

These raise two concerns in my mind

- Is it appropriate for Professor Kress to emphasise the interests of her university like this?

[sub task (a) = 40%]

- What would be the risks to the Board and to KCTP from a lower share price if AgroBiz carries out its threat?'

[sub task (b) = 60%]

Jules Malano Chair	Asked for feedback on the proposal to buy NuFoods

Sanjay Gupta CEO	Reported adverse response by AgroBiz, a large animal feed firm with a subsidiary on KCTP's campus and which owns 10% of KCTP's shares.
Shereen Peros, FD	Stated that AgroBiz is able to reduce share price of KCTP by threatening to sell its shares and end its tenancy at the park
Helga Kress NED	Objected to intervention by AgroBiz and emphasised that KCTP as set up by Capital City University to encourage research and that supporting NuFoods is consistent with this aim. It also helps benefit the university if its faculty develop commercially successful products.
Rick Adamson NED	Reminded the Board of its ethical duties to investors and expressed dislike of allowing AgroBiz to influence decisions.

Solutions to Activities

15.1 Couchweb dealing with inappropriate conduct by a star actor

Sub section (a)
What are the ethical aspects of this situation? (45%)

Ethical issues in the mistreatment of staff within Couchweb
Ethics concerns the moral rightness or wrongness of decisions and actions. 'Good' behavior involves doing things that are ethical and avoiding actions that are not. 'Bad behavior is doing unethical things, or not stopping others from doing them.

The Issues
Staff have complained of bullying behavior by managers and colleagues employed by Couchweb. There are also allegations of discrimination against women and inappropriate sexual conduct

Ethical aspects
Bullying staff is a breach of the principle of <u>professional behavior</u> by the managers and persons responsible. It is a breach of standards and brings Couchweb into disrepute. As managers they should know better than to do this.
Sexual discrimination is also a breach of <u>professional behavior</u> but also, it's a breach of the principle of <u>objectivity.</u> Treating people differently on the basis of their gender is biased and prejudiced behavior because in the work situation there is no reason to assume one gender is less able or especially qualified compared to another

The senior management have a duty of <u>integrity</u> and of <u>professional competence and due diligence</u> to get to the bottom of these allegations and to deal with them to stop such behavior happening

Consequences
This sort of behavior will lead to loss of good staff and the underperformance of the ones that remain, especially if women are precluded from roles that they could do well

It may also be hard to attract good staff to work for Couchweb if this behavior becomes known and associated with working for us. It may also render Couchweb liable for legal action from victims of this behavior.

Getting a poor reputation for staff management and discrimination will also deter some subscribers and broadcasters from taking Couchweb programming. It may also deter some investors, particularly those funds with ethical or Corporate Social Responsibility (CSR) objectives.

By stopping this behavior, Couchweb would set a good example in the industry and gain an improved reputation. It could also have a positive impact on its share price if investors had confidence in Couchweb's management's action.

Counterarguments
There are no effective counterarguments against bullying and sexual discrimination being ethically bad.

There may be an argument that this behavior is not confined to Couchweb but rather is prevalent in the media industry generally, and also in other industries. Therefore, it is wrong to highlight Couchweb in particular. This means that a wider solution is needed to stop this behavior which Couchweb will support by its own actions.

Sub section (b)
What roles did the Board fail to perform in allowing this to develop and what should the Board do to prevent it happening in future? (55%)
Approach: note that this sub section has two elements: roles that the Boards failed to perform AND what it should do to prevent this occurring in the future. Questions on the role of the Board are frequent in this examination. Usually they avoid giving opportunities to recite learning about Board structures and instead assess understanding of what Boards really do and the sorts of issues they face. In answering this task there is no need to discuss the make-up of the board or the principles of good corporate governance. It is asking about what the Board of Couchweb failed to do in the past and what it needs to do better in the future.

Roles that the Board failed to perform
The Board of Couchweb is appointed by shareholders to act as their agents to exercise stewardship over their investment in Couchweb. The allegations and bad publicity that has enveloped Couchweb suggests that the Board has neglected to safeguard the interests of investors as much as they should

In some countries the codes of Corporate Governance also impose a duty on the Board to engage with the concerns of stakeholders beyond shareholders. This means considering and preserving the interests and welfare of staff too. The
<u>Board has responsibility for corporate governance of Couchweb</u>
The Board is the 'controlling mind' of Couchweb and therefore responsible for what happens in the business. It is responsible and accountable to investors for this.

The Board should set the strategy for Couchweb, but also a system of internal controls to manage the financial performance of Couchweb, its operations and its risks. The Board should also ensure proper control of the management team of Couchweb which has authority delegated to it from the Board

There seems to have been a breakdown in this control over management behavior and action.

<u>Alleged behaviour shows a lack of internal control</u>
Internal controls are systems and processes within Couchweb that are intended to ensure that that right things happen. These internal controls include supervision, management, and personnel

Supervision seems to have broken down because the Board was unaware that these actions were taking place and managerial behavior was not monitored or reported up to Board members. However, it is the responsibility of Board members to ensure that that they do know what is happening in the company

There has also been a weakness in management because the wrong people were appointed to managerial roles and/or they were not properly inducted and trained. Personnel controls are also lacking because there seem to be no regulations in place to stop this conduct, and no systems by which staff can raise these issues without resorting to protest and the media

<u>Alleged behavior presents a risk to Couchweb</u>
These allegations will have harmed the share price. They may also lead to litigation against Couchweb. There is also a risk that staff will leave rather than work under these circumstances. This will endanger the production of new programs and the smooth conduct of operations.

This shows that the Board failed to carry out risk assessment over staffing and has therefore subjected investors to unnecessary risk.

<u>Board could be criticized for allowing this to develop</u>
The above suggests that the Board could be criticized for its failure to exercise proper corporate governance. Codes of Corporate Governance state that the Board should encourage an appropriate corporate culture to develop though publishing Values, and enforcing codes of conduct. It should also lead by example.

This criticism will come from shareholders but it could also come from the Mayland Stock Exchange and other regulators. These regulators have a duty to ensure that the firms traded on the Mayland Stock Exchange are properly and responsibly governed.

Actions to prevent it happening in future
Given the above comments, and the seriousness of the claims from staff and risk to Couchweb it is important that the Board improves the internal controls in Couchweb.

<u>Board members should provide ethical leadership to Couchweb</u>
Ethical leadership means that the Board should 'walk the talk' by exhibiting morale behavior itself

Many junior managers will assume that the behavior of the Board members is acceptable and condoned and therefore adopt it themselves. The way that the Board members deal with the allegations against Kyle Johnson and others will help set the expectations on acceptable behavior

Also, the way the Board members manage, and the gender composition of the Board will also send a message. One first step would be for the Board to undertake some awareness training itself to help ensure its members are not discriminatory or bullying in their behavior. This would make it easier to insist that junior managers undertake the same training.

<u>Board needs to investigate this fully and discipline offenders</u>
It needs to be shown that inappropriate conduct will carry penalties. A full investigation will help convince the victims of bad behavior that Couchweb's Board will stamp it out.

It may also convince the perpetrators of such behavior to question and amend their ways. This investigation might be better conducted by an independent HR advisor to ensure that the prevailing culture in Couchweb does not bias the results.

Discipline should be firm but fair. Except in extreme cases it would be wrong to dismiss staff without first giving them the opportunity to mend their ways and to make amends to their victims.

<u>Need for clear polices to outlaw bad conduct</u>
Personnel policies should be introduced to stop bad behavior. These could include codes of conduct on equal respect, but also guidelines on appropriate forms of address, personal touching and so on

As well as making staff aware of these policies there should be talks and groups to help staff explore and understand what the policies mean

<u>Need for channels for staff to raise concerns about behavior</u>
A feature of the present situation is that staff had not been able to raise these issues with management before. Couchweb needs to have a formal grievance procedure which allows staff to have their complaints heard by a senior manager.

There could also be systems of managerial appraisal that invites anonymous feedback from the subordinates and co-workers.

Finally, a 'whistle-blower' channel should be set up, perhaps to an externa agency, where staff can express their concerns confidentially and expect them to be acted on.

<u>Protection for people that raise complaints</u>
Complaints about the conduct of one's manager can often lead to penalties against the person complaining. This could be avoided by ensuring that any staff who complain have their roles protected against victimization by managers. Alternatively, all complaints should be dealt with anonymously to avoid the complainer being identified

15.2 KCTP buying NuFoods

Is it appropriate for Professor Kress to emphasise the interests of her university like this?(40%)

Professor Kress
Professor Kress is a non-executive director and is known to represent the interests of Capital City University (CCU) and is the link between KCTP and CCU.

Reasons not appropriate
<u>May be causing harm to shareholder wealth</u>
Professor Kress is endangering shareholder wealth by ignoring the risk of AgroBiz causing a sharp fall in the share price of KCTP. This includes the wealth of CCU. That would to some extent be inconsistent with the duties of any Director who should, as part of their work, be reducing risk.

Slows down decision making
The intervention of Professor Kress means that no decision has been taken on how to proceed with buying NuFoods or backing off to placate AgroBiz. Constant obstruction by Professor Kress to all decisions that may not totally suit CCU would be tiresome and could make the leadership of KCTP less effective.

Representing a minority
CCU only a 20% owner in KCTP. To a large extent the Board must act in the interests of the majority. Therefore Professor Kress would be wrong to demand that her views took priority.

Reasons why appropriate

Represents major investor
Professor Kress has a personal duty to represent those who appointed her, in this case that includes the CCU. CCU owns 20% and is a major investor, and larger than AgroBiz. It is part of the duty of all Directors to engage with all shareholders and their perspectives. Professor Kress is merely bringing CCU's perspective into the discussion.

Transparent position
Professor Kress is an NED but not classed as an Independent NED. The decision to not classify her as independent is because the Board is aware that she has a material link to an outside body and may seek to represent its interests. It retains the support of CCU and there is not secrecy around her position.

Maintains consistency with mission
Professor Kress is also reminding the Board of the mission of KCTP which other shareholders and stakeholders bought into. Buying NuFoods is consistent with the mission whereas yielding to the special interests of AgroBiz is not. She is doing her job by reminding them of this and, in this, has the support of another NED.

Board is able to filter arguments
Strong independent views and discussion are part of a healthy board. The FD is right to explain the risks of disappointing AgroBiz but this must be balanced by consideration of the impacts of any decision on other stakeholders. Professor Kress is adding to the discussion and has raised valid considerations. Ultimately the Board as a whole will take a decision and members will assess for themselves the strength in Professor Kress's argument. If Professor Kress had not spoken out the decision would not have considered the views of CCU and could have had worse consequences such as a sale of CCU's 20% of shares and withdrawal of its scientific collaboration with KCTP.

Conclusions/advice
Professor Kress has not behaved inappropriately. She has merely expressed a view.

The Chair should permit this sort of discussion but avoid it becoming heated. The Board needs to find a solution that it can agree on.

What would be the risks to the Board and to KCTP from a lower share price if AgroBiz carries out its threat?(60%)

Note the pivotal use of 'and' to distinguish the two groups of effects.

Risks to Board from lower share price

The Board should not take this decision based on the effects on its members. This would breach its duty of Objectivity, which requires it to ignore the self-interest of Board members.

Acquisition of KCTP if undervalued
A lower share price would present a raider with the opportunity of acquiring assets worth over A$7.26bn for a lower price. This could lead to acquisition for break up or perhaps to further a strategy of investing in science and technology parks. For the Board this would mean the end of some of their jobs. Once acquired there would be no regulatory requirement for NEDs, and it is unlikely that the buyer would wish to retain the CEO or FD, preferring to replace them with its own management team.

Criticism of the Board
Board members will be blamed for the share price falling. This is because one the main duties of a Board is to control risk and this includes risk to the value of the shares. Ultimately this criticism could lead to demands for changes on the Board and/or refusal to reappoint some of the directors next time (student note: the UK Corporate Governance Code 2018 specifies that directors should subject to re-election every year, and not every 3 years as in previous editions of the Code)

Harms their careers
The personal careers of Board members will be adversely affected if it is known that the share price of KCTP fell sharply during their time on the Board. This may make it hard for them to be appointed to new Board positions in the future, particularly if shareholders or the nomination committee do some research into the history of firms they have managed.

Risks to KCTP from lower share price

Higher cost of raising finance
KCTP needs A$800m to buy NuFoods and further funds to invest in developing the product. If this capital is to come from the issue of shares (or from allotting shares to the founders of NuFoods) it will require a greater number of the reduced-price shares to raise the funds. This makes the acquisition less attractive.

Also, if the share price does fall, finance for future investments will also be more expensive.

Diversion of profit to dividend to stabilise price
If the share price does fall the Board may try to stabilise it by offering a greater dividend. Given that dividend in 2019 was over 50% of profits, and that as a result KCTP borrowed additional loans that year to meet its need for funds, any increase in dividend would severely limit the cash available to maintain KCTP and to grow NuFoods.

Less appetite for innovation
If this attempt to innovate is seen to lead to a fall in the share price it will make the Board more cautious of suggesting innovation in the future. This will harm the long term of KCTP because it is faced with challenges that will require innovation and change in is business, ranging from new opportunities in technology (quantum computing, 3D printing and simulation), through ensuring the security of its IT systems against new threats, to upgrading its own assets to meet the regulations associated with reducing the impacts of climate change (such as thermal insulation, switching to sustainable energy sources). KCTP may fall obsolete by falling behind the curve.

Chapter 16
Risk Identification, Scaling and Management

1. Risk: definition and classification

Definition
Risk can be defined as 'the potential that future outcomes will be different from what was anticipated or planned'

Terminology of Risk

'the potential that future outcomes will be different from what was anticipated or planned'
- Risk = Impact x Probability
- *Upside risk*: things turn out more favourably than was anticipated or planned
- *Downside risk*: things turn our worse than was anticipated or planned

- *Risk*: something that can be anticipated and quantified and which can be pre-empted by management action
- *Uncertainty*: a situation where lack of data or understanding means that it is not possible to make any reliable assessment of its impact or probability

Classifications of risk
The SCS assessment will focus on the risks of the industry and business in the pre-seen material. And this may already include a risk register for the firm.

The *Business Encyclopedia* gives the following definitions and examples of risk
1. A probability or threat of damage, injury, liability, loss, or any other negative occurrence that is caused by external or internal vulnerabilities, and that may be avoided through preemptive action.

2. Finance: The probability that an actual return on an investment will be lower than the expected return. Financial risk is divided into the following categories: Basic risk, Capital risk, Country risk, Default risk, Delivery risk, Economic risk, Exchange rate risk, Interest rate risk, Liquidity risk, Operations risk, Payment system risk, Political risk, Refinancing risk, Reinvestment risk, Settlement risk, Sovereign risk, and Underwriting risk.

3. Food industry: The possibility that due to a certain hazard in food there will be a negative effect to a certain magnitude.

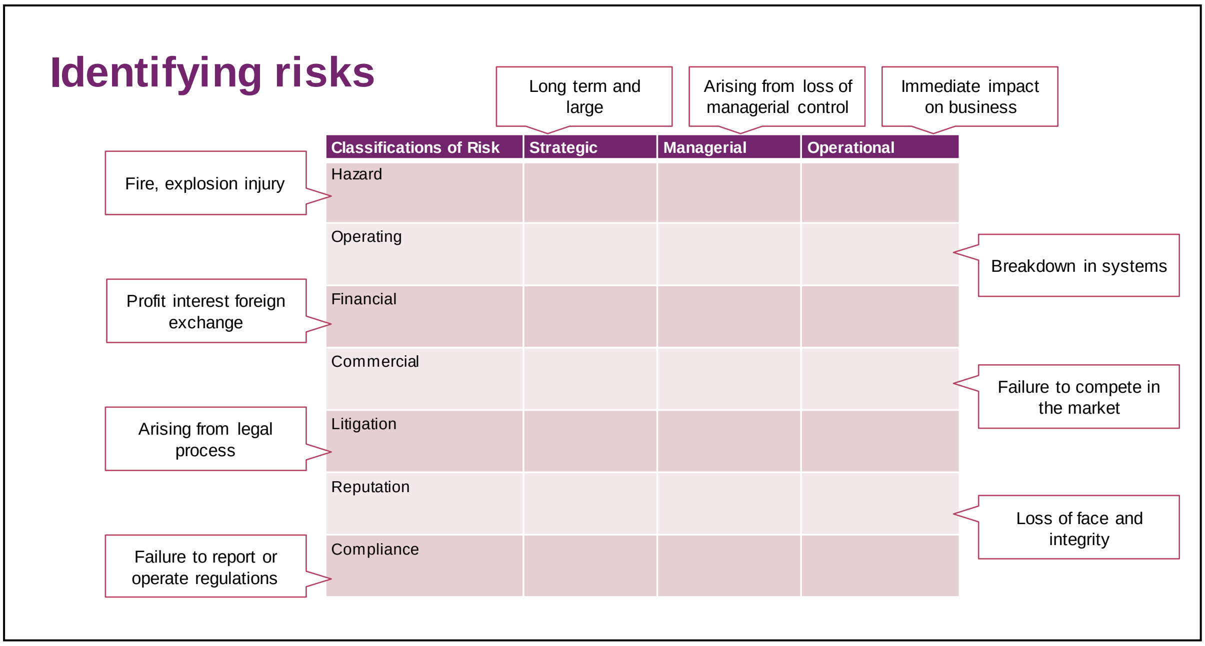
4. Insurance: A situation where the probability of a variable (such as burning down of a building) is known but when a mode of occurrence or the actual value of the occurrence (whether the fire will occur at a particular property) is not. A risk is not an uncertainty (where neither the probability nor the mode of occurrence is known), a peril (cause of loss), or a hazard (something that makes the occurrence of a peril more likely or more severe).

5. Securities trading: The probability of a loss or drop in value. Trading risk is divided into two general categories: (1) Systemic risk affects all securities in the same class and is linked to the overall capital-market system and therefore cannot be eliminated by diversification. Also called market risk. (2) Nonsystematic risk is any risk that isn't market-related or is not systemic. Also called nonmarket risk, extra-market risk, or unsystemic risk.

6. Workplace: Product of the consequence and probability of a hazardous event or phenomenon.

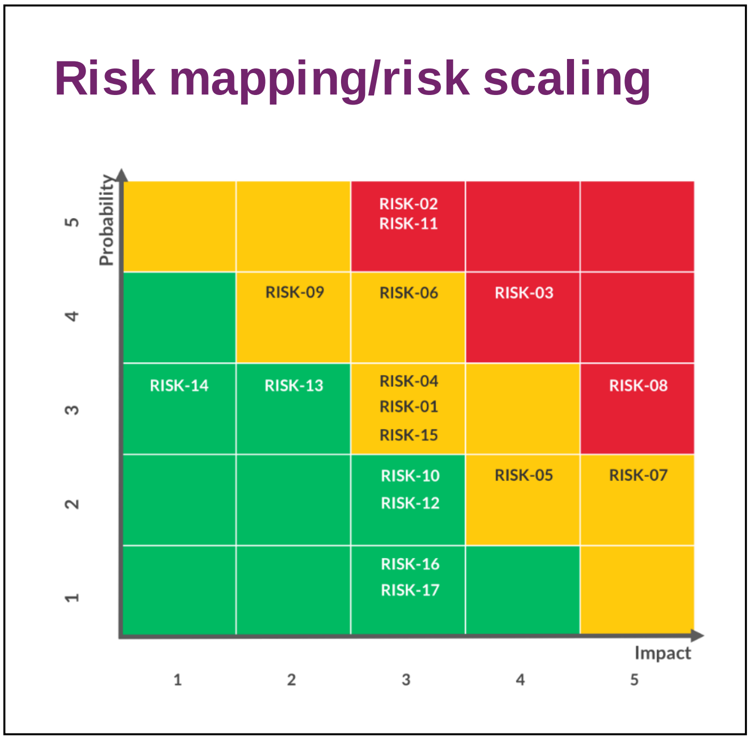
Ethical risks: these are not on this list. Were discussed in an earlier chapter.

Levels of risk
The following terms are often used. They lack precise or agreed meanings:
- *Strategic risk*: long term and major in its impact on the organization and its business model. Some of these risks come from external factors that management believes it is not able to influence or mitigate (or be blamed for) and for that reason does not include it in a risk register
- *Managerial risk*: arising from poor control of the business by management due to lack of competence, information, or control
- *Operational risk*: something arising from, or impacting on, the day to day business of the organization. Often these are accepted as one of the consequences of doing business.



2. Risk scaling
Using the formula *Risk = Impact x Probability* management will classify risks.
The diagram shows a *heat map* that classifies risks into Red, Amber and Green. The numbers for each risk refer to their location in the *risk register* of the organization which will detail the risk and the controls in place to manage it.

Exam point: it is unrealistic to expect you to scale risks in the exam because you do not know the size of potential impacts not the quality of internal controls that detects and prevent the event. Most questions ask you to identify potential risks, which you answer by stating an event and explaining its potential impact.



Risk register

A document, usually set up as a spreadsheet and showing

- Description of risk
- Risk scaling
- Controls being used
- Responsible persons

Risk	Likelihood 1=Unlikely 5=Very Likely	Severity 1=Minor impact 5=Major impact	Risk factor (L*S) Low Risk 1-8 Medium Risk 9-14 High Risk 15-25	Control	Responsibility
Operational					
Problems with product safety.	2	4	8	Constant quality checks are conducted at each stage in the manufacturing process.	Production manager
Supplies of raw materials may be interrupted.	3	4	12	Fizz takes pride in selecting only high quality and dependable suppliers.	Chief buyer
Failure of key IT system	3	5	15	Fizz maintains backup systems.	IT manager
Disruption of delivers	3	5	15	Fizz works closely with all relevant authorities and monitors factors such as the weather.	Logistics manager
Commercial					
Intellectual property rights may be compromised.	3	3	9	Fizz invests in trademark and other forms of protection. Legal enforcement is used when necessary.	Board

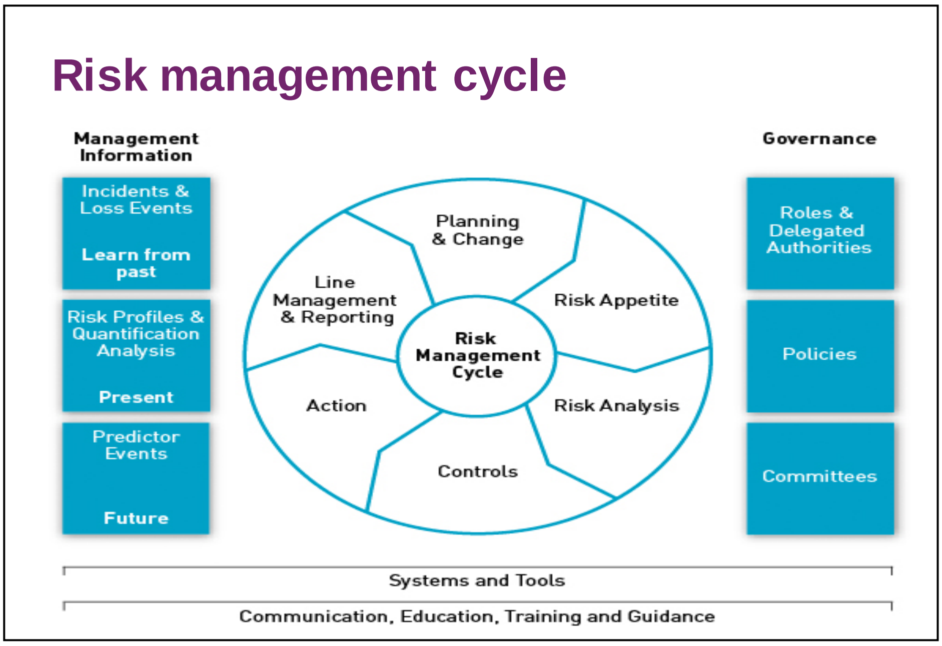
Risk registers may be drawn up for each unit, or branch, of the business and then provided to senior management for review

Risk Register v Statement of Principal Risks
The exam has featured several questions testing whether candidates can distinguish the roles of these two different documents.

Risk register is
- Internal document that is not published
- There will be separate risk registers for each area of the business
- It is a managerial control because it records the risks, the controls and the persons responsible for managing the risk
- It is a document that will be reviewed by management and by the Audit Committee and Risk Committee to ensure there is a policy of risk management in place

Statement of principal risks is
- Published in the annual statement to investors
- Helps readers to assess the *viability* of the business (effectively its sustainability)
- Is aimed as investors who wish to understand the risks they will be exposed to if they buy shares
- Focuses on the *material* risks (the ones that could influence an investment decision)
- May be assessed as a guide to the control/stewardship being exercised by management

3. Enterprise Risk Management Cycle
This refers to the states or elements in a risk management strategy.
CGMA provides a schematic diagram to summarize the elements
Understanding this schematic is difficult because it tries to encompass the wide variety of risks that feature in organizations.



For example a bank will, like all firms, have fire risks, risks of loss of key staff, risks of reputation damage, risk of robbery or fraud, risks of IT systems breaking down, risks of capital expenditure on project being allowed to overrun, litigation risk from the bullying or harassment of staff, and of course the banking risks of its borrowing, lending and investing decisions. All of these risks, and more, must be managed using this risk management cycle.

Without a formal risk management system, enforced by management, these risks would be overlooked by managers and staff because they are tiresome, and managing them seems to get in the way of getting the job done and making money. But the ‘getting the job done’ and preserving shareholder wealth can be totally undermined by the crystallization of a serious risk.

The schematic identifies the main steps in a risk management strategy

1. *Planning*: this means an initial senior management decision to set up the risk management process, and the establishment of a project with a time frame for this. A *Risk Committee* at corporate board level may be established to oversee the process and to receive reports. Sometimes this plan may prioritize particular risk areas for management, perhaps following the arrival of new laws or regulations, or the crystallization of a major risk in the industry.

2. *Risk appetite:* refers to risks the organization is willing to pursue, such as the products it will make and the markets it will pursue. *Risk tolerenc*e refers to the amount of risk the organization will accept. CGMA examiner tends to regard risk appetitite and risk tolerance as the same thing.

   It is often formulated as a statement.

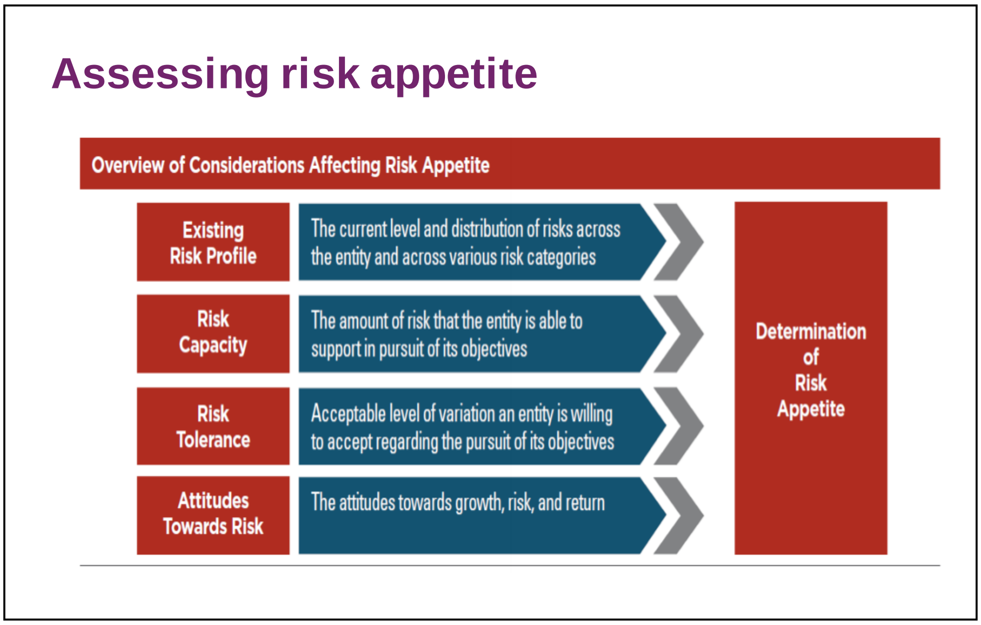
   Example of a risk statement
   Rail infrastructure provider, NR

   ‘NR has no appetite for safety risk exposure that could result in injury or loss of life to public, passengers and workforce. Safety drives all major decisions in the organization. All safety targets are met and improved year-on-year

   In pursuit of its objectives NR is willing to accept in some circumstances risks that may result in some financial loss or exposure, including a small breach of the loan limit. It will not pursue additional income generating or cost saving initiatives unless returns are probable.

   The company will only tolerate low-to-moderate gross exposure to delivery of operational performance targets including network reliability and capacity and asset condition, disaster recovery, and succession planning, breakdown in information systems or data integrity.

NR wants to be seen as best in class and respected across the industry. It will not accept any negative impact on reputation with any of its key stakeholders, and it will only tolerate minimum exposure i.e. minor negative media coverage, no impact on employees, and no political impacts.'



Establishing a risk statement often resolves into the balance between financial growth, risk, and return. This examination has featured questions on how to assess risk appetite.

- The *existing risk profile* can be derived from risk mapping and it provides a benchmark of the levels of risk that owners and managers are already tolerating.
- The *risk capacity* should be considered as a practical constraint on the maximum level of risk that can be accepted, whether or not management want a higher level of risk.
- Assessing *risk tolerance* should be done by scaling the risks of proposed changes to the business model and discussing these with investors, managers and others to assess their willingness to tolerate the higher risks. The assessment of the firms Cost of Equity may give a guide to the shareholders attitude to risk.
- Assessing *attitudes to risk* refers to assessing the *risk culture* of the organization.

Risk culture means the set of shared attitudes, values and practices that characterize how the organization (i.e. its managers and staff) consider risks in daily activities. For example, contrast the likely risk cultures of a public children's education service (risk-avoiding) with a small private equity fund (risk-taking)

A 2017 exam task required candidates to explain what could be inferred about the culture of a business partner from the actions of one of its staff members in providing someone in our firm with a piece of unlicensed software. This task was assessing candidates' appreciation of the risk culture and the quality of internal controls within the business partner. Good answers would have explained it pointed to a lack of a healthy control culture around IT applications and security and perhaps also have said it could point to a more general laxness in risk culture and internal control in other areas by the partner, and suggested the sources of these. The answer might also have criticized controls in our own firm for having accepted and installed this free software without adequate checks on its provenance (the task that followed in that exam featured our firm fearing being sued for breach of software licenses and trying to audit the extent of the problem).

Risk culture usually derives from

- the rewards or sanctions the organization offers for risk-taking or risk-avoiding behavior

- the leadership of the business and the example they set
- the extent of formal controls over risks in the organization and how strictly they are enforced
- the urgency of the situation and/or the pressure on management and staff to get results
- the characteristics and backgrounds of the employees of the organization
- the consequences of things going wrong

3. *Risk analysis* is the process of identifying events and scaling their impact and likelihood to arrive at a risk rating. The wide breadth of possible risks means this cannot simply be conducted by senior management. They will not know all the risks. It is a process that must be undertaken throughout the organization in each business unit, branch and business function. This is part of the *governance procedures* for risk management (roles, delegated authorities, committees and so on). Having *management information* about past events, or events elsewhere in the industry, will help broaden sensitivity to the potential risks that may occur. If risk management is being set up for this first time this will require training and coaching for the staff involved in the process, and the use of outside experts in complex areas (fire risks, IT security risks, risks of discrimination in the workplace, operating risks in particular countries and so on)

4. *Controls:* this is assessing the present internal controls and contingency plans for adequacy and improving them. New controls may need to be developed. These include the introduction of new policies (reporting of foreign exchange expose, purchasing and tendering processes, staff policies on corporate ethics and personal behavior, IT security processes and access controls and so on)

5. *Action:* this is the implementation of the risk management controls. Education and training will be widespread (fire evacuation procedures, safe working practices, legal compliance training, reporting incidents, and so on) and there will need to be action to change the risk culture towards an appreciation of the need for the new policies and procedures.

6. *Line and management reporting:* this is reporting on the implementation of controls and any risk incidents

The Risk Management Cycle is continuous. It requires periodic revisiting of the effectiveness of the controls and changes to the risk management process following re-assessment of each of the stages. Typically, this is done every three months as circumstances develop.

Responsibility for managing risk
This has been the subject of many exam questions

The Code of Corporate Governance states

- 'The Board should establish procedures to manage risk, oversee the internal control framework and determine the nature and extent of the principal risks the company is willing to take in order to achieve its long-term strategic objectives'
- The main roles of the Audit Committee include 'reviewing the company's …internal control and risk management systems, unless expressly addressed by a separate board risk committee composed of non-executive directors, or by the board itself'

The following are reasonable assumptions for the exam

- Management of operational risk are primarily the responsibility of the managers of the processes or activities (as stated in the risk register)
- The Board remains responsible overall for risks but this is mainly a responsibility for setting a risk culture and for ensuring that there are enterprise risk management systems in place

- The Board as a whole can delegate the duty of overseeing to a risk committee or to the audit committee, but individual executive directors remain responsible for risks in the areas they manage (e.g. CFO for financial risks, CIO for data security and so on)
- It is not reasonable to hold the Board, or risk committee, for adverse operational events because they are too remote and bad stuff is inevitable in business sometimes unless it can be shown to have been the result of a poor control culture (e.g. pressure from the board to cut corners, poor culture, lack of emphasis on risk assessment and control)
- The Board does remain responsible for assessing and monitoring the strategic risks facing the company, and for disclosing principal risks in the annual report

Big Data and risk management
Some specific applications of Big Data Analytics to provide management information include:

1. Fraud, bribery and corruption risks
   This is *forensic data analytics (FDA)* in which management and internal audit controls can be strengthened by the use of data from general ledger and transaction data, email communications, free-text searches in databases or online, telephone and data communications

2. Compliance and control risks
   Regulations and rules are encoded by computers using *cognitive computing* and the financial transactions and reporting systems, the emails and other communications of the organization are ingested and analysed to identify or to predict possible breaches of regulations and rules. This is sometimes termed *full fidelity analytics*

3. Market risk analysis
   This is the use of analytics to monitor and predict changes in the values of interest rates, exchange rates, commodity prices, and the firm's own equity price (or movements in the prices of equities and other investments it holds)

4. Information security
   The big data bank can ingest and analyze the logs of applications use to identify departures from normal behavior. It can also assess unusual events, log-ins, alerts, and user and network activity. It can also monitor external systems, such as wikis, blogs and file-sharing sites, to identify when the firm's intellectual property has been leaked

5. Commercial and operational risks
   Big data can monitor external media and financial information about suppliers, customers and other counterparties to predict corporate failure or bad-debts. It can monitor internal transactions to identify poor quality, late delivery and bottlenecks. It can identify and predict low sales and slow inventory.

6. Strategic risks
   Big data can monitor competitive activity, potential country risks, and public attitudes to the firm and its products from economic data, media and blog posts, and trending posts and topics on social media

ACTIVITY 16.1

Context
Couchweb is a global provider of Subscription Video on Demand (SVoD) television programs. One of its reality series, The Cavern, is losing popularity.

Task
You have received an email from the Finance Director of Couchweb
To: Senior Manager
From: Ed Lanson, Finance Director
Topic: The Cavern

Please review the document attached. It's written by Alana Duprey, our Creative Director.

I assume this document is the source of the rumours that appeared in the Daily News last week about us sinking The Cavern.

No actual decision has been taken by the Board on this yet, although Denny White is giving it his support.

I need to understand the risks of this proposal better and how we might reduce them.

(Sub section = 55%)

Reference Material
Refreshing 'The Cavern'
Audience research shows that the popularity of The Cavern has declined noticeably between its second and third series.

We are losing audiences to shows like 'Eating in the Outback' and 'Castaways'. Viewer focus groups tell us they enjoy seeing our contestants arguing and co-operating during their 14 days underground, but they get depressed by the same dark background and the need to constantly focus on the discussions between the contestants.

I'm suggesting that we keep the basic concept of The Cavern, but we move it to a new exotic location. A submarine near a tropical island.

I have located a decommissioned navy submarine which can still sink and surface okay. We can use it instead of The Cavern. It will cost us US$550,000 to buy which saves us the equivalent of three series worth of rent for the existing cavern. I've attached a specification for it. Denny liked it so the sellers have provisionally accepted my offer for it.

There is also a tropical island about 2 hours flying time the nearest airport that we can use. It is a paradise. Virtually unexplored and deserted. No no-one will know or care what we do there.

My idea is that we sink the contestants in the submarine at the start of the show, but each day it surfaces and two contestants have to go and get food from natural sources on the island to cook and bring back as a meal. If they fail the submarine submerges again and leaves them stranded. Of course, we'll arrange some surprises for them, challenges and games, attacks from natives with spears and cooking pots, scary noises and so on.

And I want to introduce more love interest. Have couples apply to be contestants but choose couples that are not totally happy with each other and then see if we can encourage some of the contestants to stray, on camera. That will be good for ratings.

ACTIVITY 16.2

Context
Runnabout is a listed company that hires electric hoverboards to commuters and tourists. It provides its users (members) with a smartphone app to use to may payments via credit cards but which also contains other personal information and which tracks their movements both on and off the hoverboards.

Runnabout has recently discovered that its main systems may have been hacked and data on members copied.

Task
You have been called into the office of Geo Pataros: CFO of Runnabout
'Here's a copy of the findings from the report of the consultants into our data breach (he hands you a report). It's not as bad as we had feared but there has been a data leak.

Our CEO is considering making a statement to the Geeland stock exchange and to the media explaining the extent of the data loss and also the controls we will put in place to stop it happening again. She intends to blame this on the Risk Committee of Runnabout and to make clear that she cannot be held responsible for the failure of the committee to manage this risk because she didn't appoint the non-executive directors on it.

This raises a concern in my mind

- Please advise me on how appropriate it is for the CEO to blame the Risk Committee of non-executive directors for failing to manage this risk'

[sub task (a) = 40%]

Potential Data Breach
Findings of the independent consultants

To: Board of Runnabout Group
From: Shaun McDougall: IT Director

Background
Cyber security experts Limehouse Consulting were commissioned by Shaun McDougall, IT Director to assess the causes and extent of the suspected data breach of Runnabout's systems

Findings
The main systems compromised were the GPS and hoverboard tracking systems of Runnabout. There is no evidence of penetration or compromise of the personal or financial data held by Runnabout

The penetration was sophisticated and seem to have been the work of professional cyber criminals

The method was a social engineering fraud, or 'human hacking' in which a member of staff in the marketing team was misled into responding to an external email from the private email account of a colleague who claimed to have lost their password and log-in credentials and asking to borrow the member of staff's credentials to allow them complete some work from home during the weekend. In fact the criminals had hacked the colleagues email address and the request was a phishing email

It is of concern that the member of staff willingly disclosed these credentials and that they gave the criminals access to almost all areas of Runnabout's systems

Solutions to Activities

16.1 Couchweb refreshing The Cavern with a submarine
I need to understand the risks of this proposal better and how we might reduce them (55%)

Risks of proposed new show
[Strategic]
<u>The new format may fail to attract viewers</u>
This is very different from the existing The Cavern. Couchweb may lose existing viewers for The Cavern without gaining new ones for the submarine/island show. This will be enhanced by the fact that the existing name will be hard to transfer to the new concept show

This would leave Couchweb owning a submarine with no series to use it on

<u>Reputation Risk from 'love interest'</u>
The 'love interest' is tantamount to encouraging couples to cheat and behave inappropriately. Many viewers will be annoyed by this because they will think it is irresponsible and obtrusive. Many partners will decide to not adopt a program with a bad reputation and the broadcasting authorities in some countries may ban it. This would ensure that the program has low numbers and is not renewed in future

<u>Ethical risks from treatment of contestants and the island</u>
This program exploits the contestants by exposing them to risk and ridicule for little financial gain. This is increasingly unacceptable to the viewing public. The program concept is not sustainable.

The use of an unspoiled island for hosting this show is likely to draw criticism from ecologists. The Couchweb series if bound to leave an adverse footprint on the island.

[Managerial]
<u>Distance to medical care</u>
The show involves activities that could lead to illness and injury of contestants. The two-hour flight may not be quick enough to save a contestant or crew member with a poisonous bite, injury, or extreme food allergy. The format of the show makes contestants subject to these hazards

<u>Cost and profit control</u>
There does not seem to be a budget for this show. The costing is not discussed. There is a danger than the costs will be excessive, compared to The Cavern, and that the earnings from the sale of the show will not recover these

<u>Risks of the submarine and island</u>
The management team of Couchweb has no experience of these two crucial elements of the program. There is a risk that these may pose hazards that the management cannot deal with (storms at sea, breaching of the hull of the submarine, dangerous animals on land and so on)

Inappropriate content
In addition to the 'love interest' there is a risk to indulging in the stereotypes of 'men with spears and cooking pots'. These images can be too-easily attached to outmoded ideas on racial groups and lead to embarrassment for Couchweb and loss of subscribers

Reducing the risks of the proposed series
Pilot episode
Before investing in a submarine and a whole series there should be a pilot episode made. This episode could be set in a studio, made to look like a submarine to test viewer responses. The outdoor scenes could be filmed on an island location. This will help confirm the attractiveness of the concept and the likely interest of subscribers in it.

Rent don't buy the submarine
The investment of US$500,000 is too great to justify on an unproven series. It is an asset specific investment which Couchweb has no alternative use for. However, a rental company will be able to use their submarine for many purposes. This means the risk can be more economically borne by a rental firm

Program costings and budgets
Before agreeing to the concept there should be costings and budgets prepared. These can be used to assess the viability of the new show. An assessment can also be made of the sensitivity of the financial return to the show and hence the financial risk involved

On site medics
To reduce the risk to crew and contestants there should be on-site medics at the location with appropriate equipment. There should also be a contingency plan for emergency evacuations such as helicopter on standby and agreements with ER facilities on the mainland

Adjust show format
Many of the risks can be reduced by changing the show format. The submarine could be replaced by a much less risky 'house' such as used in the Big Brother shows. There is no reason to hold the show in a deserted island. A tropical setting, or other interesting setting, can be obtained in mainland locations. The 'love interest' aspect can be removed to avoid the reputation risk

16.2 Hacking of Runnabout – is the Risk Committee responsible?

Please advise me on how appropriate it is for the CEO to blame the Risk Committee of non-executive directors for failing to manage this risk (40%)

Appropriateness of blaming risk committee
The Risk Committee is a Board level committee responsible for establishing a risk policy and overseeing the adequacy of risk management in the business. In Runnabout the Risk Committee is comprised of non-executive directors (NEDs).

Clearly the risks of data breach were not completely managed at R/T and this has led to the loss of data and reputation damage to R/T.

Reasons for holding the risk committee responsible include:

<u>It is responsible for reviewing risk procedures</u>
The Board has delegated risk management to the risk committee. There has been a failure in the internal controls over cyber risk at R/T. The controls are have been inadequate to deal with the social engineering fraud discovered here

Therefore the risk committee has failed to scrutinise controls sufficiently and ensure they are adequate to deal with the risks facing R/T.

<u>It may help to deflect criticism from general management of R/T</u>
Regardless of true responsibility, there may be a presentational benefit in blaming the risk committee because the alternative is to blame the executive management of R/T with the consequence that it will put pressure on them, particularly the IT Director, to consider whether to take the blame and resign. This might be very unhelpful whilst R/T tries to deal with the IT breach and to restore confidence in the stock market and with regulators.

Drawbacks to holding the risk committee responsible include the following:

<u>Primary responsibility for control lies with some members of executive management</u>
The design, operation, and reporting on internal controls in their area remains the responsibility of executive management. Here it seems to be the job of the IT Director. The risk committee merely scrutinises these. Blaming the risk committee seems to exonerate members of the executive management, and the people they appointed to run the IT systems, of their responsibility and accountability. This would encourage, not reduce, the poor risk culture that led to this data breach.

<u>Risk committee cannot be expected to know detail of each system and control</u>
The risk committee has to rely on the reports from functional managers and risk committees to assess the adequacy of risk controls. This data breach was a single operative responding to a phishing attack. It is not reasonable to blame NEDs for the failing of a single member of staff.

<u>Looks like senior management refusing to face up to its failings</u>
The investment markets and investor media will be aware that executive management is primarily responsible and therefore as a tactic this will fail to shift blame.

This will undermine the integrity of management and further reduce trust in the Board of R/T

<u>Will cause greater concern over adequacy of other control by NEDs</u>
Blaming the risk committee is blaming the NEDs for not ensuring adequate scrutiny, their primary function and role. This could lead investors to believe that the NEDs are not diligent in other areas such as setting senior pay, participation in the Audit Committee, ensuring independent perspectives are heard in board meetings This could harm the share price of R/T further.

It would also suggest that the CEO and the NEDs are in a state of conflict and that the CEO wishes to use this as a pretext to encourage investors to change them. Suggesting conflict on a board is never good for investor confidence.

Conclusion
The CEO should be advised that it is not a good idea to blame the Risk Committee for this control failure.

Chapter 17
Internal Controls

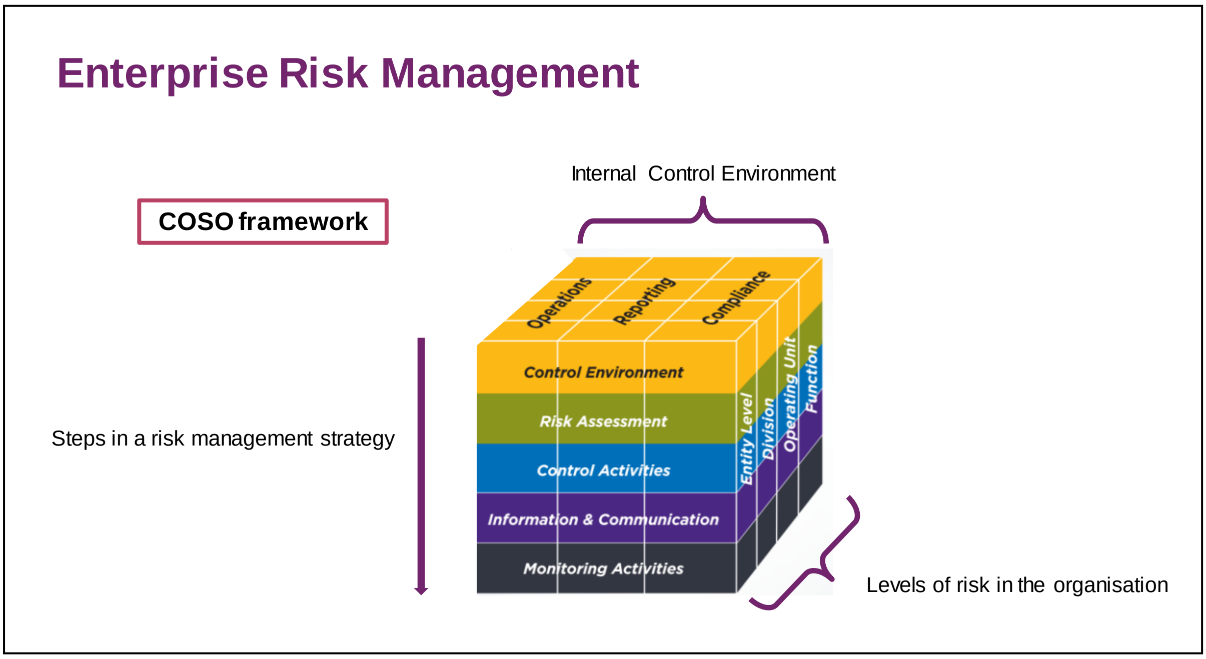
1. Enterprise Risk Management (COSO)

Developed in the USA in the wake of a series of major financial scandals that highlighted the inadequacies of the present regulatory and reporting framework and originally published in 2004 and updated several times since then, most recently in 2017

Called the COSO framework (Committee of Sponsoring Organizations of the Treadway Commission) it was developed for COSO by PwC. AICPA is a founding member of COSO.

The model performs a similar function to the CGMA Enterprise Risk Management Cycle reviewed earlier.

Each segment of this Rubik's Cube (say a particular business unit) needs to follow the steps in the strategy for all of the risks it is subject to. These are similar to the steps in the CGMA Risk Management Cycle (above).



COSO internal control environment

This is the context of the risk management process. It seeks to control the means the organization uses to reach its business objectives. Put simply, what stops it from creating risks by cutting corners?

- The organization's commitment to integrity and ethical values
- The independence of the Board of Directors from day-to-day management of the business and its ability to exercise oversight and internal control
- The quality of management below the Board and its structure of authority, responsibility and reporting
- The ability of the firm to attract, develop and retain the right quality of people
- The extent to which the firm holds individuals to account for their internal control responsibilities

These will be familiar to you from earlier discussions of corporate governance.

This may help to identify the *risk culture* of the firm which depends on four elements (according to the Financial Stability Board – an international body that monitors risk in global financial system)

1. *Tone from the top*: the values the Board encourages, how Board members behave, and the mechanisms used to hold staff to account for their behavior
2. *Accountability:* the extent to which staff understand the core values of the firm and know that they will be rewarded or punished for their behavior
3. *Effective challenge*: are there constructive challenges to inappropriate behavior and risk-taking, or does no-one speak up? This includes controls such as a complains procedure, whistle-blowing, and staff engagement processes
4. *Incentives*: how is compensation and career progression awarded and does this encourage or discourage risk-taking

Risk Assessment
This ensures that management is aware of risks and how they change as a consequence of strategic decisions or of changes in its environment/ecosystem

- Clarity in specifying objectives: these are clear enough to allow the identification of the potential risks involved
- Processes of risk identification and scaling
- Clear assessment of the potential for fraud in what the firm does
- Assessment of changes that could impact on controls and weaken them

Control Activities
These are the processes and procedures to reduce risks. COSO requires

- The organization to select control activities that mitigate risks.
- The firm has general controls over the technologies it uses to run its business. This is a reference to cyber risk
- There are clear policies in place for establishing expectations and procedures to put policies into action. For example, policies to stop bribery, bullying, discrimination, poor health and safety and so on

Information and Communication

- There should be an information system to monitor the quality of internal controls
- There is clear communication of control policies together with the giving of responsibility for each control (for example the risk register described above)
- The organization communicates its risks and controls to external parties, for example risk statements in annual reports

Monitoring Activities

- Development of ongoing and/or separate evaluations of internal controls. The Internal Audit function is an obvious example of this. So are other monitoring activities such as diversity monitoring, health and safety inspections, and quality assurance audits
- The organization evaluates and communicates internal control deficiencies in a timely manner. This includes reporting to the process owner and to the Audit and Risk Committees of the firm

2. Types of internal control

Internal controls are merely a big name for a simple thing. They are the mechanisms that management uses to make things go right in the business.

Consider a restaurant: it has a lot of customers, lots of staff, money being exchanged, and lots of hot and dangerous equipment in the kitchens. How does management arrange things to stop food poisoning, accidents in the kitchen, unsatisfied customers, wasted meals, and the restaurant operating at a loss?

You would say
- Trained chefs and serving staff
- Good recipes
- The motivation of gratuities from customers to help ensure good service
- A watchful Head Chef and Restaurant Manager
- Budgetary control over costs

These are all internal controls.

Your assessment will ask you to recommend suitable internal controls for the business in the pre-seen case. There will be no points available for reciting a list of internal control categories.

Examination technique
The frameworks of risk culture, internal controls and TARA are merely checklists to give you ideas to suggest controls that management can use to reduce risk in the tasks in the exam. Generally you will asked to either identify where there were control weaknesses that led to a problem (such as a cyber breach, reputation damage, unlawful activity, abuse of bonus systems) and/or you will be asked to suggest improvements to the internal controls. Do not mechanically run down the lists of controls to prove you have read them. Just suggest five good ideas and explain them and how they would work.

Two popular systems of categorizations of internal controls are provided below. But in the exam, these should merely be used as checklists to stimulate ideas.

Internal controls
These may be familiar to you from studying auditing, or from your own managerial experience. The mnemonic DAMP SOAP can be used in the SCS assessment as a check list of controls to identify weaknesses and/or recommend improvements.

Internal controls

- Division of responsibilities
- Arithmetic and accounting
- Management
- Personnel
- Supervision
- Organizational (structure, procedures, processes)
- Authorization and responsibility
- Physical (custody, protection)

- *Division of responsibilities*: ensure that transactions or decisions pass through more than one set of hands. This reduces the risks of error or poor decision-making from relying on one person. One aspect of this is to have independent *Internal Audit* functions so that those who designed and operate the system are not also the same people who assure the quality of the systems

- *Arithmetic and accounting:* ensure adequate transactions systems and reconciliations of all data. This includes the quality of management accounting information such as cost identification and financial forecasts

- *Management:* ensure management are competent for the roles they hold and sufficient in number
- *Personnel*: ensure staff are competent for the roles they hold and sufficient in number
- *Supervision:* ensure the ratio of staff to supervisor is not excessive and that supervision techniques and systems are adequate
- *Organizational*: ensure the organizational structure is not a source of risk and that the procedures for doing the work and monitoring and rewarding performance are not likely to create excessive risk
- *Authorization and responsibility:* this refers to the levels in the organization at which decisions can be taken
- *Physical*: this refers to the custody of assets and data, and the levels of protection given to the assets, customers, employees and visitors

Other approaches to risk management
The CGMA syllabus contains a number of risk management frameworks. They are reviewed below.

Risk management

Components of a risk management approach
1. Information systems to monitor risks
2. Preventative steps to reduce risks
3. Contingent steps to respond to risks that crystallise

The TARA Mnemonic
- Transfer
- Avoid
- Reduce
- Accept/absorb

A simpler model to help remember is the three-component model
1. Information systems to detect risks
   - Big Data monitoring transactions and trends
   - Risk officers and risk committees
   - Security patrols

2. Preventative steps to reduce risks
   - Internal controls
   - TARA
   - Internal audit

3. Contingent steps to respond to risks
   - Fire drills and evacuation processes
   - Stakeholder information and PR plans
   - Penalty clauses and contractual terms

The TARA mnemonic
The TARA mnemonic suggests four risk management strategies
1. Transfer risk
   - Outsource the activity
   - Insure the activity
   - Ensure legal documentation places risk with counterparty
   - Hedging financial exposures

2. Avoid risk
   - Don’t do the risky thing

3. Reduce the risk
   - Reduce the factors that may cause the risk (i.e. reduce the probability)
   - Reduce the impact of the risk

4. Accept/absorb
   - So-called 'residual risk' because it's what management has not Transferred, Avoided or Reduced
   - Self-insure

Terminology
The examiner has set tasks that use unusual terminology. Be aware of the following:

Gross Risk (also called *inherent risk*): this is the risk that is present if there are no controls or the internal controls have all failed. We might say that oil exploration has high gross/inherent risk because it requires enormous capital investment for uncertain outcomes, oil explodes, most oil-producing areas are politically or environmentally challenging, and oil prices are very volatile.

Net Risk (also called *residual risk*): this is the level of risk when all the controls work.

ACTIVITY 17.1

Context
Finch News Group (FNG) is a newspaper publisher. Digital disruption in the news industry means that FNG has moved from printed newspapers towards digital news sites. However, the need for immediate content, and the need to respond to stories breaking on other digital media, has weakened the traditional internal controls of the news industry. Many journalists, and readers, can posts articles and comments in the pages of FNG's online publications without going through any editorial process or fact checking.

This has led to the increased risk to the reputation of FNG from 'fake news'. Fake news means articles that are substantially untrue or exaggerated. By publishing such articles FNG can suffer litigation and damages awards from people it defames.

Task
Den Rice, Chief Finance Officer of Finch News Group has called you into his office

"Have you ever heard of a man called Rob Ridout?

No? Well, apparently, he was an intelligence officer working for the Monderlorian government and he was involved in corrupting a number of local politicians in Borland.

It must be true, because we printed it in several of our newspapers.

But he doesn't exist and the corruption never happened. Lots of news sites and newspapers were tricked by this rumor, we are not the only ones. We just realized this an hour ago

Already the non-executive directors and shareholders are demanding explanations and assurances.

I need your views on how to stop fake news appearing in FNG publications in the future

(Sub section = 55%)

Solutions to Activities

17.1 Stopping fake news at Finch Media Group

I need to understand the risks of this proposal better and how we might reduce them (55%)

Approach: this is a question on internal controls. It may be helpful to remember the main internal controls and also the TARA mnemonic to help stimulate ideas. This question was clearly indicated in the pre-seen material for the exam and therefore the program materials contained some of the technical approaches referred to below

Stopping fake news in future

<u>Transfer risk to journalists</u>

Journalists do not take significant risk, other than to their jobs, from writing articles containing fake news. The risk is presently taken by FNG (litigation, reputation and so on)

If these risks were transferred to journalists, they would be more likely to ensure that articles and stories were verifiable. This could be done by contracts of employment which indemnify FNG for legal damages arising from fake news

Or it could be done by using self-employed journalists who vouch that stories are genuine

<u>Train journalists and staff to eliminate fake news</u>

There are procedures for identifying and avoiding fake news (checking sources, looking for supporting sources)

If all staff are trained in these it would reduce the likelihood that fake news items could be published

<u>Retain editorial review for the most suspect stories</u>

Many stories are unlikely to be subject to falsification, such as sports results, public announcements and so on. Others would be of limited importance if they were fake news and so pose little risk

Dubious stories are ones which, if true, could benefit or harm certain people and which run against the conventional wisdom about the people or subjects involved (for example are the politicians of Northland known to be corrupt?)

This exercise would probably require expertise in the subject, such as specific desks in the newsroom than specialize in business, politics, celebrity gossip and so on to whom the stories can be shown before publication

<u>Institute fact-checking processes</u>

Fact-checking is a desk-based research process that can be done by relatively low-grade staff and interns. This can be assisted by special services such as the International Fact Checking Network (USA) or Decodex (France) that monitor websites and social media posts and to spot dubious stories and statements which are then flagged as fake on the social media site and this flag is carried across if the stories are hared across social media.

Some algorithms can pre-identify suspect sites and social media accounts (for example by whether they were created recently, are registered in dubious locations, or the person behind the site is anonymous or know to be a source of fake news)

These initiatives are better done as part of a collaborative network between news firms where FNG is willing to share its findings on suspect sources with other media firms

Delays and reviews at the time of publication
It is not clear when the digital content department obtains the stories from FNG news teams. If they are obtained before the process of editorial review by the news desks is complete this can lead to uncritical publishing of stories that turn out to be fake news

Therefore, there should be a policy of avoiding publication of major stories until the news team have cleared it for publication in the local newspaper

Digital content should also be reviewed before being uploaded, or at least recent uploads should be monitored and removed if they are suspect

Chapter 18
Internal Audit

1. Introduction

Internal Audit (IA) has a number of alternative names that help us to appreciate how it have moved beyond its historic focus on financial accounting systems:

- Compliance
- Business Assurance
- Risk Assurance

Internal Audit (IA)

Internal business function to conduct *independent* reviews and investigations and report to management

- Accounting and internal control systems
- Financial and operating information
- *Effectiveness*, *Efficiency*, and *Economy* of operations
- Compliance with laws and regulations
- Safeguarding of assets
- Implementation of corporate objectives
- Risk management and significant business risks
- Special investigations

The IA function acts as the eyes and ears of the Board. Its primary role is to give assurance that the internal controls are adequate and working.

In the SCS Assessment there are frequently questions that require candidates to recognize the roles of IA in such things as

- Supply-chain improvement
- Assuring compliance with CSR requirements
- Post-investment audits following projects or acquisitions

- Due diligence of target company during a merger or acquisition

Types of Audit

Types of Audit	Description
Compliance audits	• Check firm's complying with external statutory or regulatory requirements
Social and environmental audits	• Assess objectives and ways of meeting firm's CSR agenda • Assessing understanding of safeguard to environment and the efficacy of the means used to ensure safeguards
Management audits	• Assess effectiveness of management • Includes targets, decision making, competence, delegation, relationships, communication, and support systems
Post-completion review (or *post-investment audit*)	• Assessing the success of a project to learn lessons • Can involve assessing the success of an acquisition

2. Management of IA

IA conducts *independent* reviews and investigations and report to management
This means is should be

- Independent of the systems and internal controls it audits: IA doesn't actually design or operate them
- Independent of management that operate the systems: questionable because its team are employees. But it will report to the Audit Committee at Corporate Board which is comprised of NEDs. This improves its independence.
- Independent of the culture of the systems it audits. Most IA departments are separate teams with their own offices and careers outside of the finance team and other teams.

IA may be outsourced to external partners, including the firm's external auditors. This is helpful in a number of cases:

- For special areas where expertise is needed
- For remote territories where it is not cost effective to have an internal audit function
- Where greater assurance of independence is needed

Governance of the internal audit function

The UK Corporate Governance Code states the following:

- The Audit Committee should use the internal audit function as its means of assurance
- If the company does not have an internal audit function the annual report should explain why not and through what alternative mechanisms assurance is maintained

Management standards for IA

- Proper organizational structure
  The Head of IA should be appointed by Audit Committee of business (NEDs). Its management and staffing should be independent of executive management. The Head of IA may report to an Executive Director for operational reasons, and to gain authority, but also should direct access to Chair and to Audit Committee

- Proper management
  Internal audit plan for year should be established based on the risks of the business. The Audit plan should be reviewed annually and submitted to Board, or its Audit Committee, for approval.

- Proper conduct
  The IA function should be professional. This means it should abide by ethical standards such as Ethical Code for Accountants. It should identify, analyze, evaluate and record sufficient information on which to base its opinions and findings. It should communicate results to system owner, senior management and the Board

Audit approaches

Systems based auditing
- – maps the processes, systems and controls and assesses whether they
  - are likely to be effective in reaching the goals of the organization
  - are efficient and economic in operation

Risk based internal auditing
- – approach which devotes audit resources to areas and processes with the greatest *inherent risk*.
- – may make use of the risk scaling activities and the risk register

Substantive procedures audit approach
- – used where the internal processes are not regarded as effective
- – auditor will jump to checking the processes and procedures of greatest *materiality*

3. Approaches to IA tasks

The assessment often features sections that require you to plan an audit or show how you would approach an audit. These are often non-standard situations, such as investigating allegations of insider trading by a NED, or of complex processes, such as auditing a firm's supply chain to ensure exploited labour isn't being used.

Simple approach to deploying IA resources

- Ensure knowledge and competence in area to be audited
- Research and record existing procedures and system
- Assess adequacy of existing systems
- Assess operation of existing systems
- Communicate findings to management

Here is a simple approach to these tasks

1. Ensure knowledge and competence in area to be audited.
   For audits beyond normal scope of IA such as first-time audits where knowledge is lacking. This may require recruitment or secondment of expertise.

2. Research and record existing procedures and system
   Information gathering by desk research of documents or by visiting the process and conducting interviews, observations, experiments
   The audit team may produce flow charts and narrative explanations of the systems being used.
   If there is no obvious system in place it can be concluded that there is a lack of internal controls

3. Assess adequacy of existing systems
   This is a *systems audit* and it seeks to identify areas of weak internal controls or inadequate systems.
   IA may compare the firm's systems to best practice in the industry

4. Assess operation of existing systems
   The purpose of this stage is to assess whether the system is being operated as it was designed or intended. This is a *transactions audit.*
   The weaker the system the greater the amount of transactions that must be checked.
   If there are no effective systems this will lead to a *substantive procedures audit*. This means the IA team will be assessing the true situation, and so the degree of risk, in the absence of effective internal systems of control
   The *materiality* principle will guide the deployment of audit resources to the items that are most
   - Material in *amount*: they pose the greatest financial risk
   - Material in *nature*: if this is wrong it indicates something fundamental is wrong
   - Material in *effect*: it is something that, if wrong, would fundamentally change stakeholders' attitudes to the firm

5. Communicate findings to management

Deploying IA resources
Recent exams have featured questions asking candidates to express an opinion on whether it would be appropriate to instruct IA to investigate on something, or do something to improve controls.

There are four factors to mention in tasks like these:

1. <u>Governance</u>: the IA function is independent of managers like the CFO. Therefore deploying IA is ultimately a decision for the Audit Committee

2. <u>Independence</u>: there are two aspects to this:
   - IA function should not design, nor instruct on, the internal controls it will later audit. This avoids it feeling unable to criticise those controls at a later date.
   - There will always be a suspicion that IA's judgement will be influenced by its experiences in the firm and/or by what it assumes senior management wants to hear, so that it may not have clear judgement

3. <u>Competence:</u> IA will be competent in auditing the areas it routinely audits. It may not have the competence to audit the systems of new things, such as insider trading, cyber security, environmental management and so on, without extra training or the secondment of extra expertise to its team.

4. <u>Capacity:</u> IA will work to an annual audit plan focusing on the areas, systems and controls of greatest potential risk to the firm. Instructing IA to undertake additional enquiries will

stretch its resources with the effect that some areas of risk will not be audited and so the risks from them will be increased

ACTIVITY 18.1

Context
Runn is a global Courier Express Parcel business, like DHL and UPS.

Runn is considering acquiring a direct rival firm, Fronnt, by exchanging its shares for Fronnt's shares.

Task
You have been called into the office of Chen Yanyun: Director of Finance for Runn
"It seems that the Board of Fronnt Group is willing to recommend to its shareholders that they take Runn shares in exchange for their Fronnt shares. Providing 50%+ of shareholders agree to do this, Runn will then take control of Fronnt.

The Directors of Fronnt are suggesting that their present shareholders will be likely to do this if they see members of the Fronnt Board serving on the Board of Runn. They think this will also help to keep the morale of Fronnt staff high. Our CEO seems to agree with this, but I have my doubts:

Firstly, what issues should we consider before we appoint Fronnt Directors to the Board of Runn?

(Sub section (a) = 20%)

Secondly, are there any ethical issues in promising Board places to the Directors of Fronnt before the takeover is complete?

(Sub section (b) = 20%)

If this takeover is agreed by Fronnt's Board then Runn will need to do some due diligence into Fronnt before finalizing our offer. This is too protect Runn from risks arising from the acquisition. Our Internal Audit team understands the CEP industry very well and I think they should undertake this due diligence. But I would like your opinion on this.

Finally, what factors should the due diligence look into and are there any problems if we use Runn's Internal Audit team to conduct this due diligence into Fronnt?

(Sub section (c) = 60%)

Solutions to Activities

18.1 Acquisition of Fronnt by Runn

This activity involves a lot of topics from previous chapters on ethics and corporate governance, and ends with a substantial question on planning a due diligence audit and the role of IA in this.

Firstly, what issues should we consider before we appoint Fronnt Directors to the Board of Runn? (20%)

Issues to consider before appointing Fronnt directors to Runn's Board
<u>Impact on harmony of Runn Board</u>

Appointing new Directors will potentially harm the group dynamics of the Runn Board. The Runn Board has been assembled since 2016 [*data in pre-seen case*] and is functioning well. Introducing new members will make each director will be suspicious that the new appointees may be rivals for their jobs or for their chance to become CEO
This could lead to loss of Board members and conflict and harm Runn's performance at a time when it needs to sustain its profits as well and integrate Fronnt within it

<u>Process of appointment</u>
It is not consistent with the principles of good corporate governance for the executive directors of the firm to nominate or appoint new executive directors. This should be done by a nomination committee comprising a majority of NEDs

Allowing them to be appointed without shareholder approval means that any new directors would be inflicted on shareholders without any right of veto if they are unsuitable or appointed for the wrong reasons. Ideally any new directors should be appointed by shareholders based on the recommendations of the nomination committee.

Appointing Fronnt directors without proper consultation with shareholders would take Runn away from the principles of good corporate governance. This will deter investors from owning Runn shares and could harm its share price

<u>Potential roles for new directors</u>
Directors generally have areas of responsibility and team to manage. Runn already has executive directors managing the crucial functions of Runn (finance, HR, IT and so on). It is hard to see what the Fronnt appointees would add to this and they run the risk of becoming superfluous directors (such as Director of Special Projects and so on)

Alternatively, they may try to carve a role for themselves such as Director of Regions or Director of Fronnt subsidiary which would create tangles and extra costs and further detract from the integration process

<u>Effect on process of integrating Fronnt into Runn</u>
Runn needs to integrate Fronnt within it as quickly and smoothly as possible to gain the synergies.

It is possible that having a Runn Board member from Fronnt's previous Board may ease the integration by symbolizing to Fronnt staff that they may get good jobs in Runn and will be looked after

Alternatively, the Fronnt appointees may prolong the independence of Fronnt within Runn in order to keep their jobs longer because they know they are redundant once Fronnt has been integrated into Runn.

Their knowledge of Fronnt and their ability to keep it running during the transition is essential and if there is a danger they might leave if they lost Board roles it may be a good idea to appoint them

<u>Impact on costs of governance</u>
Extra Board members increases the costs of governance. These appointees will expect to retain their Board level pay and benefits.

Also, under the 50/50 rule there would need to be an equivalent extra number of NEDs appointed

This would increase the costs and slow down decision making (harder to arrange meetings, longer meetings due to more people needing to speak, greater compromise in decisions and so on)

It is hard to see that the extra costs would be rewarded by a big improvement in decision making

Secondly, are there any ethical issues in promising Board places to the Directors of Fronnt before the takeover is complete? (20%)

Ethical issues in promising positions to Fronnt's directors
Ethics concerns the moral rightness or wrongness of decisions and actions. 'Good' behavior involves doing things that are ethical and avoiding actions that are not. 'Bad behavior is doing unethical things, or not stopping others from doing them.

Ethical issues
<u>Compromise the objectivity of Fronnt directors</u>
The Directors of Fronnt have a duty to advise the shareholders to accept or to reject Fronnt's offer based on what is best for the shareholders. Making promises of jobs presents them with a conflict of interest which could lead them to recommend accepting a low price for Fronnt because the Directors want the new jobs at Runn.

<u>Affects the integrity of the process of acquisition and of appointment</u>
It is unlikely that Runn will be truthful about the reasons for appointing these new Directors, i.e. to make it easier to get the takeover accepted. It is also unlikely that the Directors of Fronnt will tell their shareholders that they have the promise of a job at Runn.

<u>Breach of professional competence and due care</u>
There is no reason to believe that these Directors are suitable to join the Board of Runn. Having unsuitable Directors could harm Runn

Consequences
<u>Could lead to a bargaining process with directors</u>
Making an offer will encourage Fronnt's directors to make their recommendation of the takeover dependent on being promised more senior positions at Fronnt. It would not be surprising if the Fronnt CEO demanded to made joint-CEO of Runn as a condition for his support.

Failure to offer adequate roles might lead the directors to recommend rejection of Runn's offer

<u>Loss of trust of shareholders</u>
Shareholders do not like to see managers rewarding themselves and each other without benefits to the shareholders. Providing the Fronnt directors with roles at Runn will erode the trust of the remaining Fronnt shareholders (in the case of an equity offer) and Runn's shareholders. This is an example of the impact of the principal/agent problem.

<u>Seen as rewarding failure</u>
Runn's Board must argue that it can run Fronnt better than Fronnt's Board can. Therefore, it would look odd to appoint Fronnt directors to the Board of Runn. This may discourage shareholders about the value of the acquisition and the prospect of realizable synergies from it

Counter arguments

May be necessary to get deal

Despite the issues above, it is very common for takeovers to involve gaining the co-operation of the target's Board by offering jobs to those that wish to stay. If the takeover is in the longer-term interest of Runn's shareholders these job offers may be essential to achieving the takeover

Retains managerial talent

Runn will need the help and knowledge of Fronnt's managers to run Fronnt in the months or years before it is integrated into Runn's operations and systems.

Without good jobs to go to there is a danger that the executive directors of Fronnt will look for new careers elsewhere. Offering them roles in Runn may be essential to keeping them in the short to medium term

Finally, what factors should the due diligence look into and are there any problems if we use Runn's Internal Audit team to conduct this due diligence into Fronnt? (60%)

Note that this sub-section contains two questions, separate by 'and'. You should assume its 30% per part.

Factors for due diligence to look into

Due diligence means the investigation and exercise of reasonable care before entering into a transaction. Fronnt is buying the net assets of Fronnt and it is buying it future incomes and skills in the form of goodwill. It needs to ensure that these exist and are going to come to Fronnt as anticipated

Ownership of assets

Fronnt claims to have a large number of depots and vehicles as well as receivables and cash These total G$24.8bn [*data in pre-seen material*]. Runn should take steps to ensure that these assets physically exist, are of the value stated, and that it will gain ownership of them.

This may require physical inspection of the vehicle fleet, inspection of leases over property and letters of confirmation from its bankers. There should be investigation into the intangible assets of Fronnt to establish what they are and how they have been valued and how are they being amortized

Quality of managers and operations

Runn is buying a going-concern and will rely on its income and the quality of its systems in the short to medium term. These should be reviewed to establish whether they are reliable

A management audit could be conducted by visiting depots, review of HR records and interviews. A review of operations could involve discussions with customers and staff, observation of processes, and review of any incident logs of client complaints

Quality of client relationships

Fronnt may have some key clients and business partners. The contracts with these could be reviewed to assess whether they are long term or whether they could be cancelled on acquisition

Interviews with key clients could give indications of how satisfied they are with Fronnt, whether they are willing to use Runn after the acquisition, as well as their likely volume of consignments over the coming years.

There should be communication with Fronnt's large clients to reassure them of service continuity after the takeover of Runn.

Pension liabilities
Fronnt may have a staff pension scheme which it is required to pay into or perhaps to make payments to retired staff. Due diligence should assess whether the pension fund is sufficient to meet future liabilities, and what payments Runn may need to make to staff in the future.

Contingent liabilities
These are financial costs that are not presently clear. Examples are pending legal cases against Fronnt and pending tax judgements that could go against it. These should be included in the liabilities of Fronnt but often are not because management is confident of winning the case

If they crystallize, they will become the responsibility of Runn to pay

Review of current legal actions pending should be reviewed to assess whether the value of contingent liabilities in Fronnt's accounts are realistic or possibly undervalued.

Problems in using Internal Audit to conduct due diligence
Internal Audit is a review and oversight process which is conducted by persons appointed by and reporting to the management of the firm. This is contrasted with external audit which is carried out by independent contractors and which usually reports to NEDs and shareholders.

There are a several problems in relying on IA for conducting due diligence into this acquisition

Lack of independence
IA is employed by, and reports to, the management of Runn. If the Board of Runn wants to buy Fronnt the IA team may be unwilling to bring bad news to the Board about what they find

There is a danger that IA would do a cursory assessment and feel under pressure to approve the suitability of Fronnt to please their bosses.

Lack of experience
Due diligence requires consideration of a broader range of issues that would be covered by the experience of the conventional IA. For example, assessing the quality of management, or the attitudes of key clients, requires expert HR and Marketing skills that IA may not possess. This could reduce the effectiveness of the due diligence investigation.

Implicated by things they miss
The IA function is a permanent function, unlike external audit where it is preferred to have a rotation of audit staff and periodic change of audit firm. If IA conducts due diligence into Fronnt and misses something important it may be tempted to assist in covering this up in future years to avoid embarrassment. For example, if particular assets are not as valuable as they are reported, but IA misses this, it may be willing to leave them overstated in subsequent years.

Reduces ability to audit Runn in meantime
Presumably IA is fully-employed in conducting its own program of IA activities in Runn. If it is diverted into the major and time-consuming work of conducting due diligence into Fronnt it cannot assure adequate control in Runn in the meantime. This could lead to undetected control weaknesses in Runn

This might also increase the annual audit fee for Runn because the external auditors feel they cannot rely on Runn's internal controls to the same extent as before and have to assess them at a cost to Runn

<u>Biased by experience of Runn</u>
IA will be very familiar with the way that Runn operates. This may lead it to assume things about the way Fronnt works and not actually check them. This could mean that weaknesses are not detected

IA may also believe that Runn represents best practice in the CEP industry and therefore if Fronnt does things differently IA may see this as a weakness. This could lead to unfair assessment of Fronnt

Having external teams involved in the due diligence means there would be the benefit of fresh eyes to assess Fronnt

Chapter 19
Cyber Risk Management

1. Approaches to cyber risk management

Basic terminology
Information Systems (IS) refers to the information capabilities of the organization. Big Data, e-commerce, Enterprise Resource Management, transactions systems, email and teleconferencing are examples of information systems

Information Technology (IT) refers to the hardware, software, and communications technologies used to deliver the Information Systems.

Approaches to assessing risk
Cyber risk management is the subject of a number of frameworks, including ISO/IEC27005, Information Security Forum's Information Risk Assessment Methodology (IRAM) and national systems such as the USA's NIST system (National Institute of Standards and Technology), or the UK's National Cyber Security Centre assurance standards.

These adopt broadly two elements to assessing the risks in the firm's cyber systems:

Approaches to cyber risk assessment

Component Approach
Considers each element of the system separately
- Impact
- Vulnerability
- Threat

System-based approach
Looks at ability of system to deliver goals of the business

1. Component-based approach
   Focuses on each application and assesses the risk it is subject to. Typical components include:
   - Hardware (computers, servers and so on)
   - Software
   - Communications systems
   - Personal information
   - Business critical information

   Assessing component risk involves assessing three things:
   1. *Impact*: the consequences of the risk being realized such as lost sales, litigation, loss of life and so on

2. *Vulnerability*: assessment of the weaknesses in the component that might lead the risk to be realized. This might include the quality of the existing hardware/software, the people that operate it, its physical location and so on
3. *Threat*: the individual, group or circumstance that could cause the threat to be realized. This could include lone hackers, state actors, simple staff errors, adverse weather and so on

This allows management to set priorities to fix the components with the most severe risks first.

This approach is useful for identifying specific vulnerabilities such as the need to upgrade a particular application. It seeks to protect the existing information systems. It does not address the broader issue of whether the systems are doing what they need to do to support the business.

2. Systems- based approach
   This assesses the goals of each system, the goals it serves and what it seeks to achieve, rather than focusing on each component. Then the factors that could stop it from achieving these goals can be identified by working backwards from the goals.

   The approach will assess the *losses* to the business if the system does not deliver the goals that were intended.

   This accepts that all the components of a systems may be functioning as intended planned but still the overall firm is at risk from a failure of its information systems to deliver what is needed.

   This is better for complex systems where the interactions between separate applications may be the source of risk.

The CGMA syllabus on cyber risk
The syllabus treats cyber risk as the danger of information systems being vulnerable to attack.

2. Sources of cyber threats

Source	Examples	Controls
Malware	• Viruses • Ransomware • Trojan horses • Worms • Spyware • Scareware • Phishing	• Anti-virus software • Firewalls • Anti-spam filters • Updates/patches • Blockers • Multi-factor authentication • End-user device controls
Application attacks	• Cross-site scripting • SQL injection • Buffer overflow • Automated threats • Botnets	• Input validation • Data sanitization • Escaping data • Vulnerability scanning systems

You are required to recognize the following threats:
1. Malware
   'Malicious software' intentionally designed to cause damage to an information system.
   Examples of malware:
   - Viruses: hidden inside another program and then copies and grows inside the system to destroy data or render the system unusable

- Ransomware: locks the screen or system and demands payment of a fee to release the system
- Trojan horses: an innocent program that, once installed, acts to destroy the system or to provide a 'backdoor' to allow hackers to enter the system
- Worms: programs that reproduce through the system and use-up processing and storage capacity and can sometimes be used to transmit data, such as bank details, back to the originator of the worm
- Spyware: this gathers information about the user of the system. Adware and Cookies are examples, although the user normally consents to these first. Malign spyware operates without permissions and includes key-loggers that transmit keystrokes, and web beacons that tell the originator when the user accesses certain sites
- Scareware: pranks the user by causing anxiety or panic leading them to pay money, reveal information, install further malware, or delete parts of the operating system of their computer
- Phishing: 'phony fishing' fraudulent emails to elicit information or money

Malware can be defended against by security controls such as:

- Anti-virus software: blocks installation, alerts user and removes malware
- Firewalls: present a barrier between the trusted internal information system and the untrusted external system such as the internet. It may block connection or it may monitor the data traffic between them to look for suspicious items
- Anti-spam filters to protect against junk mail that may be phishing or contain malware
- Installing Regular updates to software and anti-virus protection ('patches')
- Blockers installed against pop-up advertisements and to limit size of downloads
- Regular changes to passwords
- Multi-factor authentication processes: requires the user to present password plus one or more additional pieces of information (including facial or fingerprint recognition, software protection dongles/keys, one-time codes and so on)
- End-user device protection: locking-down of PC's to prevent uploading or downloading or connection to untrusted networks, policy bans on connecting personal devices and IoT devices to business systems

2. Application attacks

Information systems use applications to access and manage data. Examples include web browsers, operating systems, and particular software applications. These may contain 'vulnerabilities' which can be attacked and used by hackers to gain access to the corporate database or to disrupt the business.

Examples include:

- Cross-site scripting (XXS attack): malicious code written across from a trusted website that creates a program within the business system
- SQL injection (SQLi): a piece of machine code in a Structured Query Language is inserted and gives the originator administration rights over the database of the business
- Buffer overflow: the memory of an application is deliberately over-burdened by trivial data and this causes it to crash or to overwrite valuable data
- Automated threats: software inserted to perform heavy amounts of repetitive tasks. These can scrape data, corrupt data, or result in a Denial of Service (DoS) because the system is slowed down by the application attack
- Botnets: a network of robot computers under the control of a malicious third party. Malicious software is inserted into each device to give control to the originator who then uses the network for the gathering of data, or sending phishing or virus-laded emails, or posting messages to social media. Distributed Denial of Service Attacks

(DDoS) occur when the botnet is used to bombard the system with traffic such as queries, postings, or data

Defenses against application attacks include
- Input validation: automated system that prevents importing of pieces of code that contain applications (like email systems that don't automatically download pictures)
- Data sanitization: converts all received data into harmless data (like removing active links from emails)
- Escaping data: received data is quarantined and inspected automatically before the user can view it or save it to the business systems (like a junk mail folder)
- Vulnerability scanning systems: proprietary systems that analyze scripts, check for bots and so on

3. Social engineering attacks
These are the hacking of people though impersonation rather than direct hacking of systems. They can include fake phone calls or emails that elicit passwords or log-in details, infected emails that when open infect systems, and fake surveys and application forms that obtain bank details and other information

4. Hackers
A person who can subvert cyber security systems
- Ethical hacker: person who conducts penetration testing to identify weaknesses and help improve systems
- Gray hat: a hacker that may break ethical or legal standards to penetrate systems but does so without malicious intent
- Unethical hacker ('cracker'): a hacker who penetrates systems for personal gain or from malice

Four main motives for hacking:
1. Criminal financial gain: for example, stealing credit card and bank account numbers
2. Improving reputation in the hacking community
3. Corporate espionage: stealing information on products, services or clients
4. State-sponsored attack: weaken other nations and steal defense secrets

3. Consequences of cyber vulnerabilities

Academics at Oxford University identified 57 adverse consequences of cyber vulnerability classified under 5 types. The key ones are summarized below.

1. Physical/digital
   - Damage and infection of systems and cost of detection and repair
   - Identity theft
   - Reduced performance
   - Bodily injury/pain or loss of life
   - Abuse and mistreatment of individuals
   - Prosecution for poor cyber security

2. Economic
   - Disruption to operations and profits
   - Fall in stock price
   - Theft of finances
   - Extortion payments
   - Compensation payments
   - PR response costs

- Investigation costs

3. Psychological (suffered by customers, staff etc.)
   - Confusion and discomfort
   - Anxiety and depression
   - Shame and embarrassment

4. Reputational
   - Reduced public perception and corporate goodwill
   - Damaged relationship with suppliers and customers
   - Reduced business opportunities
   - Media scrutiny
   - Loss of certifications, licenses and accreditations
   - Loss of key staff and inability to recruit suitable staff
   - Reduced credit scores/increased risk

5. Social/societal
   - Negative social perception of the technology
   - Fall in internal morale of the business
   - Negative impact on the nation

   https://academic.oup.com/cybersecurity/article/4/1/tyy006/5133288

4. Summary of cyber risks and responses

CGMA published a cyber security tool
https://www.cgma.org/resources/tools/cgma-cybersecurity-tool.html

This is summarized below. Note that it explicitly includes the business risks from having the wrong systems or wrong management approaches in addition to cyber vulnerabilities discussed above.

Types of IT risk

Risk	Description	Controls
Strategic IT risks	• Failure of IT/IS to align with the business and competitive needs of the organisation	• Appointment of Chief Information Officer (CIO) • Develop information strategy • Maintain senior management awareness of development and potential of IT/IS
Economic risks	• Expenditure on IT/IS excessive and outweighs the business benefits • Failed IT/IS implementations	• Decentralise IT/IS expenditure to business managers • Base investment on clear deliverables and review progress periodically • Utilise successful project management (e.g. PRINCE2) and systems development methodologies • Outsource projects and operations to transfer financial risks
Availability risks	• Inability to access IT/IS systems needed for business due to technical failures • Overload of systems • Malicious Denial of Service attacks (DoS)	• Ensure testing of all hardware and software before introduction • Planned maintenance and service outages • Ensure physical security of systems • Back up sites and disaster recovery procedures • Outsource to third party provider • Load testing to assess capacity • Intrusion Prevention Systems (IPS) to automatically monitor messages to check they are legitimate, or to watch for anomalous traffic flows

Types of IT risk

Risk	Description	Controls
Security risks	• Loss of data confidentiality • Loss of data integrity due to manipulation/hacking • Loss of processing due to modification or destruction of systems	• *Protection approaches*: access identification and authentication, authorisation levels, data encryption and certificates, lock-down of personal device functionality, screening of staff and contractors • *Detection approaches*: event monitoring, intrusion detection, threat monitoring and threat intelligence (monitor the experiences of the IT community), user reports • *Response approaches*: Computer Incident Response Team (CIRT) to reduce losses, restore operations, manage crisis communications, provide decision support, and support investigations
Compliance risks	• Failure to follow laws and regulations	• Close monitoring of legal and regulatory environment • Systems audits and compliance tests • Use of Computerised-Assisted Audit Tools and Techniques (CAATTs)

Types of IT risk

Risk	Description	Controls
Performance risks	• Reduced productivity/profits due to slow access to IT/IS by staff and customers	• Appropriate supplier/vendor selection process • Clear Service Level Agreements with in-house or outsourced IT/IS providers • Uniform operating procedures across systems • Steering groups to report problems and concerns • Perfective systems maintenance and development • Screening/prioritisation of user (e.g. key customer identification • Adopt distributed processing (e.g. Cloud processing)

5. Cyber risk reporting

Cyber risk dashboard

Boards have a responsibility to monitor the controls over cyber risk. They will need reports from the CISO (Chief Information Security Officer) that answer these questions:

- How much cyber risk does the firm have?
- Which are the main risks
- Is the firm's cyber risk position improving or degrading?
- Is the level of cyber security spending too great or too little?
- What the cyber risks associated with a new initiative

A cyber risk dashboard is a graphical display that summarizes the cyber risks of the business. These are provided by consultancies and they differ from each other. Typical contents include

- A risk heat map showing the impact and probability of each cyber risk
- A financial analysis showing the potential costs of each risk against the firm's present expenditure on mitigating it
- The status of risk analysis and investigation activities
- The status of action plans to reduce identified vulnerabilities
- Status of systems against compliance benchmarks

ACTIVITY 19.1

Context
Denby Healthcare is a listed company that provides hospital and health consultancy services in the country of Keeland.

The market is competitive due to the existence of three other private competitors and a large state-funded service, the Keeland Heath Service (KHS) which is provided free to patients.

Denby's commercial success depends on having shorter waiting times than the KHS and being able to provide a higher quality service.

Denby is considering investing in smart hospital technologies to match a similar initiative by the KHS than has been announced. The main features of these smart hospital technologies are:

- Reduce reliance on referrals from the local clinic service of the KHS by creating its own virtual clinic service using online consultations with patients
- Increase the use of Artificial Intelligence (AI) in the diagnosing of patient symptoms and also in the interpretation of results from scans and blood tests.
- Significantly increase the data it takes from patients to help it to utilize Big Data and Analytics to improve the effectiveness of diagnosis and treatments and reduce reliance on the judgement of senior doctors
- Improve patient aftercare by real time monitoring of their health and behavior using wrist bands and encouraging patients to submit on-line personal medical diaries

Task
You have joined the Smart Hospitals Working Party at Denby Healthcare. The task of the Working Party is to introduce several of the initiatives that have been under discussion

Chantelle Murphy Coordinator *(Turning to you)*	I want to return to the risks and problems of what we are introducing. This is all a new step for Denby. What risks are there in investing in these new AI and Big Data applications? (sub section (a) 25%)
Shiv Mathur Technology Consultant	Don't worry, lots of firms like Denby show signs of fear when they start to use advanced IT, but it comes out okay. The firms you are using to install this and your outsource partners know what they are doing
Chantelle Murphy Coordinator *(Turning to you)*	Thanks Shiv, but I think we do need an independent view too How can our Internal Audit function help reduce the risks of this project to invest in AI and Big Data applications? (sub section (b) 25%)
Mike Tucker Clinical Staff Lead *(Turning to you)*	I have to tell you that doctors, nurses and administrators are very unhappy about these new applications Why are clinical staff so resistant to these new technologies and how can we change their behavior to make them more willing to use them? (sub section (c) 50%)

Solutions to Activities

19.1 Investment in smart hospital technology by Denby Healthcare

This activity involves topics from previous chapters on change management and internal audit as well as cyber risk.

What risks are there in investing in these new AI and Big Data applications? (sub section (a) = 25%)

Risk in new AI and Big Data applications

Risk refers to the tendency for outcomes, including financial outcomes, to be different from what was anticipated. Risk is assessed as impact of event x probability of event. Risk can be positive (better than expected) or negative (worse than expected).

<u>Strategic risks of investing in the wrong systems</u>

The investment is based on the assumption that AI and Big Data and Analytics will modernize and improve Denby. Many of the AI applications are in early stages of development and deployment and by locking itself in one form, say wrist bands, Denby may lock itself out from using another form, such as smart contact lenses, that might be better

It also assumes that patients and health professionals will welcome the use of AI to improve treatment, whilst there could be a backlash against it as patients decide to seek 'traditional' medical care as a better alternative to the technology-enabled approach of KHS and Denby.

This wrong-turn would harm Denby and take years to recover from, as well as wasting investment funds

<u>Economic risks</u>

There is a danger that the investment in AI and data would not make a good return to investment

It would also potentially reduce the funds available for other projects, such as building new hospitals or installing new capital equipment.

Many complex IT/IS projects run over-budget as unexpected barrier are found, or that the project is poorly managed and allowed to expand and extend without control.

Such excess expenditure could disappoint shareholders and threaten Denby's share price.

<u>Availability risks</u>

Using AI to conduct diagnostics means that the work of the hospital would be seriously affected by a breakdown in the systems. These breakdowns could come from overload, technical failures, or deliberate malicious denial of service attacks. Research by Neustar, a US IT security consultancy, reports a 17% rise year on year in the number of attacks on healthcare sector IT systems often with the purpose of blackmail but also malicious and intended to end patient's lives

Such breakdowns would seriously harm Denby's reputation, and lead to compliance and litigation risks.

<u>Security risks</u>

Patient and marketing data can be compromised by accidental loss, by hacking, or by inappropriate distribution by poorly trained staff. Loss of patient data would bring huge reputation harm to Denby

For example, in 2015 the US healthcare firm Anthem Blue Cross reported that the health records of 78.8m patients had been stolen. These contained personal health details but also information on credit card numbers and Social Security numbers. It has been estimated that the cost of losses from this and of rectification, to each patient, exceeded US$2,500.

<u>Compliance risks</u>
The use of Big Data and personal data is coming under increased legislative pressure, especially in the wake of revelations about Facebook's lenient policies on sharing users' data with third parties

For example, the GDPR of the European Union (General Data Protection Regulations – law from April 2018) that imposes considerable fines for poor governance of personal data

Failing to comply with rules will be a source of high fines or possibly closures of systems and denial of licenses (as has increasingly happened to Uber). But compliance may impose very high costs on Denby too as the regulations evolve.

<u>Performance risks</u>
Introduction of the new systems will require considerable staff training, which represents an expensive use of front-line staff. Staff will also be constrained to work at the pace of the technology, for example waiting for results.

This could increase costs and frustrate patients

How can our Internal Audit function help reduce the risks of this project to invest in AI and Big Data applications? (sub section (b) = 25%)

Role of IA in reducing risks of AI and Big Data project
Internal Audit (IA) is an independent function that monitors, assesses, and reports on the effectiveness, efficiency, and economy of the systems of the firm

<u>Gain specialist expertise and knowledge of the project area</u>
These new technologies, the practices they require to support them, and the compliance issues they raise are all largely new to Denby's management. The present IA function is therefore unlikely to possess the expert knowledge it needs.

This could be overcome by additional recruitment and training but is probably better dealt with by outsourcing the internal audit to expert external consultants initially.

This will enable early involvement in monitoring the project to avoid it going too far wrong.

<u>Review the processes of the project</u>
IA should use inspection, interview and observation to identify the systems being used to develop AI and Big Data and Analytics in Denby and to assess their quality. This would involve reviewing the project management processes being used and also the supporting documentation being generated.

It would also require IA to review the business case for the project, including whether the investment was based on clear and feasible deliverables

The financial controls on the project would also be reviewed, such as how costs are assigned and where sign off of next-stage funding occurs

The process by which consultants and hardware and software will be chosen should be assessed before any appointments or purchases are made. Background checks into the quality of the consultants and proposed technologies should be undertaken.

Any weaknesses in these areas should be noted and brought to the attention of the project managers and the Board

<u>Conduct testing of the project</u>
Having assessed the *intended* (or professed) systems that will be used to manage and deliver the AI and Big Data solutions, the next task of IA is to establish the extent to which they are being applied

IA will use inspection (of documents and files), interview (with participants) and observation (of what is going on). This will give confirmation that the systems are being applied as intended

This would take place at various stages in the project, from the selection and creation of the project team, through testing of the operations of the software and hardware, to the actual results of the AI and Data systems when they are rolled out.

<u>Assess the compliance procedures</u>
IA would be particularly focused on the compliance procedures used by Denby because these are the source of greatest potential risk from data theft and litigation. This assessment would be a combination of assessing impact (what the damage could be) and probability (where the controls are likely to be weak based on observation, history, or the intrinsic nature of what is being done) (*student note: this is an application of 'risk-based auditing' which means to prioritize limited IA resources, or address urgent problems, by identifying where the greatest risk lies).*

<u>Report findings</u>
The IA function will report the findings from each area of investigation to the Audit Committee of Denby, and to the management of the project and possibly the managers of each process

Conclusion
The ability of the IA function to reduce risks depends on the power of these reports and the degree of senior management support and respect for its work. For example, a contributory factor in the 2008 global financial crisis was the unwillingness of senior banking management to empower IA and respond to such warnings as came forward because senior management was not willing to disrupt the process of making money

Why are clinical staff so resistant to these new technologies and how can we change their behavior to make them more willing to use them? (sub section (c) = 50%)
Note that this sub-section contains two questions, separate by 'and'. You should assume its 25% per part.

Reasons for staff resistance to new information technologies
Resistance to change refers to when the staff do not adopt the behaviors required to make the change successful. For the AI and Big Data strategy to be successful it will need medical and administrative staff to ensure data is entered into the system and also to use its outputs in their work

This resistance can come from several sources:

<u>Resistance to de-professionalization of their role</u>
Many of the applications of AI are presently being conducted by trained professionals. For example, trained radiographers will assess the results of scans and alert doctors to areas of

concern. The doctors too will use their professional skills in interpreting results. Trained professionals have invested years in gaining the skills needed and are part of a community of practice in which their experience is valued.

They will resent the use of AI and will probably seek to cast doubt on its effectiveness and will ensure they highlight its failings. This could lead to a general suspicion with Denby about whether patient diagnosis and care is still to the highest professional standard.

Fear for their role security
One purpose of the investment is to make Denby more efficient and staff will conclude that this will mean fewer jobs in the areas mechanized by AI. This could lead to staff losses as staff seek for secure roles elsewhere, or management may find itself boxed into making concessions and promises to staff to avoid strikes or other actions.

These staff losses and concessions may limit the effectiveness of the way the technology is used

Suspicion of the reliability of the technology
Medical staff will have professional ethics that require them to avoid reliance on technologies they fear could injure patient health. The quality of AI may be suspect and therefore staff may feel professionally obliged to refuse to rely on it.

This would cause the investment to be entirely wasted.

Misunderstanding of requirements of the technology
Resistance is not always opposition. If staff are not sure of how to use the technology, what it can do, of what it demands of them they will not be able to ensure that best use is made of it.

This would reflect failures of communication or problems in staff training.

Resentment at loss of status or control
This initiative is being led by commercial rather than medical needs. The clinical staff of Denby will resent this intrusion on their area of power and practice.

It will be seen in political terms as a 'grab for power' by the commercial function and perhaps may be accompanied by a push-back from the medical functions.

Changing behavior of clinical staff to support introduction of new technology
The Lewin Ice Cube Model provides a helpful framework to discuss an orderly change process

Unfreeze
Educate and communicate the changes to the clinical staff
Many staff will be concerned when they hear of the proposed changes and there is a danger that rumors will overstate the negative impacts. Management must control the message of the changes by emphasizing the benefits

It would probably be advisable to tell each team of staff how specific technologies will benefit their work. This is better than announcing a broad re-engineering of the business because most staff will not care about changes that don't affect them, and it could increase unease at the greater risks from a broad project.

Announcements should be accompanied by timescales for change and the proposed training

Change
Encourage participation and involvement

Creating working parties for each function, or an overall project steering group for the project will help reduce the feeling of loss of control. It will also enable management to develop the systems using the knowledge and input from the teams that will use it or rely on it.

Keeping all staff informed of progress will also create a positive motivation around it and help them to adjust to the fact that the project is progressing.

Facilitate and support the transition to new ways of working
Change is problematic for staff, especially those who have limited understanding of IT or its uses

Denby could employ change agents to assist with the transition to the new systems. The change agents would take the role of intermediaries and experts who are able to listen to staff concerns and to represent them to management.

Negotiate and agree changes with clinical staff
If there is resistance from key clinical groups, such as Senior Doctors or perhaps diagnostic technicians, these groups should be addressed individually

Negotiation over minimum staff numbers, pay and conditions, or joint agreement before the systems are allowed to go live could help resolve these objections.

Negotiation may delay the project, or make it cost more, but if the staff are high-value human assets there is no alternative if Denby wants any changes to take place.

Refreeze
Publicize the successes of the project
Refreezing involves gaining widespread acceptance of the resulting change which turns it into the new ‘business as normal’. One way to do this is to ensure that staff are fully aware of the gains from the new systems, for example improvements in diagnosis, or improvements made to the jobs of the staff involved.

These can be communicated in a blog, a newsletter, or a monthly email from the CEO

Chapter 20
Foreign Exchange Risk

1. Sources of foreign exchange risk

1. Transactions risks
   These occur where the organization is making or receiving payments in different currencies
   - Buying components in a foreign currency creating an obligation to pay in that currency
   - Selling products in a foreign currency therefore receiving revenues in that currency
   - Borrowing funds in a foreign currency creating an obligation to pay interest and principal back in that currency

   A rise (appreciation) in the value of a foreign currency will make payments cost more in terms of domestic currency. This may squeeze profits and harm cash flows.

2. Translation risks
   These occur when the organization has assets denominated in foreign currencies (e.g. cash, inventories, premises). A fall (depreciation) of the foreign currency reduces the value of these assets when expressed in terms of the domestic currency. This will be treated as a reduction in shareholder equity/profit

3. Economic risks
   Changes in the exchange rate affect the competitiveness of the exports of firms in different countries.

   A rise (appreciation) of a firm's home currency will
   - Make its exported products more expensive to buyers in foreign countries. This will reduce demand for them
   - Make imported rival products cheaper and so intensify competition in the firm's home market

   To the extent that the organization sources its materials and components from a different country a fall in its domestic currency will make these imported materials more expensive and reduce margins or force price increases that will reduce demand for its products

2. Exchange rate determination

Despite this exam being promoted as a practical examination some past sittings have featured technical questions on theories of exchange rate determination and whether they are reliable for forecasting exchange rate movements

The best approach to these questions is to describe the theory but to add that other real work political and trade factors can disrupt the theories in the short to medium term which, after all, is the period that the firm need to understand in order to hedge its cash flows and understand the likely impacts on its sales and costs.

More recently the examiner has included information about the inflation rate or interest rates of the foreign economy in the task and has expected candidates to recognize the likelihood that the exchange rate will rise, or fall, and the implications of this.

Economics approaches to predicting exchange rates

Approach	Description
Relative inflation rates (Purchasing Power Parity – PPP)	• 'law of one price' means identical goods will cost the same regardless of the currency in which they are sold • If goods are generally cheaper in Country X, due to its low inflation, there will be *arbitrage* as traders buy them and sell them at a profit in Country Y (high inflation) • The high *demand* for currency of Country X, to pay for the goods, will raise its exchange rate relative to Country Y which is *supplying* its currency to pay for the goods and flooding the foreign exchange market • The exchange rate of Country X will continue to rise until its goods cost as much for a trader to buy as identical goods made in Country Y and so eliminate the arbitrage possibility • Therefore the rate of decline of the currency of Country Y will be equivalent to the percentage by which its inflation rate exceeds the inflation rate of Country X
Interest rates (Interest Rate Parity Theory – IRPT)	• Based on the rule that investors will invest their money to get the highest interest rate • If USA interest rates are higher than European interest rates then investors would convert their € into $ to invest in the USA, say for a year • But the extent to which the investor is actually better-off will depend on the rate at which they can convert their $ back to € at the end of the year • If the exchange rate of the € is expected to weaken compared to the $ then European countries will need to offer a higher interest rate to investors in its currency to compensate for this • Therefore a high interest rate denotes a currency who's value is expected to weaken

Economics approaches to predicting exchange rates

Approach	Description
International Fisher Effect (inflation again)	• Based on the Fisher Identity: nominal interest rate = real interest rate + inflation rate • Investors in all countries will require the same real rate of return on their investment (to compensate them for loss of liquidity, risk and so on) • Therefore the differences in interest rates between countries is solely due to differences in expected inflation rates • This means that differences in nominal interest rates can predict inflation and therefore, using PPP, future exchange rates

External imbalances	• This is sometimes referred to as 'the balance of payments'
	• An external deficit (less received from abroad than is spent abroad) creates an excess supply of the domestic currency seeking to buy foreign currency
	• This causes the price of the domestic currency to depreciate
	• Causes of external imbalances
	    • Adverse movements in the terms of trade, such as fall in price of major export or rise in price of major import (e.g. oil)
	    • Excessive demand in domestic economy drawing in imports and diverting exports
	    • Long-term competitive weakness of the domestic economy

Other factors impacting exchange rates

Approach	Description
Government and political factors	• Governments may intervene to manipulate exchange rates to stimulate exports and deter imports • Governments may deliberately peg their currency to the currency of another country • Changes of government may be good or bad for external balance, inflation and interest rates depending on their policy priorities • Governments may have reputations for being responsible or irresponsible with state finances and the exchange rate
World commodity prices	• Prices of key minerals and energy sources are volatile and this will affect demand for some currencies • Countries dependent on key exports will be affected by problems in production (weather, labour disputes)
Speculation	• Foreign exchange dealers and fund managers will seek to make a profit from betting on movements in a currency • Speculation will respond to any rumours about all the other factors mentioned above

3. Hedging exchange rate risk

Foreign exchange hedging

Method	Advantages	Disadvantages
Price sales and purchases in home currency	• Cheap to arrange • Assures profit margin • Assures cash flow in home currency (assuming settled in home currency)	• May put firm at competitive disadvantage if domestic exchange rate rises • Deters customers/suppliers unwilling to assume exchange risk
Leading and lagging invoices Pay when exchange rate favourable	• Cheap to arrange	• Only sort term hedge to cover currency volatility • Requires advance knowledge of exchange rate movements • May lose discounts and goodwill from suppliers
Matching/real hedging - Make purchases in same currencies as sales - Develop debt liabilities in same currency as sales - Develop new lines of business to generate earnings in currency of outgoings of existing business	• Cheap to arrange • Can give upside risk benefits from favourable movements to exchange rate	• Unlikely to give perfect hedging • May lead to unsatisfactory suppliers being chosen in order to get currency advantage • Exposed to counter-party risk: what happens if a customer doesn't pay? • New businesses may be beyond core competences of business

Foreign exchange hedging

Method	Advantages	Disadvantages
Foreign currency money market account - X$ liability: lodge funds in X$ now - X$ revenue: raise loan in X$ and convert to home currency. Pay loan off with X$ revenue	• Cheap to arrange	• Relies on credit rating of borrower • Interest costs payable on loans (but theoretically same as cost of a forward exchange contract) • Exposed to counter-party risk: what happens if a customer doesn't pay? • May require corporate treasury experience
Forward exchange contracts Arrange agreed rate of buy/sell with a bank for future	• Flexible amount and timing • Easy to arrange because traded Over the Counter (OTC)	• Obligations that can't be unwound (if a fixed option contract) • Exposed to counter-party risk: what happens if a customer doesn't pay? • Costs a premium/discount
Currency futures Buying a contract with obligation to exchange within a period at a set rate	• Flexible – date of exercise is at election of holder • Protected by exchange rules	• Not available for all currencies • Can't be tailored • Imposes an obligation to settle • Involves market and broker costs • Downside risk – must buy/sell currency at the agreed rate regardless of spot rate

Foreign exchange hedging

Method	Advantages	Disadvantages
Currency options Buying a contract with option to exchange within a period at a set rate	• Risk limited to value of the premium paid • Protected by exchange rules • Can be tailor-made (OTC or negotiated options)	• Must be paid for as soon as bought • Premium can be high if exchange rate volatile
Currency swaps Agreement between two counterparties to exchange payments on different terms (eg one has variable exchange rate obligation and one has fixed rate obligation)	• Easy to arrange (through broker) • Can be unwound by swapping with another counterparty • Can absorb excess liquidity – firm with too many X$ can create access to funds in another currency	• Default by partner on their obligation (counterparty risk) • Arrangement fees of broker • No market or exchange protection

ACTIVITY 20.1

Context
Runn is a global provider of Courier Express Parcel services (CEP) like UPS, DHL and Fedex. It collects and delivers parcels across 200+ countries. It has depots and operations in most of these countries. It reports its profits in its home currency, the Geeland Dollar (G$)

Task
You have received an email from Chen Yanyun: Director of Finance for Runn
To: Senior Finance Manager
From: Chen Yanyun: Director of Finance
Topic: Foreign exchange risk

The Board is very concerned about the present falling exchange rate and the possibility of Central Bank intervention to raise it.

As you know about 40% of our shipping is within Geeland but Runn's acquisition of several courier firms outside Geeland means 40% of our business takes place in our foreign subsidiaries. The remaining 20% is cross border courier work of which the majority comes from Geeland customers sending parcels to foreign addresses.

The growth in cross-border parcels from Geeland seems to be a consequence of the G$ depreciating over the past two years.

What I need you to do:
Evaluate the foreign exchange risks Runn is subject to and evaluate the potential methods available to manage these

(sub section (a) = 55%)

Advise on methods to forecast exchange rate changes and the likely effects of interest rate changes on our business

(sub section (b) = 45%)

Reference Material
(See the next page)

Geeland Investor

Geeland Central Bank to Act

Continued political turmoil in Geeland has forced the Central Bank to announce its intention to swoop into the foreign exchange market to support the G$. This will almost certainly involve an increase in the domestic interest rate.

Geeland's present government is a coalition of free-market and pro-business political groups. But its popularity is at an all-time low due to its program of raising taxes and cutting government spending to reduce price inflation in Geeland.

The financial markets are showing increasing concern that in the forthcoming election may be won by the Reform Party, led by Wat Tyler. Tyler has promised policies of high state spending and a protectionist policy against imports from outside Geeland.

The threat of a Reform Party victory has led to a 20% fall in the exchange rate for the G$ in the past 9 months.

Solutions to Activities

20.1 Foreign exchange risk for Runn couriers

Sub section (a)
Evaluate the foreign exchange risks Runn is subject to and evaluate the potential methods available to manage these (55%)
Approach: Note the two elements of the task, separated by the 'and'. You must answer both parts. The requirement to 'evaluate' foreign exchange risk' has appeared several times. Its meaning is not clear. It would be impossible to measure the risk due to lack of data in the question. However, in her solution the examiner interpreted it to mean 'assess the expose of the firm to the three potential types of foreign exchange risk'.

Please note that the term 'international customers' and 'international subsidiaries' is used in this solution to denote customers and subsidiaries based outside Geeland. This seems preferable to terms like 'foreign' or 'overseas'.

The second part of the task also requires an evaluation of the methods of managing exchange rate risk. This means to stress the good and bad aspects of each.

This practice task is probably longer than you would face in a real exam. It has been developed to ensure you have a full understanding of this entire area.

Student note: CGMA tends to adopt the UK convention of describing indirect quotation of exchange rates. For example, saying that G$1 = €2 means that it requires €2 to buy G$1. A rising or strengthening Geeland exchange rate means that the rate moves to G$1=€3, so it now it is worth €3 because it will require €3 to buy G$1. The price of the G$ has gone up from €2 to €3. You may be more familiar with direct quotation which, in the above two examples, would be a shift from G$0.50 = €1 to G$0.33 = €1. This can be read wrongly as a fall in the value of a unit of G$.

Evaluating foreign exchange risk
The foreign exchange rate is the price of one currency, say the Geeland currency, G$, expressed in terms of the prices of other currencies. If there were a strengthening exchange rate it would mean that a G$ could buy a greater value of another currency than before and that the G$ was rising in value.

Foreign exchange risk refers to a volatility in the value of transactions or the value of a business arising from changes in the foreign exchange rate.

Transactions risk
This is the risk that the G$ cost of payments may rise, or the G$ value of receipts may fall, on transactions denoted in other currencies if the G$ weakens. If Runn charges for its services in G$ there would be no transactions risk on revenues from parcels sent to Geeland by international customers

However, Runn will need to pay international currencies to cover the costs of transporting and delivering parcels outside Geeland, and it may buy some packaging materials, equipment and vehicles from outside Geeland and potentially it may pay for aircraft landing fees and fuel in foreign currency and would suffer a transactions risk

The weakening of the G$ will have raised these costs of buying outside Geeland would rise and lead to a fall in the profits on its G$ denominated business

Translation risk
This refers to the value of assets or liabilities denominated in foreign currencies.

For financial reporting purposes a firm's assets and liabilities that are denominated in a foreign currency must be converted to G$ for the purpose of reporting Runn's financial position at the prevailing exchange rate *(called the spot rate*) and so the G$ value of these assets or liabilities will rise and fall and be recorded as gains or losses in the profit and loss account

Runn owns international subsidiaries which will conduct their operations and record their assets, liabilities, and profits in a foreign currency (*student note: this is called the subsidiaries' 'functional currency').* The net value of the subsidiaries must be converted to G$ for the purpose of reporting Runn's financial position (*student note: this is its 'reporting currency')* at the prevailing exchange rate and so the G$ value of these subsidiaries will rise and fall and be recorded as gains or losses in the foreign currency translation reserve, a part of Runn's equity

Therefore, Runn has significant translation risk would be on the value of subsidiaries, and its receivables and payables denominated in foreign currency

Economic risk

This is a type of risk that refers to the loss of competitiveness of Runn's services or, more likely, in the competitiveness of the exporting clients on whom Runn depends for some business

If the G$ strengthens it will cost international customers more of their foreign currency to buy a given amount of G$ to pay for Geeland exports and to pay Runn to deliver their parcels in Geeland (assuming that Runn charges for these in G$)

Managing foreign exchange risk
(Student note: the description below is very full and describes the workings and risks of each hedging technique. This is to ensure that you can revise it to understand this complex area of financial management)

Most hedging techniques are used to reduce transactions risks.

<u>Price Geeland services in G$</u>
This will ensure the G$ value of the revenue received from an international delivery (one delivered outside Geeland or one collected from outside Geeland and delivered in Geeland) is fixed and not subject to changes in the G$ exchange rate

However, it does have the effect of causing the economic risk described above

<u>Price international services in foreign currency</u>
Runn could adopt the policy of airlines and hotels and charge customers in a non-Geeland currency

To some extent this would remove economic risk on overseas customers, providing the currencies chosen to invoice them were not appreciating against the customer's domestic currency (i.e. agreeing to charge Asian customers in US$ rather than G$ could be attractive unless the US$ started to appreciate too)

However, this would create transactions and translation risk because Runn's G$ revenues, and the value of its receivables at year end, would now vary according to movements in the G$

<u>Matching</u>
Matching of costs and revenues in the respective currencies. This may be part of the reason why Runn has acquired international subsidiaries

These would charge for their services in the local currency and pay the costs of operations and deliveries in the local currency. This could reduce economic risk but it would lead to transactions risk and translation risk

The translation risk arises from the fact that Runn now has foreign currency denominates assets. The transactions risk would mainly arise from the repatriation of profits from the foreign subsidiary back to Runn in Geeland because the exchange rate would influence the G$ value of these profits

But developing subsidiaries outside Geeland involves incurring many additional risks (political, social, technological and economic) which have to be considered beside the reduction of economic risks on international business

<u>Time payments to coincide with favorable exchange rates (leading and lagging)</u>
Runn could make payments in foreign currency when the G$ is stronger against foreign currencies This would reduce the amount of G$ needed to pay them

One obvious drawback is that suppliers are unlikely to be willing to wait for payment and would probably start to charge interest on overdue payments or cease supply. Receipts should be advanced whilst the G$ is low.
However, clients might require discounts to bring forward payments. If the G$ is expected to fall, this would mean that Runn would seek to pay ahead of time and this would disrupt its cash flows. It also suffers from the drawback that Runn is very unlikely to forecast future exchange rate movements with any certainty

Runn could charge and pay in foreign currencies and then neutralize the transactions risk using a hedging technique from one of the following financial hedging techniques.

Foreign currency money market account
Runn could charge foreign currency prices for its deliveries and collections (say US$ to customers from the USA). It could then raise loans in US$ to the value of its expected revenues in US$

This money from this loan could be converted to G$ at the present foreign exchange rate to give a fixed and certain amount of G$

When the revenues are received in US$ they can be used to repay the loan (*student note: this is a rather simplistic explanation but suitable for this exam. In practice the corporate treasury division of Runn would already have a loan accounts set up in a variety of currencies and would draw down and repay these loans as the net surplus or deficit of Runn in each currency changed)*

Forward exchange contracts
This is an agreement with a bank to buy, in the future, a given amount of foreign currency for an agreed amount of G$.

This would allow Runn to charge for its services in foreign currencies and use the forward exchange contract to hedge the transactions risk. The bank assumes the foreign exchange risk on this contract

This is a simple process for Runn and would not require special expertise in its corporate treasury function

The risk created by a forward exchange contract would come if Runn fails to generate the anticipated revenues and therefore ends up with an insufficient amount of the foreign currency to pay across to the bank. It would need to buy extra amounts of the foreign currency on the foreign exchange market and therefore be exposed to exchange rate risk again

Currency futures
Runn could charge for its Geeland services in a foreign currency to eliminate economic risk and the manage transactions risk through buying ('taking') a currency future which gives it the obligation to sell a given value of foreign currency in exchange for a fixed value of G$ and a fixed date in the future

Without the future, if the foreign currency fell in value against the G$ during this period it would mean that Runn's earnings from sales would translate to a lower value of G$ than had been anticipated. The currency future means that Runn can convert the foreign currency to G$ at the contracted rate and so avoid this loss

Alternatively, if the G$ has risen above the value in the currency future, Runn could sell this future to a third party because it will have a market value, in G$, because it allows the new

owner to convert their receipts of the foreign currency to G$ at a better rate than the spot rate. The profit Runn makes on selling the future will compensate it for the fall in earnings from the sales due to the falling value of the foreign currency
Futures carry two main risks: firstly the risk that if the foreign currency strengthens instead of falls, Runn still must sell its foreign currency earnings at below the new spot price because it need to close out its futures position (i.e. buying the future deprives Runn of a potential upside gain); secondly if Runn does not receive the anticipated foreign currency payments it will be forced to seek for the foreign currency on the spot market to close its futures position, incurring further exchange rate losses.

Futures are only available on certain currencies, and futures for obscure currencies may not be available so it may not be able to eliminate risk on customers from some countries where there are not futures contracts written for the currency

<u>Currency options</u>
Currency options operate in a similar way to futures except that there is no obligation on buyer/taker of the option to sell the currency.

Currency options require Runn to pay a ‘premium’ to buy the option and the benefits from the option are forfeited if Runn does not use the option. However, if the G$ moves advantageously for Runn (for example it weakens so that a given amount of foreign currency can buy a greater amount of G$ than before) the option is ‘out of the money’ and Runn will let it lapse and use the spot rate instead

Unlike futures, which are standard contract sizes, options can be tailored to Runn’s foreign exchange exposure. Options can be sold/assigned to others by Runn. Therefore, if they are ‘in the money’ (i.e. exercise price for foreign currency is more favorable than the market price) they will have a market value.

<u>Currency swaps</u>
Runn could arrange currency swaps with other companies who have the opposite problem, such as firms in Geeland that buy things from overseas and therefore need to make payments in foreign currency. Both parties to the swap could be put into a win/win position

However, this approach may not be advisable because Runn will not be certain of the value of foreign currency sales it will receive from overseas patients nor whether these patients will actually pay their bills

Sub section (b)
Advise on methods to forecast exchange rate changes and the likely effects of interest rate changes on our business (45%)
The Examiner has set questions on forecasting demand, legal matters and the general state of the economy in past exams. She has also set questions on factors that are believed to determine exchange rates, such as inflation and interest rates. This task combines these into one question.

Again, it is to be hoped that the Examiner will keep her questions simpler than this, and the solution is written to ensure you are fully prepared for any question on exchange rates.

Methods of exchange rate forecasting
Forecasting means making assessments of the likely future behavior of something. In this case, the exchange rate

A difficulty in forecasting will be that the G$ has a different exchange rates for each currency, for example a G$:US$ rate, a G$:€ rate, a G$:RNB rate and so on. This means that a

strengthening of the G$ against the US$ may actually be caused by a weakening of the US$ for its own reasons and in fact the G$ is not strengthening against other currencies
A distinction should be made between precise prediction of the future exchange rate and forecasting its general direction and likely range of values

A foreign currency trader tries to predict very precise values over very short periods of time in order to make speculative profits on buying and selling the currency

Runn is a CEP firm, not a bank, and its interest in exchange rates will focus on the likely trend of exchange rates over several months or years so that it can make assessments on how demand for its services may change and decisions on whether to open subsidiaries in new countries

It may also wish to forecast whether the government might try to stabilize the exchange rate and the consequences of it doing this, for example by raising interest rates

The corporate treasury function of Runn must not try to use forecasting to determine its policy on foreign exchange exposure because this will subject it to risk, it should one of the management and hedging techniques described above (*student note: corporate treasury should try to hedge, which means to adopt a position that will eliminate gains and losses resulting from foreign exchange movements but it should not to seek to profit from them because that would also incur risk. The currency dealer is different because they are guessing future rates in order to profit from the positions they take. In currency deals there are inevitable winners and losers under each contract)*

Forecasting relies on understanding the factors that determine interest rates *(student note: in 2018 and 2019 the examiner set tasks that required explanations of all of the theories below and therefore these explanations have been inserted in italics)*

<u>Relative interest rates (Interest Rate Parity Theory)</u>
This refers to forecasting the difference between spot and forward rates of exchange

- A spot rate is the price of the currency if bought or sold immediately
- A forward rate is the price of the currency if bought or sold at some time in the future, usually measured in days or months

The IRPT enables us to predict the forward rate for the currency <u>at a particular point in time</u> once we know the interest rates of the respective countries because IRPT predicts that the difference between spot and forward rates is determined by differences in the interest rates between the two countries

Analysis of interest rates would provide this forecast

Explanation: If the interest rate on US$ is higher than the interest on G$ funds will transfer from G$ into US$ to gain more interest. The forward rate on US$ offered by currency dealers will depreciate, meaning that the dealers will adjust the spot rate using a premium or a discount in order to buy or sell U$ at a poorer rate than the spot rate.

Dealers do this to protect themselves from losses because their banks will hedge their forward position by keeping funds deposited in G$ at low interest ready to supply when they are required to buy back dollars in exchange for G$ in the future. This has the effect of limiting the bank's exposure by stopping further flows of G$ towards US$ because the gains from the higher interest rate in the USA will be wiped out by the poorer exchange rate when the investors seek to convert their US$ back to G$

Student note: given the importance of international banks rather than sovereign states in this process it's probably more accurate to say differences in the interest rates available on deposits denominated in one or other currency
Student note: most text books make clear that forward rates do not predict future spot rates. They are the forward rates available *<u>today</u>* *but they will change tomorrow if the spot rate changes for some reason. However, the text books on which the Examiner relies seem to ignore this and use phrases like 'the country with the higher interest rate will have a falling exchange rate' which is a poor explanation of what IRPT says. IRPT simply says that the country with the higher interest rate will find that its currency trades at a discount on the forward rate compared with the spot rate. I am not confident that your Examiner fully appreciates this distinction and she may believe that a rise in interest rates will cause the spot rate to fall, which is the opposite to established economic theory)*

<u>Relative inflation rates</u>
The Purchasing Power Parity effect predicts that one currency will weaken against another currency in the same proportion as its domestic rate of inflation exceeds the inflation rate of the other county. Therefore, forecasting the future value of the exchange rate over a few years can be partly based on quantitative analysis of the differences in inflation rates

Explanation: this is because if prices were rising faster in Geeland than in other countries it would increase demand for imports by Geeland people because imports would look relatively cheap. This increased import demand would lead to a greater value of G$ seeking foreign currencies, with the effect that the value of G$ would fall on the foreign exchange markets

<u>The International Fisher Effect</u>
This states that differences in nominal interest rates can be used to predict the future of the spot rate.

The nominal interest rate incorporates the financial markets' expectations of inflation in each country and so a country with a higher nominal rate of interest will have higher expectations of inflation and therefore, using PPT above, its spot rate will be depreciating. So, to forecast future spot rates Runn should assess the trend of inflation rates by using the quantitative analysis or modelling of universities and professional advisors.

Explanation: Do not confuse this with IRPT above where interest rate differences are used to predict the difference between spot and forward rates. Fisher developed the equation Nominal Rate of Interest = Real Rate of Interest + Inflation Rate. The Real Rate of Interest is assumed to be the same for all investors and reflects the minimum return they require for investing their money anywhere.

Therefore, in countries with high inflation rates the value of bonds will be at a significant discount in order to find willing buyers who will have realized that the money they invest will fall in value over the years due to inflation and will need higher yields to compensate for this. Higher interest rates may be the only policy that governments can use to maintain the value of their currency if there is high inflation

<u>External imbalances</u>
The exchange rate is set by the balance of demand and supply of the currency. A deficit on the balance of payments will cause the exchange rate to fall

Runn could inspect balance of payments figures, or use the models of universities and banks, to forecast the deficits or surpluses on the balance of payments in order to forecast the trend of the exchange rate

Explanation: If Geeland has a balance of payments deficit on its current account, due to buying a greater value of imports than it earns from selling exports, there will be a net surplus of G$ on the foreign exchange markets which will have the effect of causing the G$ to weaken over time
Another imbalance might be on the capital account of the balance of payments in which investors are liquidating G$ denominated investments and exchanging G$ for foreign currency to buy assets overseas

Capital account imbalances are partially caused by interest rates being out of line with other countries, but also could be capital flight such a due to worries about future government policies

Expectations of the market
News, or rumors, about changes in inflation rates, interest rates, and balance of payments, will cause speculation on the currency and this will make dealers change the rates at which they deal

Also, there will be self-fulfilling prophesies such as where there is an expectation that exchange rates may depreciate that leads to a selling of the currency which causes the exchange rate to fall as expected. Modelling expectations for forecasting is beyond the skill of a CEP firm like Runn, although some awareness of the comments in the investment media and of the general economic and political outlook may provide some indication of direction of exchange rates

Political factors
The policy aims of the government, its political strength, and investment market opinions of the government will also have an influence on the future of the exchange rate

Forecasting these political factors may require expert judgement, and the consideration of a wide range of variables

Explanation: the Geeland government is saying that its Central Bank intends to intervene in the currency markets. Governments can use direct buying and selling of the currency and interest rate policy to manipulate the exchange rate directly. They can also try to address the underlying factors causing changes in the exchange rate, such as curbing excessive demand for imports by protectionist policies or by deflating the economy, or reducing inflation by wage policies or higher taxation (or in some cases may deliberately depress the exchange rate to stimulate export-led growth

Effects of interest rate changes on Runn
Increased costs of variable interest debt
Runn has G$14.578bn of debt at September 30th 2019 [*data in pre-seen material*] This requires interest of about 3.07% (*finance costs in 2019 were G$433m on an average debt balance of G$14,099m: G$14,578+G$13,620/2).* This means a rise in the interest rate of 1% would increase finance costs by 32.6% (*1/3.07 x 100%),* an increase of finance costs of G$141m

This would reduce profits and cash flows.
Fall in Runn's share price
Runn's share price would fall for two reasons:
Firstly, the rise in its finance costs would increase its cost of capital and so lead its future free cash flows to be discounted more. This assumes the Efficient Market Hypothesis (EMH) applies

Secondly shares prices in general will fall in Geeland as investors liquidate some of their equity holdings to buy bonds with improved yields.

Potential reduction in business demand
Higher interest will reduce domestic aggregate demand in the Geeland economy (*student note: domestic aggregate demand is the total value of demand from Geeland households, firms and government for goods and services).* This is because it will increase the costs of borrowing and this will deter purchasing and investment

This lower overall demand will reduce the number of parcels that need to be delivered, for example by reducing online buying

Improved earnings from treasury cash balances
Runn had G$3.09bn of cash balances at 30th September 2019 [*data in pre-seen material*]. Presumably corporate treasury has this invested in money market accounts and short-dated bonds and bills

Initially the rise in interest rates will reduce the value of the bills and bonds (*student note: the central bank will press the interest rate up by increasing the supply of treasury bills and bonds and by repurchasing them at higher rates of discount. This makes bond prices fall*)

Ultimately the rate of return on cash balances will increase

Rise in Geeland exchange rate
Higher domestic interest rates should increase the exchange rate of the Geeland dollar. This is because the higher rates of yield will attract international investment into Geeland bonds. The international investment flows will increase the demand for the G$ as foreign currency is exchanged for G$ to buy the bonds

However, this impact may be offset by the markets noting that trade imbalances and higher inflation rates may have prompted the higher interest rates